



GE Medical Systems

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Book 1 of 2
Pages 1 - 142

GE Medical Systems **HiSpeed QX/i Installation Manual**

Chapters 1-3
(Mechanical Installation)

Retain this installation manual permanently with your CT System. Any other revisions of this installation manual may not exactly match your system.

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IMPORTANT PRECAUTIONS

LANGUAGE

WARNING

- THIS SERVICE MANUAL IS AVAILABLE IN ENGLISH ONLY.
- IF A CUSTOMER'S SERVICE PROVIDER REQUIRES A LANGUAGE OTHER THAN ENGLISH, IT IS THE CUSTOMER'S RESPONSIBILITY TO PROVIDE TRANSLATION SERVICES.
- DO NOT ATTEMPT TO SERVICE THE EQUIPMENT UNLESS THIS SERVICE MANUAL HAS BEEN CONSULTED AND IS UNDERSTOOD.
- FAILURE TO HEED THIS WARNING MAY RESULT IN INJURY TO THE SERVICE PROVIDER, OPERATOR OR PATIENT FROM ELECTRIC SHOCK, MECHANICAL OR OTHER HAZARDS.

AVERTISSEMENT

- CE MANUEL DE MAINTENANCE N'EST DISPONIBLE QU'EN ANGLAIS.
- SI LE TECHNICIEN DU CLIENT A BESOIN DE CE MANUEL DANS UNE AUTRE LANGUE QUE L'ANGLAIS, C'EST AU CLIENT QU'IL INCOMBE DE LE FAIRE TRADUIRE.
- NE PAS TENTER D'INTERVENTION SUR LES ÉQUIPEMENTS TANT QUE LE MANUEL SERVICE N'A PAS ÉTÉ CONSULTÉ ET COMPRIS.
- LE NON-RESPECT DE CET AVERTISSEMENT PEUT ENTRAÎNER CHEZ LE TECHNICIEN, L'OPÉRATEUR OU LE PATIENT DES BLESSURES DUES À DES DANGERS ÉLECTRIQUES, MÉCANIQUES OU AUTRES.

WARNUNG

- DIESES KUNDENDIENST-HANDBUCH EXISTIERT NUR IN ENGLISCHER SPRACHE.
- FALLS EIN FREMDER KUNDENDIENST EINE ANDERE SPRACHE BENÖTIGT, IST ES AUFGABE DES KUNDEN FÜR EINE ENTSPRECHENDE ÜBERSETZUNG ZU SORGEN.
- VERSUCHEN SIE NICHT, DAS GERÄT ZU REPARIEREN, BEVOR DIESES KUNDENDIENST-HANDBUCH ZU RATE GEZOGEN UND VERSTANDEN WURDE.
- WIRD DIESE WARNUNG NICHT BEACHTET, SO KANN ES ZU VERLETZUNGEN DES KUNDENDIENSTTECHNIKERS, DES BEDIENERS ODER DES PATIENTEN DURCH ELEKTRISCHE SCHLÄGE, MECHANISCHE ODER SONSTIGE GEFAHREN KOMMEN.

AVISO

- ESTE MANUAL DE SERVICIO SÓ LO EXISTE EN INGLÉS.
- SI ALGÚN PROVEEDOR DE SERVICIOS AJENO A GEMS SOLICITA UN IDIOMA QUE NO SEA EL INGLÉS, ES RESPONSABILIDAD DEL CLIENTE OFRECER UN SERVICIO DE TRADUCCIÓN.
- NO SE DEBERÁ DAR SERVICIO TÉCNICO AL EQUIPO, SIN HABER CONSULTADO Y COMPRENDIDO ESTE MANUAL DE SERVICIO.
- LA NO OBSERVANCIA DEL PRESENTE AVISO PUEDE DAR LUGAR A QUE EL PROVEEDOR DE SERVICIOS, EL OPERADOR O EL PACIENTE SUFRAN LESIONES PROVOCADAS POR CAUSAS ELÉCTRICAS, MECÁNICAS O DE OTRA NATURALEZA.

ATENÇÃO

- ESTE MANUAL DE ASSISTÊNCIA TÉCNICA SÓ SE ENCONTRA DISPONÍVEL EM INGLÊS.
- SE QUALQUER OUTRO SERVIÇO DE ASSISTÊNCIA TÉCNICA, QUE NÃO A GEMS, SOLICITAR ESTES MANUAIS NOUTRO IDIOMA, É DA RESPONSABILIDADE DO CLIENTE FORNECER OS SERVIÇOS DE TRADUÇÃO.
- NÃO TENHA TENTAR REPARAR O EQUIPAMENTO SEM TER CONSULTADO E COMPREENDIDO ESTE MANUAL DE ASSISTÊNCIA TÉCNICA.
- O NÃO CUMPRIMENTO DESTA AVISO PODE POR EM PERIGO A SEGURANÇA TÉCNICA, OPERADOR OU PACIENTE DEVIDO A CHOQUES ELÉTRICOS, MECÂNICOS OU OUTROS.

AVVERTENZA

- IL PRESENTE MANUALE DI MANUTENZIONE È DISPONIBILE SOLTANTO IN INGLESE.
- SE UN ADDETTO ALLA MANUTENZIONE ESTERNO ALLA GEMS RICHIEDE IL MANUALE IN UNA LINGUA DIVERSA, IL CLIENTE È TENUTO A PROVVEDERE DIRETTAMENTE ALLA TRADUZIONE.
- SI PROCEDA ALLA MANUTENZIONE DELL'APPARECCHIATURA SOLO DOPO AVER CONSULTATO IL PRESENTE MANUALE ED AVERNE COMPRESO IL CONTENUTO.
- NON TENERE CONTO DELLA PRESENTE AVVERTENZA POTREBBE FAR COMPIERE OPERAZIONI DA CUI DERIVINO LESIONI ALL'ADDETTO ALLA MANUTENZIONE, ALL'UTILIZZATORE ED AL PAZIENTE PER FOLGORAZIONE ELETTRICA, PER URTI MECCANICI OD ALTRI RISCHI.

警告

このサービスマニュアルには英語版しかありません。

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注意:

本维修手册仅存有英文本。

非 G E M S 公司的维修员要求非英文本的维修手册时，
客户需自行负责翻译。

未详细阅读和完全了解本手册之前，不得进行维修。
忽略本注意事项会对维修员，操作员或病人造成触
电，机械伤害或其他伤害。

DAMAGE IN TRANSPORTATION

All packages should be closely examined at time of delivery. If damage is apparent write "Damage In Shipment" on ALL copies of the freight or express bill BEFORE delivery is accepted or "signed for" by a GE representative or hospital receiving agent. Whether noted or concealed, damage MUST be reported to the carrier immediately upon discovery, or in any event, within 14 days after receipt, and the contents and containers held for inspection by the carrier. A transportation company will not pay a claim for damage if an inspection is not requested within this 14 day period.

Call Traffic and Transportation, Milwaukee, WI **(262) 827-3408** immediately after damage is found. At this time be ready to supply name of carrier, delivery date, consignee name, freight or express bill number, item damaged and extent of damage.

Complete instructions regarding claim procedure are found in Section S of the Policy And Procedures Bulletins.

14 July 1993

CERTIFIED ELECTRICAL CONTRACTOR STATEMENT

All electrical Installations that are preliminary to positioning of the equipment at the site prepared for the equipment shall be performed by licensed electrical contractors. In addition, electrical feeds into the Power Distribution Unit shall be performed by licensed electrical contractors. Other connections between pieces of electrical equipment, calibrations and testing shall be performed by qualified GE Medical personnel. The products involved (and the accompanying electrical installations) are highly sophisticated, and special engineering competence is required. In performing all electrical work on these products, GE will use its own specially trained field engineers. All of GE's electrical work on these products will comply with the requirements of the applicable electrical codes.

The purchaser of GE equipment shall only utilize qualified personnel (i.e., GE's field engineers, personnel of third-party service companies with equivalent training, or licensed electricians) to perform electrical servicing on the equipment.

IMPORTANT...X-RAY PROTECTION

X-ray equipment if not properly used may cause injury. Accordingly, the instructions herein contained should be thoroughly read and understood by everyone who will use the equipment before you attempt to place this equipment in operation. The General Electric Company, Medical Systems Group, will be glad to assist and cooperate in placing this equipment in use.

Although this apparatus incorporates a high degree of protection against x-radiation other than the useful beam, no practical design of equipment can provide complete protection. Nor can any practical design compel the operator to take adequate precautions to prevent the possibility of any persons carelessly exposing themselves or others to radiation.

It is important that anyone having anything to do with x-radiation be properly trained and fully acquainted with the recommendations of the National Council on Radiation Protection and Measurements as published in NCRP Reports available from NCRP Publications, 7910 Woodmont Avenue, Room 1016, Bethesda, Maryland 20814, and of the International Commission on Radiation Protection, and take adequate steps to protect against injury.

The equipment is sold with the understanding that the General Electric Company, Medical Systems Group, its agents, and representatives have no responsibility for injury or damage which may result from improper use of the equipment.

Various protective materials and devices are available. It is urged that such materials or devices be used.

LITHIUM BATTERY CAUTIONARY STATEMENTS



CAUTION Risk of Explosion

Danger of explosion if battery is incorrectly replaced. Replace only with the same or equivalent type recommended by the manufacturer. Discard used batteries according to the manufacturer's instructions.



ATTENTION Danger d'Explosion

Il y a danger d'explosion s'il y a remplacement incorrect de la batterie. Remplacer uniquement avec une batterie du même type ou d'un type recommandé par le constructeur. Mettre au rebut les batteries usagées conformément aux instructions du fabricant.

OMISSIONS & ERRORS

Customers, please contact your GE Sales or Service representatives. GE personnel, please use the GEMS CQA Process to report all omissions, errors, and defects in this publication.

Revision History

Revision	Date	Reason for change
0	05/17/02	Initial release of document.
1	07/18/02	Corrected the information about the Global Console.
2	09/26/02	Added a picture showing the SCIM and SCSI tower cable tags in Sec 4.3 of Chap 2 - Power, Ground & Interconnect Cables. Corrected the OBC Board Slot Assignment figure in Sec 2.0 - mA Meter Check With Verification of Chap 10. Added a Note for Scan Protocol in Sec 5.1 of Chap 11 - System Tests.
3	01/30/03	Chapter 1: Sec. 1.4 Pre-Installation Template Sec. 9.0 Seismic Mounting (9.1 Console) Instructions for brackets. Chapter 2: Sec. 2.0 System Interconnect Diagram Added Table "GEMS Supplied Cables (Long Run)". Sec. 4.3 Connecting the SCSI Tower Added a picture of Optional Seismic Mounting Bracket. Chapter 11: Sec. 2.1.1 Scanning with Service Protocols Corrected the steps. Added information for NGPDU.
4	05/07/03	Updates to GOC2 information. Chapter 1: - Updated Hand Tools list, in Section 1.6 - Added Sect 2.1.1 - System Transportation - Temperature Extremes Appendix C: Corrected graphic in Cradle Characterization procedure.
5	11/21/03	Chapter 1: Added a Notice for GRE console documentation.
6	01/29/04	Updates to NGPDU front block diagrams. Chapter 1: - Added Laser Level to Hand Tools list, in Section 1.6 - Added Console Seismic Kit P/N to Section 9.0 Chapter 10: Removed ref to "factory supplied HHS Data Sheets" (sect 2.1)
7	03/23/04	Chapter 2: Added notes for installing DASM II (Sec. 6.8). Chapter 6: Changed the note so that it states the reference to the correct LFC manual (Sec. 3.6). Corrected "Important Precautions".

List of Effected Pages

PAGES	REVISION	PAGES	REVISION
1 through 298	07		

List of Effected Pages

Table of Contents: Book 1

Preface

Publication Conventions 19

Section 1.0

Safety & Hazard Information 19

- 1.1 Text and Character Representation..... 19
- 1.2 Graphical Representation 20

Section 2.0

Publication Conventions 21

- 2.1 General Paragraph and Character Styles..... 21
- 2.2 Page Layout..... 21
- 2.3 Computer Screen Output/Input Character Styles 22
- 2.4 Buttons, Switches and Keyboard Inputs (Hard & Soft Keys) 22

Chapter 1

Position Subsystems 23

Section 1.0

Install Table/Gantry Introduction 23

- 1.1 Skill Set Required 23
- 1.2 Floor and Room Preparation 24
- 1.3 Overview..... 24
- 1.4 Pre-Installation Template..... 25
- 1.5 Installation Alignment Tool Kit 25
- 1.6 Required Common Tools and Supplies 26

Section 2.0

Delivery and Inventory Procedure 27

- 2.1 Delivery Procedure 27
 - 2.1.1 System Transportation - Temperature Extremes..... 27
 - 2.1.2 Working with the Mover 28
 - 2.1.3 Floor Protection 28
 - 2.1.4 Equipment Delivery Route 28
 - 2.1.5 Removing Gantry Dollies and Covers..... 28
 - 2.1.6 Check for Shipping Damage 28
 - 2.1.7 A1 Breaker..... 29
- 2.2 Mechanical Block Diagram 30

Section 3.0

Install and Level Table/Gantry 31

- 3.1 Establish the Room Layout..... 31
- 3.2 Position the Gantry 32
- 3.3 Level the Gantry 34
- 3.4 Position the Table 36
- 3.5 Install Alignment Bars 37
- 3.6 Level the Table 39
 - 3.6.1 Before You Begin..... 39
 - 3.6.2 Procedure 39

3.7	Table-Gantry Mechanical Alignment	41
3.7.1	Alignment Conditions	41
3.7.2	Alignment Specification	41
3.7.3	Procedure	41
3.7.3.1	Cradle Center Procedure	42
3.7.3.2	Measuring Cradle Center Procedure	45
3.7.3.3	Cradle/Table Parallel Movement.....	45
3.7.3.4	Determining Table to Gantry Perpendicularity	46
3.8	Tighten the Lock Rings	46
3.9	Drill the Anchor Holes	47
3.10	Gantry & Table Redundant Anchor Holes.....	50
3.11	Install the Anchors.....	50
3.12	Inside Levelers	52
Section 4.0		
Rear Entry Cable Box (Site Specific)		54
Section 5.0		
Install Table Footswitch Assembly		55
Section 6.0		
Remove Gantry Tilt Bracket.....		56
Section 7.0		
Position the Power Distribution Unit		57
7.1	Compact PDU	57
7.2	NGPDU	59
Section 8.0		
Install Operator Console		62
8.1	Unpack Console	62
8.2	Remove Console Covers	64
8.3	Adjust Table Top Height and Position Console.....	65
8.4	Attach Keyboard Table Top	66
8.5	Monitor and SCSI Tower Placement.....	66
Section 9.0		
Seismic Mounting		67
9.1	Console	67
9.2	Power Distribution Unit.....	68
Section 10.0		
Install Options.....		69
10.1	Camera (Filming Device)	69
10.2	Injector	69
10.3	Advantage Windows Workstation (AWW).....	70
10.4	Uninterruptible Power System (UPS).....	70
10.5	Other Options.....	70

Chapter 2

Power, Ground & Interconnect Cables 71

Section 1.0

Scanner Configuration Variations 71

1.1 PDU Varieties 71

1.2 Console Varieties..... 72

Section 2.0

Introduction 72

2.1 System Component Identification..... 73

2.2 Cable Color Identifiers 73

Section 3.0

System Interconnect Diagrams..... 75

3.1 System Configured with NGPDU & Global Console - Octane2 75

3.2 System Configured with NGPDU & Global Console - Linux 76

Section 4.0

Cable (Run) Specification..... 77

Section 5.0

Contractor Connections 79

Section 6.0

Console Connections 81

6.1 Accessing the Computer..... 81

6.2 SCIM, Keyboard, Trackball & Mouse Installation 81

6.3 Connecting the SCSI Tower 83

6.4 Connecting the Monitor..... 84

6.5 Power Panel Connections 84

6.6 Linux PC Connections 85

6.7 LAN Connections..... 87

6.8 Installing a DASM (Hardware Option)..... 87

6.9 Bulkhead Connections..... 88

Section 7.0

Gantry Cable Connections 89

Section 8.0

Option Cables 90

Section 9.0

Table Connections 91

Section 10.0

PDU Cable Connections & Configuration 92

10.1 Compact PDU..... 93

10.1.1 A3 Panel - 380 - 480VAC Mains "A1" Input Power Connection..... 94

10.1.2 A3 Panel - Circuit Breakers 95

10.1.3 A5 Panel - Transformer (480VAC) Taps..... 95

10.1.4 A2 Panel - HVDC Connection..... 96

10.1.5 A3 Panel - 440V Connection 96

10.1.6 A4 Panel - Gantry & Console Power Connections 97

10.1.6.1	Console Power Cable Re-termination.....	97
10.1.6.2	Gantry Power Cable Re-termination	98
10.1.7	A4 Panel – PDU Control Cable	98
10.1.8	A4 Panel – System Ground Connection	98
10.1.9	A4 Panel – X-Ray Light & Door Interlock Connections	98
10.1.9.1	X-Ray Light Configuration & Connection	98
10.1.9.2	Door Interlock Connections.....	99
10.2	NGPDU	100
10.2.1	Panel - 380 - 480VAC Mains “TS1” Input Power Connection	101
10.2.2	Panel - Circuit Breakers	102
10.2.3	Transformer (480VAC) Taps	102
10.2.4	HVDC Connection.....	103
10.2.5	440V Connection.....	103
10.2.6	Gantry & Console Power Connections.....	104
10.2.6.1	Console Power Cable Re-termination.....	104
10.2.6.2	Gantry Power Cable Re-termination	105
10.2.7	PDU Control Cable.....	105
10.2.8	System Ground Connection	105
10.2.9	X-Ray Light & Door Interlock Connections.....	105
10.2.9.1	X-Ray Light Configuration & Connection	105
10.2.9.2	Door Interlock Connections.....	106

Section 11.0		
System Ground Connections		108

Chapter 3

System Continuity & Ground Checks	111
--	------------

Section 1.0		
System Continuity & Ground Checks (Mechanical Contractor).....		111
1.1	Tools Required	111
1.2	Procedure.....	111
1.2.1	Compact PDU	111
1.2.2	NGPDU	114

Section 2.0	
Site Ground Continuity Check.....	117

Section 3.0		
Mechanical Installation Completion Section.....		118
3.1	Clean Up	118
3.2	Remove Trash.....	118
3.3	Return Dollies.....	118
3.4	Install Covers	118
3.5	Clean the Equipment.....	118
3.6	Touch Up Nicks.....	118
3.7	Complete Paperwork.....	118

Appendix A

Removal & Installation of Covers	119
---	------------

Section 1.0	
Gantry Side Covers	119
1.1 Side Cover Removal	119
1.2 Side Cover Installation	120
Section 2.0	
Gantry Top Covers	120
2.1 Top Cover Removal	120
2.2 Top Cover Installation	120
Section 3.0	
Gantry Front Cover	121
3.1 Original Front Cover Dolly Setup	121
3.2 Redesigned Front Cover Dolly Setup	123
3.3 Removal	124
3.4 Installation	129
Section 4.0	
Gantry Rear Cover	131
4.1 Removal	131
4.2 Installation	131
Section 5.0	
Gantry Scan Window	133
5.1 Remove Scan Window	133
5.2 Install Scan Window	133
Section 6.0	
Gantry Base Covers	134
Section 7.0	
Foot-switch Covers	135
 Appendix B	
Pictorial Representation of Required Tools	137

Preface

Publication Conventions

Purpose: This section means to inform the reader on publication conventions used. So that the reader can identify safety and general material that is considered important by its format. This includes the interpretation of computer screen text as either input or output. There are a number of specific text and paragraph styles/conventions used within this section to accomplish this task. Please become familiar with the conventions used within this publication before proceeding.

Section 1.0 Safety & Hazard Information

1.1 Text and Character Representation

Within this publication, different paragraph and character styles have been used to indicate potential hazards. Paragraph prefixes, such as hazard, caution, danger and warning, are used to identify important safety information. Text (Hazard) styles are applied to the paragraph contents that are applicable to each specific safety statement. Words describe the type of potential hazard that may be encountered and are placed immediately before the paragraph it modifies. Safety information will normally include:

- Type of potential hazard
- Nature of potential injury
- Causative condition
- How to avoid or correct the causative condition

EXAMPLES OF HAZARD STATEMENTS USED

A few examples are provided that have been adapted from GEMS' global document standard (2119696-100). They include paragraph prefixes and modified text styles.



CAUTION
Pinch Points
Loss of Data
Sharp Objects

Caution is used when a hazard exists that can or could cause minor injury to self or others if instructions are ignored. They include for example:

- Loss of critical patient data
- Crush or pinch points
- Sharp objects



DANGER
EXCESSIVE
VOLTAGE
CRUSH
POINT

DANGER IS USED WHEN A HAZARD EXISTS THAT WILL CAUSE SEVERE PERSONAL INJURY OR DEATH IF INSTRUCTIONS ARE IGNORED. THEY CAN INCLUDE:

- ELECTROCUTION
- CRUSHING
- RADIATION

WARNING
ROTATING
EQUIPMENT
BARE WIRES

WARNING IS USED WHEN A HAZARD EXISTS WHICH COULD OR CAN CAUSE SERIOUS PERSONAL INJURY OR DEATH IF INSTRUCTIONS ARE IGNORED. THEY CAN INCLUDE:

- Potential for shock
- Exposed wires
- Failure to Tag and lockout system power could allow for un-command motion.

NOTICE
Equipment
Damage
Possible

Notice is used when a hazard is present that can cause property damage but has absolutely no personal injury risk. They can include:

- Disk drive will crash
- Internal mechanical damage, such as to the x-ray tube
- Coasting the rotor through resonance.

It's important that the reader not ignore hazard statements in this document.

1.2 Graphical Representation

Important information will always be preceded by the exclamation point  contained within a triangle, as seen throughout this chapter. In addition to text, several different graphical icons (symbols) may be used to make you aware of specific types of hazards that could possibly cause harm.

ELECTRICAL



LASER



MECHANICAL



HEAT



RADIATION



PINCH



Some others make you aware of specific procedures that should be followed.

AVOID STATIC ELECTRICITY



TAG AND LOCK OUT



WEAR EYE PROTECTION



Section 2.0

Publication Conventions

2.1 General Paragraph and Character Styles

Prefixes are used to highlight important non-safety related information. Paragraph prefixes (e.g., Purpose, Example, Comment and Note) are used to identify important but non-safety related information. Text styles are also applied to text within each paragraph modified by the specific prefix.

EXAMPLES OF PREFIXES USED FOR GENERAL INFORMATION

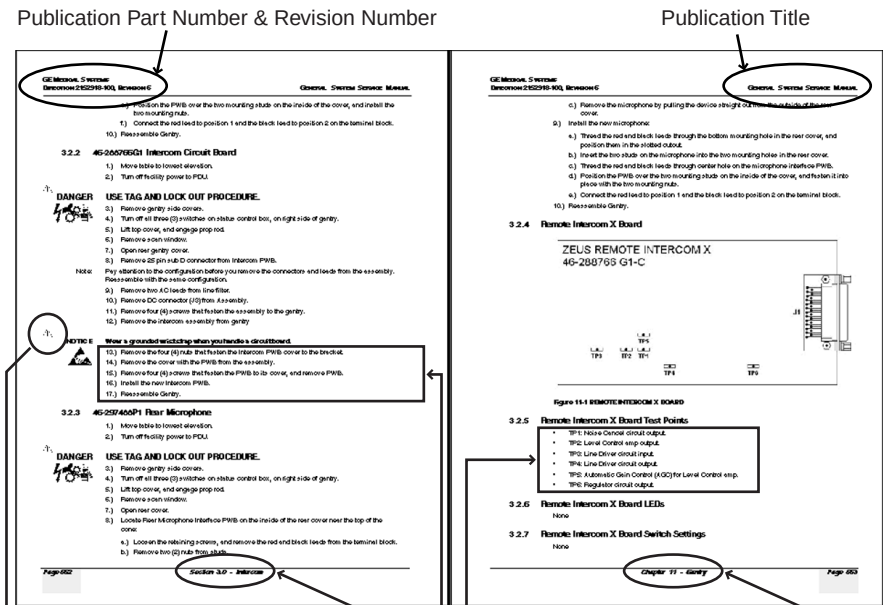
Purpose: Introduces and provides meaning as to the information contained within the chapter, section or subsection, such as used at the beginning this chapter for example.

Note: Conveys information that should be considered important to the reader.

Example: Used to make the reader aware that the paragraph(s) that follow are examples of information possibly stated previously.

Comment: Represents “additional” information that may or may not be relevant.

2.2 Page Layout



The current section and its title are always shown in the footer of the left (even) page.

An exclamation point in a triangle is used to indicate important information to the user.

Paragraphs preceded by Alphanumeric (e.g. numbers) characters is information that must be followed in a specific order.

The current chapter and its title are always shown in the footer of the right (odd) page.

Paragraphs preceded by symbols is (e.g. bullets) is information that has no specific order.

Headers and footers in this publication are designed to allow you to quickly identify your location. The document's part number and revision number appears in every header on every page. Odd numbered page footers indicate the current chapter, its title, and current page number. Even page footers show the current section and its title, as well current page number.

2.3 Computer Screen Output/Input Character Styles

Within this publication different character styles are used to indicate computer input and output text. Character (input, output, and variable) styles are used and applied to the text within a paragraph so as to indicate directions. Computer screen output and input is also formatted using mono (fixed width) spaced fonts.

Example: This paragraph denotes computer screen fixed output. It's output is fixed
Fixed Output from the sense that it does not vary from application to application. It's the most commonly used style used to indicate filenames, paths, and text.

Example: *This paragraph denotes computer screen output that is variable. Its output
Variable Output varies from application to application. Variable output is sometimes found placed between greater than and lesser than operators. For example:*
<variable_ouput>

Example: This paragraph denotes fixed input. It's typed input that will not vary
Fixed Input from application to application. Fixed text the user is required to supply as input.

Example: *This paragraph denotes computer input that can vary from application to
Variable Input application. Variable text the user is required to supply as input. Variable input sometimes is placed between greater than and lesser than operators. For example: <variable_input>. In these cases, the (<>) operators are dropped prior to input. Exceptions are noted in the text.*

2.4 Buttons, Switches and Keyboard Inputs (Hard & Soft Keys)

Different character styles are used to indicate actions requiring the reader to press either a hard or soft button, switch, or key. Physical hardware, such as buttons and switches, are called hard keys because they are hard wired or mechanical in nature. A keyboard or on/off switch would be a hard key. Software or computer generated buttons are called soft keys because they are software generated. Software driven menu buttons are an example of such keys. Soft and hard keys are represented differently in this publication.

Example: A power switch **ON/OFF** or a keyboard key like **ENTER** is indicated by applying a character style
Hard Keys that uses both over and under-lined bold text that is bold. This is a hard key.

Example: Whereas the computer MENU button that you would click with your mouse or touch with your hand
Soft Keys uses over and under-lined regular text. This is a soft key.

Chapter 1

Position Subsystems



NOTICE
Potential for
Data Loss and/
or Equipment
Damage

To prevent potential data loss and equipment damage, please do the following:

- Record data collected from procedures in this chapter into Form F4879 when directed, located in [Chapter 13](#) of this book.
- Only use the Installation manual that arrives with your system. Any other revisions of this manual may not exactly match your system.



NOTICE
GRE Console

For systems with GRE console, use the following documentation:

- GRE Xstream Service Information CD-ROM (2384451-200)
- GRE Software Installation Procedures (2378261-100), contained in the CD-ROM above.
- GRE Upgrade Illustrated Parts List (2394950-100)

Section 1.0

Install Table/Gantry Introduction

Note: This chapter describes how to mount, position, and level the HiSpeed QX/i subsystems. Before you start the installation, make sure the site preparation complies with conditions and instructions found in the pre-installation manual. Failure to comply, will result in excessive installation delay and potential increased, unrecoverable installation costs. This product is designed to meet mechanical installation standards, and these standards should be reviewed prior to installing this system.

1.1 Skill Set Required

Only people experienced in the installation of a GE CT Scanner should perform the work described in this document. A person doing the installation should have:

- Completed three LightSpeed Ultra CT installations as a helper, prior to leading an installation.
- Must have reviewed the LightSpeed Plus Installation Video, Safety Video and Installation Training CD, and successfully passed all related tests. (Refer to [Table 1-1](#).)

Media	Part Number
LightSpeed Installation CD	2289471-100
LightSpeed Installation Video	2289468-100
LightSpeed Installation Training CD	2289779-100
GE Safety CD	2290198-100
GE Safety Video	2290199-100

Table 1-1 Installation and Training Media

- A working understanding of the mechanical installation process, prior to beginning.
- Completed environmental health and safety (EHS) training, understand and apply lockout/tagout processes, and have taken Blood Borne Pathogens training.

- Basic measurement skills, system layout skills, and the ability to read blue prints.

1.2 Floor and Room Preparation

PREPARATION

Consult your local GE Sales and service representative about your specific needs. It is the purchaser's (buyer) responsibility to provide an approved support structure and an approved method of mounting. General Electric is not responsible for any failure of the support structure or method of anchoring.

The HiSpeed QX/i system has a total floor load of approximately 6660 lbs (3021 kg). About 5050 lbs (2291 kg), including patient (450 lbs (204 kg)), is concentrated in the table-gantry assembly. Refer to the *HiSpeed QX/i Pre-installation* manual for more information.

FLOORING

Do not place the scanner on any resilient flooring. Resilient tile or carpeting may slowly yield over a period of time and disturb the alignment of the table to the gantry. Refer to the floor template to determine locations where resilient flooring material should be removed.

Limitations include:

- No part of the floor surface within the table, gantry, or the two interface areas between table and gantry, should be higher than the support areas for the table and gantry.
- The floor structure must withstand the occupied weight of table and gantry, as well as the individual contact area loading of these components.
- The method and placement of anchors, or through bolts, must not reduce the structural strength of the floor.

If you have to remove the gantry covers in order to move the gantry into the room, refer to [Appendix A](#), for the cover removal procedure. Please read the caution statement on [page 119](#) before removing the gantry covers.

1.3 Overview

Procedures in this chapter provide detailed instructions to position, level, and anchor the HiSpeed QX/i gantry and table securely for operation. The HiSpeed QX/i system uses adjustable leveling pads to support the gantry and table. The gantry has four (4) primary leveling pads and six (6) auxiliary "leveler" pads located just inside the gantry base, and two each on the front and rear support frame. The auxiliary pads are sometimes referred to as "Jig-feet" and/or "inside levelers". The table has five (5) pads used for leveling it.

The process you will be following is:

- 1.) Use the room-layout template to determine the general position of the gantry and table.
- 2.) Move the gantry into position.
- 3.) Level gantry.
- 4.) Use the three alignment bars to position the table relative to the gantry.
- 5.) Level the table to the gantry, and anchor the system.

Use the template to position the system; however, use the gantry and table to locate and drill the anchor holes. Drill the anchor holes with the system in place. Refer to [Section 3.0, on page 31](#) and [Figure 1-16, on page 38](#), for an example of this procedure. This HiSpeed QX/i system installation procedure requires the items listed in [Section 1.5, on page 25](#) and [Section 1.6, on page 26](#).

1.4 Pre-Installation Template

Always use the room-layout template, P/N 2265473, (in two pieces), during installation. The gantry and table will not be properly aligned if existing holes are used. The template shows the location of the gantry and table anchor holes.

This template is shipped with the system. It is enclosed in a tube that is shipped in the box with the cover dollies.

1.5 Installation Alignment Tool Kit

The installation personnel provide the installation tool kit, P/N 2110953. The tool kit contains the following special tools (See [Figure 1-1](#)). **This installation alignment toolkit requires yearly calibration and must have a current calibration sticker.**

- **Guide Pin (Bolt) Tool** included in the tool kit (not needed).
- **Long Alignment Bar** (in two pieces), P/N 2106369: use to position the table, relative to gantry
- **Two Short Alignment Bars**, P/N 2106368: use to position the table, relative to the gantry
- **Adjuster Tool**, P/N 2107863: use to turn the adjusters, which in turn level the system.
- **½" Drill Bushing**, P/N 2106205: use to center and guide the bit when drilling the anchor holes
- **Vacuum Attachment (funnel)**, P/N 2110944: use to vacuum debris out of the anchor holes
- **Spanner Wrench**, P/N 2110003: if necessary, use to turn the adjusters after anchoring system
- **Anchor Seating Tool**, P/N 2247965: used to properly seat anchors

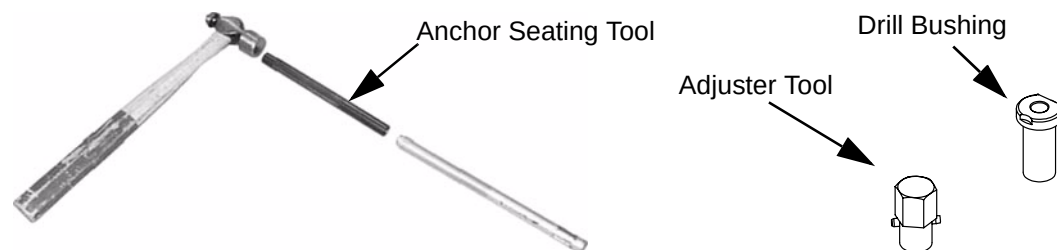


Figure 1-1 Installation Alignment Toolkit Accessories



Figure 1-2 Gantry/Table Alignment Toolkit

1.6 Required Common Tools and Supplies

The following tools and supplies are required for installation of the scanner. Please refer to [Appendix B, on page 137](#), for pictorial descriptions of the tools and supplies.

WRENCHES

- Standard and Metric combination wrench sets
- Standard and Metric Hex Key (Allen wrench) sets
- ½" and 3/8" drive torque wrench: 0-100 N-m (0-100 ft.-lb.) Must be calibrated yearly.

SOCKETS AND EXTENSIONS

- 3/8" and ½" drive ratchet wrenches
- ½" drive 3" & two 6" long extensions
- 3/8" drive 12" long extension
- Standard & Metric 3/8" drive socket sets
- ¾" deep well socket 3/8"
- 1", 1-1/8", 1-¼" & 1-½" sockets for ½" drive
- 3/8" drive universal joint
- Metric hex bit set ¼" or 3/8" drive, including:
 - 14mm hex bit 3/8" or ½" drive (14mm ball hex helpful)
 - 10mm hex bit 3/8" drive

SCREW DRIVERS

- Phillips screwdriver set (small, medium, and large)
- Straight blade screwdriver set (small, medium, and large)

DRILL BITS

- Complete set of standard (U.S.) drill bits
- Metric tap set
- ½" masonry bit, min. 8" long USA – 18" optional (for rear table hole)
- 13mm masonry bit, min. 203mm long – 457mm optional (for rear table hole)
- 3" (76mm) hole saw with 1/4" (6mm) masonry bit (to remove flooring)

POWER TOOLS

- 3/8" or ½" drill, cordless or electric
- Reciprocating Saw (Sawzall or equivalent) and assorted blades.
- Hammer Drill
- Sears 17740 Shop vacuum or equivalent, with "HEPA" or dry wall dust filter (Sears part number 17918) or equivalent
- 25' Extension power cords

HAND TOOLS

- Ball-Peen Hammer (1lb or 2lb)
- Tongue & Groove Pliers (large)
- Diagonal Cutting Pliers, Large (to cut 1/0 ground)
- Framing Square (e.g., Empire 16" x 24" aluminum square)
- Diagonal Cutting Pliers, Small
- Large pry bar
- 4', 2' & 9" torpedo levels (see [Table 1-2: Recommended Levels](#))
- Laser level (see [Table 1-2: Recommended Levels](#))

9"	Johnson Magnetic Level, model 7500M
	Johnson Magnetic Level, model 4500
	Stanley Magnetic Level
2'	Johnson Professional Box Beam Level, model 9624
	Empire Titan Professional Box Beam Level, model 900 series
4' (nominal)	48" Johnson Professional Box Beam Level
	42" Stanley Contractor Grade Level
	48" Empire Titan Professional Box Beam Level, model 900 series
	48" Stabila Aluminum Box Beam Level, Kit 24816
Laser**	Sears Laser Level Tool (\$40)

* Preferred levels
** Laser level is required for table cradle alignment. Using this tool reduces the alignment procedure by 1 man-hour. New tables have a cradle center line to be used with the laser level.

Table 1-2 Recommended Levels

ELECTRICAL TOOLS

- DVM
- Continuity tester

PERSONAL SAFETY EQUIPMENT

- Safety shoes*
- Safety glasses*
- Gloves
- LOTO Kit (supplied)*
- Hearing Protection*
- 6' & 8 ' Step ladder

* These PPE items are absolutely required for every installation job, with NO exceptions.

Note: A box labeled Installation Support Kit is shipped with each system. It contains paint, masking tape, cleaners, towels, and other materials needed to install this CT scanner.

Section 2.0
Delivery and Inventory Procedure

2.1 Delivery Procedure

2.1.1 System Transportation - Temperature Extremes



NOTICE Component Freezing occurs if CT system is exposed to temperatures below 0° F (-18° C) for a period longer than two days.
Allow a minimum of 12 hours for the CT system to adjust to ambient room temperature, prior to installation.

When transporting the CT system, ensure that the system is not exposed to temperatures or humidity outside the following specifications.

Temperature: 0° to +120° F (-18° to +49° C)

Humidity: 20% to 80%

2.1.2 Working with the Mover

Follow the instructions provided by your installation specialist for the equipment movers. Help direct movers on where to place equipment and which items you need first.

The mover's inventory list reflects only those boxes actually delivered, *NOT* what was ordered.

2.1.3 Floor Protection

It is suggested that you use floor protection. Most equipment movers can provide floor protection during the equipment delivery. Installers should provide floor protection for the room.

2.1.4 Equipment Delivery Route

Prior to equipment delivery, review the delivery route with the movers. Refer to the installation specialist for any additional delivery instructions.

2.1.5 Removing Gantry Dollies and Covers

Refer to [Appendix A](#), for the dolly and cover removal procedures. Please read the caution statement on [page 119](#) before removing the gantry covers.

2.1.6 Check for Shipping Damage

Check for the following damage, which may have occurred during shipping:

- Equipment damage: paint scrapes and cover damage.
- Damage to hospital property: floors, door frames and walls.

If damage is found—or items are missing in shipment—notify the appropriate service personnel:

- For item(s) missing in shipment or short shipped, contact the install specialist to have the item(s) shipped.
- For damaged items, report to Traffic and Transportation, at (262) 827-3408. For further details, see ["DAMAGE IN TRANSPORTATION," on page 7](#).

2.1.7 A1 Breaker

Lock-out and tag-out the A1 breaker now.



Figure 1-3 Sample A1 Breaker

2.2 Mechanical Block Diagram

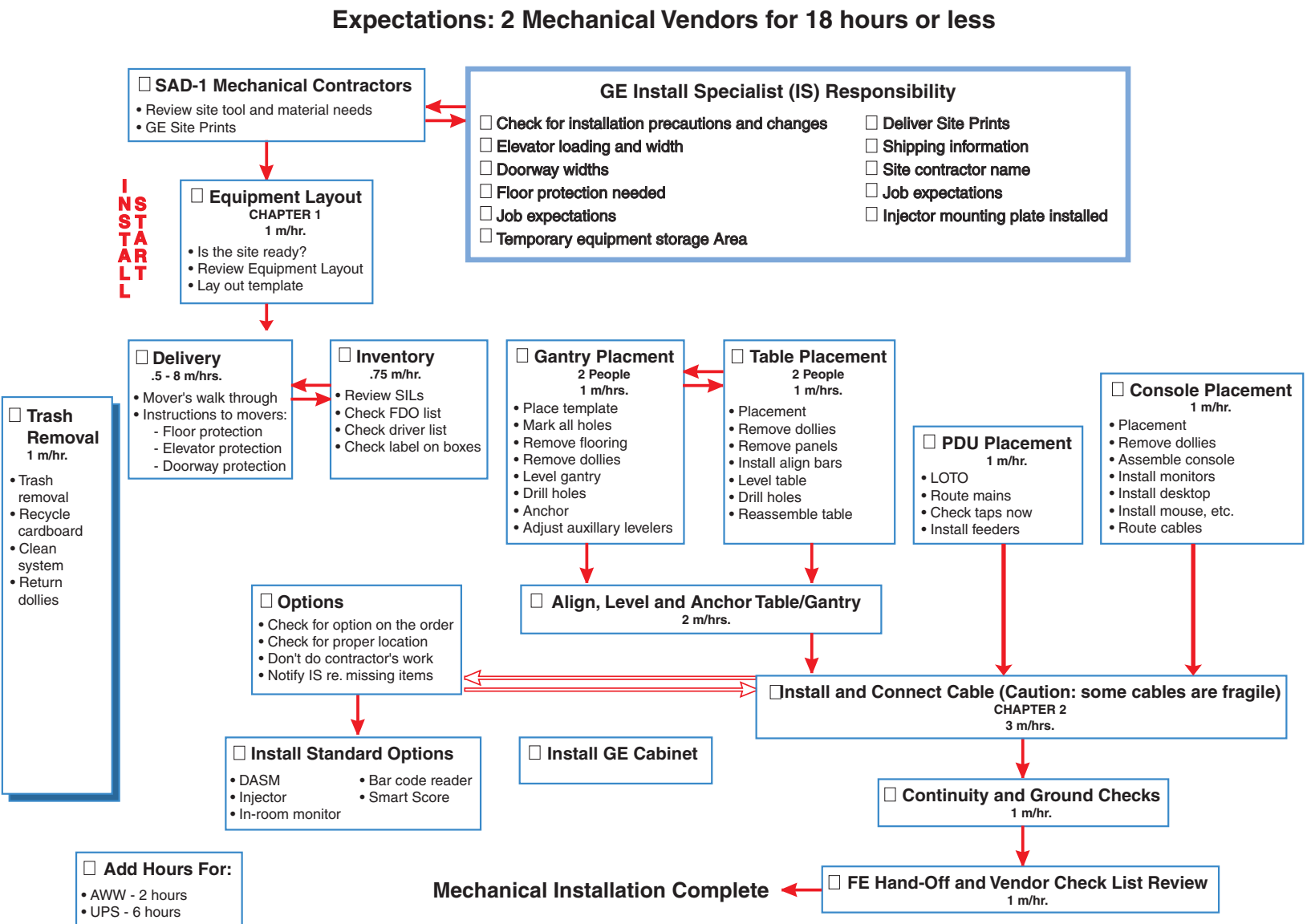


Figure 1-4 Mechanical Installation Block Diagram

Section 3.0

Install and Level Table/Gantry

3.1 Establish the Room Layout

Using the GE print (developed for your site) to establish the room layout, make sure all the operating and service clearances shown on the print are observed. Using the supplied template, locate the anchor holes. Make sure they clear structural interferences in the floor.

Clean the area. Free the mounting surface of any material that may interfere with the positioning and leveling of the system.

- 1.) Lay out the 2 floor templates.
- 2.) Start with the Gantry template—align per the GE print.
- 3.) Place the table template over the top of the Gantry template. Align the scan and table center-lines and secure the templates to the floor. Make sure there are no potential clearance issues. HiSpeed QX/i service clearance areas are different than LightSpeed QX/i.
- 4.) Now, check the level of the floor (See [Figure 1-5](#)) across the templates.
- 5.) Scribe a mark (e.g., use a center punch) at each of the gantry's mounting hole locations (there are four (4) of these). Also scribe marks for each of the gantry's six (6) leveler locations.
- 6.) Using a center punch, mark the three (3) table mounting hole and two (2) leveler locations.

Note:
Clearances
differ by model.



NOTICE
Positioning
requires
cutting 15
holes in the
floor.

Before you drill or cut any flooring, make sure the appropriate hospital personnel have approved the location of the table/gantry.

- 7.) Cut tiles (or other resilient flooring) around all 15 holes punched in the template for the gantry and table. Use a utility knife with a heat gun, a 3" hole saw with a 1/4" masonry bit or other adequate tools to cut the flooring.
- 8.) Some sites may require sealing of the 15 floor penetrations after the flooring is removed.

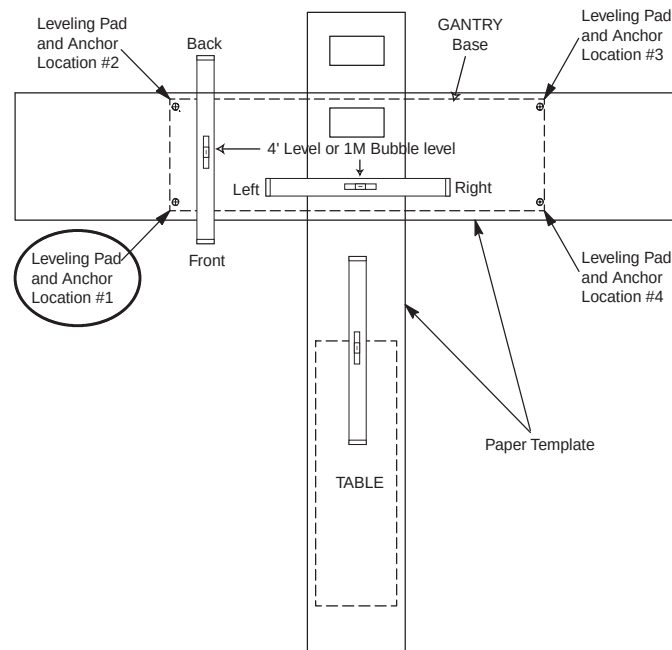


Figure 1-5 Hole Locations

3.2 Position the Gantry

- 1.) Remove all the transportation packaging, except for dollies, from the gantry.

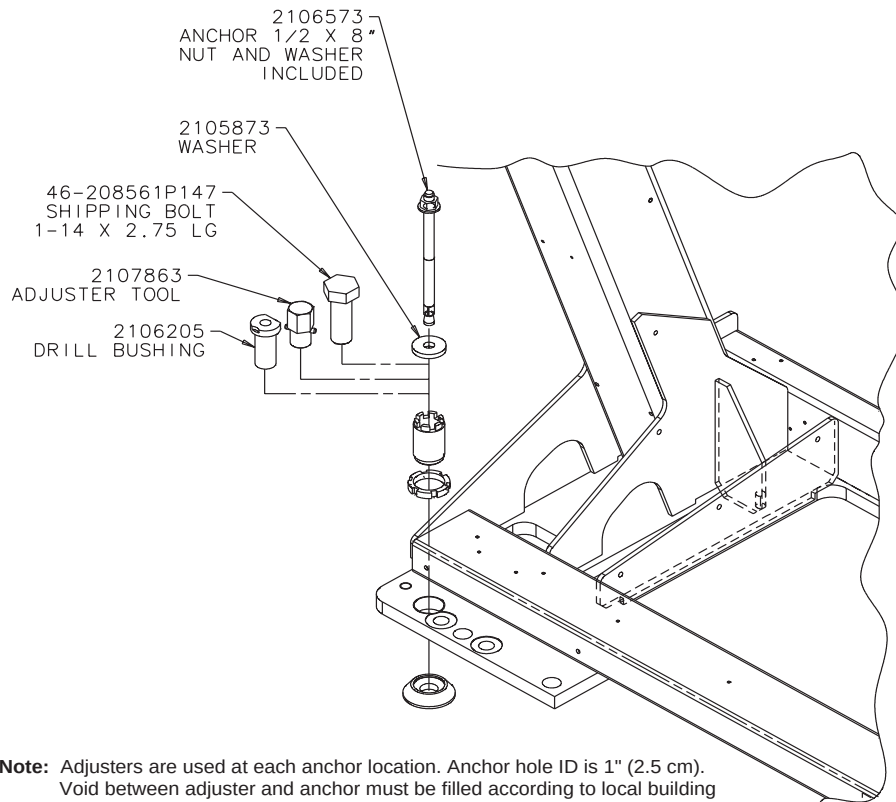


Figure 1-6 Table Base Installation Hardware

- 2.) Position the gantry over the template appropriately.

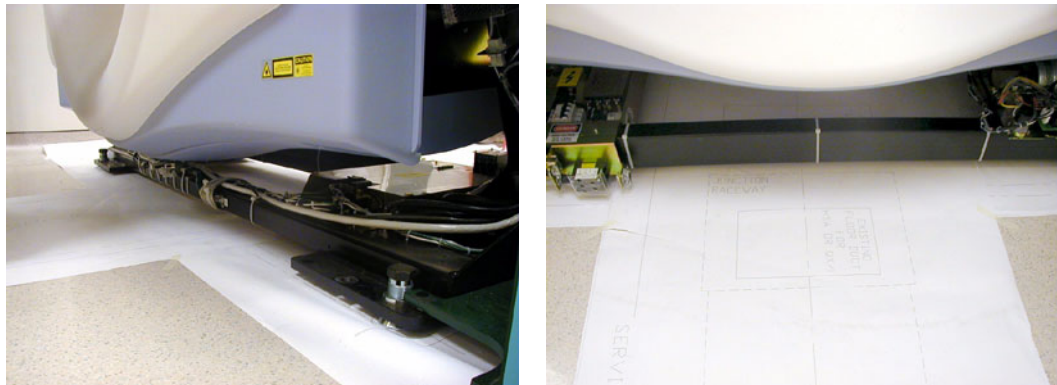


Figure 1-7 Gantry and Table Paper Templates

- a.) Locate the four (4) leveling pads, and position each of them beneath its associated adjuster.
- b.) Use the dollies to evenly lower the gantry, until it's just off of the floor (approximately 1/2" or 12.7 mm). Use a 1/2" ratchet to raise and lower the dollies.
- c.) Carefully rotate the gantry into the correct position over the template.

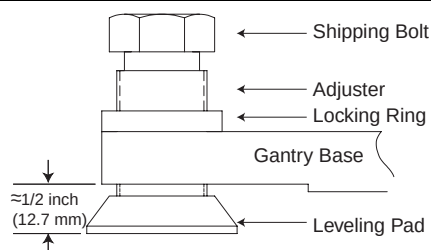


Figure 1-8 Gantry and Table Base Leveling Pads (Starting Positions)

- 3.) Remove the paper templates from the floor and discard of properly.
- 4.) Loosen the locking rings and shipping bolts so you can fine tune the leveling pads to compensate for slight variations in the floor surface.
- 5.) Position the gantry so that the adjusters are centered over their respective holes scribed earlier into the floor.
- 6.) Using a 1/2" ratchet, gently lower the gantry until it rests on the floor, over the marked areas.
- 7.) Using a 14mm hex socket, remove the dollies from the gantry by removing the three dolly bolts found at both ends of the gantry ([Figure 1-9](#)).



Figure 1-9 Gantry Dolly Bolts

- 8.) Remove the (4) gantry shipping bolts, using a 1 1/2" socket.

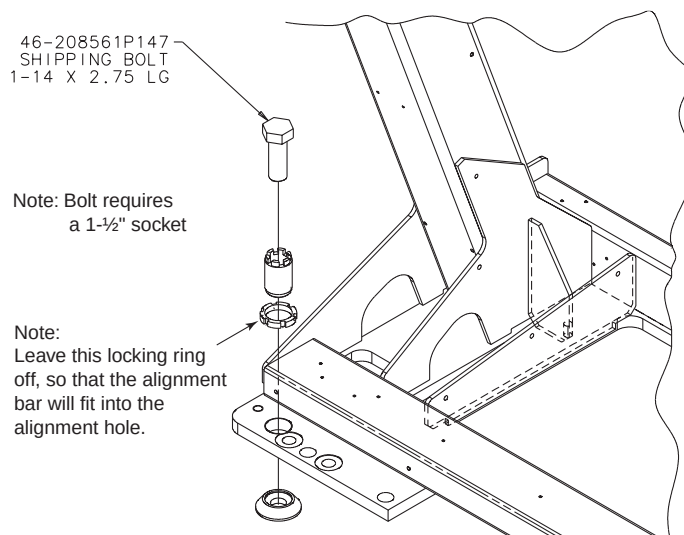


Figure 1-10 Gantry Shipping Bolts

3.3 Level the Gantry

The gantry uses 2 bubble levels that are permanently mounted to machined surfaces on the stationary base to tell when it's level.

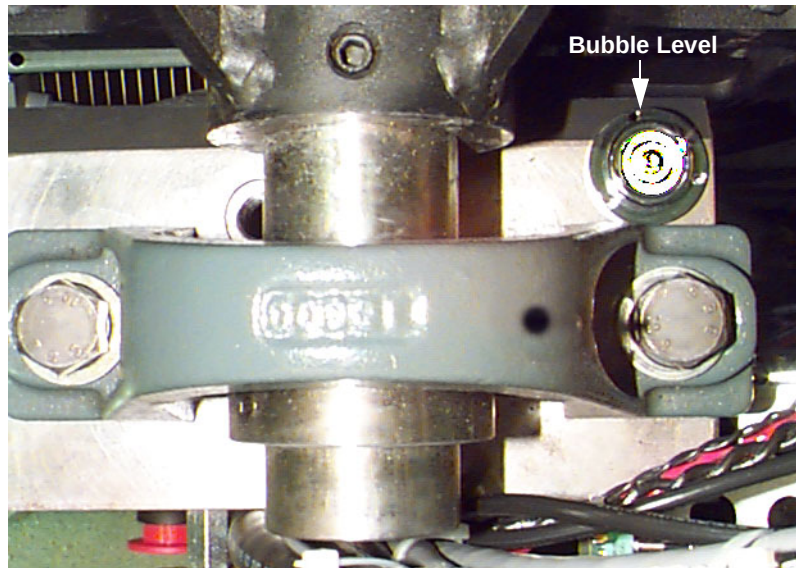


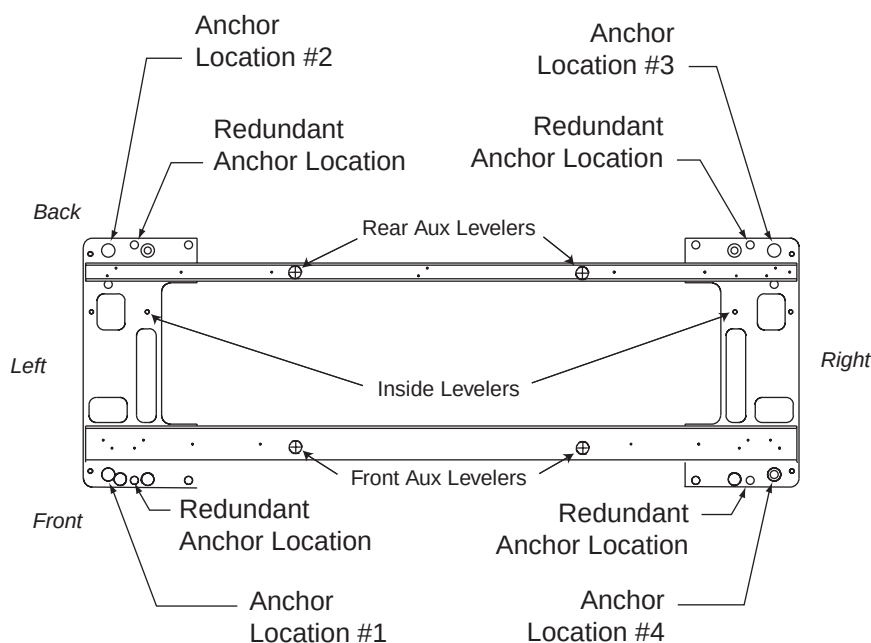
Figure 1-11 Gantry Bubble Level

Bubble levels are located on both ends of the gantry stationary base. They're located on the stationary base near a point where the rotating structure pivots mount to the base structure. (See [Figure 1-11](#).) The gantry is properly leveled when the bubble is centered. (See [Figure 1-13](#), on page 35.)

- 1.) **Remove adjuster #1 lock ring** and only loosen all the other (#2, #3 and #4) adjuster lock rings. This can be done by using a spanner wrench 2110003 (the spanner wrench can be found in the alignment toolkit). Check to make sure auxiliary levelers do not cause interference.
- 2.) Systematically turn each of the gantry's adjusters (locations 1, 2, 3 and 4 in [Figure 1-12](#)) until both bubble levels are centered left to right, and front to back.
 - Begin by turning each adjuster no more than 1 turn at a time.
 - Use the adjuster tool, 1¹/₈" socket, and the ½" drive ratchet to turn each adjuster. (Refer to [Figure 1-6](#), on page 32.)

Systematic Procedure for Leveling gantry follows:

- 1.) Level the left side from front to back by turning adjusters #1 and #2.
- 2.) Level the right side from front to back by turning adjusters #3 and #4.
- 3.) Level the side (right or left) that is higher with respect to the other side. Turn both adjusters on a side equally until that side is level. The side should now also be level.



Note: Adjusters are used at each anchor location. Anchor hole ID is 1" (2.5 cm).
Void between adjuster and anchor must be filled according to local building codes for seismic application.

Figure 1-12 Gantry Base "Adjuster" Locations —Top View

Note:
"Levelers" are
addressed later

The use of the **levelers**—shown in [Figure 1-12](#) (above)—will be addressed later (see [3.12](#), on [page 52](#)).

- 3.) When the bubble levels are centered ([Figure 1-13](#)), each of the four (4) leveling pads should be carrying a portion of the gantry weight. Distribution of the gantry weight prevents the base frame from rocking during normal operation. **DO NOT leave any adjuster un-loaded or floating.**

Correct level is 100% of bubble within small circle
Incorrect Level is less than 100% of bubble within small circle

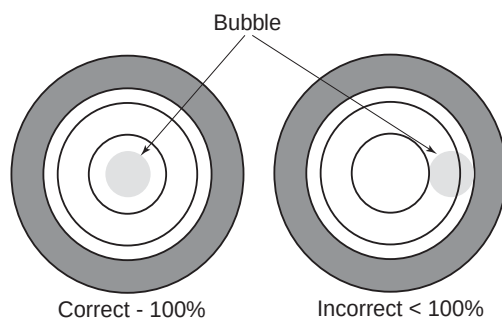


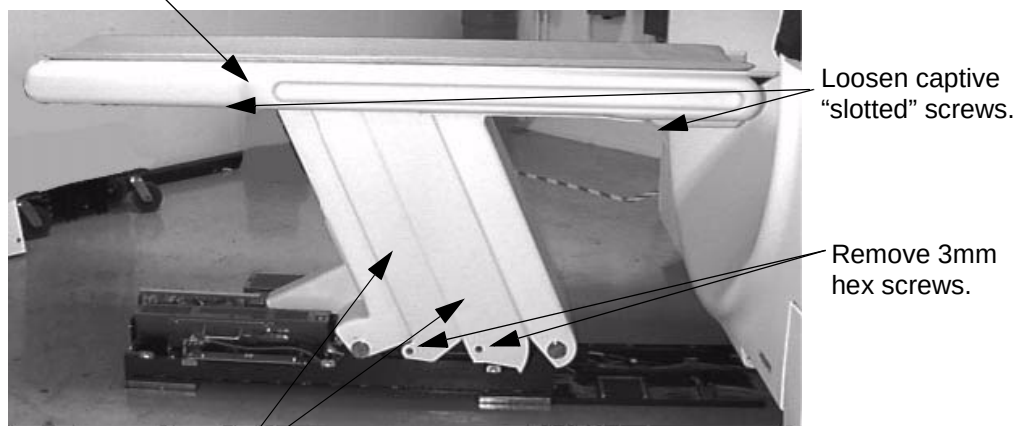
Figure 1-13 Bubble Level Centering

3.4 Position the Table

- 1.) Remove all the transportation packaging and boxes, except dollies, from the table.
- 2.) Wheel the table to its approximate position relative to the gantry, using the marks made earlier.
 - a.) Locate the table leveling pads and position them against the base of the table, using the adjusters with a 1½" socket and ½" ratchet.
 - b.) Use the dollies to evenly lower the table until it rests on the leveling pads using an ½" ratchet.
- 3.) **Loosen adjuster shipping bolts and locking rings "one turn" before you lower the table to the floor.** Loosen the locking rings and shipping bolts so you can **fine tune** the leveling pads to compensate for slight variations in the floor surface. Use a 1½" socket with 6" extension to loosen the shipping bolts.
- 4.) Remove the table left and right upper side covers (refer to [Figure 1-14](#)).

Note: The captive screws have a tendency to fall out occasionally.

Remove left and right side covers.



Remove the two panels, on both sides.

Figure 1-14 Table Cover Removal

- 5.) Remove the four side panels, using a 5mm hex key. Without disconnecting the ground straps at the z-channel, carefully lay the side panels on top of the padded cradle.



NOTICE
Potential for
Equipment
Damage

Take extreme care to prevent damaging the ETC and CPU printed wire assemblies. Use some packing material to protect these components.

- 6.) Remove the table base cover center support bar. (See [Figure 1-15.](#))

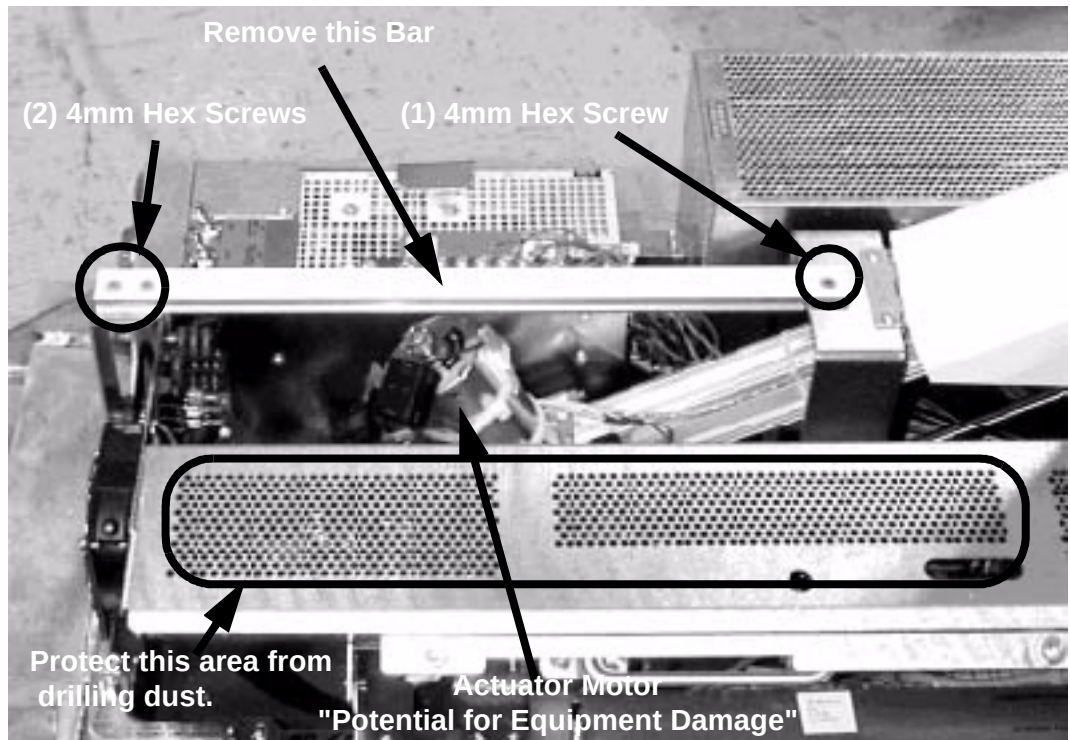


Figure 1-15 Center Support Bar

3.5 Install Alignment Bars

- 1.) Assemble the two (2) halves of the long gantry/table alignment bar, #3:
Adjuster locking ring at location #1 (see [Figure 1-16](#)) must be removed to properly position the alignment bars. Also, observe the following:
 - Bars should be at room temperature ($70^{\circ}\text{F} \pm 1\text{ degree}$ or $21^{\circ}\text{C} \pm 1\text{ degree}$).
 - The alignment bar must be calibrated. They must have a valid calibration sticker to be accurate.
 - Do not use or trust bars that have a missing locking pin or loose end cams.
 - a.) Insert the pin and rotate the handle to tighten the joint. If the joint remains loose, tighten the tip of the expanding pin to increase its diameter and try again.
 - b.) Check the serial numbers on both halves of the alignment bar; make sure the serial numbers match.
- 2.) The short alignment bars, #1 and #2, are identical and interchangeable. Refer to [Figure 1-16](#):
 - a.) Install the short alignment bars #1 & #2 first. Bars should fit snugly.
 - b.) Pivot the far end of the table until the long alignment bar #3 fits into place.
 - c.) Adjust the position of the table until the three alignment bars fit into the $\frac{3}{4}$ " alignment holes in the gantry base and the $\frac{1}{2}$ " alignment holes in the table base.
 - d.) Fully seat all three bars on their mating surfaces.

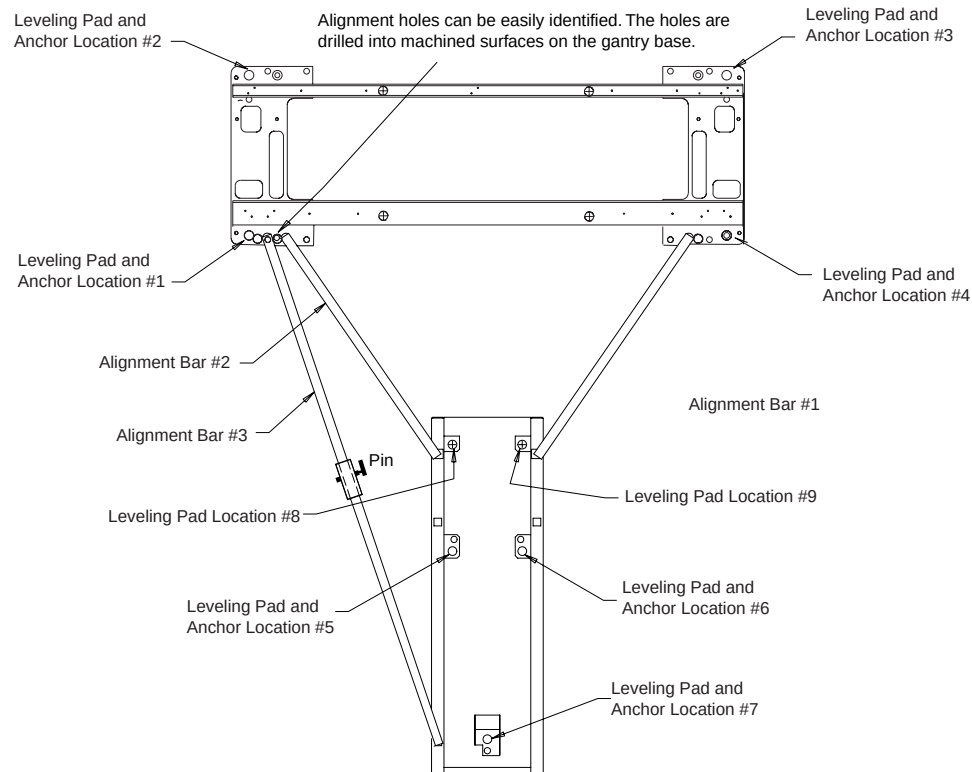


Figure 1-16 Table Base Leveling Pad and Anchor Locations

- 3.) After installing the alignment bars, lower the table to the floor and remove the transportation dollies (Figure 1-17).



Figure 1-17 Table transportation dollies

- a.) Temporarily remove alignment bar #1, to remove the front table dolly.
- b.) Replace alignment bar #1 after you remove the front dolly.
- 4.) Remove the five adjuster shipping bolts and loosen the lock rings.
- 5.) Use the 1½" socket and 6" extension to remove the shipping bolts.

3.6 Level the Table

3.6.1 Before You Begin

- Only adjusters should be used to level the table.
- Use the adjuster tool, $1\frac{1}{8}$ " socket, and $\frac{1}{2}$ " drive ratchet to turn each adjuster.
- Place a mark on the adjuster to aid in counting adjuster rotation.

3.6.2 Procedure

- 1.) Loosen and unload adjusters #5 and #6. (Refer to [Figure 1-18](#).) Do not remove shipping bolts.
- 2.) Using a 4' level, measure the levelness of the table at locations A, B and C.
 - Place the level on the machined surfaces of the alignment bars.
 - Do NOT rest the level on the radiused surface on alignment bar #3. (See [Figure 1-16](#).)

Because the leveling pads rest against the bottom of the base, you have to raise the table from its lowest position to get it leveled.
- 3.) Map out the adjustments needed to level all 3 locations.
- 4.) Remove the shipping bolts from adjusters 7, 8 and 9 only.
- 5.) Systematically level locations A, B, and C (refer to [Figure 1-18](#)) by turning each adjuster. We suggest a maximum of 1 turn at a time.



NOTICE

Always use the adjuster tool to raise or lower the table.

- Use adjuster #8 to level Location A.
- Use adjuster #9 to level Location B.
- Use adjuster #7 to level Location C.
- Use adjuster #8 and 9 to level Location D.

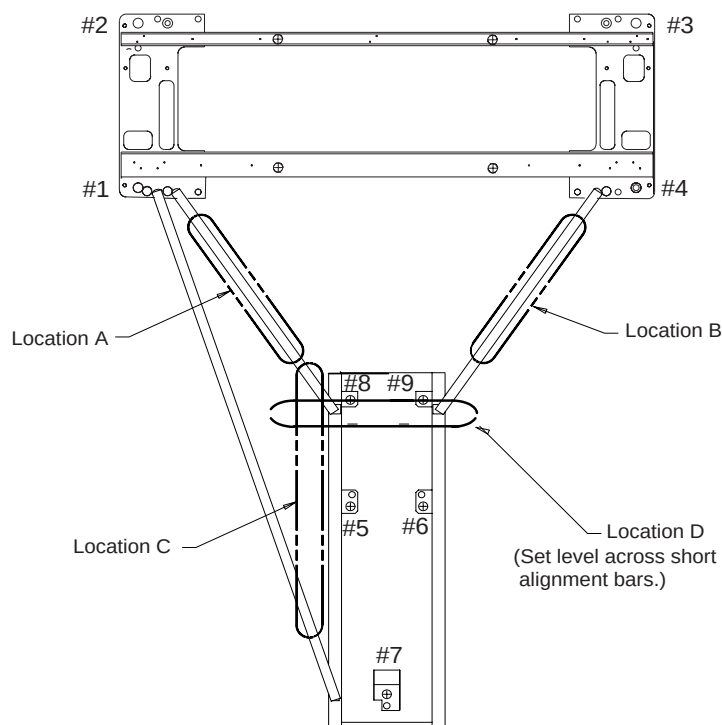


Figure 1-18 Table Base Gantry/Table Level Locations

- 6.) Continue to adjust until Locations A, B, and C are centered at the same time.
- 7.) Check location D. Fine tune only.
- 8.) Screw down the adjusters at table Locations #5 and #6, until each rests tightly against its leveling pad. Tighten each adjuster a little more than hand tight, but take care not to unload any of the other table adjusters. It's important that locations #5 - #9 (Figure 1-18) all bear part of the table's load.
- 9.) The table base covers need a minimum of $27\text{mm} \pm 3\text{mm}$ between the bottom of the table base and the floor for the covers to fit correctly. Check the distance between the table base and the floor, to ensure the covers will fit properly. See Figure 1-19 and Figure 1-20.

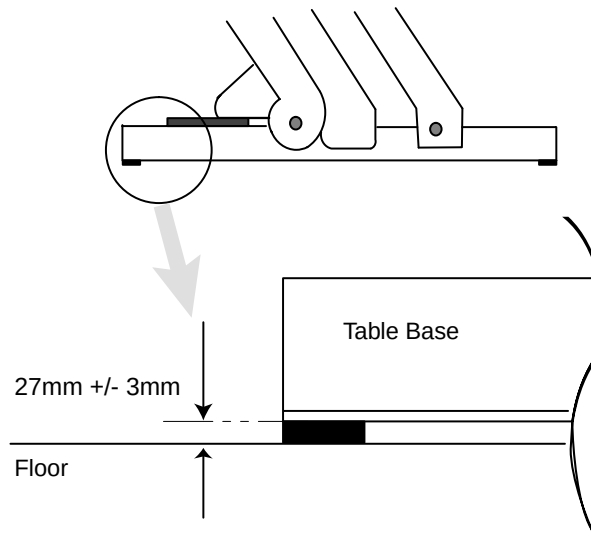


Figure 1-19 Typical Table Base Clearance

*Comment:
You should fill
the gap if
excessive.*

If excessive clearance cannot be avoided, fill the gap. For example, you can use ½" round pipe insulation.



Figure 1-20 Excessive Table Base Cover Clearance

3.7 Table-Gantry Mechanical Alignment

3.7.1 Alignment Conditions

- 1.) Complete all sections of this procedure before drilling any anchor holes.
- 2.) Alignment tool was used to align and level the table gantry.
- 3.) Alignment tool MUST have a current calibration sticker.

3.7.2 Alignment Specification

Mechanical alignment must be within ± 3 mm

Table cradle travel must be parallel

Table must be perpendicular to the gantry

3.7.3 Procedure

- 1.) Remove gantry scan window. (Refer to [Figure 1-21.](#))



Figure 1-21 Removing gantry scan (mylar) window

- 2.) Do not remove the alignment bars unless told to do so in the procedure
- 3.) Do not place any object near or into the x-ray opening. This slot must not be nicked or marred (refer to [Figure 1-22](#)).

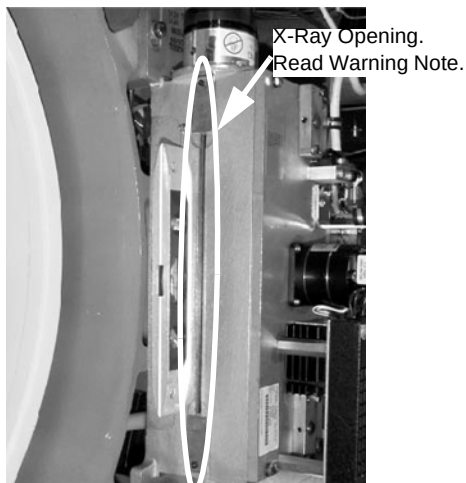


Figure 1-22 X-Ray Opening

- 4.) Rotate gantry by hand until collimator face plate is at 3 o'clock (refer to [Figure 1-23](#)). Use 9-inch level to verify plate is vertical. With power off, the gantry movement is tight.

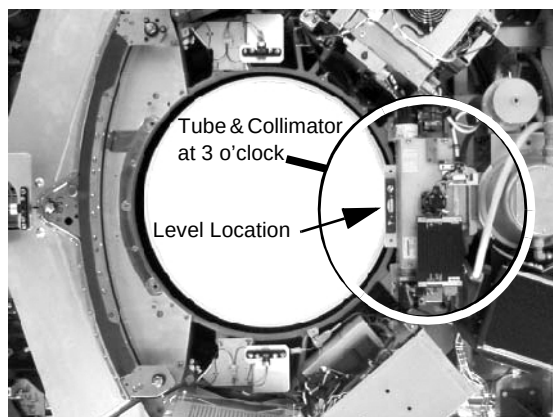


Figure 1-23 Three o'clock position

- 5.) Remove the cradle shipping bolt, cradle right side cover. Cradle shipping bolt is located under the table/cradle assembly at the foot end of the table. It has a RED TAG attached to it.
- 6.) Push the plunger on the right rear of the table to unlatch the cradle (refer to [Figure 1-24](#)).

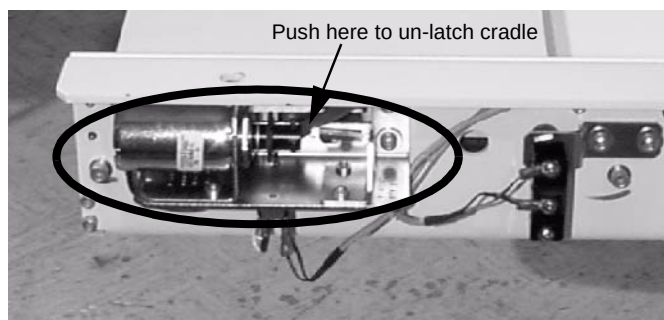


Figure 1-24 Unlatch Cradle

3.7.3.1 Cradle Center Procedure

- 1.) Center the cradle on the drive rollers assembly by pushing the cradle into the gantry to its maximum position (1400 mm) and back to just before the latch position six times. The cradle should be centered as shown in [Figure 1-25](#).



Figure 1-25 Centered Cradle

- 2.) Adjust cradle as needed, using the six bolts located on the back of the cradle ([Figure 1-26](#)).

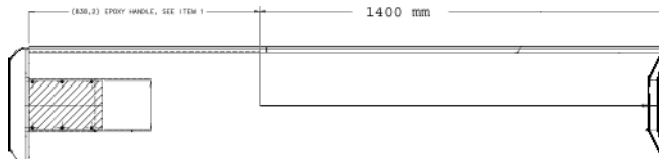


Figure 1-26 Use bolts to adjust cradle

- 3.) Unlatch and move the cradle towards the gantry a few inches until the front of the cradle is near the gantry opening.
- 4.) Use a 2ft level to check cradle level from side to side. Level the cradle using the table front level adjusters #8 & #9. (See [Figure 1-27](#).)

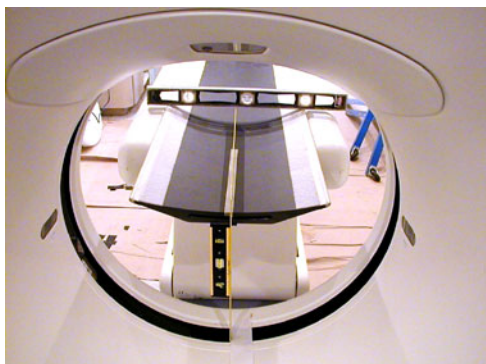


Figure 1-27 Place level across cradle

- 5.) Place the level in the center of the cradle and check for level (see [Figure 1-28](#)). The width and the length of the cradle must be level when finished. Re-tighten all adjuster lock rings.

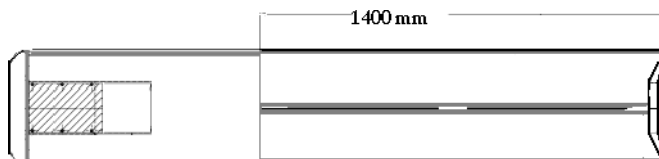


Figure 1-28 Place level along the length of the cradle

- 6.) Find the table center line marked on the front edge of the cradle. If your system had no center line mark, complete the following procedure to make a cradle center line.
- 7.) Using a metric tape measure or other suitable measuring device:
 - a.) Using supplied masking tape, place a 380mm (15") strip on the front of the cradle. See [Figure 1-29](#).



Figure 1-29 Masking tape on the front of the cradle

- b.) Determine the cradle center-line by measuring the cradle width 421.1mm (210.5mm center) Use a square or straight edge to determine the center.

Note:

The white edges on the cradle may not always be parallel and a second measurement is needed. See [Figure 1-30](#).

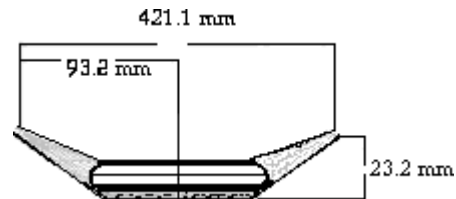


Figure 1-30 Front of Cradle

- c.) Find the center of the inside slot. Width of the inside slot is 186.4 mm (93.2 mm center.) Use a square or straight edge to determine the center. See [Figure 1-31](#).

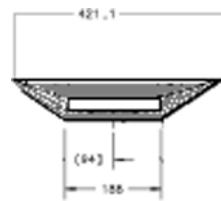


Figure 1-31 Determine slot center

- d.) Use these two dimensions to determine the table center line by connecting the two marks. Measure back 300mm, then measure across the cradle and place a second cradle center mark on the tape. Use a square or straight edge to connect the two marks. See [Figure 1-32](#).



Figure 1-32 Use a square to connect marks

3.7.3.2 Measuring Cradle Center Procedure

- 1.) Carefully measure from cradle center line to collimator faceplate. It may be helpful to use a square or straight edge to help determine the measured distance. See [Figure 1-33](#).

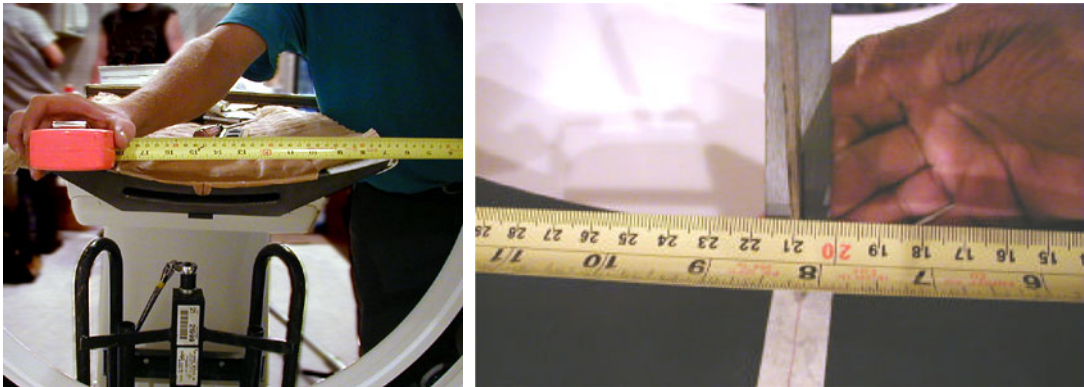


Figure 1-33 Measure Distance

Record this measurement as A on the worksheet ([Table 1-3](#)).

	FRONT		
	A 3 o'clock	B 9 o'clock	SPEC $ A-B \pm 1\text{mm}$
Start Position			
# 2			
# 3			

Table 1-3 Alignment Worksheet

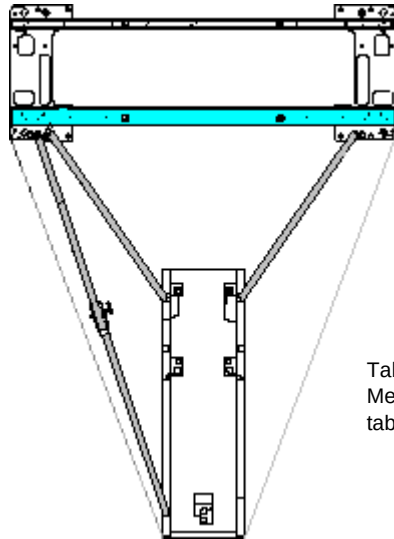
- 2.) Carefully rotate the gantry by hand until the tube and collimator is in the nine o'clock position.
- 3.) Use a bubble level to precisely position the collimator faceplate at the nine o'clock position.
- 4.) Carefully measure from cradle center mark to collimator faceplate. It may be helpful to use a square or straight edge to help determine the measured distance. See [Figure 1-33](#).
- 5.) Record this measurement as B on the worksheet.
- 6.) Perform the equation in the SPEC column to determine if the table should to be moved. If so move the table half the distance to correct the error.
- 7.) The table and gantry are in alignment when both measurements meet the specification.
- 8.) If A and B are not equal, move the table and repeat until measurements meet the specification ($\pm 1\text{mm}$).

3.7.3.3 Cradle/Table Parallel Movement

- 1.) Extend the cradle into the gantry 1400 mm (see [Figure 1-28](#), on page 43).
- 2.) Measure from the collimator face plate to the cradle edge.
- 3.) Rotate the gantry and re-measure each side until the front measurements match the extended measurement. Visually inspect the table / cradle for perpendicular movement. Actual perpendicular movement will be determined later with x-ray.

3.7.3.4 Determining Table to Gantry Perpendicularity

Measure from each corner of the gantry base to the corner of the table as shown in [Figure 1-34](#). Each dimension should be equal when finished.



Table/Gantry Perpendicular Check:
Measure from gantry base corner to
table base corner.

Figure 1-34 Table to gantry perpendicularity

3.8 Tighten the Lock Rings

- 1.) Re-check bubble levels and loading first. Refer to [Figure 1-18, on page 39](#).
 - a.) Re-check that bubbles levels at locations, A through D on the table.
 - b.) Re-check that bubbles levels at both ends of the gantry.
 - c.) Re-check that adjusters at Locations 1, 2, 3, 4, 7, 8 and 9 carry a portion of the load.
 - d.) Re-check that gantry "inside" levelers carry a portion of the load.



CAUTION



Eye protection is required when using a hammer and chisel.

- 2.) Tighten the lock rings at Locations #1 through #9 with the spanner where possible. Use a hammer and chisel to tighten the lock rings only where you can not use the spanner.

3.9 Drill the Anchor Holes

- 1.) Make certain that the gantry scan window is in place, to prevent dust from entering the gantry.



Figure 1-35 Gantry Scan Window

- 2.) Use a piece of tape to mark the drill bit depth of 190 mm ($7\frac{11}{16}$ ") from the tip of the 13mm ($\frac{1}{2}$ ") masonry drill bit.



NOTICE
Potential for
Equipment
Damage from
Dust

You must cover all electronic assemblies in the table base prior to drilling to prevent damage due to the dust created during drilling.

- 3.) Use the 13mm ($\frac{1}{2}$ ") bit to drill all seven (7) anchor holes (refer to [Figure 1-38](#) for locations.)
 - Review [Figure 1-36](#), prior to drilling.

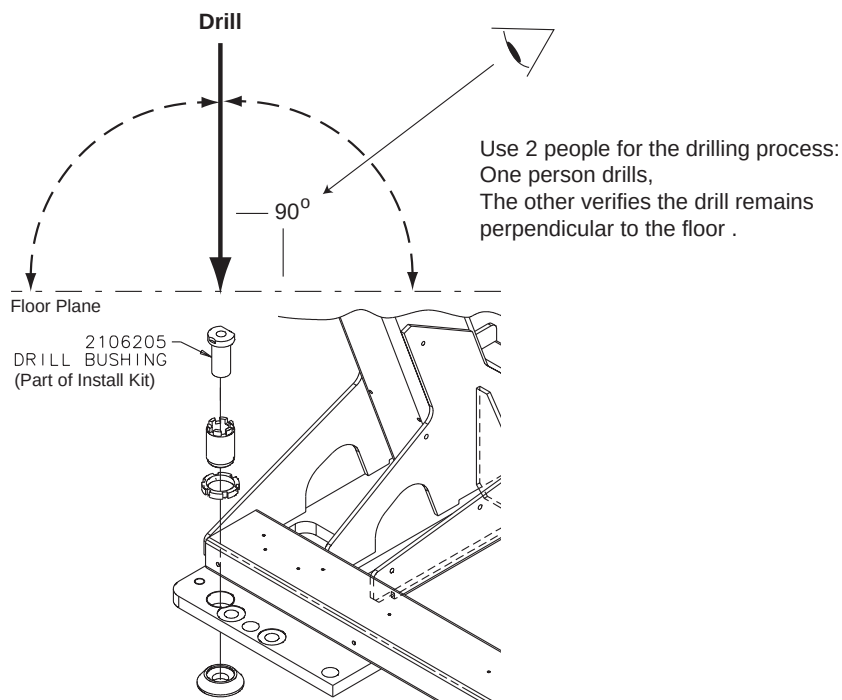


Figure 1-36 Drilling Position

- **Important** - Drill the holes perpendicular to the floor. While one person drills the holes, position a second person to watch the relationship between the drill bit and floor. Make sure the bit remains absolutely perpendicular to the floor throughout the drilling operation. Always use the mechanical guide supplied with the alignment kit.
- Stop drilling every 15 or 20 seconds and clear the hole of debris. This helps to prevent binding of the drill bit.
- a.) Place appropriate protection to prevent damage and dust contamination to electronic assemblies. Refer to [Figure 1-15, on page 37](#), for placement of dust protection materials.
- b.) Drill each hole until the mark on the drill bit is even with the top of the drill bushing. (195mm)
- c.) Place the drill bushing inside each adjuster, to keep the hole vertical, and centered within the adjuster.
 - * Use the drill bushing to center the anchor holes in adjuster Location #5, #6, and #7, to provide maximum lateral alignment capacity when you center the cradle on isocenter during subsequent system testing.
 - * Take care not to injure yourself on the gantry cover brackets.
 - * Take care to avoid the actuator motor lead when you drill the Location #7 anchor hole. Remove the lead if necessary.

Note:

An 18" bit will make drilling anchor location #7 easy, and will prevent component damage (see [Figure 1-37](#)).

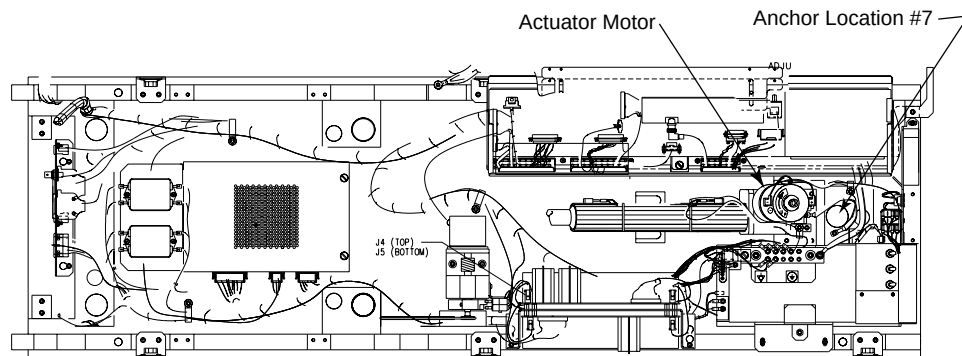


Figure 1-37 Table Assembly Showing Actuator Motor and Anchor Location #7

- d.) Stop the drill every 15 or 20 seconds and clear the hole of debris to prevent binding of the drill bit.
- 4.) Vacuum all debris from the inside of the hole:
 - a.) Place the funnel tip inside the hole; place the vacuum hose in the funnel.
 - b.) Continue to vacuum while you drill, if you can, to keep gantry and table as free of dust contamination as possible.
 - c.) When you finish clearing the anchor hole, vacuum the debris from the surrounding area.
 - d.) Stop drilling frequently to let the drill bit cool.
 - e.) All holes must be a minimum of 108 mm deep (see [Figure 1-40, on page 51](#)). Recheck all holes.

Note:

A drywall dust filter should be used on the vacuum.

- 5.) Do not drill anchor holes at Location #8 and #9. These locations help support the weight, but do not anchor the table into place.

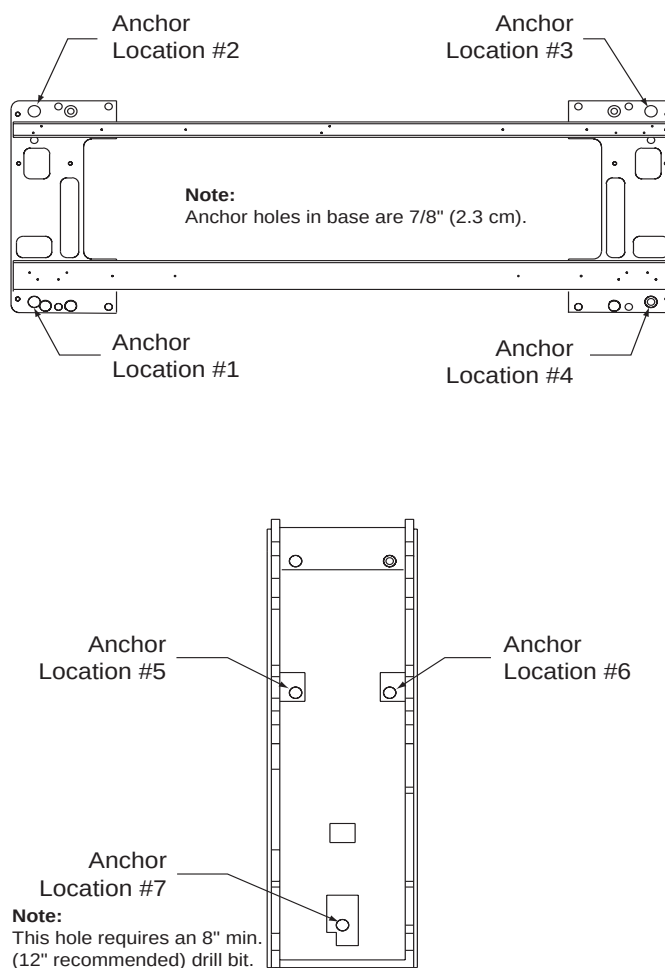


Figure 1-38 Anchor Locations (standard)

3.10 Gantry & Table Redundant Anchor Holes

If you cannot use one of the adjuster anchor holes due to structural interference, such as reinforcement bars in the concrete, you must use one of the redundant anchor locations, as shown in [Figure 1-39](#).

- The gantry requires a minimum of four (4) anchors, one (1) in each corner.
- The table requires a minimum of three (3) anchors, one (1) at location #5, #6 and #7.

If you must use a redundant anchor hole in the gantry, you must remove the gantry covers. See [Appendix A](#), for gantry cover removal.

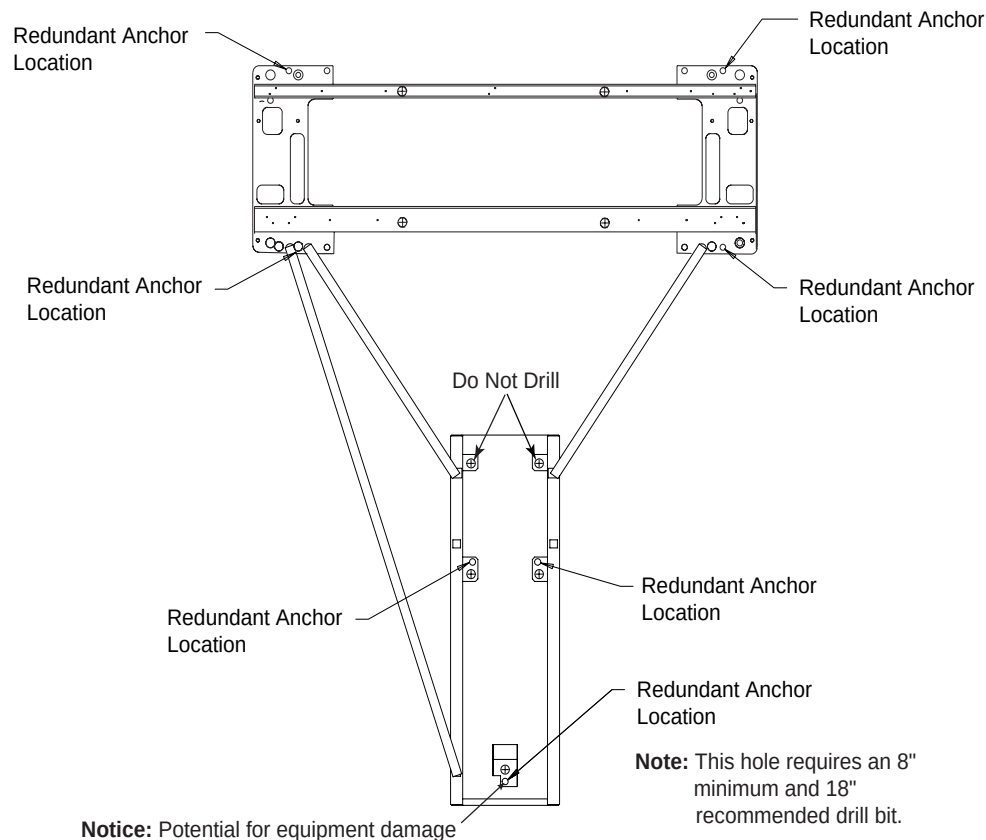


Figure 1-39 Table Base Redundant Anchor Locations

Mark the 13mm ($\frac{1}{2}$ ") drill bit with a piece of tape 170 mm ($6\frac{11}{16}$ ") from the tip of the bit, and follow the instructions in [Section 3.9](#), on [page 47](#).

3.11 Install the Anchors

Recommended - Use "Hilti Kwik-Bolt II" anchors for this procedure. P/N 2106573 ($\frac{1}{2}$ " diameter by 8" long).

- 1.) Remove the three alignment bars and repack the install kit.
- 2.) Refer to [Figure 1-40](#). Assemble the anchors before you install them:
 - a.) Remove the nut and washer from the anchor.
 - b.) Add a $\frac{1}{4}$ " thick washer (PN 2105873) under the regular anchor washer
 - c.) Reassemble the anchor washer and nut and position nut so top is flush with threads of anchor before hammering into hole (refer to [Figure 1-40](#)).

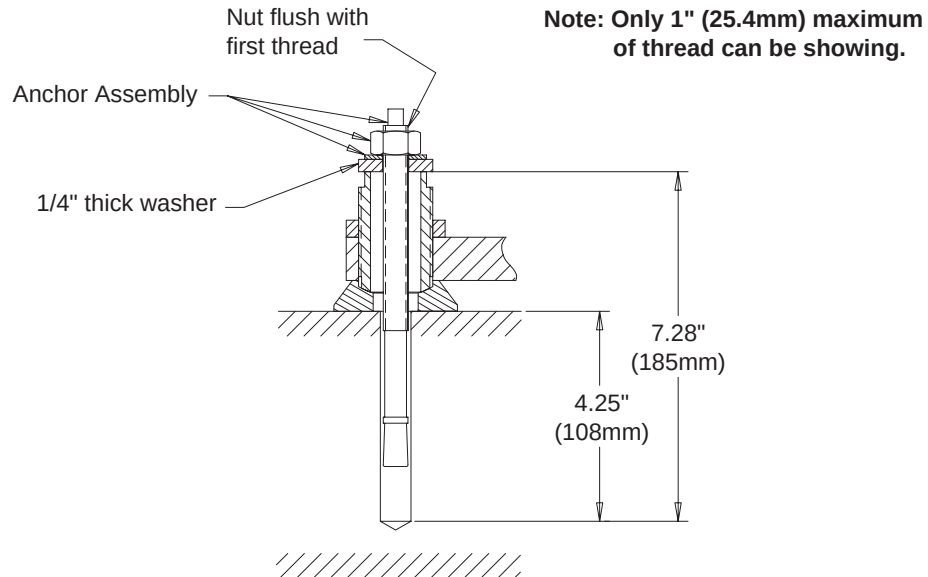


Figure 1-40 Table Base Anchor Assembly



Figure 1-41 Measure Table Base Anchor Thread

- 3.) Fixed site Installations must be centered under mounting holes in table base, with 3mm of space visible all of the way around the tapped bolt hole. [Figure 1-42](#).

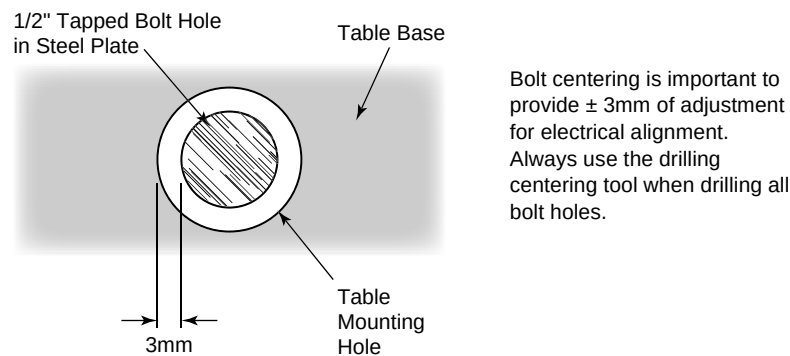


Figure 1-42 Center tapped holes under mounting holes in table base



CAUTION



Eye protection is required when using a hammer.

- 4.) Refer to [Figure 1-40](#) for instruction on inserting the anchors in each hole. Using the anchor seating tool, hammer the anchor downward to *completely insert the anchor in the hole*.
- 5.) Repeat Sections [3.7.3.2](#), [3.7.3.3](#) and [3.7.3.4](#).

This procedure is designed to mechanically center the table cradle within the gantry. All steps are important to achieve final electrical x-ray alignment of 1 mm. Alignment bars are still needed to level and position the gantry. Alignment bars must be calibrated yearly.

- 6.) Once alignment has been verified, torque all mounting bolts.
Tighten the Location #1 through #7 anchors and torque to 75 ± 6 N-m (55 ± 5 ft.-lb.).
- 7.) Replace all the table parts you removed during installation, including covers, ground strap and the center support bar. Check the motor lead.

Note: If you cannot replace the lower table cover because the floor interferes, adjust all of the table and gantry levelers by $\frac{1}{2}$ turn increments to raise the table/gantry until the lower table covers clear the floor. (Refer to [Figure 1-20, on page 40](#), for table base cover clearances.) Return to [Sections 3.3, on page 34](#), [3.6, on page 39](#) and [3.8, on page 46](#) to level the gantry, level the table, and tighten the locking rings, respectively.

3.12 Inside Levelers

There are six (6) additional (auxiliary) levelers/stabilizers on this gantry. These levelers help to spread the gantry load across the base. The two “inside” levelers are located on the rear side of the gantry inside the base near adjusters #2 and #3. Two levelers/stabilizers each are located on both the front and rear cross bars. Make sure they have remained at their lowest point (i.e., against the base) during shipment and do not interfere with positioning.

Note: The levelers are removed for international shipments and must be reinstalled.

International Shipments



Figure 1-43 Detail of Inside Leveler near Gantry base adjuster

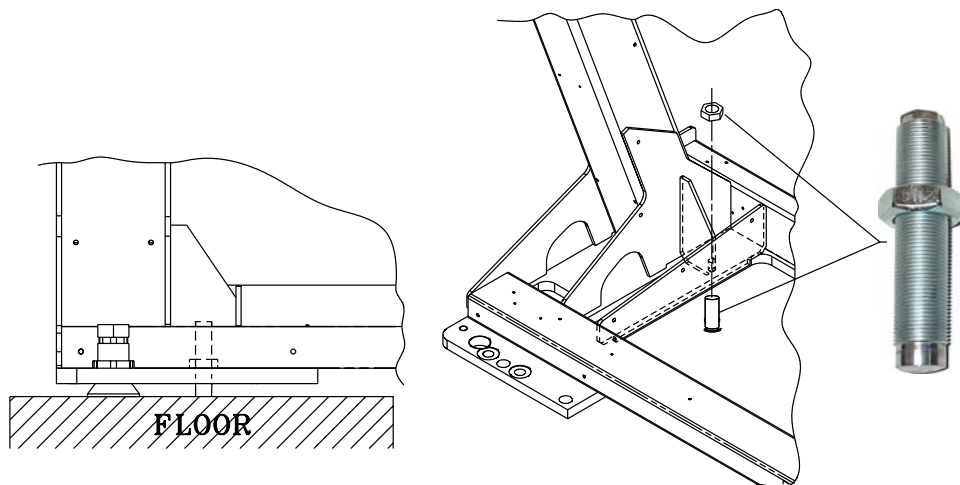


Figure 1-44 Inside Levelers

Adjust the six (6) auxiliary levelers so as to stiffen the gantry base. Torque the cross-bar levelers, in the order shown in [Figure 1-45](#) (1, 2, 3 & 4), to 10 ft.-lbs. Repeat at 20 ft.-lbs. Then torque inside levelers 5 and 6 to 20 ft.-lbs. Lock-down the levelers at all six locations, using the supplied nut.

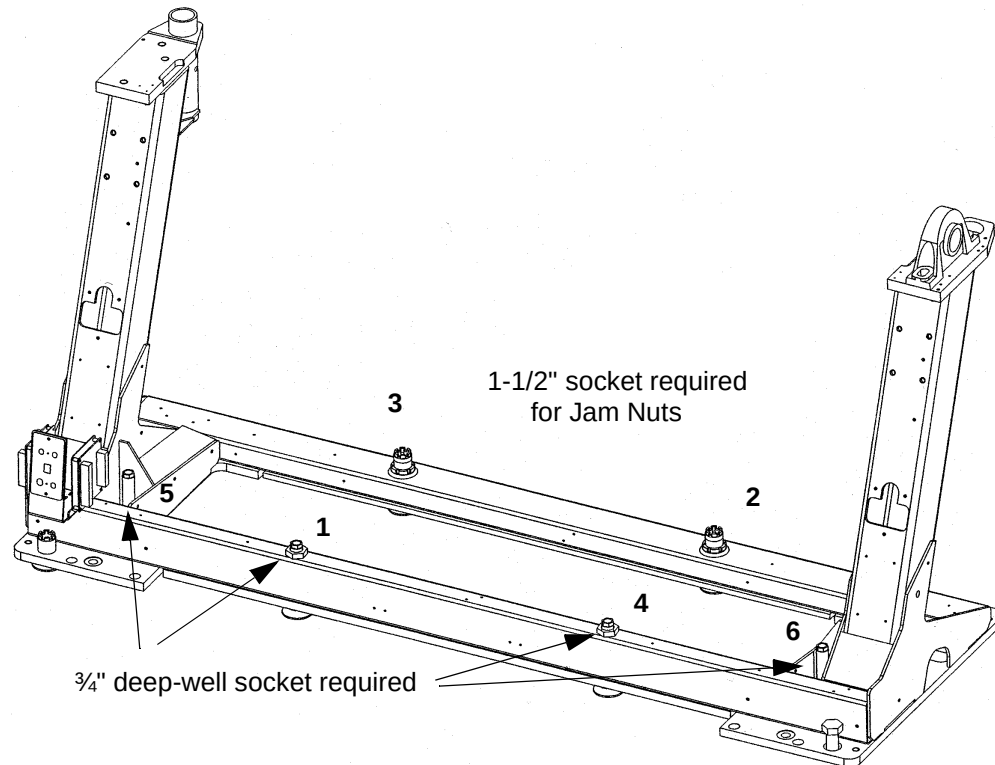


Figure 1-45 Auxiliary Leveler Layout, Rear View



Figure 1-46 Auxiliary Leveler, with Foot Pad

Section 4.0

Rear Entry Cable Box (Site Specific)

A rear entry cable box (B7850RC) is used when the cables to the gantry cannot be brought-up inside the gantry base. The box is not supplied with the system and must be ordered separately. The rear entry cable box frame attaches to the gantry base. See [Figure 1-47](#).

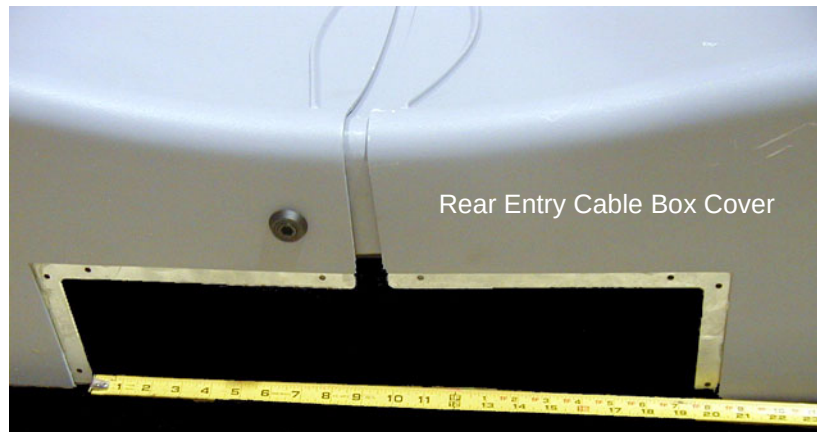
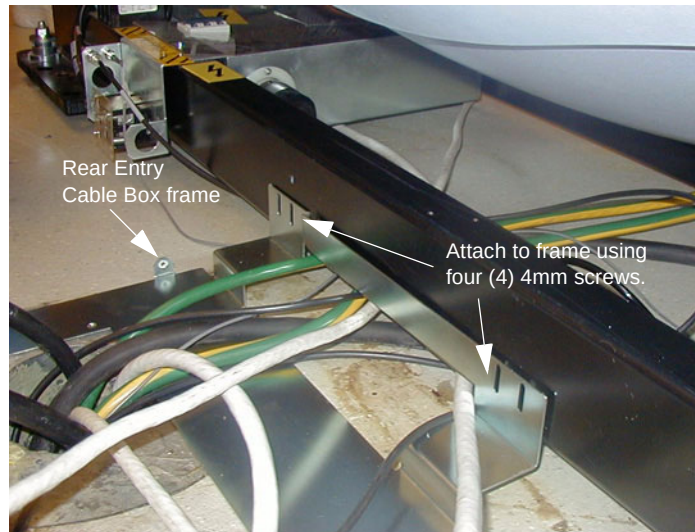


Figure 1-47 Rear Entry Cable Box

The filler plate may need to be cut in order to fit the rear cable box into the cover opening. Use the supplied white molding around the filler plate.

Section 5.0

Install Table Footswitch Assembly

Install the table foot-switch assembly as shown in [Figure 1-48](#), along with filler strip.

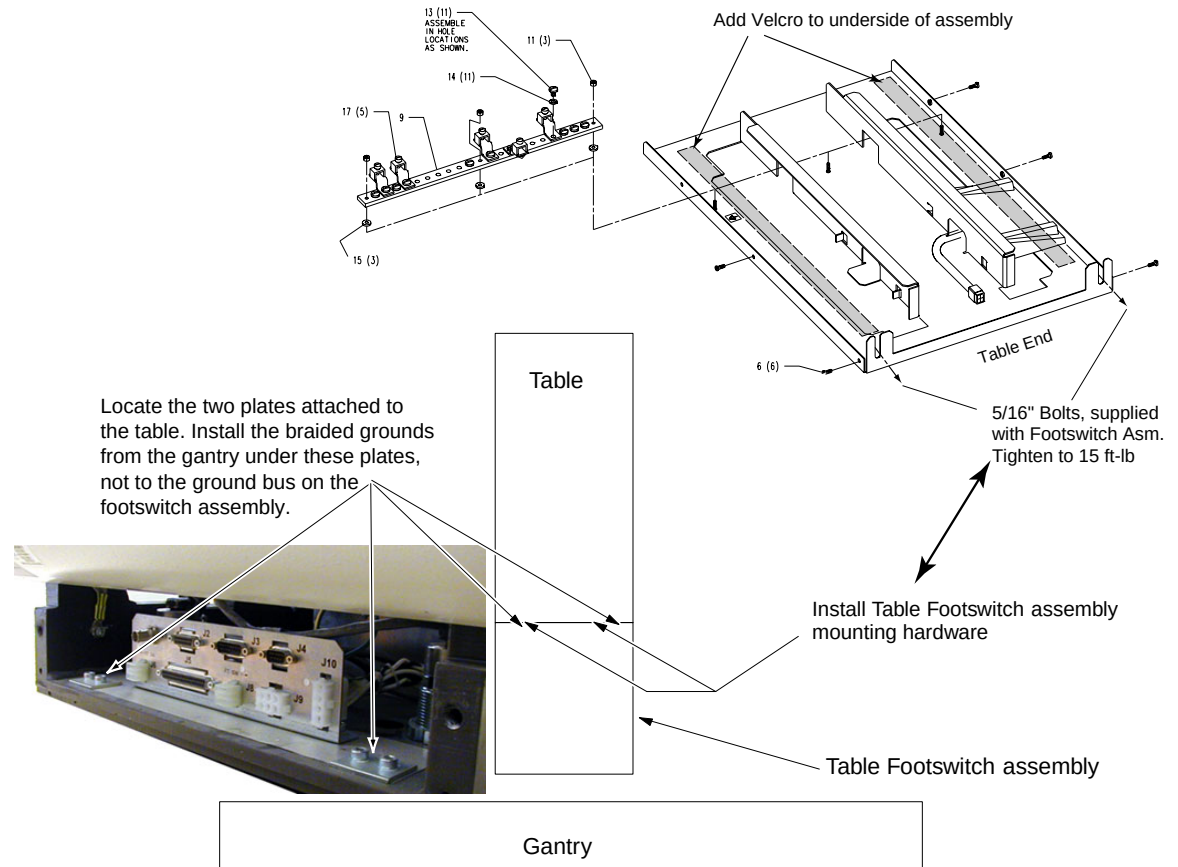


Figure 1-48 Install Table Foot-switch Assembly

- 1.) Remove foot switch assembly, and locate hardware.
- 2.) Loosen six (6) hex head screws and remove foot-switch assembly.
- 3.) Place foot-switch cover in a secure place.
- 4.) Locate the supplied 2' velcro in the install support kit, and apply two strips to the bottom of the foot-switch assembly.
- 5.) Using the two 5/16" bolts and blue loctite (242), attach the foot-switch assembly.
- 6.) Remove the 2" braided strip and gantry cable that are secured to gantry base, and place in the cable tray.

Section 6.0

Remove Gantry Tilt Bracket

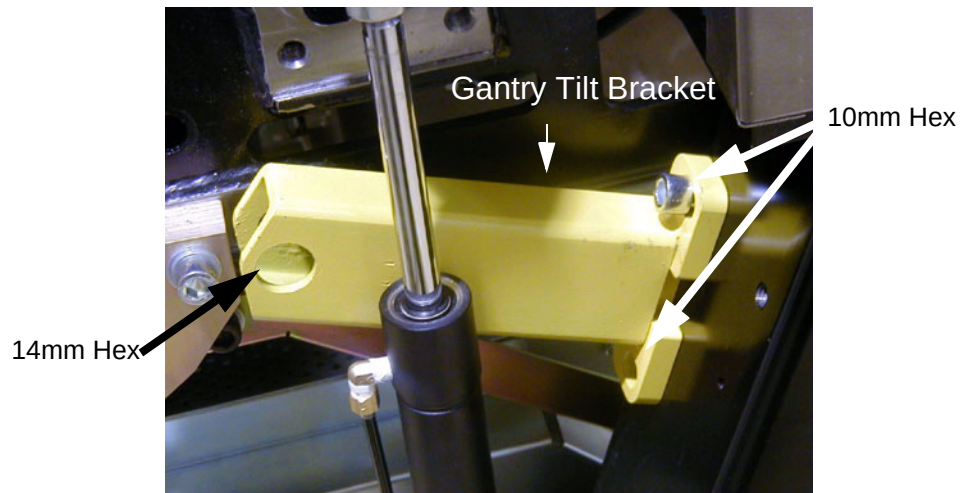


Figure 1-49 Gantry Tilt Bracket Removal

- 1.) Refer to [Figure 1-49](#). Remove the gantry shipping bolts and brackets.
- 2.) Close the gantry covers and reinstall the scan window.
- 3.) Store brackets in the gantry base.

Section 7.0

Position the Power Distribution Unit

7.1 Compact PDU



WARNING LOCKOUT/TAGOUT IS REQUIRED BEFORE PERFORMING THIS TASK.

- 1.) Roll the PDU into position on its permanently mounted casters. Leave at least 15.5 cm (6") between the PDU and back wall to allow cooling air to circulate.

Note: Connecting the primary incoming power (steps 2 through 5, below) is performed by the customer's electrical contractor.

Connection or Wall Box	AWG #	Connection From	Connection To PDU	Installed & Checked
A1	#1	4A	A3 TB1 L1	
	#1	4B	A3 TB1 L2	
	#1	4C	A3 TB1 L3	
	#1/0	GND	A3 TB1 GND (Do NOT connect anything to neutral point.)	

Table 1-4 Contractor Connections

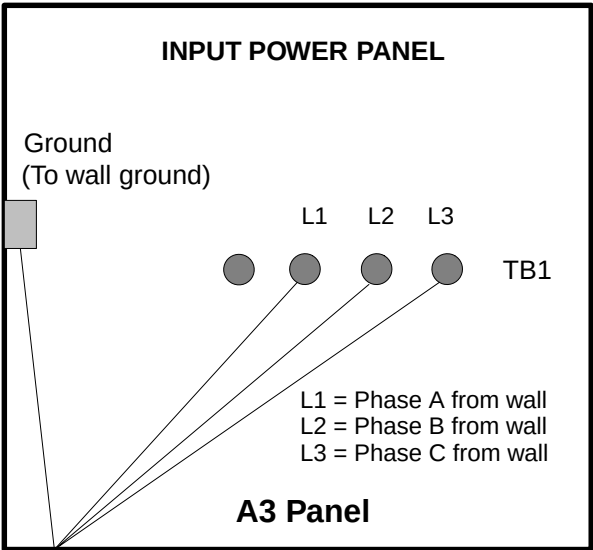


Figure 1-50 A3 Input Power Panel Connections

- 2.) Run the main input power conductors and ground through flexible metal conduit (attached between the PDU chassis and room duct-work) so you can move the PDU away from the wall during service.



Figure 1-51 Flexible Conduit for PDU Power

- 3.) Locate the hole cover plate in Box 1 and attach the flexible metal conduit to the PDU.
- 4.) Cut the three phase and 1/0 ground wire to size.
- 5.) Place cable lugs on the wires and attach as shown in [Figure 1-50](#).

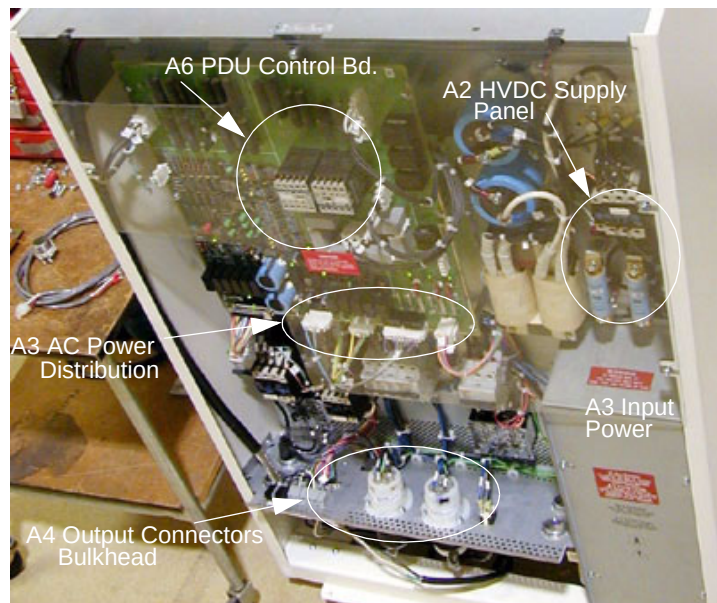


Figure 1-52 PDU Area Locations

7.2 NGPDU



NOTICE

There are two versions of the NGPDU—verify that the version shipped with your system is the correct one for the application. Use of the incorrect PDU type can result in severe equipment damage.

- Model 2326492 is intended for use with high power systems (w/ a 110kW generator)
- Model 2326492-2 is intended for use with medium power systems (w/ a 53kW generator)

	Item	2326492	2326492-2
Power Rating		Input Voltage: 380 to 480V 3 phase Line Frequency: 50 / 60 Hz Input Power: 150 kVA momentary (at 0.85 PF) 25 kVA average	Input Voltage: 200 to 240V, 380 to 480V 3 phase Line Frequency: 50 / 60 Hz Input Power: 90 kVA momentary (at 0.85 PF) 20 kVA average
Configuration	Fuse (F1-3)	200A	100A
	CB5	40A	32A
	Transformer	Secondary Winding #2: 474VAC (Part No.: 2332326)	Secondary Winding #2: 494VAC (Part No.: 2332326-2)
	Interface Board	No	Yes
Note: Wire size is #1/0 AWG both for 2326492 & 2326492-2.			

Table 1-5 NGPDU Differences



WARNING



LOCKOUT/TAGOUT IS REQUIRED BEFORE PERFORMING THIS TASK.

- 1.) Roll the PDU into position on its permanently mounted casters. Leave at least 15.5 cm (6") between the PDU and back wall to allow cooling air to circulate.

Note: Connecting the primary incoming power (steps 2 through 5, below) is performed by the customer's electrical contractor.

Connection or Wall Box	AWG #	Connection From	Connection To PDU	Installed & Checked
A1	#1	PDB-A	TS1-L1	
	#1	PDB-B	TS1-L2	
	#1	PDB-C	TS1-L3	
	#1/0	GND	Ground Block (Do NOT connect anything to neutral point.)	

Table 1-6 Contractor Connections

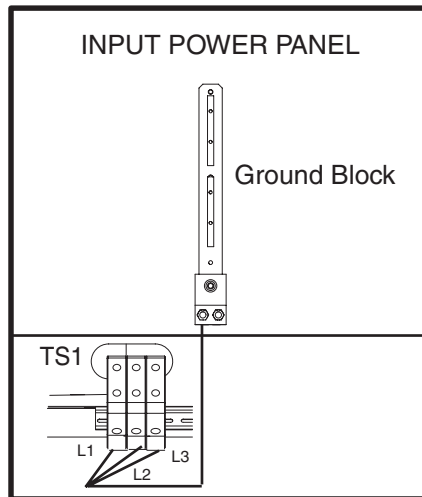


Figure 1-53 Input Power Panel Connections

- 2.) Run the main input power conductors and ground through flexible metal conduit (attached between the PDU chassis and room duct-work) so you can move the PDU away from the wall during service.



Figure 1-54 Flexible Conduit for PDU Power

- 3.) Locate the hole cover plate in Box 1 and attach the flexible metal conduit to the PDU.
- 4.) Cut the three phase and 1/0 ground wire to size.
- 5.) Attach cables as shown in [Figure 1-53](#) and [Figure 1-55](#).

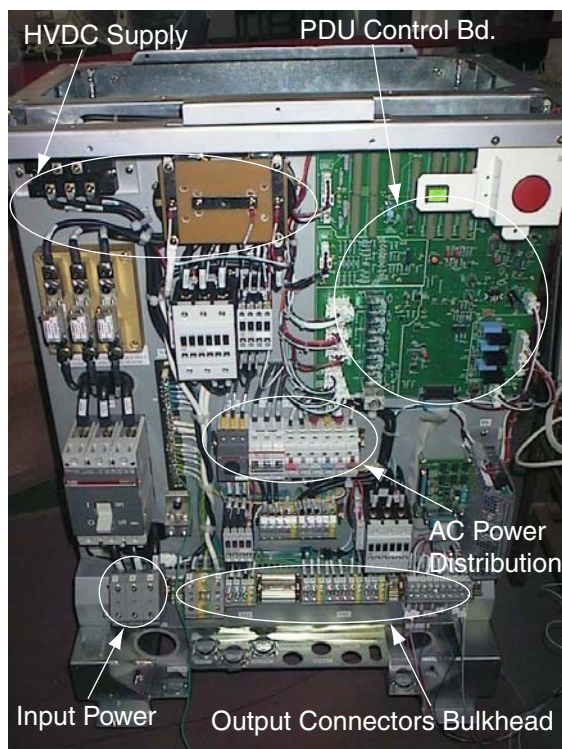


Figure 1-55 PDU Area Locations

Section 8.0

Install Operator Console

8.1 Unpack Console



Figure 1-56 Console boxed on skid.

- 1.) Remove all items from the console.
- 2.) Remove all packing materials and discard.
- 3.) Place the step-board under the front edge of the skid and step on it to raise the front edge of the skid as in [Figure 1-57](#).



Figure 1-57 Step-board used to raise front edge of skid.

- 4.) Remove the two front cushions from the bottom of the skid. Refer to [Figure 1-58](#)



Figure 1-58 Cushion on bottom of skid.

- 5.) Remove the Seismic Brackets from each side of the console. Refer to [Figure 1-59](#).



Figure 1-59 Removing the Seismic Brackets

- 6.) Lift up on the strap on the front of the step-board ([Figure 1-57](#)) to lower the skid. Remove the step-board.
- 7.) Ensure the console stabilizers are in line with the notched portion at the front of the skid. This will allow enough clearance to smoothly roll the console down the ramps. Refer to [Figure 1-60](#).



Figure 1-60 Console ready to be unloaded

- 8.) Move console to installation location.

- 9.) Attach the two pads to the console stabilizers. Refer to [Figure 1-61](#).

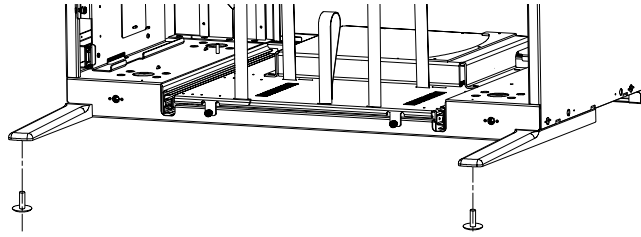


Figure 1-61 Console Stabilizer Pads

- 10.) Screw the pads all the way into the legs

8.2 Remove Console Covers

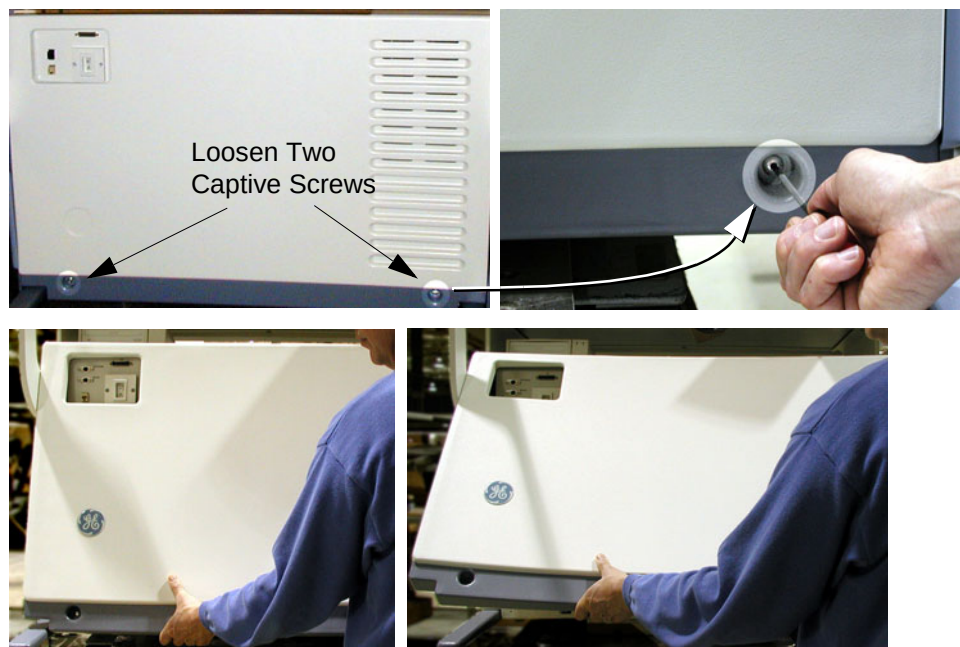


Figure 1-62 Front Cover Removal - Global Consoles

- 1.) Loosen two captive screws at bottom of console.
- 2.) Rotate bottom of cover outward and upward until it may be lifted free of the console at the top.

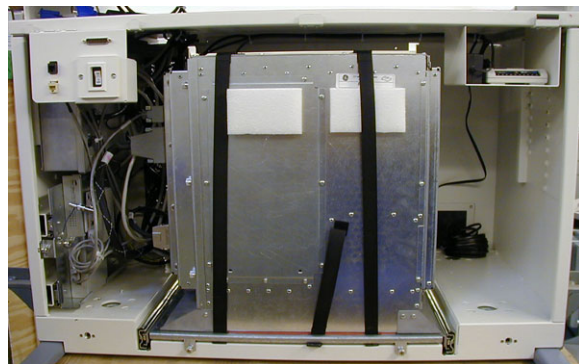


Figure 1-63 Global Console - Octane2, front cover removed

8.3 Adjust Table Top Height and Position Console

- 1.) Refer to [Figure 1-64](#). The console normally arrives with the bottom hole on the monitor top aligned with the fourth bolt from the bottom.
- 2.) For optimum operator comfort, align the bottom hole of the keyboard table top with the fifth hole from the bottom on the console.

Your site may have different requirements and you may have to adjust the monitor top and/or keyboard table top up or down from this position.

- Always select a table top height that permits the operator to see the patient on the table.
- Keep the one bolt-hole relationship between the monitor top and the keyboard table top. Fasten the keyboard table top one (1) hole lower than the console table top ([Figure 1-64](#)).



Figure 1-64 Global Console tabletop adjustments

- 3.) Install the console side covers.
Side covers are left and right specific.
- 4.) Move the console into its final position, maintaining a 6" (15 cm) minimum distance between the console and the wall.

8.4 Attach Keyboard Table Top

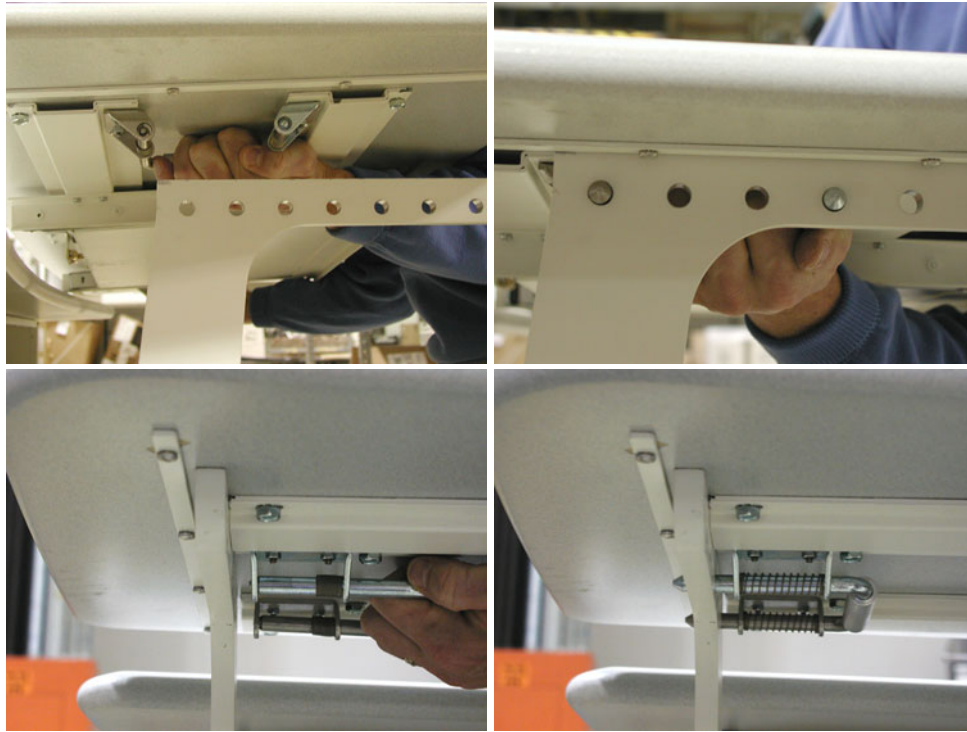


Figure 1-65 Keyboard Table Top Placement

- 1.) Locate the spring loaded latches at either end of the underside of the keyboard table top.
- 2.) Pull latches inward (toward center of table top).
- 3.) Position table on brackets so that latches align with holes.
- 4.) Release latches, making certain that the pins fully engage the bracket holes (pins should protrude beyond outside edge of bracket).

8.5 Monitor and SCSI Tower Placement

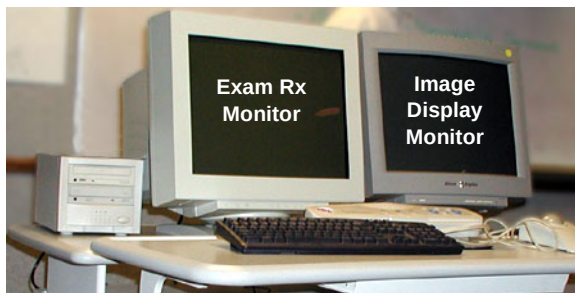


Figure 1-66 Global Console tabletop component placement

- 1.) Locate and unpack the two monitors. Be careful as each monitor weighs approximately 80 pounds (36 Kg) each.
- 2.) Carefully place the monitors on the console desktop. **This step requires two people.**
- 3.) Locate and unpack the SCSI drive tower (it ships inside the console).
- 4.) Place the SCSI drive tower on the console desktop. A suggested placement is shown in [Figure 1-66](#).

Section 9.0 Seismic Mounting

The Seismic Kit is part number R4390JC.

9.1 Console

If site specifications require seismic mounting, use ½" bolts to mount the brackets to the floor. Refer to [Figure 1-67](#) for hole placement.

Note: The Global Console is 2" narrower than the H3 console.

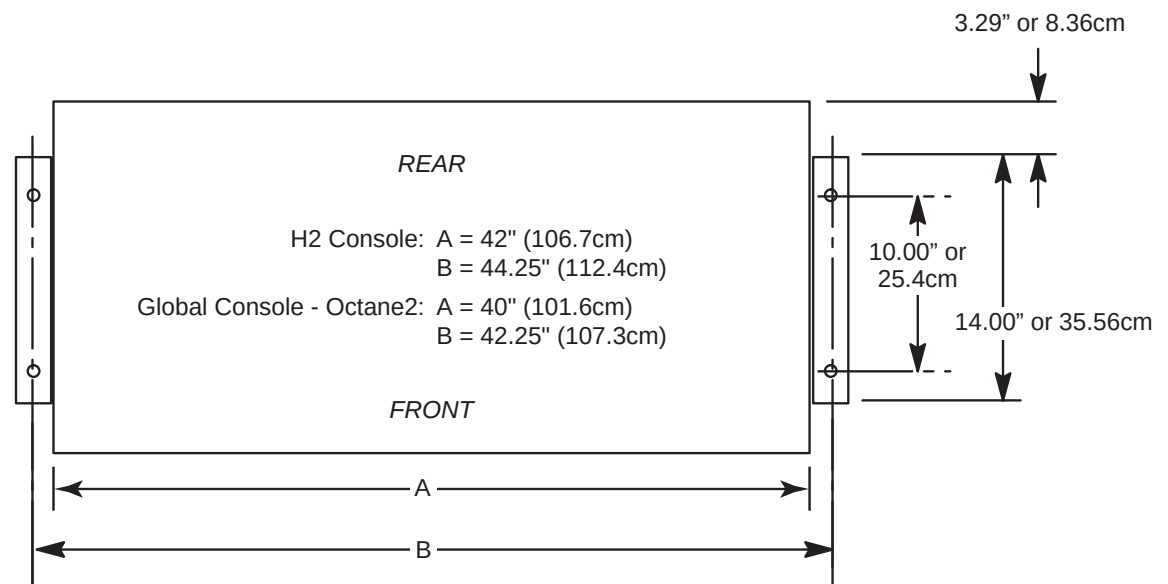


Figure 1-67 Seismic Console Mounting Hole Locations

For the Global Console - Octane2, the brackets are shipped on the shipping skid. Remove and reuse these brackets as needed.

Console seismic kit, 2336557, contains console mounting and SCSI tower mounting brackets.

9.2 Power Distribution Unit

If site specifications require seismic mounting, use the PDU seismic brackets that were shipped with the seismic kit. Refer to [Figure 1-68](#) for hole placement.

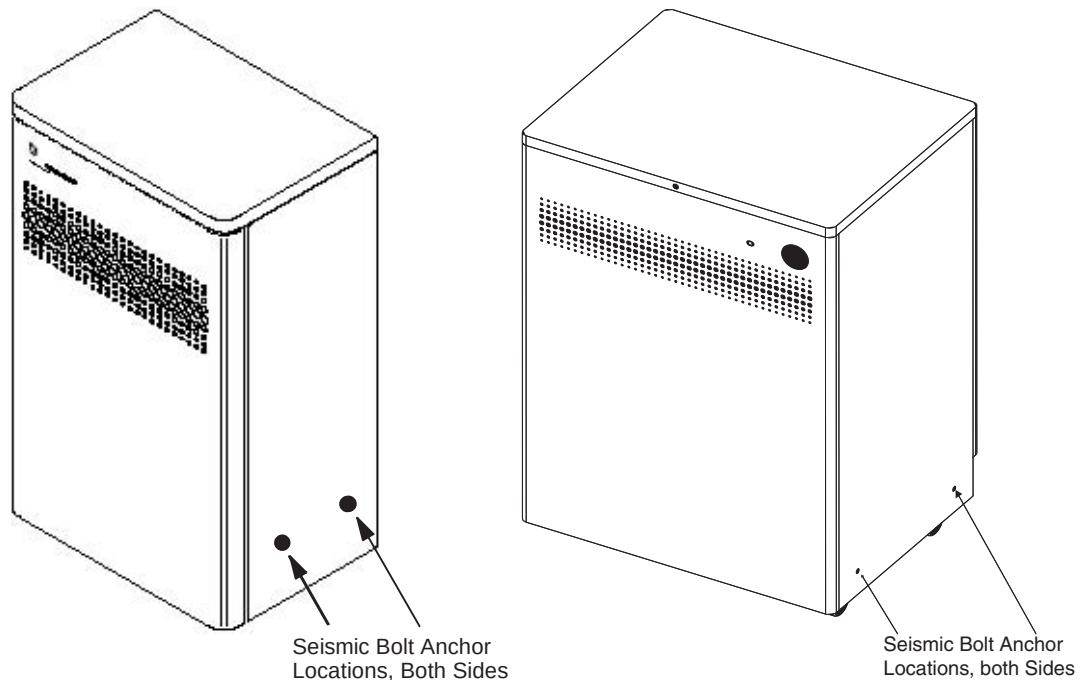


Figure 1-68 Seismic PDU Mounting Hole Locations (CPU at left, NGPDU at right)

Section 10.0

Install Options

Refer to:

- GE Prints and schematics for mechanical (physical) location of option
- FDO shipment for identification of items
- Installation Specialist for installation instructions if they differ from print
- GE Field Engineer for installation support, if needed

10.1 Camera (Filming Device)

Refer to documentation shipped with camera

- DICOM 2210573 GE Document
- DICOM Print 2152913

10.2 Injector

Refer to documentation shipped with the injector.



WARNING

COLLISION OF THE CEILING STAND WITH OTHER EQUIPMENT (SUCH AS C-ARM OR GANTRY) WILL DAMAGE THE CEILING STAND AND MAY CAUSE IT TO FALL. INSTALL THE CEILING STAND OUTSIDE THE RANGE OF MOTION OF ALL EQUIPMENT.

Mounting the ceiling stand is a critical factor in determining the usefulness and life of the unit. Consult with an architect before installation to establish ceiling or wall weight limits. Medrad is available for additional assistance. The following guidelines will ensure maximum utility and system life of the ceiling stand:

- The ceiling stand must be able to access the front and rear of the gantry of various scanners. Position the ceiling stand for optimum use with the existing system. Covers of the gantry must be removable for servicing.

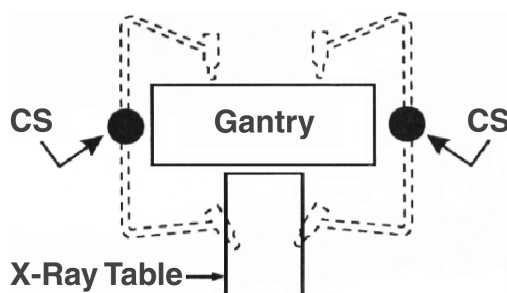


Figure 1-69 Ceiling Stand Position

- The position of the ceiling mount assembly must allow the ceiling stand movement and gantry to tilt, typically 20 degrees, without interference.

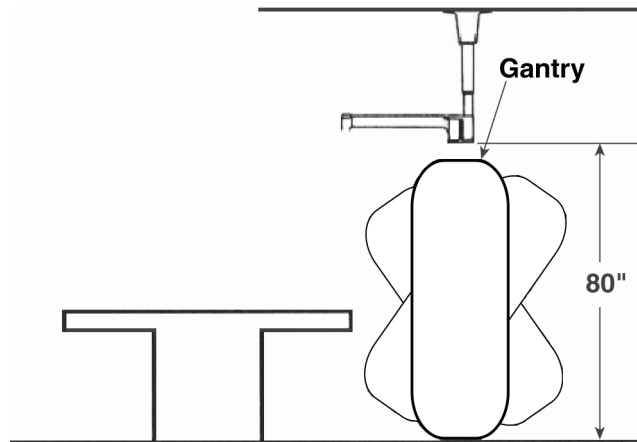


Figure 1-70 Ceiling Mount Position

10.3 Advantage Windows Workstation (AWW)

Refer to the directions provided:

- Pre-install 2111833
- Service 2111831

10.4 Uninterruptible Power System (UPS)

Refer to directions shipped with uninterruptible power system (UPS):

- Powerware B7999PS
- Liebert B7999PT
- Installation support phone numbers found in shipped documentation or in *HiSpeed QX/i Pre-Installation Manual*.

10.5 Other Options

Install all other options now.

Chapter 2

Power, Ground & Interconnect Cables



NOTICE
Potential for
Data Loss and/
or Equipment
Damage

To prevent potential data loss and equipment damage, please do the following:

- Record data collected from procedures in this chapter into Form F4879 when directed, located in [Chapter 13](#) of this book.
- Only use the Installation manual that arrives with your system for installation. Any other revisions of this manual may not exactly match your system.

Section 1.0

Scanner Configuration Variations

The HiSpeed QX/i scanner can be configured with either of two possible Power Distribution Units (PDUs) and either of two possible operator consoles.

1.1 PDU Varieties

The two variations of Power Distribution Unit that may be shipped with the HiSpeed QX/i scanner are the NGPDU and the CPDU (Compact PDU). The chief distinguishing features of the NGPDU (compared to the CPDU) are that it has a green power indicator LED and a red e-stop button on the front cover (see [Figure 2-1](#)).



Figure 2-1 NGPDU power indicator lamp



NOTICE

Proper identification of the PDU variety is crucial, as the connections differ between the NGPDU the CPDU.

1.2 Console Varieties

The two variations of Operator Console that may be shipped with the HiSpeed QX/i scanner are the Global Console - Octane2 (a.k.a. "GC-Oct2" or "GOC1") and the Global Console - Linux (a.k.a. "GC-Linux" or "GOC2"). The only external cue to the difference between these two consoles is the different configuration of the SCSI Tower, which sits on top of the console. The GC-Oct2 SCSI Tower contains a CD-RW drive and a MOD drive. The GC-Linux exchanges the CD-RW drive for a DVD-RAM drive.

As their names indicate, the key difference between these two operator consoles is the computer housed within. The GC-Oct2 uses an SGI Octane2 computer, running IRIX OS. The GC-Linux, however, uses a HP workstation, running Linux OS. With the front cover off, the GC-Linux may be readily identified by the fact that the workstation sits in front of the Recon box, while the opposite is true for the GC-Oct2.

Section 2.0 Introduction

Site use of conduit, floor duct, wall duct, or a raised computer floor, as well as the individual component layout determines the system cable sequence. If your site has floor or wall ducts that will interfere with placement of the table/gantry, it may be important to have the movers unload the cable boxes (8 & 9) first and run those cables while others unload the subsystems.

- Try to run the system cables after the contractor completes the contractor supplied wiring.
- All ground wires and other contractor wiring should be complete to the point of equipment placement.



NOTICE
Potential for
Equipment
Damage

Do not store excess cable in the bottom of the PDU or Gantry.

When possible, store excess cable length in a serpentine configuration, approximately one meter long. **(Do NOT coil excess cable.)**

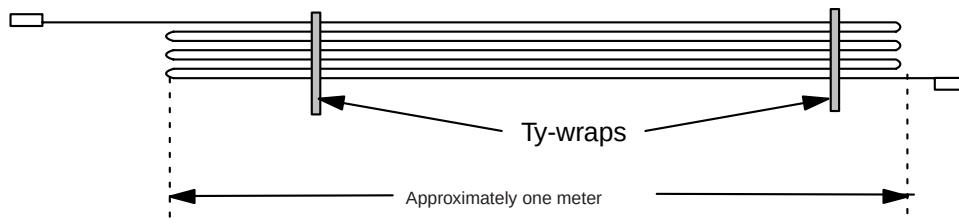


Figure 2-2 Excess Cable Storage Configuration

- Keep signal and control cables away from power cables and power wiring. When you lay cables in a raceway, locate the signal cables in a separate section of the raceway, or a separate conduit.
- Check all connections for tightness.
 - Use suitable tools and judgment.
 - Check all visible connections, especially ground connections.
- Check for reasonable cable routing.
 - Take into consideration necessary take-up distances for equipment maintenance, etc.
 - Try to complete as neat a job as possible.

2.1 System Component Identification

Identify all system cables by the system component designators listed in [Table 2-1](#). Each end of a system cable has a label, and may have a color near the connector, (refer to [Table 2-2](#)) to indicate the component and the jack identifier of the component.

DESIGNATOR	SYSTEM COMPONENT
CT2	Gantry
CT1	Patient Table
PM	Power Distribution Unit
OC	Operator Console (Console Computer)
WL	X-Ray ON Warning Light

Table 2-1 System Component Identifiers

2.2 Cable Color Identifiers

The ends of the cables may be marked with a piece of blue, yellow, red, or orange colored tape to help with the cable installation. [Table 2-2](#) lists the subcomponent, and corresponding color.

SUBCOMPONENT	COLOR
Gantry	Blue
Table	Yellow
PDU	Red
Console Computer	Orange

Table 2-2 Cable Color Identifiers

QTY	RUN DESCRIPTIONS	PART NUMBER LONG CABLE SET (B7850LL)
1 PC	PDU J5 TO CONSOLE POWER PAN	2335376
1 PC	GROUND - PDU A1 TO TABLE RACEWAY 1/0	2275844
1 PC	GANTRYPOWER PAN TO TS1 PDU HVDC	2267642
1 PC	PDU J4 TO GANTRY POWER PAN 120VAC	2266880
1 PC	GANTRY POWER PAN TO PDU A3 HVAC	2267644
1 PC	GROUND - CONSOLE #2 TO TABLE RACEWAY	2275844-2
1 PC	GANTRY STC J8 TO PDU J2	2266014
1 PC	STC J7 TO CONSOLE SCAN INTL J20	2271060
1 PC	CONSOLE LAN TO GANTRY POWER PAN	2266887
1 PC	FIBER OPTIC CONSOLE J52 TO GANTRY PAN	2117848-2

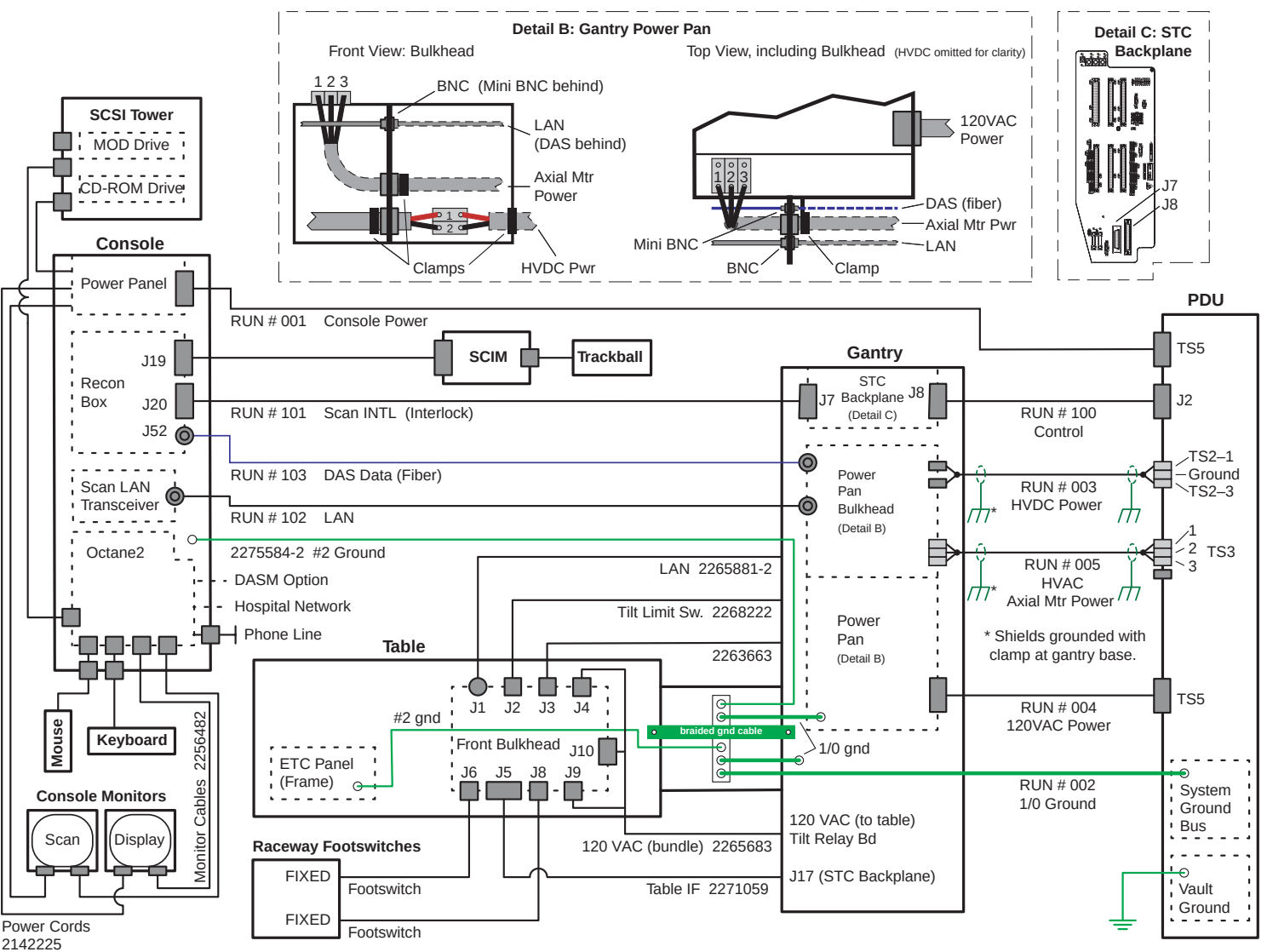
Table 2-3 Cables (CPDU used)

QTY	RUN DESCRIPTIONS	PART NUMBER
		LONG CABLE SET (B75942AC)
1 PC	PDU TS5 TO CONSOLE POWER PAN	2347378
1 PC	GROUND - PDU A1 TO TABLE RACEWAY 1/0	2275844
1 PC	GANTRYPOWER PAN TO TS2 PDU HVDC	2347513
1 PC	PDU TS5 TO GANTRY POWER PAN 120VAC	2347377
1 PC	GANTRY POWER PAN TO PDU TS3 HVAC	2347514
1 PC	GROUND - CONSOLE #2 TO TABLE RACEWAY	2275844-2
1 PC	GANTRY STC J8 TO PDU J2	2266014
1 PC	STC J7 TO CONSOLE SCAN INTL J20	2271060
1 PC	CONSOLE LAN TO GANTRY POWER PAN	2266887
1 PC	FIBER OPTIC CONSOLE J52 TO GANTRY PAN	2117848-2

Table 2-4 Cables, for use with NGPDU

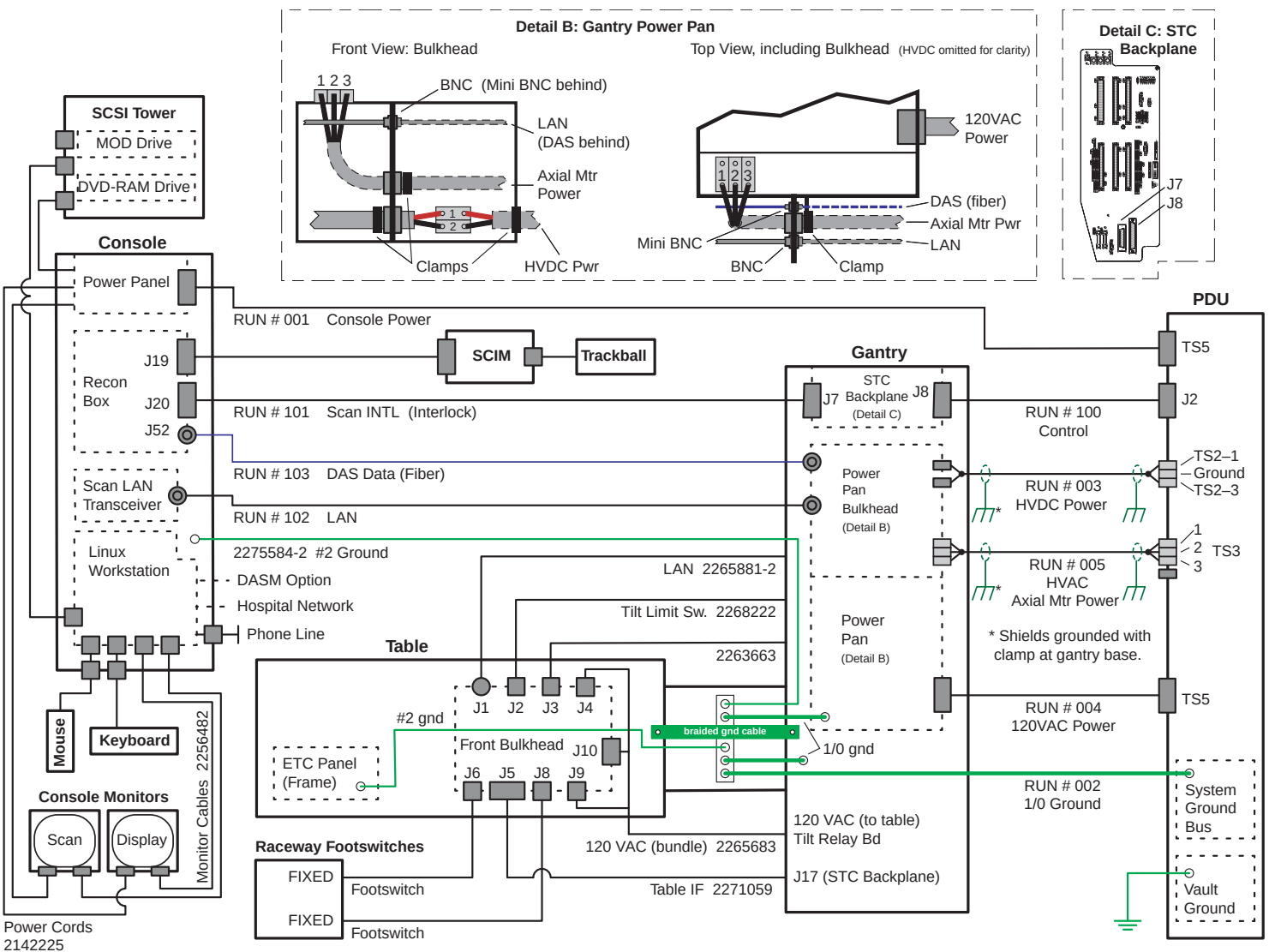
Section 3.0 System Interconnect Diagrams

3.1 System Configured with NGPDU & Global Console - Octane2



rev. 12/06/02

3.2 System Configured with NGPDU & Global Console - Linux



rev. 12/06/02

Section 4.0 Cable (Run) Specification

Run #	Length, Actual (Usable)		Part #	Description	UL Cable Information								Pull Size mm (Inches)
	ft	m			UL Style	Flam. Rating	Voltage Rating	Voltage Actual	Temp. Rating (C)	Dia. mm (inch)	# of Cond	Size AWG	
001	80 (75)	24.5 (22.86)	2335376* 2347378 ^N	Power - PDU to Console	2587 2586 ^N	FT-4	600	208Y/120	90 105 ^N	12.3 (.483)	4	10	56.4 (2.22) Dia
002	63 (55)	19.3 (16.76)	2275844	Ground - PDU to Raceway	1284	VW-1 (FT-1)	600	0	105	15.5 (.608)	1	1/0	15.8 (.62) Dia
003	63 (55)	19.3 (16.76)	2267642 2347513 ^N	HVDC - PDU to Gantry	2587 2586 ^N	FT-4	600	+ & - 350VDC	90 105 ^N	19 (.751)	2	6	19.8 (.78) Dia
004	60 (55)	18.5 (16.76)	2266880 2347377 ^N	Power - PDU to Gantry	2587 2586 ^N	FT-4	600	208Y/120	90 105 ^N	13.8 (.542)	5	8	56.4 (2.22) Dia
005	62.5 (55)	19 (16.76)	2267644 2347514 ^N	HVAC - PDU to Gantry	Flexible Motor Supply Cable 2586 ^N	FT-4 TC	1000 600 ^N	440Y/254	90 105 ^N	15.3 (.604)	4 3 ^N	12	16.8 (.66) Dia
006	83 (75)	25.5 (22.86)	2275844-2	Ground - PDU to Console	1283	VW-1 (FT-1)	600	0	105	11.9 (.467)	1	2	12.2 (.48) Dia
100	63 (55)	19.3 (16.76)	2266014	Signal - PDU to Gantry STC	UL	FT-4	300	<30VDC	80	13.3 (.525)	37	22	20 x 75 (.78 x 2.95) 20 x 51 (.79 x 2.01)
101	83 (75)	25.5 (22.86)	2271060	Signal - Console to Gantry STC	UL	FT-4	300	<30VDC	80	11.2 (.440)	25	22	17 x 58 (.68 x 2.30) 19 x 51 (.75 x 2.01)
102	80 (75)	24.3 (22.86)	2266887	Signal - LAN Console to Gantry	UL (RG-223/U)	FT-4	1900	<30VDC		5.9 (.234)	1	19	15 (.59) Dia
103	80 (75)	24.3 (22.86)	2117848-2	Fiber Optic - Console to Gantry			NA	NA			1	NA	10 (.39) Dia

* Shipped with Global Console

^N For NGPDU

Table 2-5 GEMS Supplied Cables (Long Run) - UL Information

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Section 5.0

Contractor Connections

CONNECTION OR WALL BOX	AWG #	CONNECTION FROM	CONNECTION TO PDU	INSTALLED AND CHECKED
A1	#1	4A	A3 TB1 L1	
	#1	4B	A3 TB1 L2	
	#1	4C	A3 TB1 L3	
	#1/0	GND	A3 TB1 GND (Do NOT connect anything to neutral point.)	
WL (Warning light) See Figure 2-32 , on page 99 .	#12	WL-0A (Black) (from external low voltage (<30v) power source)	A4 WL Connection 1	
	#12	Neutral (White) (from external power source)	A4 WL Connection 2	
	#12	PM-A4 WL Connection 3	0A (Black) (to external warning light relay)	
	#12	PM-A4 WL Connection 4	Neutral (White) (to external warning light relay)	
	#12	Ground (from external power source)	Ground (to warning light ground)	
Door Interlock	—	Room Door Interlock Connections	A4 TS2 - 1,2	

Table 2-6 Contractor Connections (Compact PDU)

CONNECTION OR WALL BOX	AWG #	CONNECTION FROM	CONNECTION TO PDU	INSTALLED AND CHECKED
A1	#1 #1 #1 #1/0	PDB-A PDB-B PDB-C GND	TS1-L1 TS1-L2 TS1-L3 Ground Block (Do NOT connect anything to neutral point.)	
WL (Warning light) See Figure 2-32 , on page 99.	#12	WL-0A (Black) (from external low voltage (<30v) power source)	TS6-1	
	#12	Neutral (White) (from external power source)	TS6-2	
	#12	0A (Black) (to external warning light relay)	TS6-3 (X-ray On) TS6-7 (Ready)	
	#12	Neutral (White) (to external warning light relay)	TS6-4 (X-ray On) TS6-8 (Ready)	
	#12	Ground (from external power source)	Ground (to warning light ground)	
Door Interlock	—	Room Door Interlock Connections	TS6-9, 10	

Table 2-7 Contractor Connections (NGPDU)

Note:
Add #2 ETC
ground wire.

IMPORTANT: Add AWG #2 ground wire from Table ETC frame to Table/Gantry raceway ground bar (as shown in [Figure 2-3](#) or [Figure 2-4](#)).



WORK WITH THE ELECTRICAL CONTRACTOR TO BE SURE EXTERNAL POWER SOURCE IS TURNED OFF.

Section 6.0 Console Connections

6.1 Accessing the Computer

- 1.) Loosen the two (2) captive screws, and gently pull the tray forward, using the attached strap. Refer to [Figure 2-5](#).



Figure 2-5 Console Slide Out Tray (GC-Linux shown)

- 2.) Connect the hospital ethernet cable to ethernet port on the rear of the computer, routing the cable through the console's rear access flap.
- 3.) Slide the tray back into the console and tighten the two (2) captive screws.

6.2 SCIM, Keyboard, Trackball & Mouse Installation

- 1.) Route the keyboard cable under the SCIM, as shown in [Figure 2-6](#).



Figure 2-6 SCIM control with keyboard cable routed through SCIM



NOTICE
Potential for
equipment
damage

Note:

Never connect a mouse or keyboard with the host computer powered “ON”. Doing so can destroy components within the host computer.

- 2.) The keyboard and mouse connect to the ports on the top of the console front bulkhead. The mouse connects to the port on the right, the keyboard connects to the left port (Figure 2-7).
The front cover cannot be installed with these connectors in place.

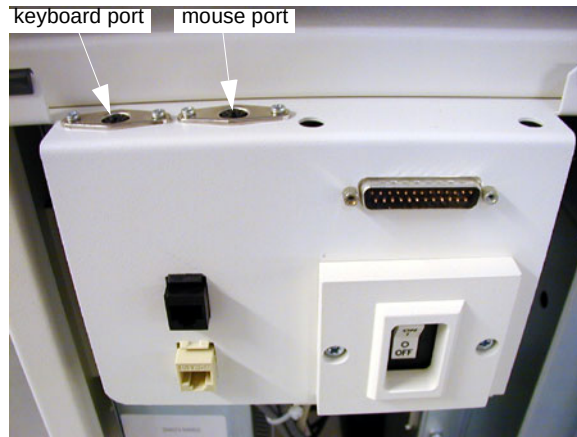


Figure 2-7 Global Console front bulkhead, showing keyboard and mouse connections

- 3.) Connect the SCIM cable to the SCIM as shown in Figure 2-8 (note the cable routing).



Figure 2-8 SCIM bottom, showing cables and keyboard mounting bracket

- 4.) Connect the trackball cable directly to the SCIM (see Figure 2-8).
- 5.) Select and install the proper overlay for your system: (1) with Tilt or (2) without tilt.
Verify that none of the buttons get caught and stuck under the overlay. Pay close attention to the prescribed tilt button on systems with the tilt feature.

- 6.) The keyboard should attach to the SCIM using the supplied Velcro strip and fit snugly against the SCIM when finished, as shown in [Figure 2-9](#).



Figure 2-9 SCIM connected to the keyboard with the US English tilt overlay installed

6.3 Connecting the SCSI Tower

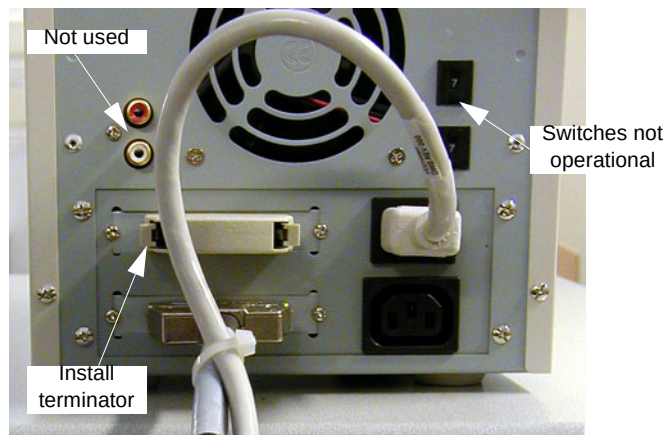


Figure 2-10 SCSI tower: Connections (left) and Optional Seismic Mounting Bracket (right)

- 1.) Connect the SCSI cable to the rear of the SCSI tower.



NOTICE
Potential for
Equipment
Damage

The connectors for the SCSI tower and the SCIM look similar. Make certain to use the proper connector (the SCSI tower cable is shorter, and is ty-wrapped to the power cable). Attaching the SCIM cable to the SCSI tower can result in serious damage to the SCSI tower components.

- 2.) Connect the SCSI terminator to the rear of the SCSI tower.
- 3.) Connect the power cable to the rear of the SCSI tower.

6.4 Connecting the Monitor

- 1.) With the monitor and computer switched off, connect the video signal cable to the monitor's video input (see [Figure 2-11](#)) and to the computer's video output.



NOTICE
Equipment
Damage
Possible

Do not touch the video signal cable connector pins as this might bend them. When connecting the video signal cable, check the alignment of the HD15 connector. Do not force the connector in the wrong way, or the pins might bend.

- 2.) Connect the power cord to the monitor, then connect it to console power panel outlet J1 or J7.

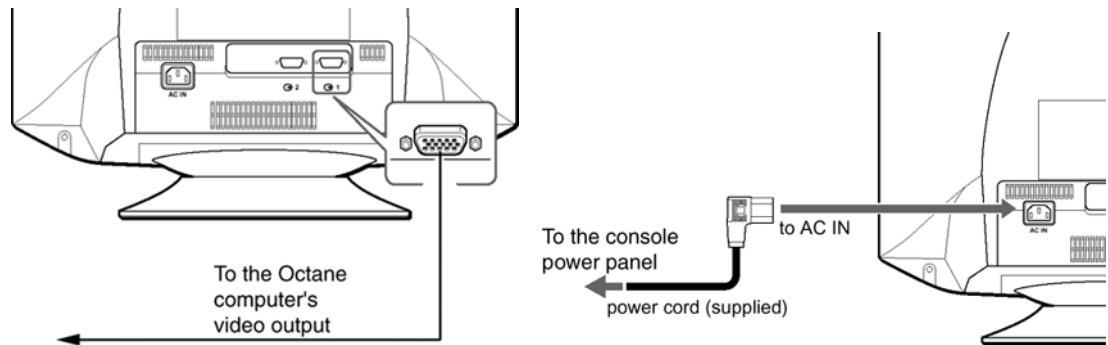


Figure 2-11 Monitor Connections (Sony Model GMD-F520 -- GE #2333243)

6.5 Power Panel Connections

- 1.) Connect the console power cable to the console power panel.
- 2.) Connect console component power cords as listed in [Table 2-8](#). ("J numbers" increment from top to bottom, left to right)

J#	DEVICE	J#	DEVICE
J1	Host Computer	J9	Display Monitor
J2		J10	Scan Monitor
J3	Remote Monitor (option)	J11	
J4	Video Splitter (option)	J12	Modem
J5		J13	SCSI Tower
J6	Fast Ethernet LAN Switch	J14	
J7	Recon Box	J15	
J8	Powered Transceiver	J16	Console Fans

Table 2-8 Power Panel Outlet Assignments - GC-Octane2

6.6 Linux PC Connections

[Figure 2-12](#) (next page) shows the connections on the rear of the Linux-based HP xw8000 computer. Note the connections to the ethernet switch (left port) and the hospital network (right port), at the bottom of the computer chassis.

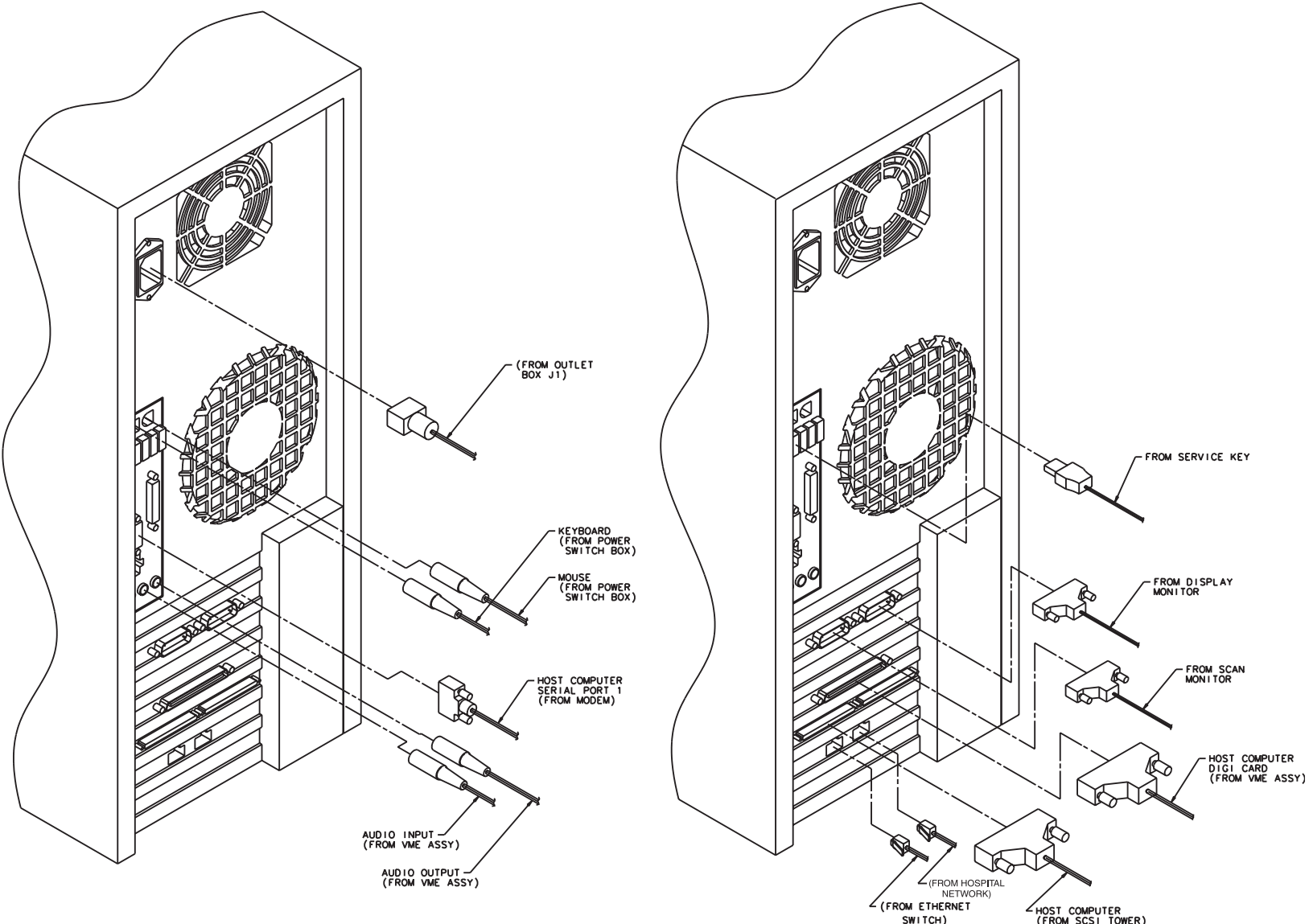


Figure 2-12 Connections on Rear of HP xw8000 (GC-Linux only)

6.7 LAN Connections

Install LAN cable (2266887) onto transceiver (route cable through console rear access flap).

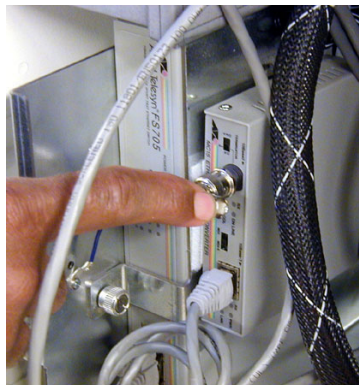


Figure 2-13 LAN Connection

6.8 Installing a DASM (Hardware Option)

If you have a DASM to install, perform the following procedure now.

Although DASMs are available in two types, digital and analog, setup is identical. Complete the following steps to connect a DASM to the computer.

Note: For GC-Linux computers (HP xw8000), **DASM II** models only can be installed. DASM I models (rectangle models) are not supported on GC-Linux computers.

Note: If you have a digital DASM II, but do not have a DASM (digital) kit (P/N: 2337592), you need to order as a minimum the following to connect it to the computer.

2324697-2 (SCSI cable)

46-269601P5 (Terminator)

(The internal termination does not work and the external terminator is required.)

- 1.) Turn the DASM over so you are looking at the bottom.
- 2.) Set the DASM SCSI ID to 1.
- 3.) Set the DASM termination to ON.

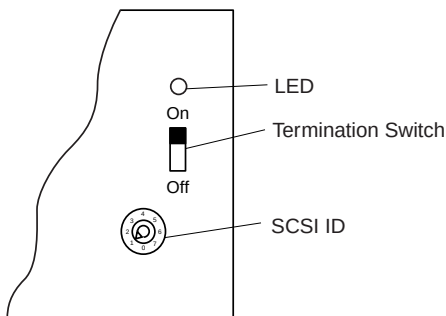


Figure 2-14 DASM Bottom

- 4.) Set the RS232/RS422 switch to the RS232 position, if you have a digital DASM.
- 5.) Attach the SCSI cable supplied with DASM kit to the back of the DASM.

6.) Attach the free end of the SCSI cable to the rear of the Host Computer (see [Figure 2-15](#)).

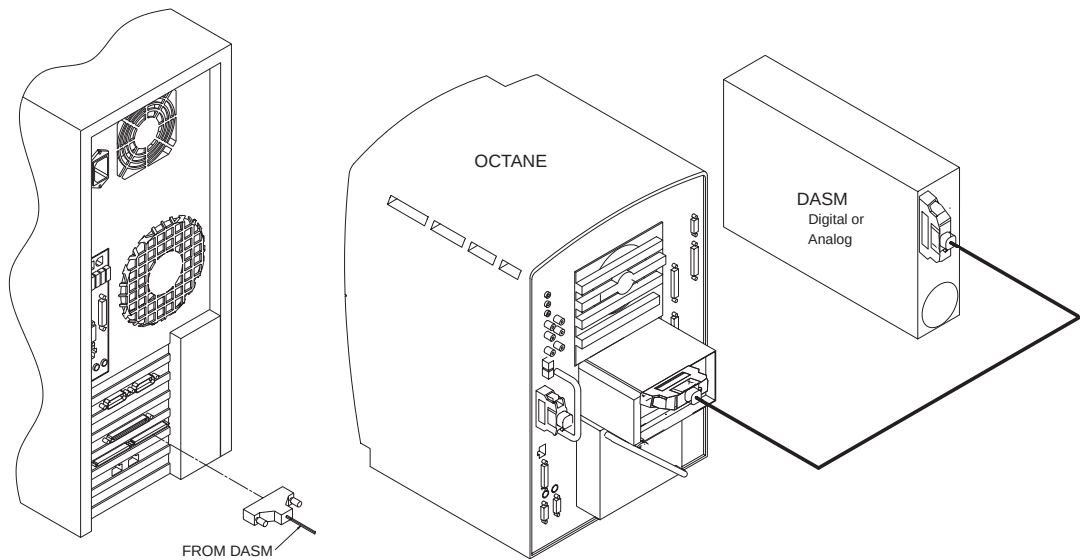


Figure 2-15 SCSI Cable Connection for DASM (HP xw8000 at left, Octane at right)

- 7.) Plug DASM power cord into J10.
- 8.) Attach remaining cable(s) from camera to DASM and install the DASM to the right of the slide-out tray.

Note: See your *Software Installation Procedures* manual for instructions on configuring cameras.

6.9 Bulkhead Connections

Make the following connections to the bulkhead on the front of the Recon Box (route the cables through one of the console’s rear service flaps):

J#	CABLE DESCRIPTION
J52	Fiber Optic Cable (2117848-2)
J20	Scan INTL Cable (2271060)

Table 2-9 Cables Connections to Recon Box Bulkhead

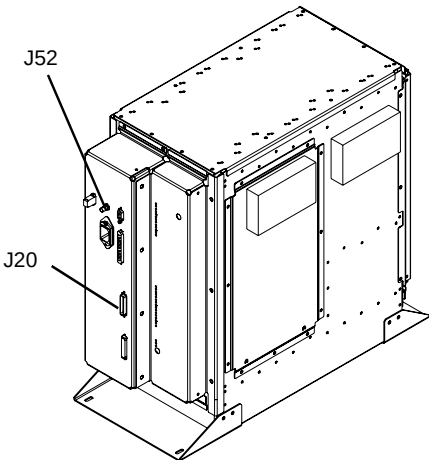


Figure 2-16 Recon Box (showing bulkhead)

Section 7.0

Gantry Cable Connections

Please refer to [Figure 2-3](#) or [Figure 2-4](#) for complete system interconnect details.

TO	FROM	CABLE DESCRIPTION
Gantry Power Pan	PDU	HVDC
Gantry Power Pan	PDU	440VAC
Gantry Power Pan	PDU	120VAC
Gantry Power Pan	Console	Fiber - <i>Take extreme care when you install the fiber optic DAS data cable. Do not step on, kink, or sharply bend this fragile DAS cable.</i>
Gantry Power Pan	Console	LAN
STC Backplane (J7)	Console	Control
STC Backplane (J8)	PDU	Control

Table 2-10 Gantry Cable Connections

- 1.) If using a rear cable entry box (B7850RC), install it now, before routing cables to gantry.
- 2.) Install the cables to the gantry power pan. The power pan is located on the rear of the gantry at its base. See “Detail B” ([Figure 2-17](#)) of the system interconnect drawing for connections.

Note: The gantry 120VAC cable may not fit under the gantry frame. Install this cable before gantry placement—or remove the power plug—to route it under the gantry.



NOTICE
Potential for
equipment
damage.

Observe correct polarity when connecting the high voltage DC power. Reversing these leads will result in serious equipment damage. The HVDC positive conductors have red insulation and are labeled “ONE.” The HVDC negative conductors have black insulation and are labeled “TWO.” Lead “ONE” must be connected to lead “ONE,” and lead “TWO” must be connected to lead “TWO.”

Observe correct phase rotation when connecting the axial motor power. Phases one, two and three should be connected left to right, as shown in [Figure 2-17](#).

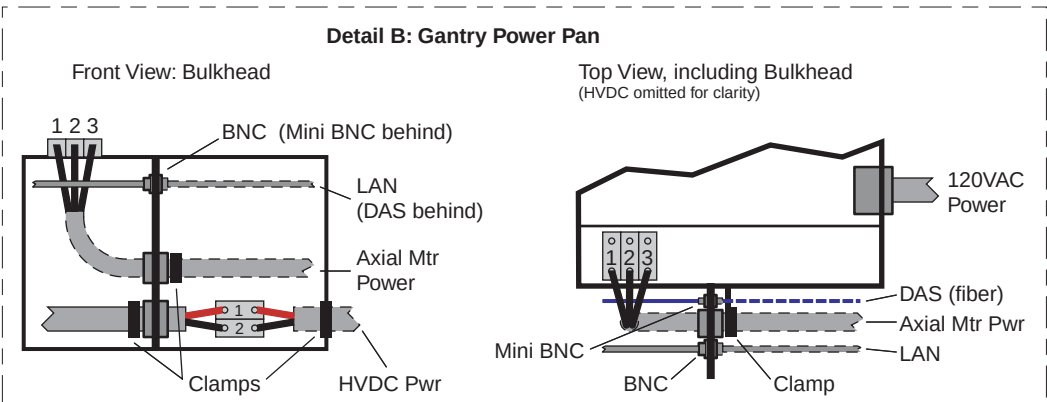


Figure 2-17 Power Pan Connections

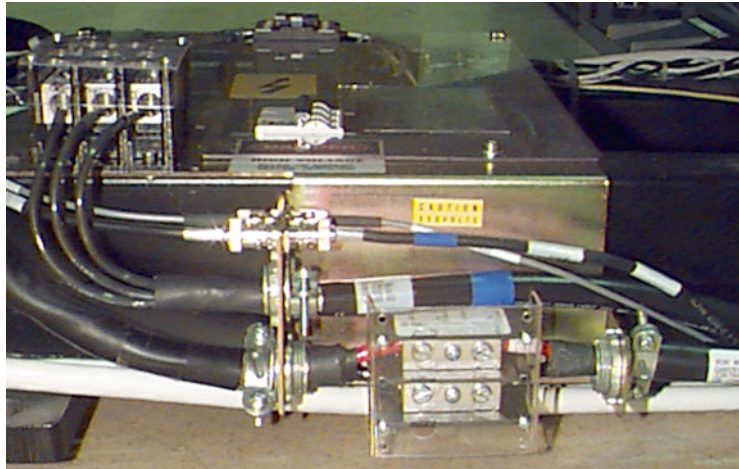


Figure 2-18 Gantry Power Pan

- 3.) Install cables to the STC backplane.

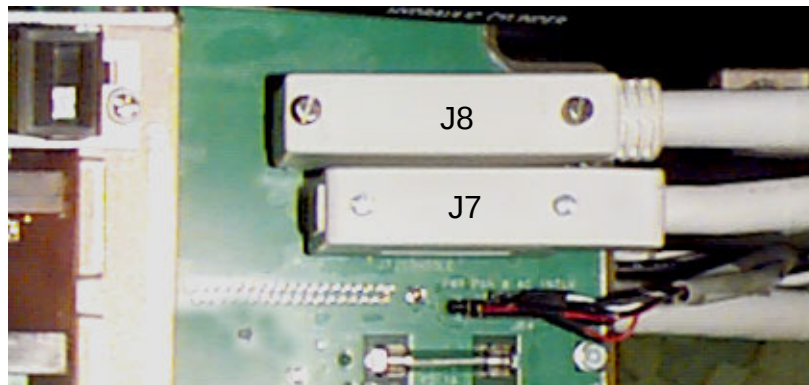
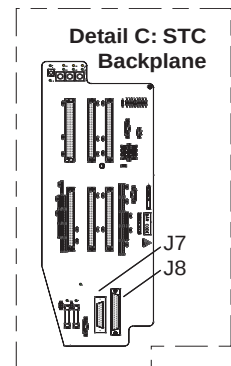


Figure 2-19 A1 STC Connections



Section 8.0 Option Cables

Run all option cables that connect to the Table, Gantry or Console.

Section 9.0

Table Connections

Pull and connect the following cables:

J#	CABLE DESCRIPTION
J1	LAN (2265881-2)
J2	Tilt Limit Switch (2268222)
J3	CAN BUS (2263663)
J4, J9, J10	120 VAC bundle (2265683)
J5	Table IF (2271059)
J6, J8	Raceway Foot switches
ETC frame	AWG #2 ground to table/gantry raceway

Table 2-11 Cables Connected to Table

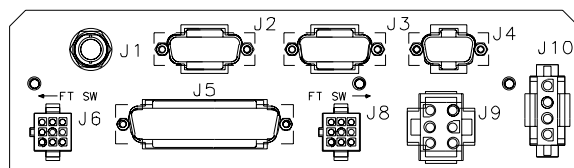


Figure 2-20 Table Bulkhead Connections

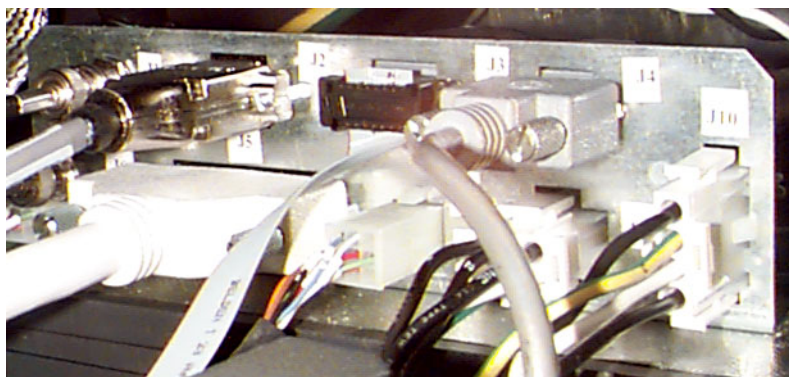


Figure 2-21 Table Bulkhead

Section 10.0

PDU Cable Connections & Configuration



CAUTION



Do not work in an energized PDU. When working on the PDU, follow this simple rule: Always tag and lock out power to the PDU at the “main” disconnect. Failure to do so can result in electrocution or death.

Do apply power to the PDU until all work has been completed and all PDU covers are in their proper place.

A number of cables must be connected to the Power Distribution Unit before it will operate properly. In addition: taps, jumpers and switches may have to be moved from their default positions. In this section you will connect cables to the PDU and verify and/or change settings to match your site's specific needs.



NOTICE

Two PDU varieties

Verify the variation of PDU supplied with your scanner before proceeding. The NGPDU has a green “Power ON” indicator and a red e-stop button on the front cover, whereas the CPDU does not (See [Section 1.0 on page 71](#) for further details).

- CDPU connections are covered in Section 10.1
- NGPDU connections are covered in Section 10.2

10.1 Compact PDU

As seen in [Figure 2-22](#), a number of cables must be installed throughout the PDU. Specific details on each connection can be found in the sub-sections that follow. Use [Figure 2-22](#) for reference. The PDU has been designed to have cables routed into the PDU from its behind and/or from beneath it.

Compact PDU (Covers Removed)

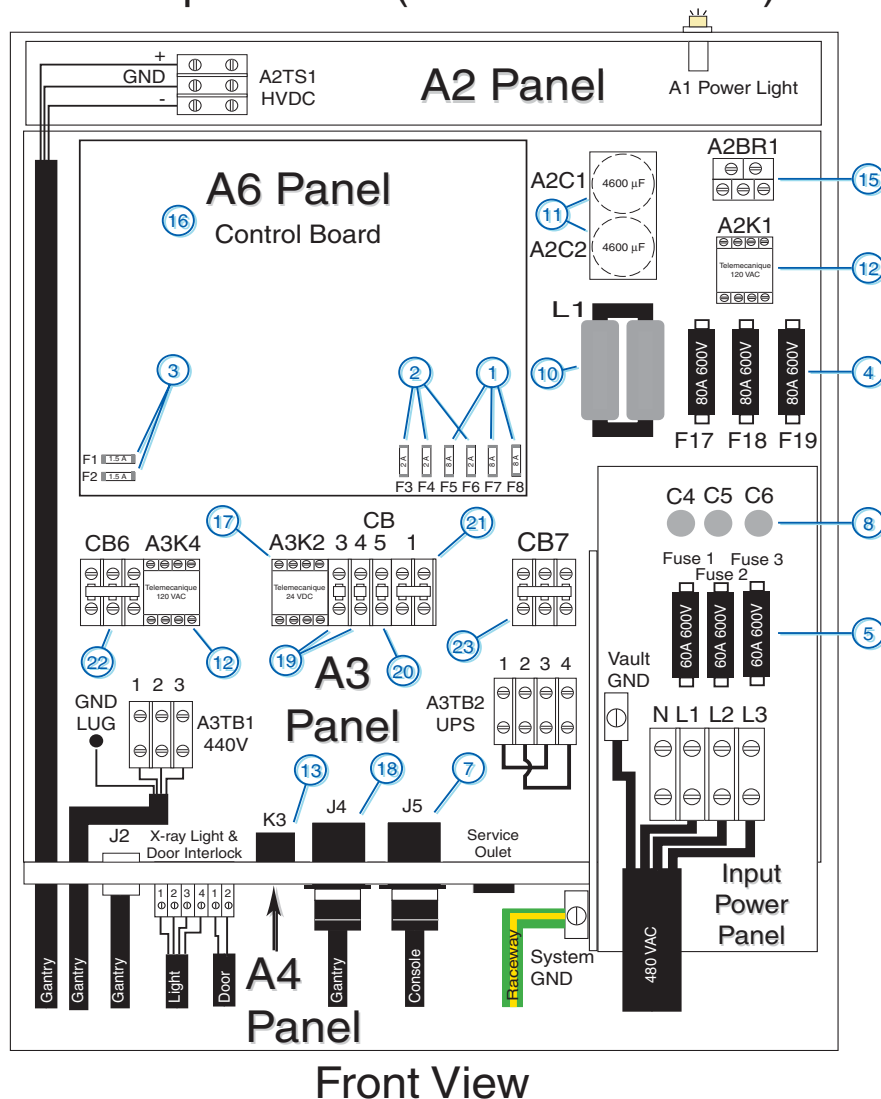


Figure 2-22 PDU Cable Connections - Front

Compact PDU (Covers Removed)

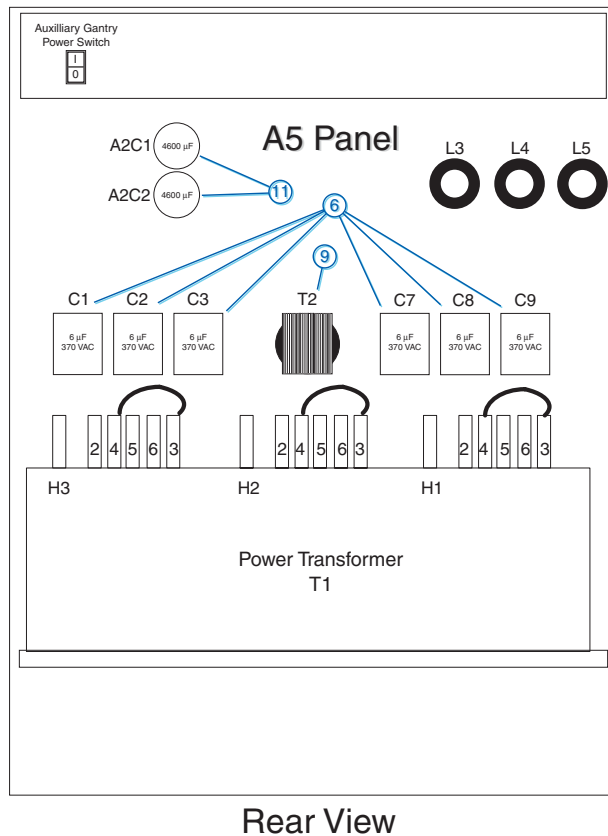


Figure 2-23 PDU Cable Connections - rear

10.1.1 A3 Panel - 380 - 480VAC Mains “A1” Input Power Connection

- 1.) Remove the A3 panel front cover.
- 2.) Strip the wires to fit securely on the power block.
- 3.) Observe incoming phases (L1, L2 and L3) and insert bare leads into power block.
- 4.) Insert "vault" ground into PDU "vault" ground lug.
- 5.) Tighten all fasteners securely and replace the A3 front panel.

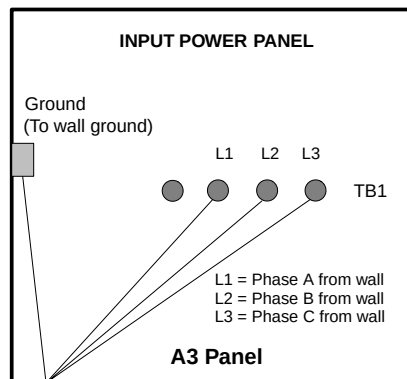


Figure 2-24 A3 Input Power Panel Connections

When Mains power is available to the PDU, the “A1” power light will be illuminated.
(See [Figure 2-22.](#))

10.1.2 A3 Panel - Circuit Breakers

Place the circuit breakers in the “off/down” position during installation, even with Mains incoming power tagged and locked out. After you have completed work on the PDU, you may return the circuit breakers to the “ON” positions.

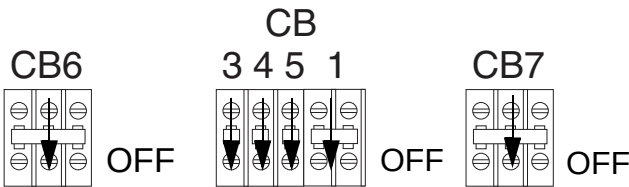


Figure 2-25 A3 Circuit Breaker Panel

By design, when CB7 is in the "OFF" position, circuit breakers 3, 4, 5, and 1 are switched "OFF". CB7 is essentially in series with these breakers.

CIRCUIT BREAKER	DESCRIPTION
CB6	Axial Drive Power, 440VAC
CB3	Service Outlets (PDU, Table and Gantry), 120VAC
CB4	24 Hour Power (STC and Table), 120VAC
CB5	Gantry Rotating Power (Cover fans and axial drive)
CB1	Console Power, 208VAC
CB7	Master Power (CB3, 4, 5 and 1), 120 and 208VAC

Table 2-12 A3 Panel Circuit Breaker Descriptions

10.1.3 A5 Panel - Transformer (480VAC) Taps

Verify that the transformer taps are set properly. The transformer taps are set to 480 VAC operation at the factory. The taps should be set as shown in [Figure 2-26](#).

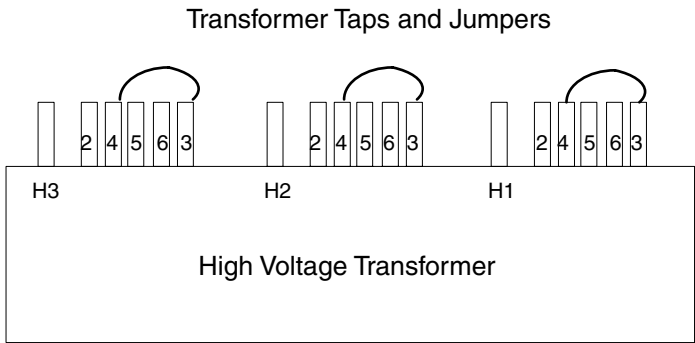


Figure 2-26 PDU Tap Positions for 480 volt operation

Adjustment of the 480VAC and these “taps” is done in [Chapter 5, Section 4.0 on page 162](#).

10.1.4 A2 Panel - HVDC Connection

Connect the internally shielded HVDC cable to A2TS1 on the A2 panel. See [Figure 2-22](#) for the location of the connector and [Figure 2-27](#) for details. Observe polarities and grounds.

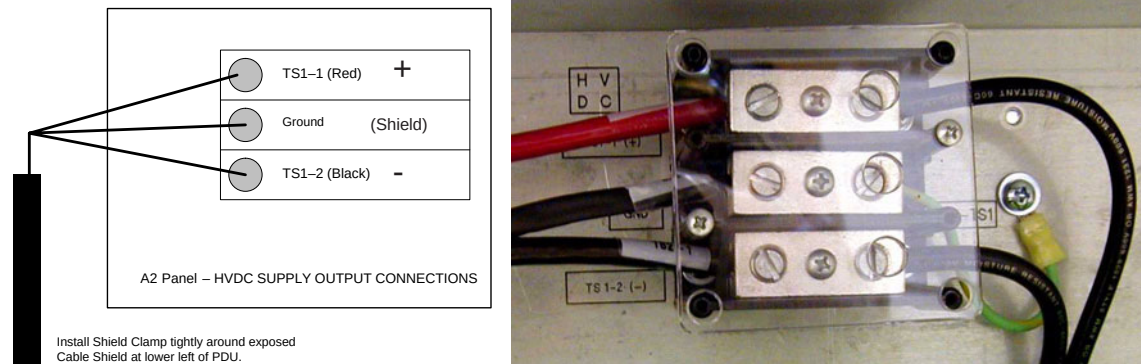


Figure 2-27 HVDC Connection

The HVDC cable can be re-terminated to eliminate excessive cable length, if necessary. Observe cable colors and polarities during re-connection. The cable's shield acts as the ground.

10.1.5 A3 Panel - 440V Connection

Connect the internally shielded 440V cable from the gantry to A3TB1 on the A3 panel. See [Figure 2-22](#) for the location of the connector and [Figure 2-27](#) for details. Observe the labels on the cable leads for proper identification and orientation.

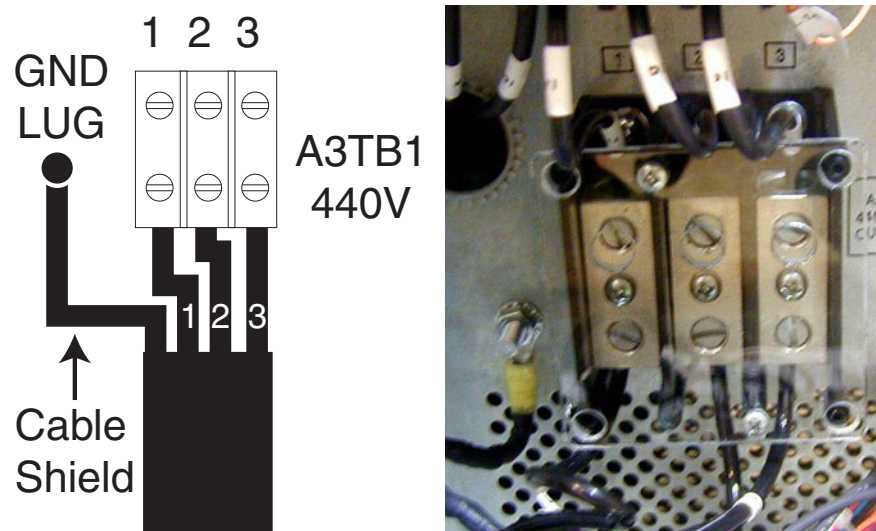


Figure 2-28 440VAC Connection

The 440V cable can be re-terminated to eliminate excessive cable length, if necessary.

10.1.6 A4 Panel - Gantry & Console Power Connections

Both Gantry and Console power cables come pre-terminated. Simply plug the Gantry power cable into “J4” and the Console power cable into “J5” as shown in [Figure 2-29](#).



Figure 2-29 Gantry & Console Power Connections

Both the Gantry and Console Power cables can be re-terminated at the PDU end, if excessive cable length prevents the proper storage.

10.1.6.1 Console Power Cable Re-termination

- 1.) Carefully remove the power plug and **record the color of the wires** in [Table 2-13](#). The terminals are labelled X, Y, W and G on the plug.

TERMINAL	X	Y	W	G
Description	Hot	Hot	Neutral	Ground
Color				

Table 2-13 Console Power Cable Termination

- 2.) Cut the cable to desired length and dress ends.
- 3.) Re-install the power plug, according to the orientations recorded in [Table 2-13](#).
- 4.) Verify that less than 1 ohm of resistance exists between the following connections:

FROM PDU PLUG	TO CONSOLE PLUG END	
CB1 -11 (Black) (J5 Phase X (Brown))	Phase X (Brown)	☐ Check box when complete
CB1-12 (Black) (J5 Phase Y (Black))	Phase Y (Black)	☐ Check box when complete
A3 Neutral Buss Bar (Blue) (J5 -13 Phase W)	Phase W (Blue) Neutral	☐ Check box when complete
A3 Ground Buss Bar (J5 -22 Ground Green)	Ground (Green) Ground Green screw	☐ Check box when complete

Table 2-14 Resistance Verification Points

10.1.6.2 Gantry Power Cable Re-termination

- 1.) Carefully remove the power plug and **record the color of the wires** in [Table 2-15](#). The terminals are labelled X, Y, Z, W and G on the plug.

TERMINAL	X	Y	Z	W	G
Description	Hot	Hot	Hot	Neutral	Ground
Color					

Table 2-15 Gantry Power Cable Termination

- 2.) Cut the cable to desired length and dress ends
- 3.) Re-install the power plug, according to the orientations recorded.
- 4.) Resistance checks will be verified later.

10.1.7 A4 Panel – PDU Control Cable

The PDU control cable comes pre-terminated and should not be re-terminated in the field. Excess cable length must be stored. Simply plug the cable into "J2" on the "A4" panel. Secure it by using the fasteners intergrated into cable's connector shell.

10.1.8 A4 Panel – System Ground Connection

Connect the ground wire (green with a yellow strip) from the Table/Gantry raceway ground bus to the system ground lug in the PDU. See [Figure 2-22, on page 93](#), and [Figure 2-30, below](#).



Figure 2-30 PDU System Ground Connection

10.1.9 A4 Panel – X-Ray Light & Door Interlock Connections

10.1.9.1 X-Ray Light Configuration & Connection

- 1.) Configure PDU control board for correct x-ray light operation. Determine whether the x-ray light should come on when high voltage is present or when x-ray is present. The board is shipped with the x-ray light turning "ON" when x-ray is present only. To change this, go to the PDU Control Board (see [Figure 2-22, on page 93](#)), and move the jumper from position "A" to position "B". See [Figure 2-31](#). In the "B" position, the x-ray light illuminates whenever high voltage is present.

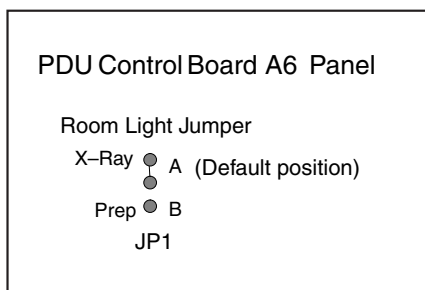


Figure 2-31 X-ray Light Configuration

- 2.) This step is site specific. The PDU by default is configured for “no” external x-ray light connection. If you have an external x-ray light, see [Figure 2-32](#) for proper connection.

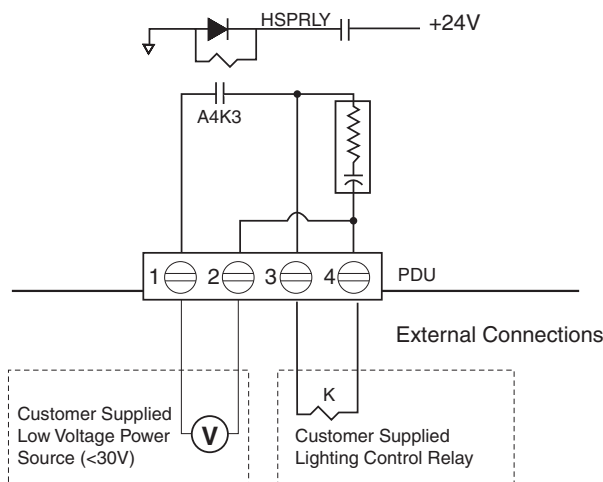


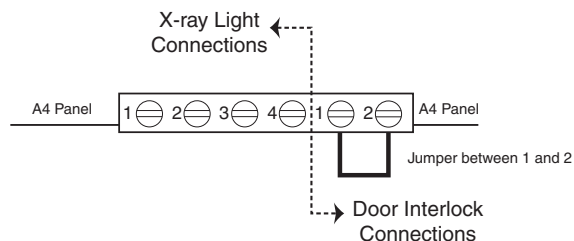
Figure 2-32 External X-ray Light Connection

It is recommended that you use the four (4) wire method of adding a X-ray warning light to a room, as shown in [Figure 2-32](#). When using this method, you:

- Minimize EMC interference.
- Increase contact life of the relay used in the PDU.

10.1.9.2 Door Interlock Connections

Door interlocks are used to prevent X-Rays from being generated when the scan room door is open. The Door Interlock circuitry in the PDU is shipped from the factory engaged. This means the system cannot generate X-ray until disengaged. A short must exist between pins 1 & 2 for X-ray to be generated. Using a small piece of wire, short pins 1 and 2 together. See [Figure 2-33](#).



If jumper is not in place, exposures will not be made. Check this jumper if you get scan interlock errors.

Figure 2-33 Without a Door Interlock

To use the system with a door interlock, wire a normally open switch between pins 1 & 2 that is attached to the interlock.

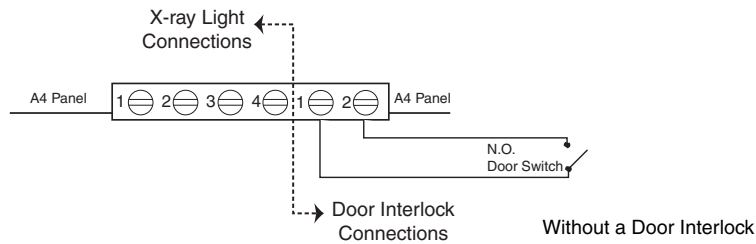


Figure 2-34 With a Door Interlock

10.2 NGPDU

As seen in Figure 2-35, a number of cables must be installed throughout the PDU. Specific details on each connection can be found in the sub-sections that follow. Use Figure 2-35 for reference. The PDU has been designed to have cables routed into the PDU from its behind and/or from beneath it.

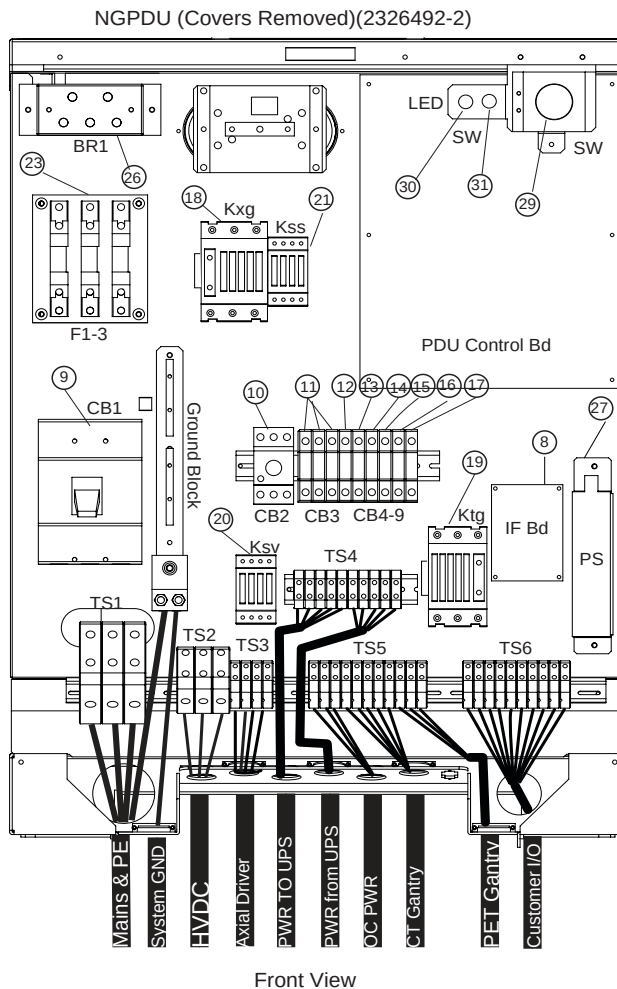


Figure 2-35 PDU Cable Connections - Front

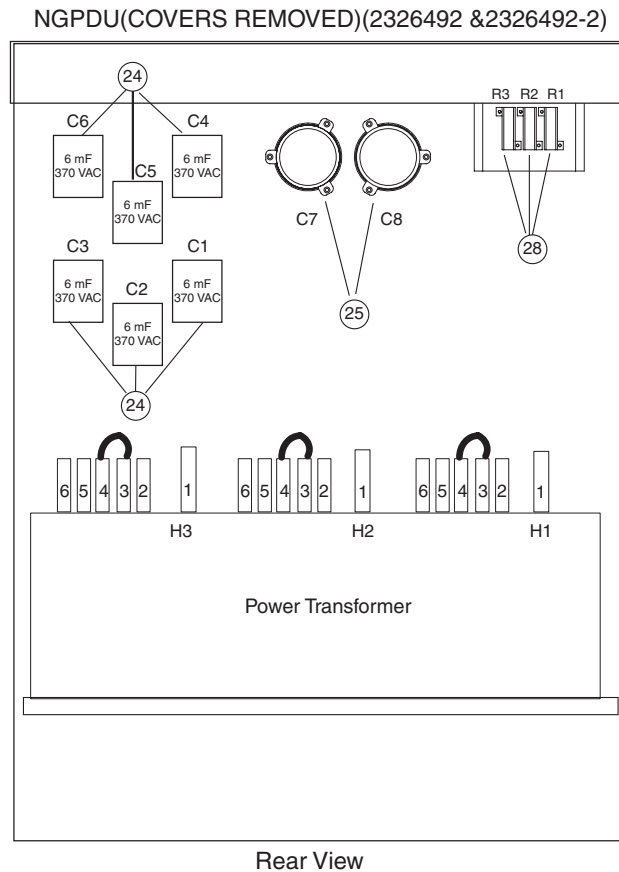


Figure 2-36 PDU Cable Connections - rear

10.2.1 Panel - 380 - 480VAC Mains “TS1” Input Power Connection

- 1.) Remove the TS3 panel front cover.
- 2.) Strip the wires to fit securely on the power block.
- 3.) Observe incoming phases (L1, L2 and L3) and insert bare leads into power block.
- 4.) Insert “vault” ground into PDU “vault” ground lug.
- 5.) Tighten all fasteners securely and replace the TS3 front panel.

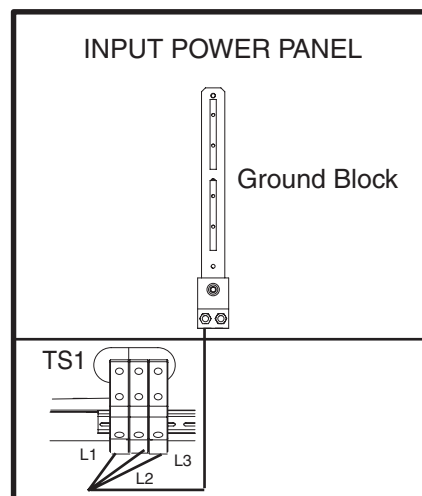


Figure 2-37 Input Power Panel Connections

When Mains power is available to the PDU, the “TS1” power light will be illuminated.
(See [Figure 2-35](#).)

10.2.2 Panel - Circuit Breakers

Place the circuit breakers in the “off/down” position during installation, even with Mains incoming power tagged and locked out. After you have completed work on the PDU, you may return the circuit breakers to the “ON” positions.

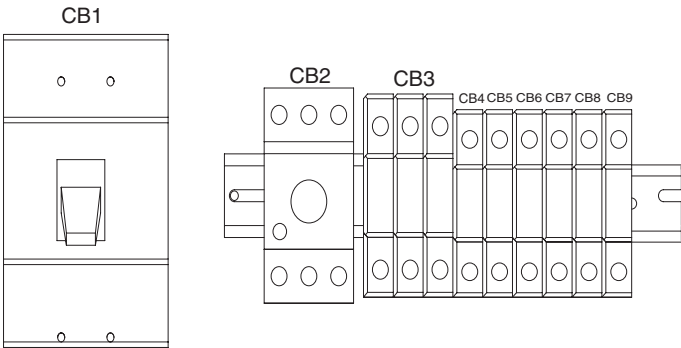


Figure 2-38 Circuit Breaker Panel

By design, when CB3 is in the “OFF” position, circuit breakers 4, 5, 6, and 7 are switched “OFF”. CB3 is essentially in series with these breakers.

CIRCUIT BREAKER	DESCRIPTION
CB3	Fully Winding Protection (Master power of CB 4, 5, 6, and 7)
CB4	CT Gantry Service Outlets
CB5	CT Gantry rotating loads
CB6	Table & CT Gantry Station Loads
CB7	Operator Console
CB8	PET Gantry
CB9	NGPDU Control Power Supply

Table 2-16 Panel Circuit Breaker Descriptions

10.2.3 Transformer (480VAC) Taps

Verify that the transformer taps are set properly. The transformer taps are set to 480 VAC operation at the factory. The taps should be set as shown in [Figure 2-39](#).

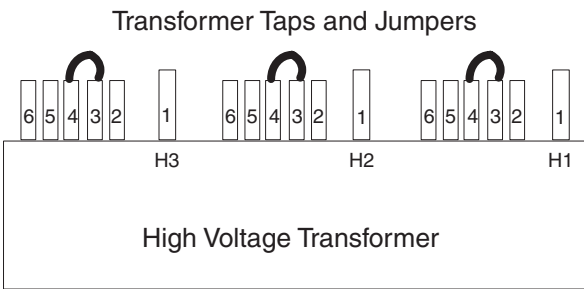


Figure 2-39 PDU Tap Positions for 480 volt operation

Adjustment of the 480VAC and these “taps” is done in [Chapter 5, Section 4.0 on page 162](#).

10.2.4 HVDC Connection

Connect the internally shielded HVDC cable to TS2 on the standing panel. See [Figure 2-35](#) for the location of the connector and [Figure 2-40](#) for details. Observe polarities and grounds.

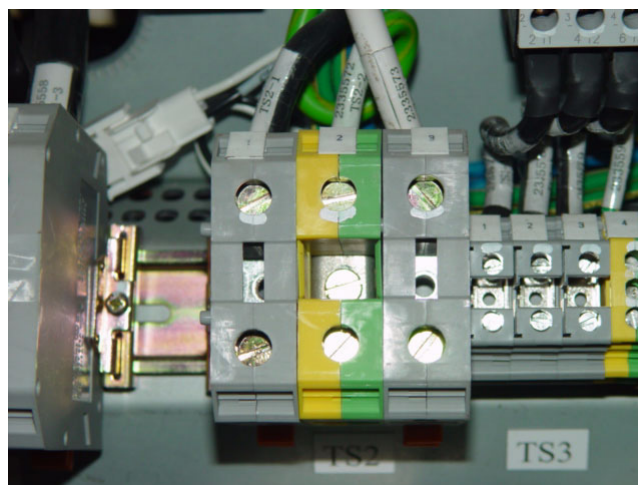
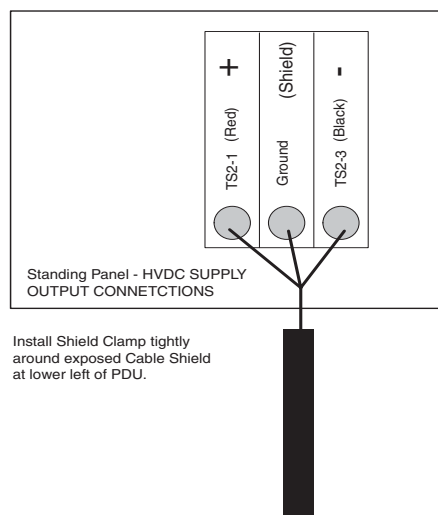


Figure 2-40 HVDC Connection

The HVDC cable can be re-terminated to eliminate excessive cable length, if necessary. Observe cable colors and polarities during re-connection. The cable's shield acts as the ground.

10.2.5 440V Connection

Connect the internally shielded 440V cable from the gantry to TS3 on the panel. See [Figure 2-35](#) for the location of the connector and [Figure 2-41](#) for details. Observe the labels on the cable leads for proper identification and orientation.

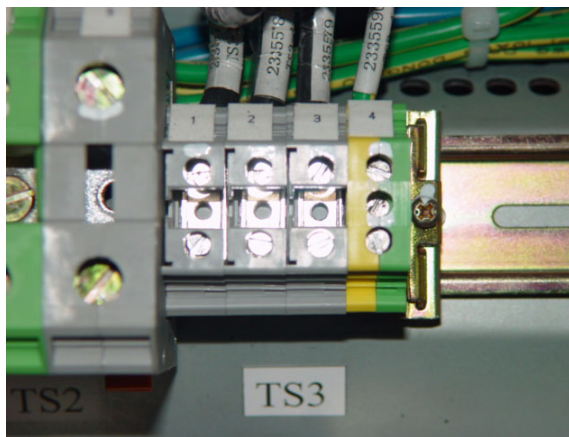
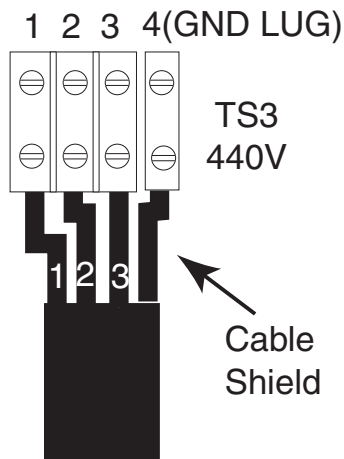


Figure 2-41 440VAC Connection

The 440V cable can be re-terminated to eliminate excessive cable length, if necessary.

10.2.6 Gantry & Console Power Connections

Both Gantry and Console power cables come pre-terminated. Simply plug the Gantry power cable into “J4” and the Console power cable into “J5” as shown in [Figure 2-42](#).

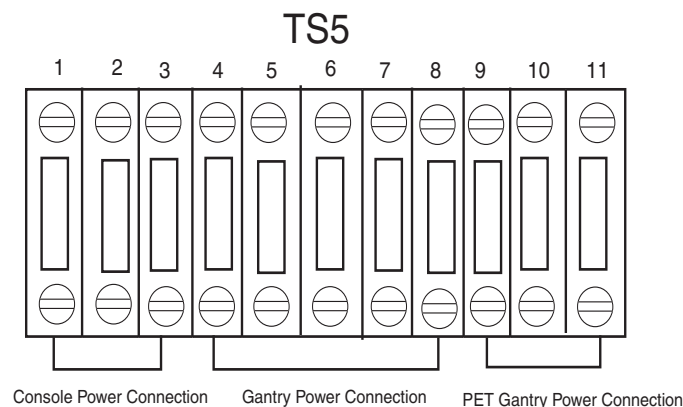


Figure 2-42 Gantry & Console Power Connections

Both the Gantry and Console Power cables can be re-terminated at the PDU end, if excessive cable length prevents the proper storage.

10.2.6.1 Console Power Cable Re-termination

- 1.) Carefully remove the power plug and **record the color of the wires** in [Table 2-17](#). The terminals are labelled X, Y, W and G on the plug.

TERMINAL	X	Y	W	G
Description	Hot	Hot	Neutral	Ground
Color				

Table 2-17 Console Power Cable Termination

- 2.) Cut the cable to desired length and dress ends.
- 3.) Re-install the power plug, according to the orientations recorded in [Table 2-17](#).
- 4.) Verify that less than 1 ohm of resistance exists between the following connections:

FROM PDU PLUG	TO CONSOLE PLUG END	
CB7-2 (Black) TS5-1, (Black)	Phase X (Brown)	☐ Check box when complete
N/G-Neutral (Blue) TS5-2, (Black)	Phase Y (Black)	☐ Check box when complete
N/G-Ground (Yellow/Green) TS5-3, (Yellow/Green)	Phase W (Blue) Neutral	☐ Check box when complete
	Ground (Green) Ground Green screw	☐ Check box when complete

Table 2-18 Resistance Verification Points

10.2.6.2 Gantry Power Cable Re-termination

- 1.) Carefully remove the power plug and **record the color of the wires** in [Table 2-19](#). The terminals are labelled X, Y, Z, W and G on the plug.

TERMINAL	X	Y	Z	W	G
Description	Hot	Hot	Hot	Neutral	Ground
Color					

Table 2-19 Gantry Power Cable Termination

- 2.) Cut the cable to desired length and dress ends
- 3.) Re-install the power plug, according to the orientations recorded.
- 4.) Resistance checks will be verified later.

10.2.7 PDU Control Cable

The PDU control cable comes pre-terminated and should not be re-terminated in the field. Excess cable length must be stored. Simply plug the cable into "J2" on the "A4" panel. Secure it by using the fasteners integrated into cable's connector shell.

10.2.8 System Ground Connection

Connect the ground wire (green with a yellow strip) from the Table/Gantry raceway ground bus to the system ground lug in the PDU. See [Figure 2-35, on page 100](#), and [Figure 2-43, below](#).



Figure 2-43 PDU System Ground Connection

10.2.9 X-Ray Light & Door Interlock Connections

10.2.9.1 X-Ray Light Configuration & Connection

- 1.) X-Ray Light is controlled by a signal from the system.

- 2.) This step is site specific. The PDU by default is configured for “no” external x-ray light connection. If you have an external x-ray light, see [Figure 2-44](#) for proper connection.

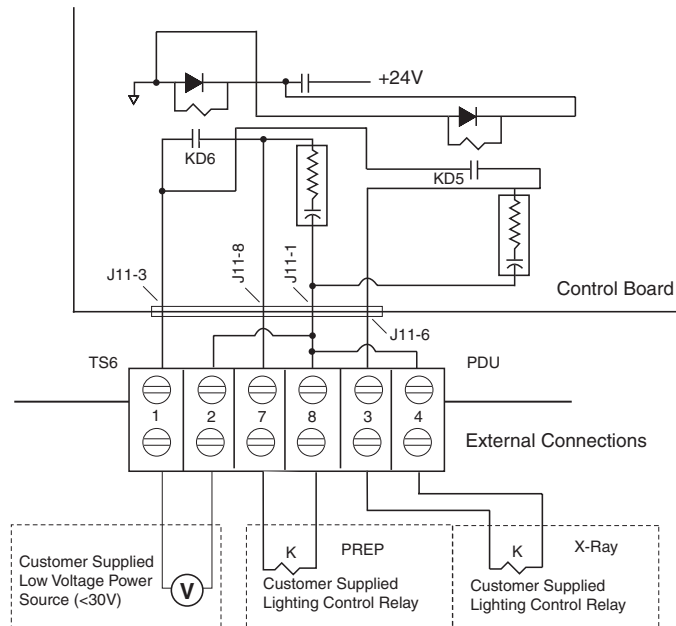


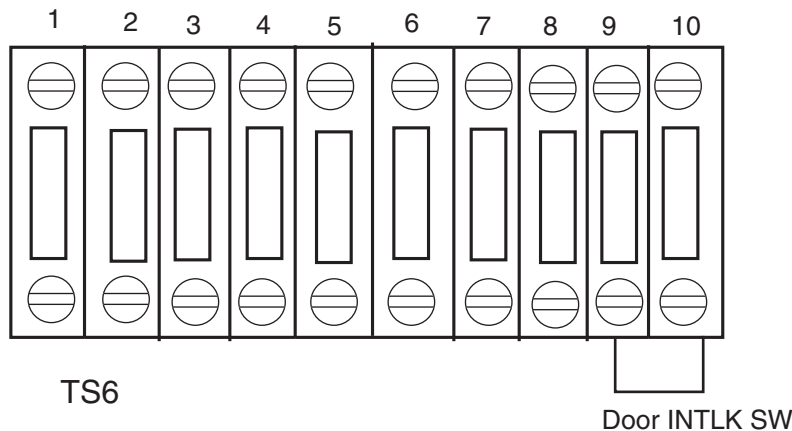
Figure 2-44 External X-ray Light Connection

It is recommended that you use the four (4) wire method of adding a X-ray warning light to a room, as shown in [Figure 2-44](#). When using this method, you:

- Minimize EMC interference.
- Increase contact life of the relay used in the PDU.

10.2.9.2 Door Interlock Connections

Door interlocks are used to prevent X-Rays from being generated when the scan room door is open. The Door Interlock circuitry in the PDU is shipped from the factory engaged. This means the system cannot generate X-ray until disengaged. A short must exist between pins 9 & 10 for X-ray to be generated. Using a small piece of wire, short pins 1 and 2 together. See [Figure 2-45](#).



If jumper is not in place, exposures will not be made. Check this jumper if you get scan interlock errors.

Figure 2-45 Without a Door Interlock

To use the system with a a door interlock, wire a normally open switch between pins 1 & 2 that is attached to the interlock.

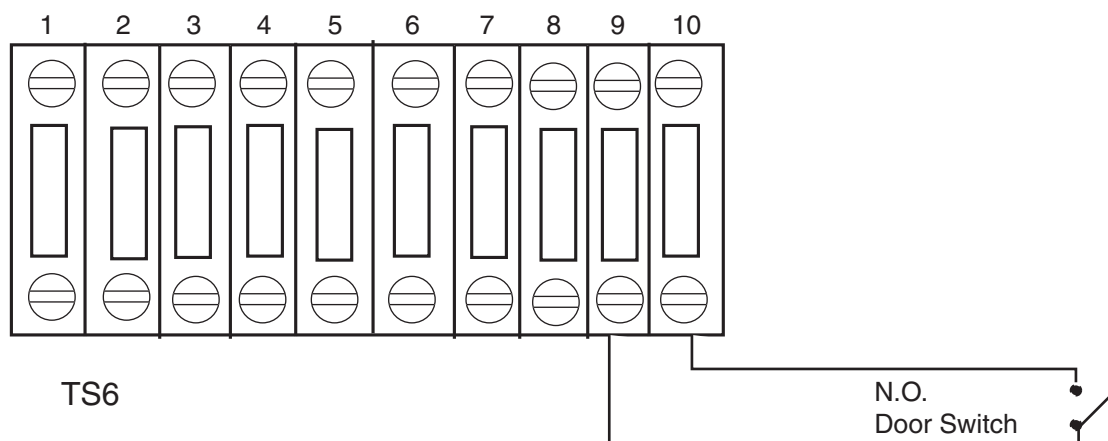


Figure 2-46 With a Door Interlock

Section 11.0

System Ground Connections

As seen in [Figure 2-47](#) or [Figure 2-48](#), the Table/Gantry raceway ground bus is used to centralize all system grounding. The system ground is tied to vault ground at the PDU, through its chassis.

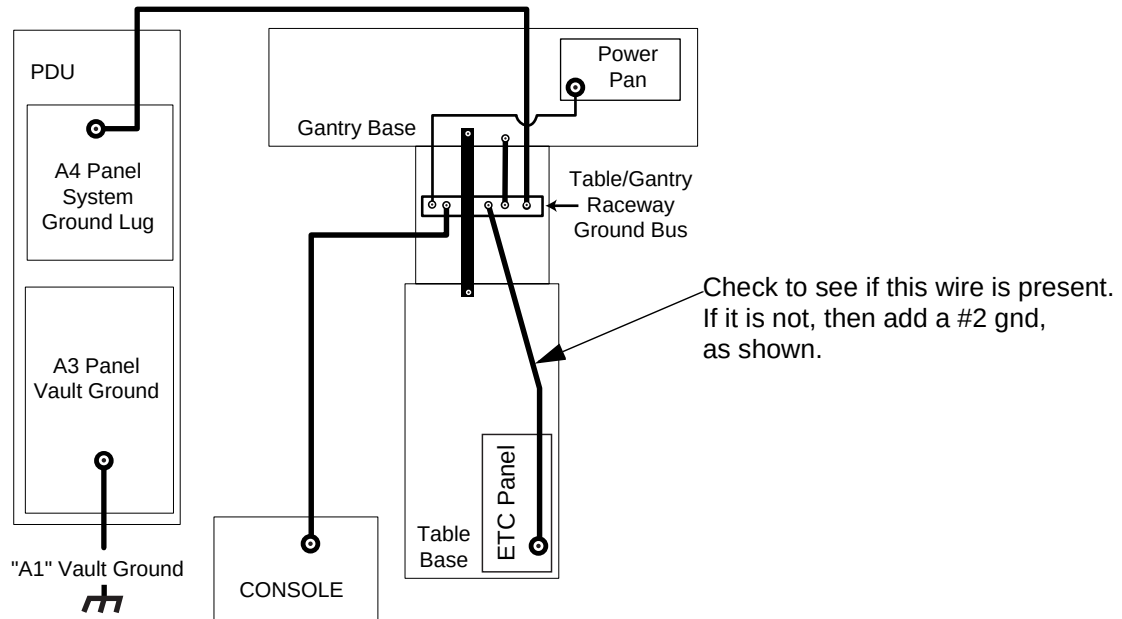


Figure 2-47 CT System Ground Connections (CPDU shown; see [Figure 2-48](#) for NGPDU)

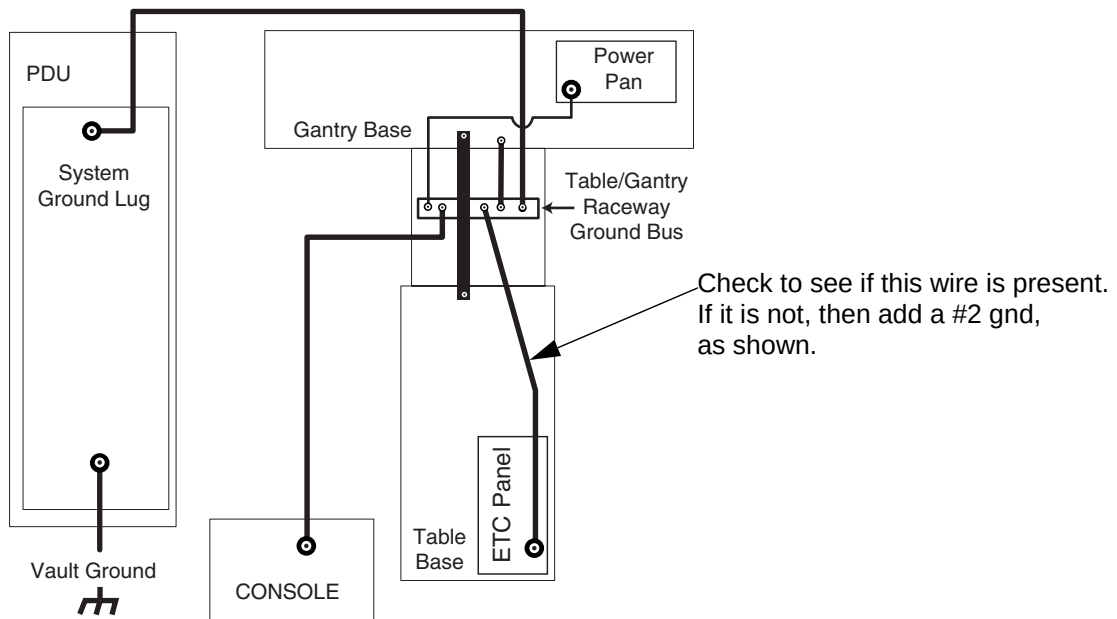


Figure 2-48 CT System Ground Connections (NGPDU shown; see [Figure 2-47](#) for CPDU)

The gantry is tied to system ground at a number of point. It is important that all of these ground connections are securely made. See [Figure 2-49](#).

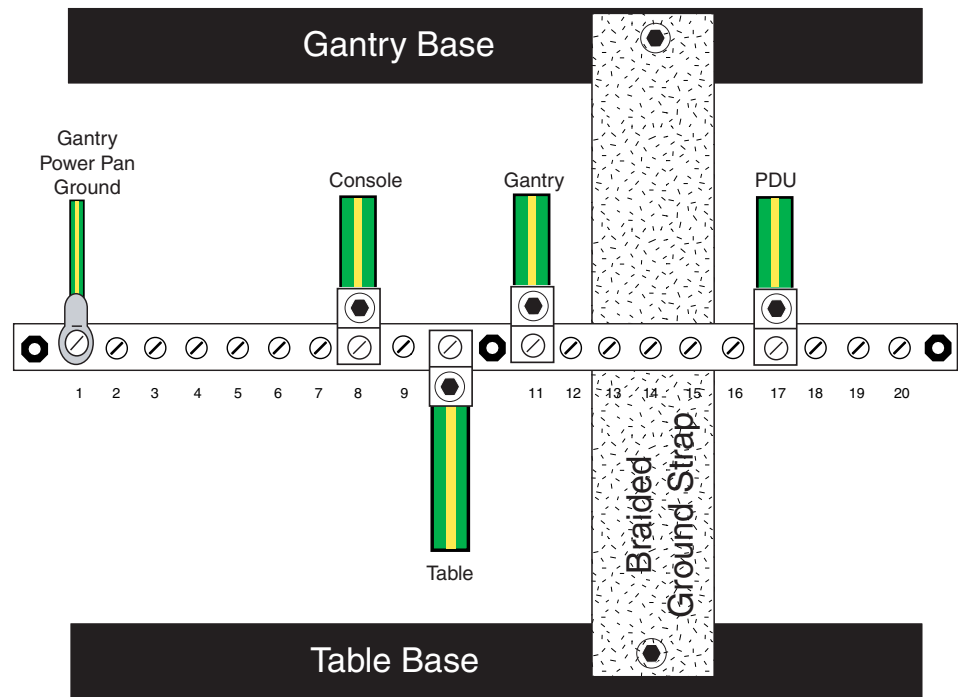


Figure 2-49 Table/Gantry Raceway Bus - Grounds

Various types and sizes of wire are used to ground the system. Please use the type and sizes specified in [Table 2-20](#), below.

AWG #	CONNECTION TO	CONNECTION TO
1/0	PDU	Power Main
#10	Gantry (Power Pan)	Raceway
#2	Console	Raceway
#1/0	Gantry	Raceway
#2	Table (ETC frame)	Raceway
#1/0	PDU	Raceway
GND Braid (2)	Table Frame	Gantry Frame

Table 2-20 System Ground Connections

Chapter 3

System Continuity & Ground Checks

**NOTICE**
Potential for
Data Loss and/
or Equipment
Damage

To prevent potential data loss and equipment damage, please do the following:

- Record data collected from procedures in this chapter into Form F4879 when directed, located in [Chapter 13](#) of this book.
- Only use the Installation manual that arrives with your system for installation. Any other revisions of this manual may not exactly match your system.

Section 1.0

System Continuity & Ground Checks (Mechanical Contractor)

Use this section to check cable and ground connections.

1.1 Tools Required

- Digital VOM
- 30 ft of #18 wire
- 600 VAC meter leads

1.2 Procedure

1.2.1 Compact PDU

Reference [Figure 3-1: Front View of PDU, with Covers Removed on page 112](#) and [Figure 3-2: Gantry Power Pan on page 113](#).

**WARNING**


USE AND FOLLOW LOCKOUT/TAGOUT PROCEDURES; LOCK OUT WALL POWER.

- Remove all System Power at the A1 Mains Disconnect Panel. Follow Lockout/Tagout procedures.
- Put the UPS in the Service Position.
- Remove the PDU A3 input power panel cover.
- Verify, with a voltmeter, that mains power is disconnected.
- Verify that less than 1 ohm of resistance exists between the following ground connections:

FROM A1 MAIN DISCONNECT PANEL	TO PDU A3 INPUT POWER PANEL	
A1 Breaker, wall ground connection	System gnd	☐ Check box when complete
L1, L2, L3	L1, L2, L3	☐ Check box when complete

Table 3-1 Mains Connections to PDU

6.) Verify that less than 1 ohm of resistance exists between the following connections:

FROM PDU	TO GANTRY POWER PAN	
A2 TS1-1 (+HVDC)	TS1-1 (RED) (+HVDC connection)	☐ Check box when complete
A2 TS1-2 (HVDC RTN)	TS1-2 (BLACK) (HVDC RTN connection)	☐ Check box when complete
A4 HVDC Cable Shield Braid Clamp	HVDC Cable Shield Braid Clamp	☐ Check box when complete
A3K4-2 T1 (Black) (A3 TB1-1)	TS2-1	☐ Check box when complete
A3K4-4 T2 (Black) (A3 TB1-2)	TS2-2	☐ Check box when complete
A3K4-6 T3 (Black) (A3 TB1-3)	TS2-3	☐ Check box when complete
HVAC 440 VAC Axial Drive Cable Shield Braid Clamp	HVAC 440 VAC Axial Drive Cable Shield Braid Clamp	☐ Check box when complete
A3 System Ground	Table Raceway (Green/Yellow)	☐ Check box when complete

Table 3-2 Resistance Verification Points

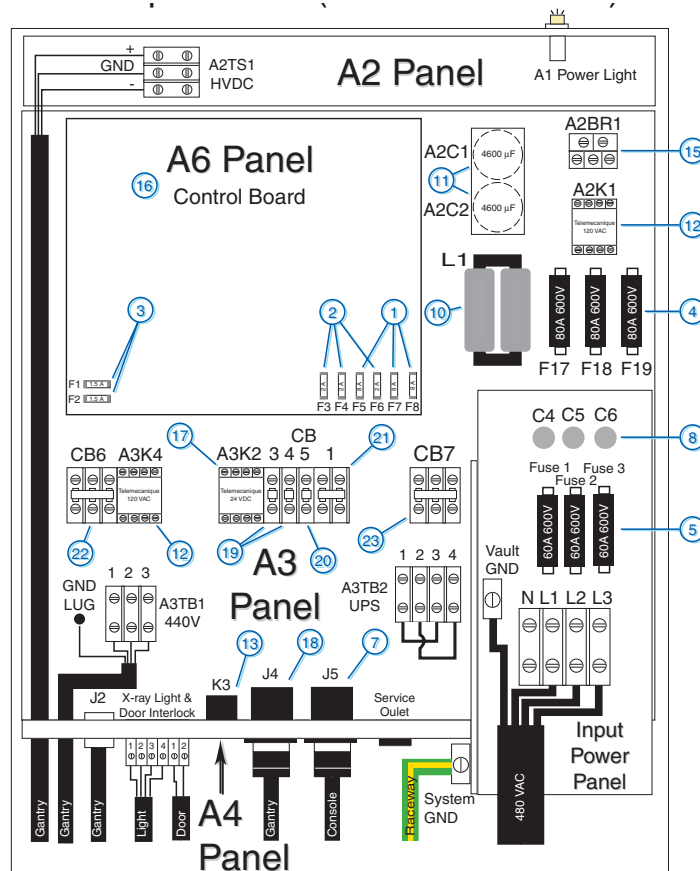


Figure 3-1 Front View of PDU, with Covers Removed

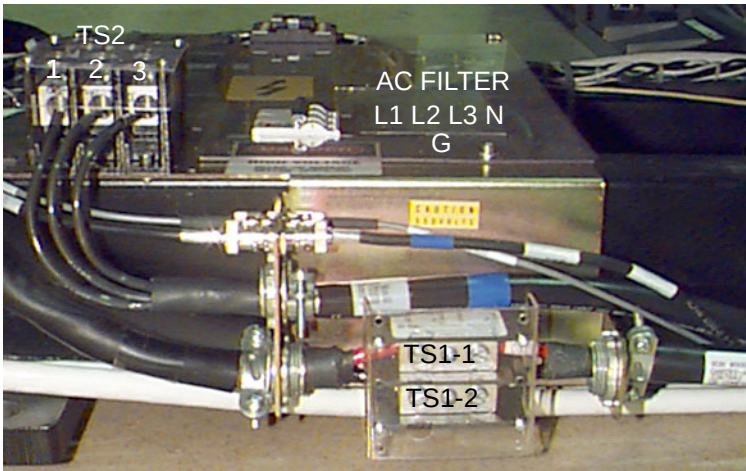


Figure 3-2 Gantry Power Pan



WARNING TURN OFF ALL PDU CIRCUIT BREAKERS.

- 7.) Set an ohmmeter to the lowest scale. Check between the following points for shorts to ground. Verify no continuity exists between the following points:

FROM PDU	TO A1 BREAKER BOX	
A2 TS1-1 (+HVDC)	vault ground	☐ Check box when complete
A2 TS1-2 (-HVDC)	vault ground	☐ Check box when complete

Table 3-3 No Continuity Verification Points

- 8.) Leave the metal cover off the PDU A3 input power panel until you complete the checks in the next section.

1.2.2 NGPDU

Reference [Figure 3-3: Front View of PDU, with Covers Removed on page 115](#) and [Figure 3-4: Gantry Power Pan on page 115](#).



WARNING

USE AND FOLLOW LOCKOUT/TAGOUT PROCEDURES; LOCK OUT WALL POWER.

- 1.) Remove all System Power at the A1 Mains Disconnect Panel. Follow Lockout/Tagout procedures.
- 2.) Put the UPS in the Service Position.
- 3.) Verify, with a voltmeter, that mains power is disconnected.
- 4.) Verify that less than 1 ohm of resistance exists between the following ground connections

FROM A1 MAIN DISCONNECT PANEL	TO PDU INPUT POWER PANEL	
A1 Breaker, wall ground connection	System gnd	☐ Check box when complete
L1, L2, L3	TS1-L1, TS1-L2, TS1-L3	☐ Check box when complete

Table 3-4 Mains Connections to NGPDU

- 5.) Verify that less than 1 ohm of resistance exists between the following connections:

FROM PDU	TO GANTRY POWER PAN	
TS2-1 (+HVDC)	TS1-1 (RED) (+HVDC connection)	☐ Check box when complete
TS2-3 (HVDC RTN)	TS1-2 (BLACK) (HVDC RTN connection)	☐ Check box when complete
HVDC Cable Shield Braid Clamp	HVDC Cable Shield Braid Clamp	☐ Check box when complete
TS3-1	TS2-1	☐ Check box when complete
TS3-2	TS2-2	☐ Check box when complete
TS3-3	TS2-3	☐ Check box when complete
HVAC 440 VAC Axial Drive Cable Shield Braid Clamp	HVAC 440 VAC Axial Drive Cable Shield Braid Clamp	☐ Check box when complete
System Ground	Table Raceway (Green/Yellow)	☐ Check box when complete

Table 3-5 Resistance Verification Points (NGPDU)

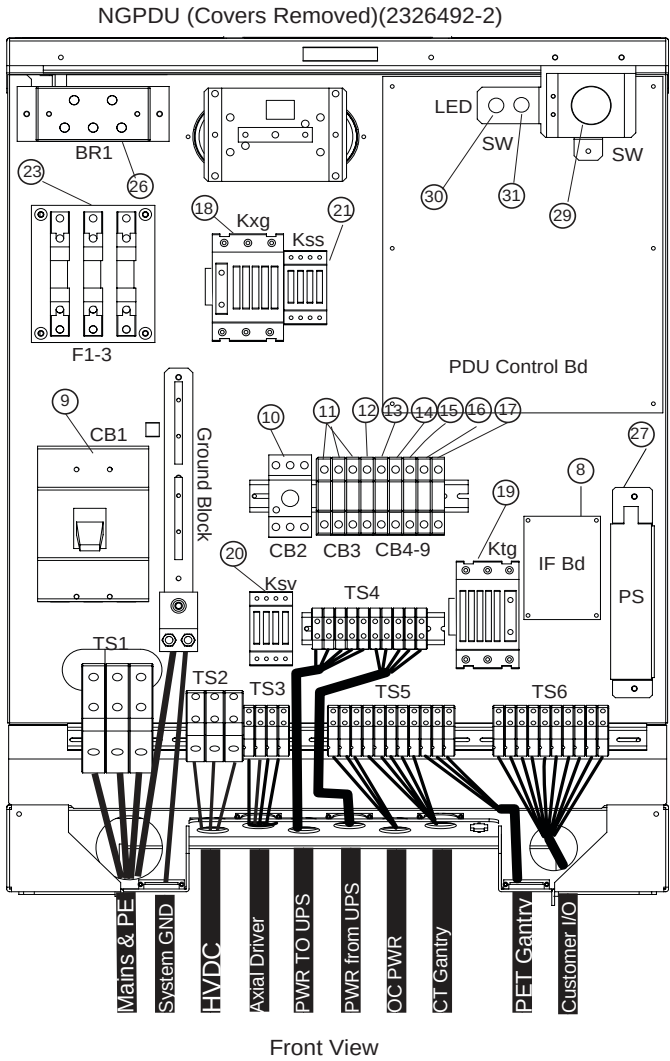


Figure 3-3 Front View of PDU, with Covers Removed

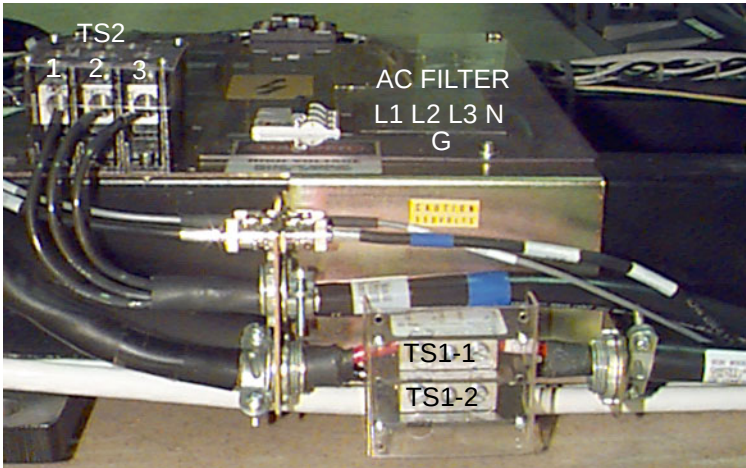


Figure 3-4 Gantry Power Pan



WARNING TURN OFF ALL PDU CIRCUIT BREAKERS.

- 6.) Set an ohmmeter to the lowest scale. Check between the following points for shorts to ground. Verify no continuity exists between the following points:

FROM PDU	TO A1 BREAKER BOX	
TS2-1 (+HVDC)	vault ground	☐ Check box when complete
TS2-3 (-HVDC)	vault ground	☐ Check box when complete

Table 3-6 No Continuity Verification Points

Section 2.0

Site Ground Continuity Check

Use an ohmmeter to verify the presence of **less than 1.0 ohm of resistance** between each of the following points:

FROM	TO	
PDU A3 System Ground	Vault Ground A1 Breaker Box	☐ Check box when complete
PDU A3 System Ground	Table/Gantry Raceway ground point	☐ Check box when complete
Table/Gantry raceway ground point	Gantry	☐ Check box when complete
Table/Gantry raceway ground point	Table	☐ Check box when complete
Table/Gantry raceway ground point	Operator Console	☐ Check box when complete
All Display or Computing Options	Operator Console	☐ Check box when complete
Any room facility patient ground points	PDU A4 Ground Bus	☐ Check box when complete

Table 3-7 Resistance Verification - Site Ground (Compact PDU)

FROM	TO	
PDU System Ground	Vault Ground A1 Breaker Box	☐ Check box when complete
PDU System Ground	Table/Gantry Raceway ground point	☐ Check box when complete
Table/Gantry raceway ground point	Gantry	☐ Check box when complete
Table/Gantry raceway ground point	Table	☐ Check box when complete
Table/Gantry raceway ground point	Operator Console	☐ Check box when complete
All Display or Computing Options	Operator Console	☐ Check box when complete
Any room facility patient ground points	PDU A4 Ground Bus	☐ Check box when complete

Table 3-8 Resistance Verification - Site Ground (NGPDU)

Section 3.0

Mechanical Installation Completion Section

Complete the Mechanical Vendor checklist:

3.1 Clean Up

- ☐ Move box labeled "Customer" to one corner.
- ☐ Move Field Engineer Material (phantoms, hoist, kits, tools) to another corner or counter top.
- ☐ Put all documentation (incl. software, HHS doc) in the purple box.

3.2 Remove Trash

- ☐ Recycle cardboard.
- ☐ Remove skids and wood.

3.3 Return Dollies

- ☐ Check the packing list found in the purple box.

3.4 Install Covers

- ☐ Recheck Gantry and Table anchor for 75ft-lb.
- ☐ PDU covers.
- ☐ Table foot-switch assembly (refer to [Figure A-31, on page 135](#)).
- ☐ Console covers.
- ☐ Gantry base covers (refer to [Figure A-30, on page 134](#)).

3.5 Clean the Equipment

- ☐ Use the contents of the cleaning kit to clean the equipment.

3.6 Touch Up Nicks

- ☐ Use the paint found in the cleaning kit to touch up nicks in the equipment.

3.7 Complete Paperwork

- ☐ Collect all Product Locators.
- ☐ Complete the Field Engineer sign-off.

Appendix A

Removal & Installation of Covers

Section 1.0 Gantry Side Covers

1.1 Side Cover Removal

CAUTION
Potential for
injury if covers
removed and
power is left
"ON".

- 1.) Lower table to home (lowest) position.

Always remove the right side cover first, and turn OFF power at the STC.

- 2.) Use an 8mm Hex wrench to unlatch the side cover from the front cover. See [Figure A-1](#).

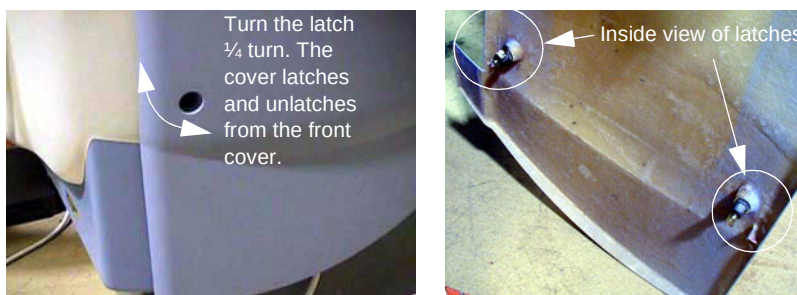


Figure A-1 Side Cover Latches

- 3.) Remove the right side cover by lifting it upward to release the two (2) latches, located on the top edge of the cover. Once removed, the STC backplane should be exposed.

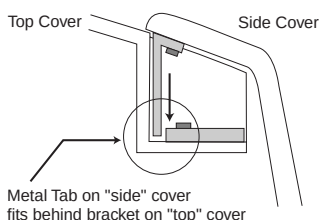


Figure A-2 Side and Top Cover Clasp

- 4.) Turn off all three (3) power switches on the STC backplane.

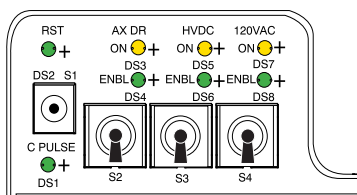


Figure A-3 STC Power Switches

- 5.) Repeat steps 1-3 for the left side cover.

1.2 Side Cover Installation

- 1.) To install a side cover, place it over the top cover and let the two (2) side cover latches slide behind the metal tabs, located on the top cover. See [Figure A-2](#).
- 2.) Use Hex wrench to secure the side cover to front cover by turning the bolts a quarter turn. See [Figure A-1](#).

Section 2.0 Gantry Top Covers

 **CAUTION** Before you remove top covers, always make sure the three (3) power switches have been turned off. (See [Figure A-3](#).)

2.1 Top Cover Removal

- 1.) Remove the associated side cover if you have not already removed it.
- 2.) Take the end of the top cover nearest to the side cover and tilt upwards.
- 3.) Slide the cover down 75 millimeters. This allows the cover's tab to disengage from the mounting bracket. See [Figure A-4](#).
- 4.) Lift the cover clear and repeat the above steps for the other cover.

2.2 Top Cover Installation

The top cover consists of two (2) pieces. Install the front and rear gantry covers, if not already installed. See [Section 3.0 on page 121](#), and [Section 4.0 on page 131](#).

- 1.) Take one of the top covers and align the tabs on the cover with its associated bracket. Lift and slide the cover into place. Position the cover to fully engage the fan interlock switch.



Figure A-4 Top cover tabs and bracket, plus fan interlock switch.

- 2.) Take the other top cover and align the tabs on the cover with its associated bracket. Lift and slide the cover into place, while being sure to engage the fan interlock switch.

Section 3.0

Gantry Front Cover



NOTICE
Potential for
front and rear
cover damage.

Front and rear cover removal and installation can be safely accomplished by one (1) person using the dollies provided with the system. Failure to use these dollies will significantly increase the likelihood of damage to the covers. Do not lean covers against walls.

3.1 Original Front Cover Dolly Setup

The front cover dollies are a folding design for easy storage. This design is stable and assembly/disassembly can be done in 30 second



DANGER



DO NOT USE DOLLIES ON UNEVEN SURFACES SUCH AS STEPS OR ELEVATOR THRESHOLDS. THE DOLLIES ARE DESIGNED TO BE USED ON FLAT LEVEL FLOORS WITHIN THE SCANNING SUITE ONLY. MISUSE CAN RESULT IN PERSONAL INJURY OR DAMAGE TO COVERS OR OTHER FACILITY ITEMS.



WARNING

Rotating arms on the stand are supposed to be stiff. If they fall freely, tighten the tensioning nuts. Loose rotating arms will reduce the stability of the dollies when supporting the front cover. Do not lubricate.

Note: Rotating arms are shown in the inverted or upside down position for clarity.

- 1.) Arrange Dolly sections for assembly. The base and stand can be assembled only one way. Refer to [Figure A-5](#).
 - The stand has a large stud and 3 smaller studs that engage the base assembly.
 - The large stud provides stability.
 - The three small studs engage the base arms preventing them from folding under if accidentally tipped.
 - The Locking pin engages the 4th base arm and prevents accidental separation when assembled.

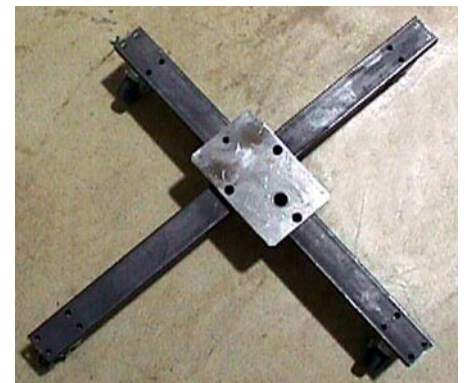
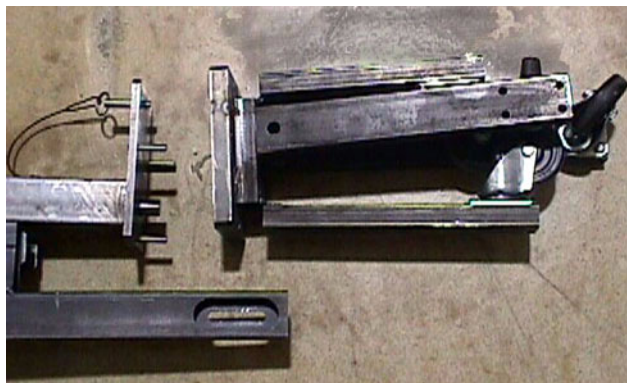


Figure A-5 Front Cover Dolly base and stand disassembled

- 2.) Unfold the base as shown in [Figure A-5](#) and place on flat surface.

- 3.) Install Stand in base, insert base locking pin, unfold stand top and secure with palm screw. Ensure additional safety bracket and washer are installed. Reference [Figure A-6](#).

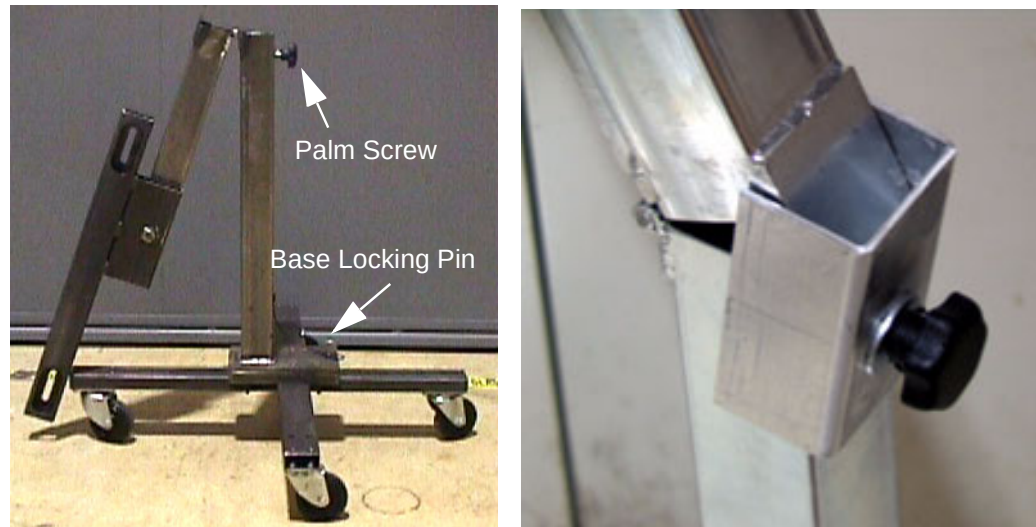


Figure A-6 Front Cover Dolly Stand and Base with additional Safety Bracket

- 4.) Assemble second dolly. Both look similar as in [Figure A-7](#). However the stand portion of the dollies are side specific. The “Right” dolly has the locking pin for cover rotation. The “Left” dolly does not have a cover rotation locking pin.
- Both dolly stands should have an “R” or an “L” indicating right or left.
 - The base assemblies are interchangeable.

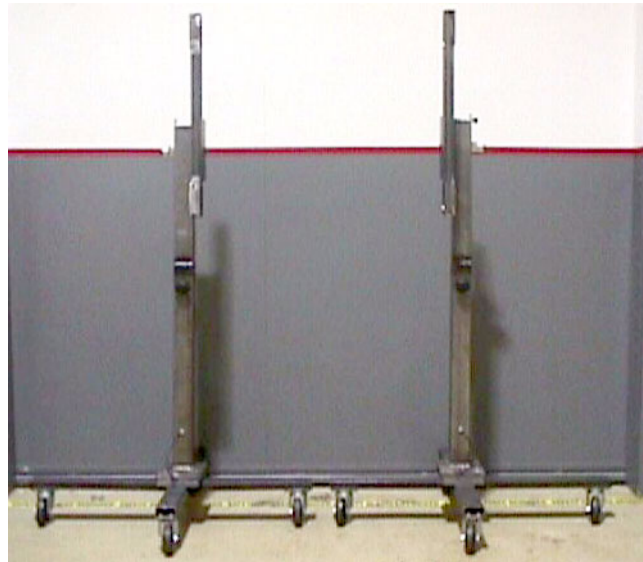


Figure A-7 Front Cover left and right Fully Assembled

3.2 Redesigned Front Cover Dolly Setup

The Front Cover Dollies were redesigned for cost reduction and potential safety concerns. The new Front Cover Dollies can be ordered as replacements for the original design.



DANGER



DO NOT USE DOLLIES ON UNEVEN SURFACES SUCH AS STEPS OR ELEVATOR THRESHOLDS. THE DOLLIES ARE DESIGNED TO BE USED ON FLAT LEVEL FLOORS WITHIN THE SCANNING SUITE ONLY. MISUSE CAN RESULT IN PERSONAL INJURY OR DAMAGE TO COVERS OR OTHER FACILITY ITEMS.



WARNING

Rotating arms on the stand are supposed to be stiff. If they fall freely, tighten the tensioning nuts. Loose rotating arms will reduce the stability of the dollies when supporting the front cover. Do not lubricate.

- 1.) Arrange Dolly sections for assembly. The base and post can be assembled only one way.
Refer to [Figure A-8](#) and [Figure A-9](#).
 - The base uses two (2) palm screws to clamp the four (4) legs in the open or usage mode.
 - The base also uses the same palm screws to prevent the legs from falling in storage mode.
 - The top post can be inserted in either base and is keyed for proper engagement.
 - The top post locking pin prevents the sections from separating during usage.

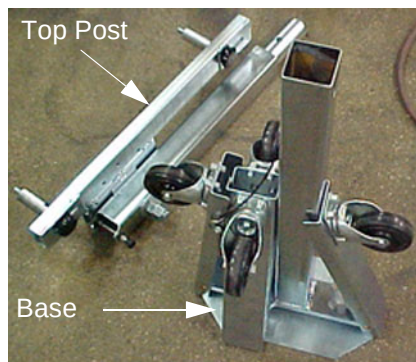


Figure A-8 Redesigned Front Cover Dolly in Storage Mode

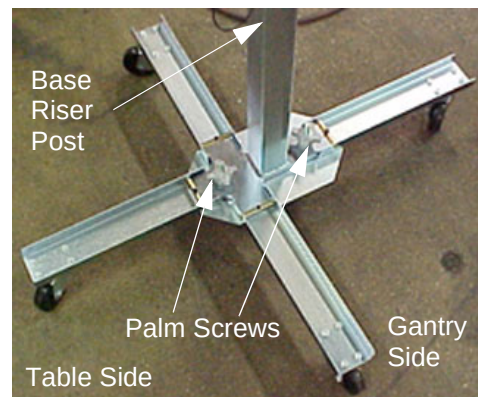
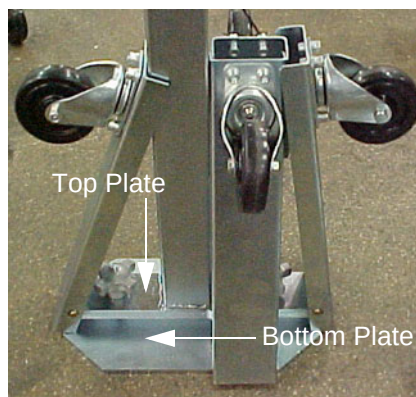


Figure A-9 Redesigned Front Cover Dolly Base Assembly

- 2.) Unfold the base legs by loosening both palm screws to the top of their travel.
- 3.) Carefully unfold the legs so that the castors touch the floor.

- 4.) Tighten the palm screws to clamp the legs between the base top and bottom plates.

Note: Lifting the base by the riser post while leaving the castors on the floor will ease palm screw tightening. Reference [Figure A-9](#).



WARNING

ENSURE BOTH PALM SCREWS ARE TIGHTENED SECURELY AND THE LEGS ARE CLAMPED TIGHTLY BETWEEN THE BASE TOP AND BOTTOM PLATES. FAILURE TO DO SO WILL RESULT IN INSTABILITY DURING FRONT COVER HANDLING.

- 5.) Insert top post into the base riser post. Align the key for complete engagement.
- 6.) Insert top post locking pin to secure both top and bottom sections.
- 7.) Reverse above steps to disassemble.

Note: For base storage only one (1) palm screw needs to be tightened. This will engage the bottom base plate and the leg ends preventing the legs from unfolding during transport and storage.

3.3 Removal

- 1.) Position the table at its lowest position.
- 2.) Remove gantry side and top covers, if you have not already done so. See [Section 1.0 on page 119](#). Make sure that the three (3) power switches have been turned off. See [Figure A-3](#).
- 3.) Assemble the front cover dolly.
 - a.) Tighten the two (2) shoulder bolts to the gantry securely. This will make cover installation easier. See [Figure A-10](#).

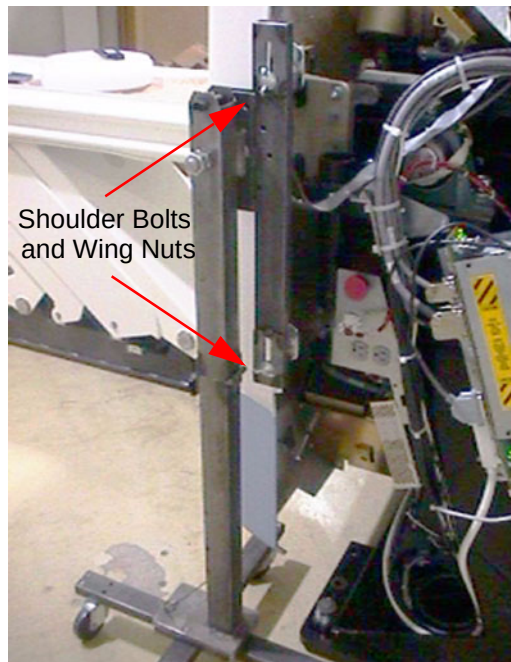


Figure A-10 Front Side Dolly

- b.) Attach side dolly to the shoulder bolts and secure assembly with two (2) wing nuts.
 - c.) Repeat steps a and b to assemble the other side dolly.
- 4.) Detach front cover J3 and J2 and front cover BKHD J1 cables.
- 5.) Remove front cover
 - a.) Disengage upper and lower cantrell brackets on both sides of the cover.
 - 1.) Using steady but firm pressure, lift each of the lower cantrell brackets from their

associated retainers. See [Figure A-11](#).

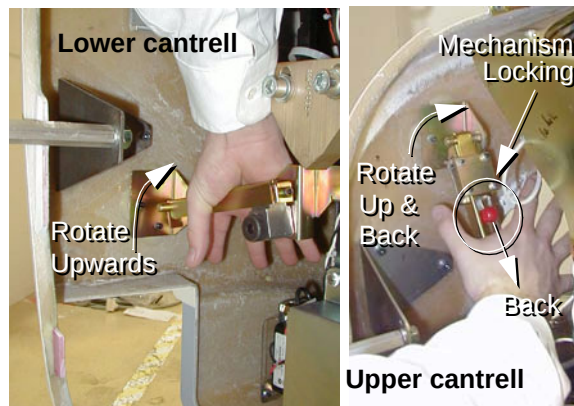


Figure A-11 Releasing cover brackets

- 2.) Disengage the locking mechanism on the upper cantrell brackets by using your thumb to slide the trigger (red lever) back. This will release the locking mechanism and allow the cantrell to be rotated upwards with steady and firm pressure.
- b.) Disengage the rubber retaining straps on both sides. You may find it helpful to lift “up” on the cover to align the stud while attaching the rubber retaining straps.
- c.) Also lift and rotate cover locking arm to unlocked position.

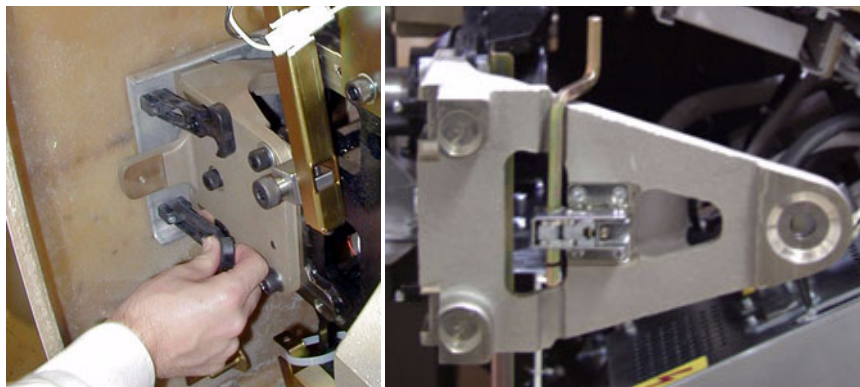
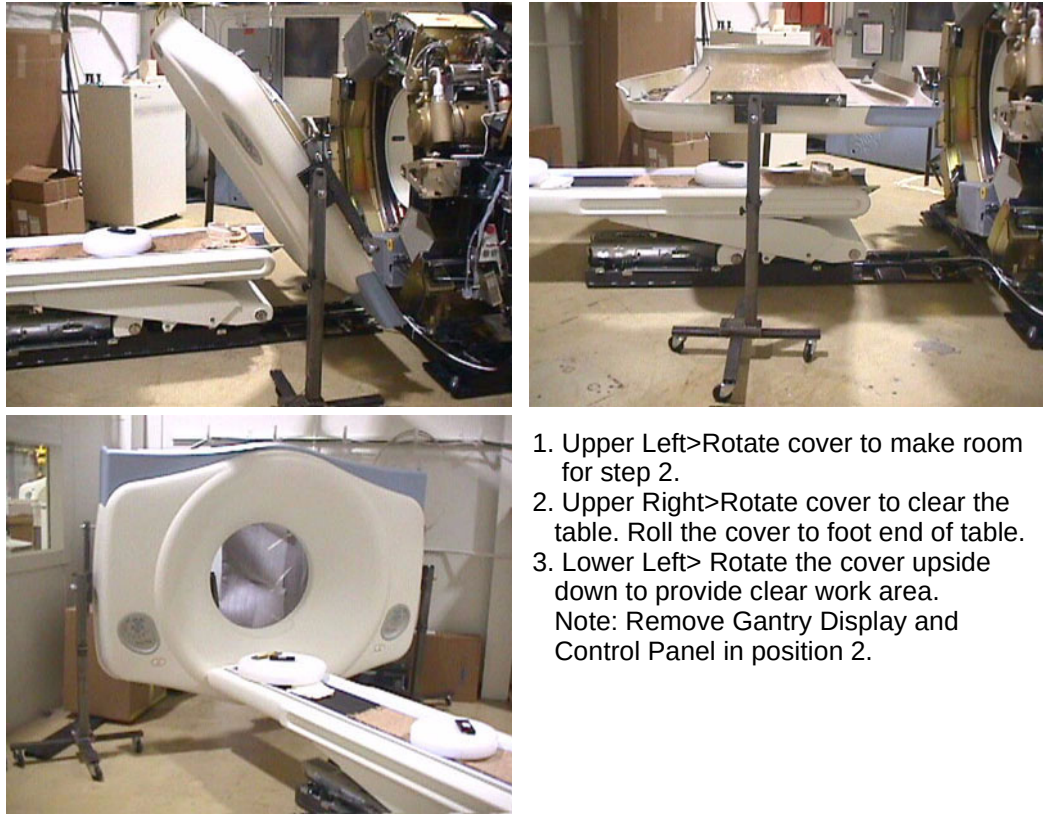


Figure A-12 Rubber retaining straps and Cover Locking mechanism

- 6.) Rotate front cover away from gantry.
 - a.) Move front cover away from gantry giving enough space (about 5 feet) between front cover and gantry.
 - b.) Pull the locking pin and rotate front cover away from gantry. Place locking pin in one of the side dolly perforations. See [Figure A-13](#).



Figure A-13 releasing Front Cover Dolly Hinge



1. Upper Left>Rotate cover to make room for step 2.
2. Upper Right>Rotate cover to clear the table. Roll the cover to foot end of table.
3. Lower Left> Rotate the cover upside down to provide clear work area.
Note: Remove Gantry Display and Control Panel in position 2.

Figure A-14 Front Cover Removal Sequence

- 7.) Rotate the cover horizontally and move it back and over the table to a safe location. Once in a safe location, you may over-rotate the cover full vertically but upside down.
- 8.) Remove the gantry display and one (1) of the cover's control assemblies, and place them into the service positions.
 - a.) Remove the gantry display and place it into its service position.
 - * The gantry display is held in place with (5) thumb screws. Use a flat-blade screwdriver to remove the Display. Reference [Figure A-15](#).
 - * Mount the Display on the Right Gantry Fan.
 - * There are (2) mounting methods. Both use the cables connected to the REAR GANTRY COVER.
 - * Disconnect the cabling at the right rear gantry cover. Only (1) cable will connect to the Gantry Display.
 - * Position "A" use one of the display T-hook to hang in the "T" slot on the side of the right Gantry Fan Assembly. Reference [Figure A-16](#).
 - * Position "B" place the Display in the cradle across the top of the right Gantry Fan Assembly. Use thumb-screw to secure display on right side. Reference [Figure A-17](#).

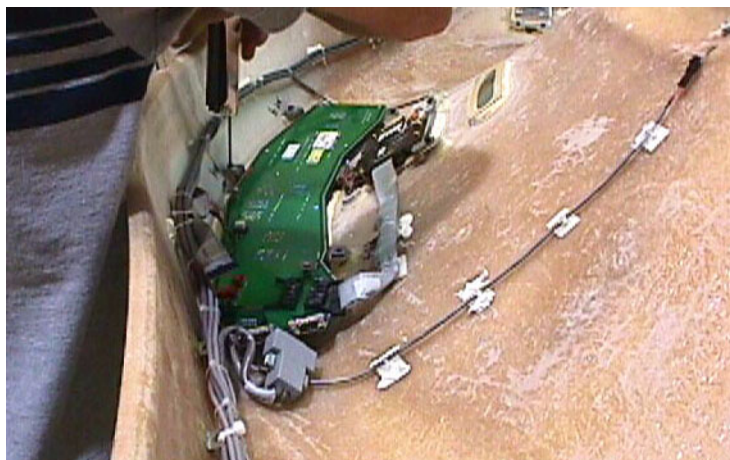


Figure A-15 Gantry Display Removal

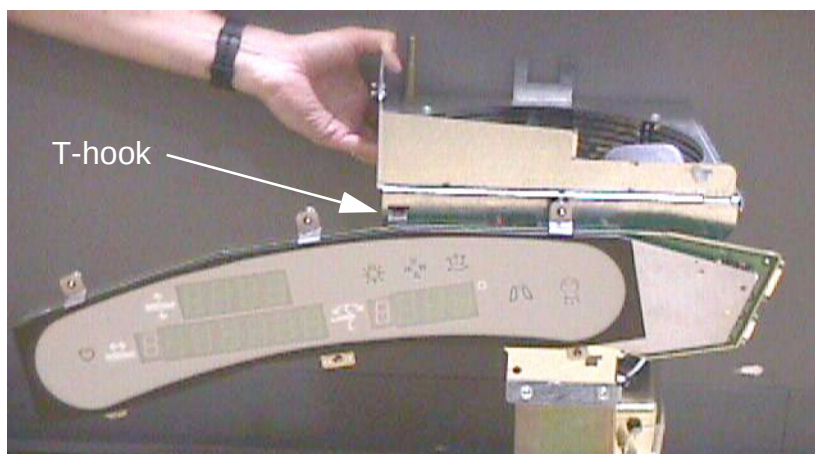


Figure A-16 Gantry Display Service Mounting Location "A"

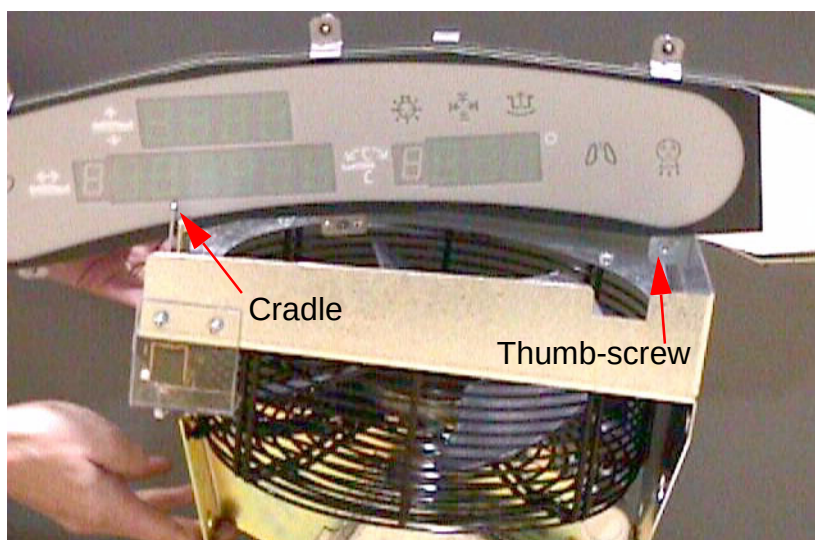


Figure A-17 Gantry Display Service Mounting Location "B"

- b.) Remove a gantry control and place it into its service position.
- 1.) Press on each ball stud until the panel is released. Keep one hand on the control panel at all times to prevent it from dropping to the floor.

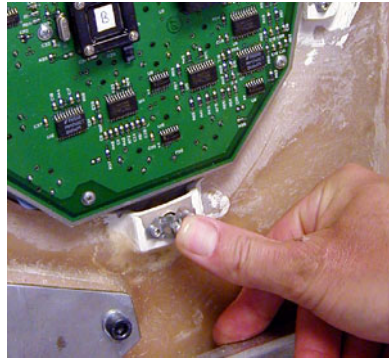


Figure A-18 Gantry control panel removal

- 2.) Align the ball studs with their associated receivers and snap into place.

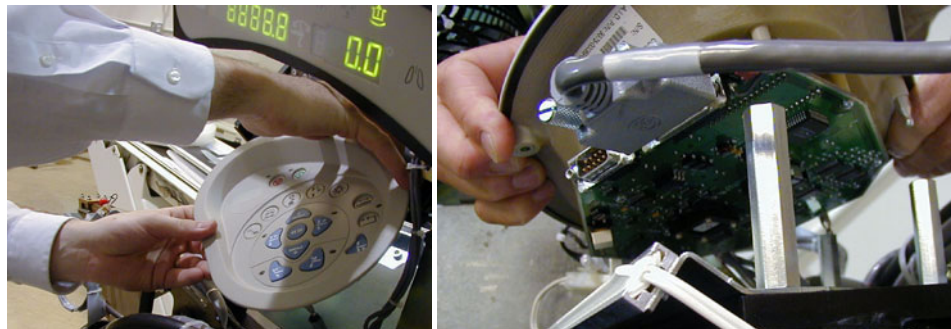


Figure A-19 Control panel service position

Note:

- 3.) Connect cable to terminator located on the cantrell arm. Reference [Figure A-20](#). There are 3 cables, each of which is unique. The ribbon cable is not used in the Service configuration. The other 2 cables will only fit in the terminator or the control panel, not both.



Figure A-20 Gantry Service Mode Cable Terminator

3.4 Installation

- 1.) Remove the gantry display and control assembly from their service positions and re-attach them to the gantry cover.
 - a.) Disconnect cables from Display and Gantry Control Panels.
 - b.) Install Gantry Display in front cover. Secure the 5 thumbscrews. With a flat-blade screw driver gently tighten past finger tight.
 - c.) Install the gantry control panel making sure the ball studs are secure within the receivers.
 - d.) Re-attach cables.
- 2.) Rotate gantry back to its vertical position.



NOTICE
Potential for
front cover
damage.

When you put rotate the gantry back to its vertical position, make sure not to scratch the front cover with the edge of the table cradle.

- 3.) Attach the front cover.
 - a.) Align the studs on both sides of the front cover with each associated receiver. Receiver is located on the gantry frame.



Figure A-21 Cover stud and Mounting bracket receiver

- b.) Insert the stud on one side into its associated receiver and attach the rubber retaining straps. Then insert the stud on the other side into its associated receiver and attach its rubber retaining straps.

You may find it helpful to lift “up” on the cover to align the stud while attaching the rubber retaining straps.

- 4.) Re-attach upper and lower cantrell brackets on both sides.
 - a.) Remove upper Cantrell brackets from service position and rotate them into position over their associated retaining pins. See [Figure A-22](#).

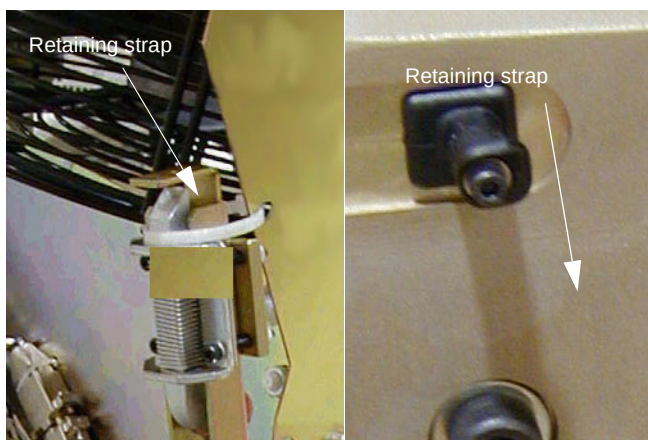


Figure A-22 Service position of upper and lower cantrell brackets.

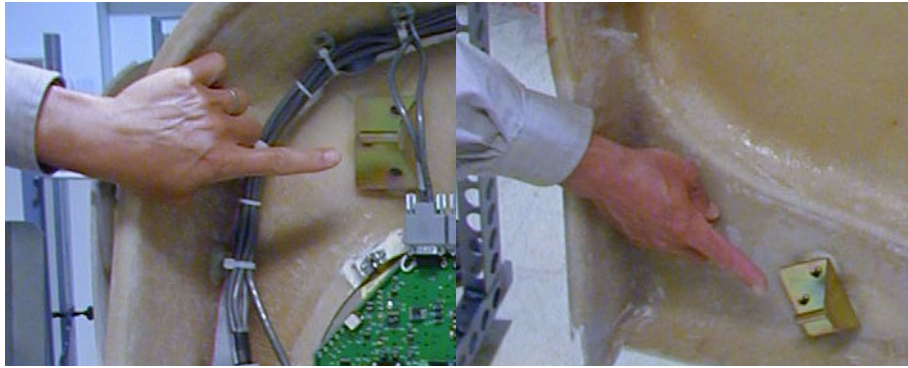


Figure A-23 Cover retaining pins (top and bottom)

Press down firmly on the bracket and snap it into place. The locking mechanism on each upper bracket should lock the bracket securely into place. Do this on both sides. See [Figure A-24](#).



Figure A-24 Locking the cover brackets into place.

- b.) Remove lower cantrell brackets from service position (see [Figure A-22](#)), and rotate them into position over their associated retaining pins. Press down firmly on the bracket and snap it into place. See [Figure A-24](#).

Note: Mis-adjustment of the cantrell brackets can cause misalignment of the top and side covers. The upper and lower cantrell brackets do not require adjustment during normal use.

- 5.) Remove dolly, disassemble and store safely away for later use.
- 6.) Re-attach cables to cover.

Section 4.0

Gantry Rear Cover

4.1 Removal

- 1.) Assemble the rear cover dolly.
 - a.) Tighten the two (2) shoulder bolts to the rear cover.

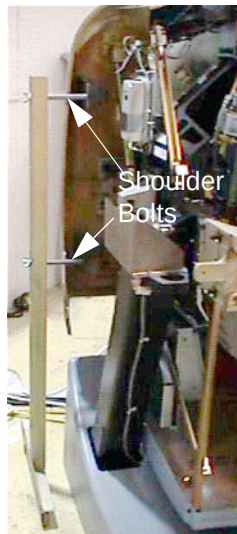


Figure A-25 One side of the Rear cover dolly

- b.) Fit side dolly through the shoulder bolts and secure assembly with two (2) wing nuts. See [Figure A-25](#).
 - c.) Repeat steps a and b for the other side dolly.



CAUTION
Potential for
injury if covers
removed and
power is left
"ON".

- 2.) Disconnect cables on the right side of the rear cover.
- 3.) Remove rear cover.
 - a.) Disengage upper and lower cantrell brackets on both sides of the rear cover.
 - 1.) Using steady but firm pressure, lift each of the lower cantrell brackets from their associated retainers. See [Figure A-11](#).
 - 2.) Disengage the locking mechanism on the upper cantrell brackets by using your thumb to slide the trigger (red lever) back. This will release the locking mechanism and allow the cantrell to be rotated upwards with steady and firm pressure.
 - b.) Disengage the rubber retaining straps on both sides.

4.2 Installation

- 1.) Position cover in back of gantry
- 2.) Attach the rear cover
 - a.) Align the studs on both sides of the rear cover with the receivers located on the gantry frame.
 - b.) Insert the stud on one side into its associated receiver and attach the rubber retaining straps. Then insert the stud on the other side into its associated receiver and attach its

rubber retaining straps.

Note: You may find it helpful to lift “up” on the cover to align the stud while attaching the rubber retaining straps.

- 3.) Reattach upper and lower cantrell brackets on both sides.
 - a.) Remove upper cantrell brackets from service position and rotate them into position over their associated retaining pins. Press down firmly on the bracket and snap it into place. The locking mechanism on each upper bracket should lock the bracket securely into place. Do this on both sides.
 - b.) Remove lower cantrell brackets from service position and rotate them into position over their associated retaining pins. Press down firmly on the bracket and snap it into place.

Note: Adjustment of the cantrell brackets can cause misalignment of the top and side covers. The upper and lower cantrell brackets do not re-quire adjust during normal use.

- 4.) Remove dolly, disassemble and store safely away.
- 5.) Re-attach cables to cover.
- 6.) Re-install the mylar (scan) window. Carefully bend the scan window and place it into the channel (groove) provided in the covers.



Figure A-26 Installing the mylar window

Section 5.0 Gantry Scan Window

5.1 Remove Scan Window

- 1.) Grab the window at the top and pull firmly downward.
- 2.) Continue to pull until the top of the scan window makes contact with the bottom portion of the scan window.
- 3.) Hold the top and bottom portions of the scan window together, grasp both sides of the scan window, move them together and lightly pull upward, until you can free the window from between the front and rear covers.



Figure A-27 Scan Window Removal

5.2 Install Scan Window

- 1.) Install the front and rear covers.
- 2.) Deform the scan window, as shown in [Figure A-28](#), and nest the scan window at the bottom of the opening between the front and rear covers, ([Figure A-29](#)) with the rivets in the 6 o'clock position installation position. Remember the rivets must be in the 12 o'clock position when the mylar window is fully installed.
- 3.) After you complete the initial seating of scan window, let the window slowly unfold, and work both sides of the window into position, starting at the bottom and finishing at the top.
- 4.) Make sure you position the window with the rivets at the 12 o'clock position, and the mylar window slit at either the 3 or 9 o'clock position.

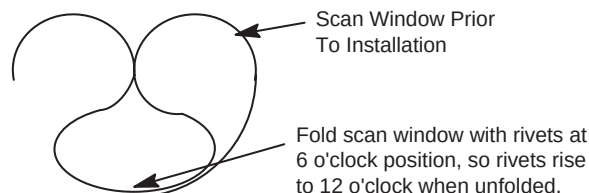


Figure A-28 Install Scan Window

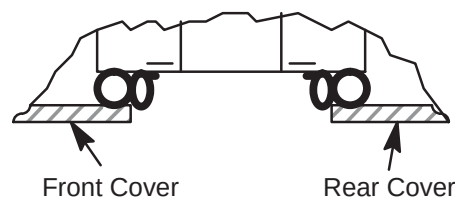


Figure A-29 Scan Window Nested Between Front and Rear Cover

Section 6.0

Gantry Base Covers

Refer to [Figure A-30](#) for the following assembly sequence.

Note: Tighten means torque to 2.3 Nm

- 1.) Assemble two (2) brackets (item 11) to front cover (item 5), using four (4) hardware items 1, 2 and 3. Align leg of brackets parallel to front edge of cover and tighten. Position cover on gantry base, with bracket slots aligned to gantry holes. Center cover left to right, and attach with four (4) hardware items 1, 2 and 3, as shown, and tighten.
- 2.) Assemble two (2) brackets (item 12) and two (2) brackets (item 13) to gantry base, using eight (8) hardware items 1, 2 and 3. Finger tighten hardware with brackets moved outward to end of slots. Install side covers (items 6 & 7) on base, pushing brackets (items 12 & 13) inward until properly aligned with front cover. Remove side covers, tighten fasteners and replace side covers, using two (2) hardware items 1, 2 and 3 and one item 4 on each cover, and tighten.
- 3.) Assemble last bracket (item 13) loosely to gantry base with two (2) hardware items 1, 2 and 3. Assemble latch (item 10) to rear cover (item 8), oriented as shown. Install rear cover (item 8) to base, properly aligned to side cover (item 6). Attach rear cover to bracket with hardware items 1, 2, 3 and 4, tightening all fasteners. Lock latch (item 10).
- 4.) Assemble two (2) latches (item 10) to rear cover (item 9), oriented as shown. Place cover on gantry base, aligned to covers 7 and 8. Lock both latches.

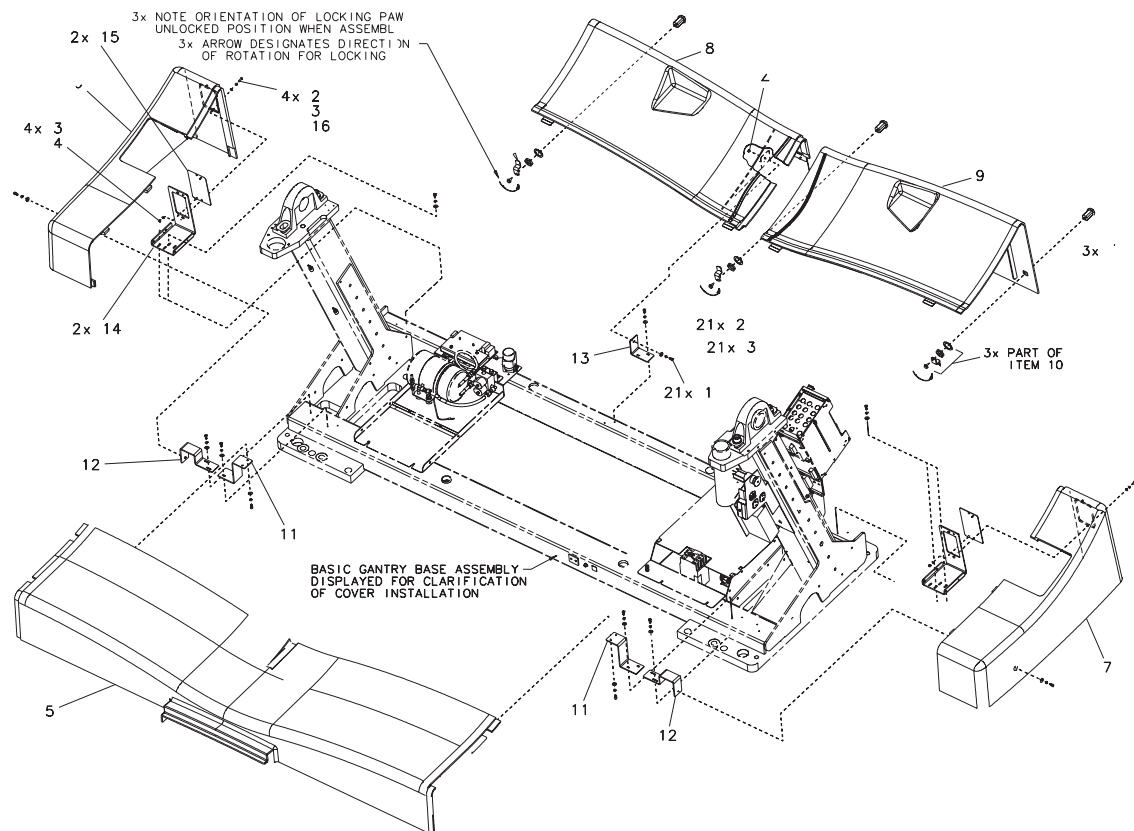


Figure A-30 Gantry Base Covers

Section 7.0

Foot-switch Covers

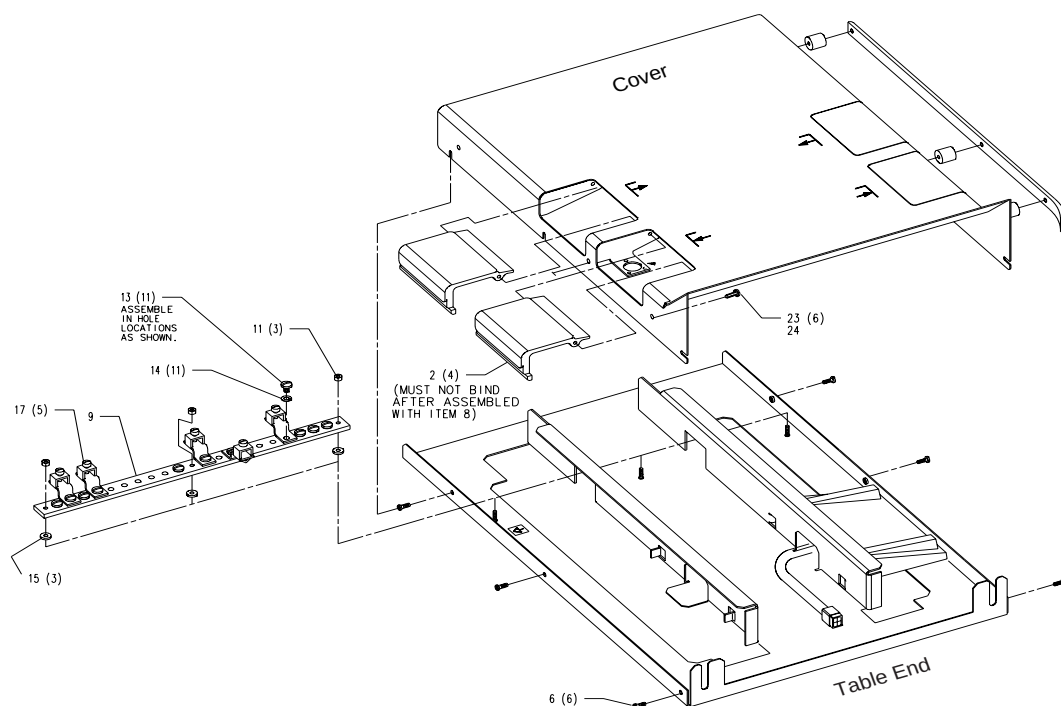


Figure A-31 Table Foot-switch Assembly

Appendix B

Pictorial Representation of Required Tools

Use the following guide as a reference, if you are unsure of a tool listed in [Section 1.6, on page 26](#).

TOOL NAME	PICTURE	EXAMPLE PART NUMBER*
Adapter		Sears Industrial: $\frac{3}{8}$ " to $\frac{1}{2}$ " (9-4258)
Ball-Peen Hammer		Sears Industrial: 1lb/2lb (9-38465)
Canned Air		Miller Stephenson: Aero Duster (MS-222N)
Clamp on Amp Meter		Sears Industrial: 9-WTAD105
Combination Wrench Set		Sears Industrial: U.S. Standard & Metric (9-44048)
Cordless Screwdriver		Sears Industrial: 9-MU65401
Deep Well Socket		Sears Industrial: $\frac{3}{4}$ " X $\frac{3}{8}$ " (included with 9-34496)
Dental Pick		
Diagonal Cutting Pliers		Sears Industrial: Small (9-45077)

* Part Numbers given for reference only. GEMs does not endorse any tool brand name.

Table B-1 Required Tools

TOOL NAME	PICTURE	EXAMPLE PART NUMBER*
Drill		Sears Industrial: $\frac{3}{8}$ " or $\frac{1}{2}$ " (9-27859)
Drill Adapter		Sears Industrial: 3" X $\frac{3}{8}$ " (9-APSZ24)
Drill Bit Set		Sears Industrial: U.S. Standard (9-66084)
DVM		Sears Industrial: 9-82028 Sears Industrial: 9-FL873
Extension for Ratchet Wrench		Sears Industrial: 3" X $\frac{1}{2}$ " (9-44133)
Gloves		Sears Industrial: Large (9-40502)
Hammer Drill		Sears Industrial: $\frac{1}{2}$ " (9-27205)
Hex Bit Set		Sears Industrial: $\frac{1}{4}$ " (9-SK45508)
Hex Key (Allen Wrench) Set		Sears Industrial: U.S. Standard (9-46284)

* Part Numbers given for reference only. GEMs does not endorse any tool brand name.

Table B-1 Required Tools (Continued)

TOOL NAME	PICTURE	EXAMPLE PART NUMBER*
Level		Sears Industrial: 4' (9-39856)
Masonry Bit		
Open-End Wrench (Thin or Standard Tappet)		Snap-on: 10mm (SRSM10) & 21mm (LTAM2124)
Ratchet Wrench		Sears Industrial: $\frac{3}{8}$ " (9-43175)
Reciprocating Saw with Blades		Sears Industrial: 9-MU650921
Safety Glasses		Sears Industrial: 9-18650
Safety Shoes		
Screwdriver Set		Sears Industrial: Phillips & Straight (9-41505)
Socket Set		Sears Industrial: Standard $\frac{3}{8}$ " (9-34496)
Sockets		Sears Industrial: 1 $\frac{1}{8}$ " X $\frac{1}{2}$ " (9-47516)

* Part Numbers given for reference only. GEMs does not endorse any tool brand name.

Table B-1 Required Tools (Continued)

TOOL NAME	PICTURE	EXAMPLE PART NUMBER*
Step Ladder		Sears Industrial: 6' (9-WN6006)
Tongue & Groove Pliers		Sears Industrial: Large (9-CL440)
Torpedo Level		Sears Industrial: 9" (9-39829)
Torque Wrench		Sears Industrial: $\frac{3}{8}$ " (9-WR3470)
Universal Joint		Sears Industrial: $\frac{3}{8}$ " (9-4435)
Vacuum Cleaner		Sears Industrial: 8 Gal (9-17780)

* Part Numbers given for reference only. GEMs does not endorse any tool brand name.

Table B-1 Required Tools (Continued)



GE MEDICAL SYSTEMS

**GE MEDICAL SYSTEMS-AMERICAS: FAX 262.544.3384
P.O. BOX 414; MILWAUKEE, WISCONSIN 53201-0414, U.S.A.**

**GE MEDICAL SYSTEMS-EUROPE: FAX 33.1.40.93.33.33
PARIS, FRANCE**

GE MEDICAL SYSTEMS-ASIA: FAX 65.291.7006



GE Medical Systems

gemedical.com

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Book 2 of 2
Pages 143 - 298

GE Medical Systems **HiSpeed QX/i Installation Manual**

Chapters 4 to End
(Electrical Installation Procedures)

Retain this installation manual permanently with your CT System. Any other revisions of this installation manual may not exactly match your system.

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Table of Contents: Book 2

Chapter 4

Electrical Introduction.....	153
Section 1.0	
Introduction	153
Section 2.0	
Review Mechanical Hand Off Material.....	153
Section 3.0	
Electrical Procedures Block Diagram	154
Section 4.0	
Required FE Common Tools and Supplies.....	155
4.1 FE Calibration and Service Tool List	155
4.2 Electrical Tools	155
4.3 Image Quality Calibration Tools.....	155
4.4 Detector Service Tools	155
4.5 Optional Tools.....	156
4.6 Safety Materials	156
4.7 Cleanliness	156
Section 5.0	
Requirements/Assumptions.....	156

Chapter 5

Electrical Power ON & Ground Checks	157
Section 1.0	
Compact PDU - Electrical Configuration.....	157
1.1 Required Tools	157
1.2 Initial PDU Configuration	158
1.2.1 Circuit Breakers	158
1.2.2 Relay Board	158
1.2.3 Power Switches	158
1.2.4 Hardware and Connection Check.....	158
1.2.5 Covers	158
Section 2.0	
NGPDU - Electrical Configuration.....	159
2.1 Required Tools	159
2.2 Initial PDU Configuration	159
2.2.1 Circuit Breakers	160
2.2.2 Relay Board	160
2.2.3 Power Switches	160
2.2.4 Hardware and Connection Check.....	160
2.2.5 Covers	160
Section 3.0	
Suite Emergency Off Checks	161

Section 4.0	
Line Transformer Settings	162
4.1 Requirements	162
4.2 Line Input Conditions	162
4.2.1 Compact PDU	162
4.2.2 NGPDU	164
Section 5.0	
System Power Up	165
Section 6.0	
Test HVDC Enable Switch.....	166
Section 7.0	
Emergency Stop Check.....	168
 Chapter 6	
Computer Integration.....	169
 Section 1.0	
Introduction.....	169
 Section 2.0	
System Configuration Data Sheets.....	170
2.1 Requirement.....	170
2.2 Manual Film Composer Options.....	171
2.3 System Network Configuration.....	172
2.4 Host Ethernet Address	172
2.5 Network Application (Image Transfer) Configuration	173
2.6 Camera Application Configuration	173
 Section 3.0	
Configuration	175
3.1 Restore System State	175
3.2 Install Customer Options.....	176
3.2.1 Software	176
3.2.2 Camera	177
3.3 Monitor Setup.....	178
3.3.1 Overview	178
3.3.2 Color Adjustments	179
3.3.3 Using Menu Buttons	180
3.3.4 Adjusting Brightness and Contrast.....	180
3.4 Shut Down Application	181
3.5 Check/Set Date and Time	181
3.6 Reconfig the OC.....	181
3.6.1 Overview	181
3.6.2 Procedure.....	182
3.7 Applications Start-Up	188

Chapter 7

Table and Gantry Integration..... 189

Section 1.0

Introduction 189

Section 2.0

Autovoice/Intercom Check 190

2.1 Gantry Speaker..... 190

2.2 Console Speaker 190

2.3 Autovoice Volume..... 190

Section 3.0

Table/Gantry Characterizations 191

Section 4.0

Interference Test 192

Section 5.0

Tilt Speed Adjustment Procedure..... 193

Section 6.0

System Checks..... 194

6.1 Alignment Lights 194

6.1.1 Turn Alignment Lights ON 194

6.1.2 Internal Axial Lights 194

6.1.3 External Axial to Internal Axial Distance 194

6.1.4 Coronal Lights..... 194

6.1.5 Turn Lights OFF..... 194

6.2 Laser Warning Labels..... 195

Section 7.0

Collimator Characterization & Flash Download 195

Chapter 8

DAS Integration 197

Section 1.0

Introduction 197

1.1 Failure Protocol..... 197

1.2 Special Handling Precautions..... 197

1.3 Anti-static Handling..... 197

1.4 Cleanliness 197

1.5 Required Tools 198

1.6 Process Overview..... 198

Section 2.0

DAS Tools 199

Section 3.0

Verify X-Ray/Signal to Noise Test..... 201

Chapter 9

System Alignments 203

Section 1.0

Introduction..... 203

- 1.1 Required Tools..... 204
- 1.2 Gantry Geometry Definitions..... 204
- 1.3 Tube Heat Definitions..... 204

Section 2.0

Plane of Rotation (POR) 205

- 2.1 Overview 205
- 2.2 Tools Required..... 206
- 2.3 Procedure..... 206
 - 2.3.1 Start the POR Software..... 206
 - 2.3.2 Measure Tube Alignment 207

Section 3.0

Beam on Window Alignment (BOW) 210

- 3.1 Overview 210
- 3.2 Cool Tube..... 211
- 3.3 Access the Software 211
- 3.4 Adjustment Procedure..... 211

Section 4.0

CBF/SAG Alignment Process 213

- 4.1 Cool Tube..... 213
- 4.2 Access the Software 214
- 4.3 Adjustment Procedure..... 215

Section 5.0

ISO Alignment..... 216

- 5.1 Overview 216
- 5.2 Cool Tube..... 216
- 5.3 Access the Software 216
- 5.4 ISO Adjustment Procedure 217

Section 6.0

HOT ISO Alignment 218

- 6.1 Accessing the Software..... 218
- 6.2 Adjustment Procedure..... 218

Chapter 10

Generator Calibration 219

Section 1.0

Introduction..... 219

- 1.1 Process Overview 219
- 1.2 Required Tool..... 219

Section 2.0

mA Meter Check With Verification 220

2.1	mA Meter Verification - HHS Requirement	221
2.2	Auto mA Cal.....	221
2.3	X-ray Lights.....	222

Chapter 11 System Tests 223

Section 1.0		
Introduction		223

Section 2.0		
Calibration Process.....		224

2.1	Reference Procedures.....	224
2.1.1	Scanning with Service Protocols	224
2.1.2	Center Phantom.....	224
2.2	Prepare the QA Phantom	224
2.3	Calibration Process Introduction.....	224
2.4	Fast Cal	224
2.5	Table/Gantry Alignment Procedure	225
2.5.1	Conditions.....	225
2.5.2	Procedure	225
2.6	Auto CT N Number Adjustment	226

Section 3.0		
System Tests		227

3.1	Introduction	227
3.2	Tomographic Plane Indication	227
3.3	System Scanning Test.....	228

Section 4.0		
Collimator Aperture Test		229

4.1	Small Aperture Test.....	229
4.2	Large Aperture Test.....	229

Section 5.0		
Image Series		230

5.1	Scan Protocol	230
5.2	Data Recording: Means and Standard Deviation	230
5.3	Term Definitions & Screens	230
5.4	48cm Phantom Image Series Image Performance Verification	232
5.4.1	Acquiring the 48cm Phantom Image Series	232
5.4.2	Brightness Uniformity and Noise	233
5.4.2.1	Image Performance Verification	233
5.4.2.2	Failure Recovery	235
5.5	20cm QA Phantom Image Series Image Performance Verification.....	236
5.5.1	Acquiring the 20cm QA Phantom Image Series	236
5.5.2	High Contrast Spatial Resolution.....	237
5.5.2.1	Image Performance Verification	237
5.5.2.2	Failure Recovery	240
5.5.3	Low Contrast Detectability.....	241
5.5.3.1	Image Performance Verification	241
5.5.3.2	Failure Recovery	243

5.5.4	QA#3 Phantom Brightness Uniformity and CT#.....	243
5.5.4.1	Performance Verification.....	243
5.5.4.2	Failure Recovery.....	246
5.5.5	QA#3 Phantom Noise	246
5.5.5.1	Performance Verification.....	246
5.5.5.2	Failure Recovery.....	248
Section 6.0		
Save System State.....		249

Chapter 12

Hospital/System Network Integration 251

Section 1.0		
Overview.....		251

Section 2.0		
Before You Start		252

2.1	Network Physical Requirements	252
2.2	HiSpeed QX/i Network Identity Information.....	252
2.3	HiSpeed QX/i to DICOM Remote Hosts Network Information	252
2.4	HiSpeed QX/i to DICOM HIS/RIS Interface Network Information	252
2.5	HiSpeed QX/i to DICOM Printers Network Information.....	253

Section 3.0		
Declaring the System on the Hospital Network.....		254

3.1	Enter Configuration Routine.....	254
3.2	Configure Network Settings	255
3.3	Initiate System Reconfiguration	257

Section 4.0		
Declaring Remote Hosts on the System.....		258

4.1	Enter Remote Host Configuration Screen.....	258
4.2	Declaring Advantage NET Remote Hosts on the HiSpeed QX/i	258
4.3	Declaring DICOM Remote Hosts on the HiSpeed QX/i	259

Section 5.0		
Declaring the System on Remote Hosts.....		261

5.1	Declaring HiSpeed QX/i on Advantage NET Protocol Devices/Systems	261
5.2	Declaring HiSpeed QX/i on DICOM Protocol Devices/Systems	261

Section 6.0		
Declaring DICOM HIS/RIS Interfaces on the System.....		262

6.1	Prerequisites	262
6.2	Loading ConnectPRO Software Option on the System	262

Section 7.0		
Declaring DICOM Filming Devices on the System		264

7.1	Prerequisites	264
7.2	Declaring DICOM Filming Devices on the System.....	264
7.3	Troubleshooting Tips.....	274

Section 8.0		
Teleradiology (Framegrabber Type) Systems & Aux. Monitors		276

8.1	Introduction	276
8.2	Auxiliary Monitors	277
8.3	GEMS Analog Filming Interface Specs (Video & Serial)	278

Chapter 13

FORM 4879 (Data Record for HiSpeed QX/i)	281
---	------------

Section 1.0

Site History	281
--------------------	-----

Section 2.0

System Tests	283
--------------------	-----

Chapter 14

Install Completion Checklist	291
---	------------

Section 1.0

CT System.....	291
----------------	-----

Section 2.0

CT Options.....	291
-----------------	-----

Section 3.0

System Cleanup	291
----------------------	-----

Section 4.0

Dollies Return.....	291
---------------------	-----

Section 5.0

Install Completion Material Checks.....	292
---	-----

Appendix C

Additional Characterization Procedures	293
---	------------

Section 1.0

Tilt Characterization.....	293
----------------------------	-----

Section 2.0

Elevation Characterization	294
----------------------------------	-----

2.1 Prerequisite.....	294
-----------------------	-----

2.2 Procedure	294
---------------------	-----

Section 3.0

Cradle Characterization	295
-------------------------------	-----

Chapter 4

Electrical Introduction



NOTICE
Potential for
Data Loss and/
or Equipment
Damage

To prevent potential data loss and equipment damage, please do the following:

- Record data collected from procedures in this chapter into Form F4879 when directed, located in [Chapter 13](#) of this book.
- Only use the Installation manual that arrives with your system for installation. Any other revisions of this manual may not exactly match your system.

Section 1.0 Introduction

Use the continuity and ground checks to verify the system power connections have not shorted to ground and that the ground and neutral connections are intact.

Section 2.0 Review Mechanical Hand Off Material

Complete the Mechanical Hand Off checklist:

- ☐ All options were installed. If not, contact your install specialist.
- ☐ Check for short ships.
- ☐ Review cable connections with mechanical team.
- ☐ Complete mechanical vendor evaluation.

Section 3.0 Electrical Procedures Block Diagram

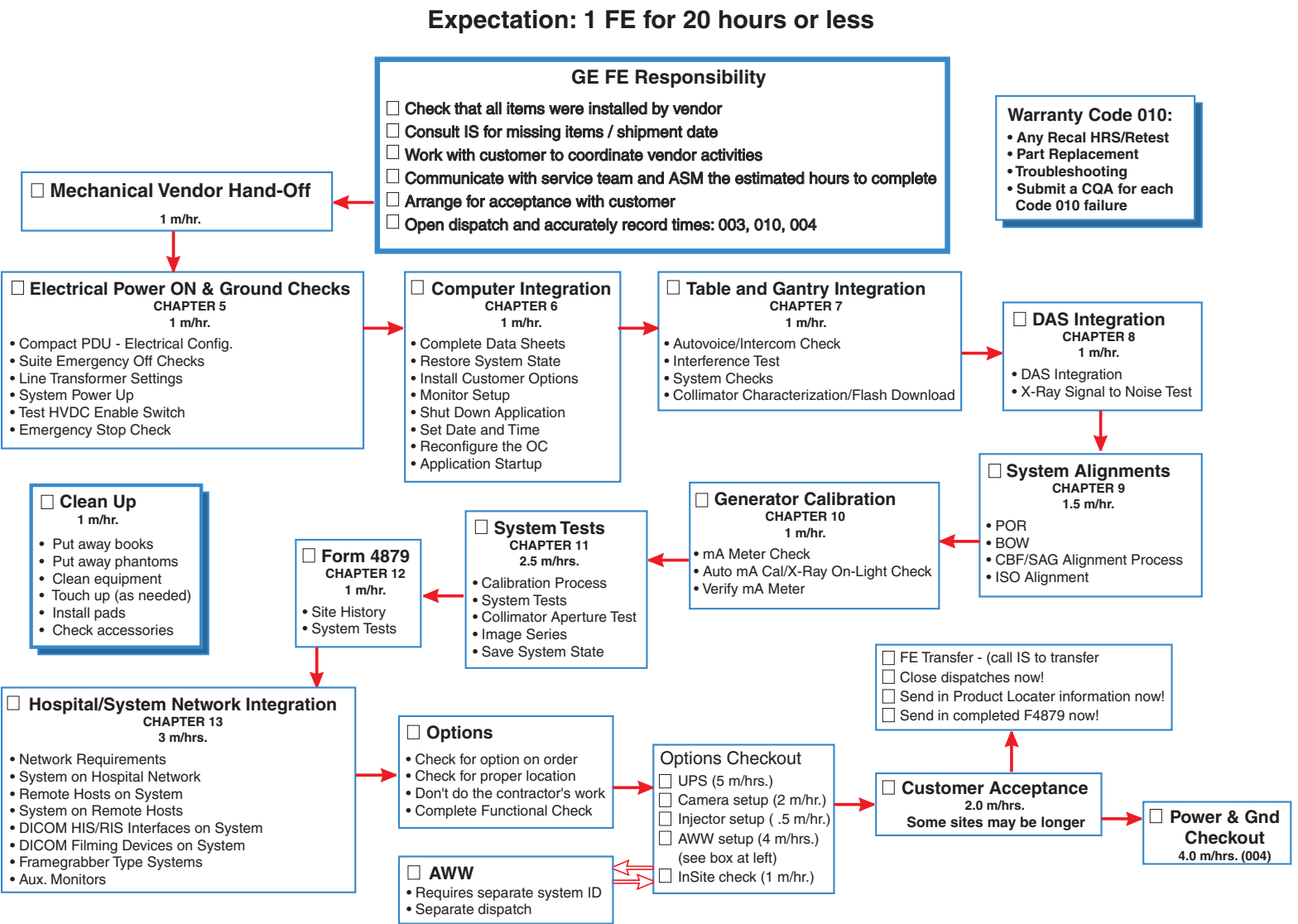


Figure 4-1 Electrical Installation Block Diagram

Section 4.0

Required FE Common Tools and Supplies

4.1 FE Calibration and Service Tool List

Special CT Tools used for mechanical alignments (found in the install support kit):

- 5mm hex bit for 3/8" drive 6" long (Snap-On FAML5E or equivalent)
- 10mm hex bit for ratchet wrench (ball end) (Snap-On FABM10E or equivalent)
- 2.5mm Allen hex bit for 1/4"
- Standard FE Tool Kit (not included in install support kit)
- 10mm open-end thin wrench (Snap-On SRSM10 or equivalent)
- 21mm open-end thin wrench (Snap-On LTAM2124 or equivalent)
- 14mm ball hex socket bit for 3/8" drive
- 14 mm hex socket bit for 1/2" drive

4.2 Electrical Tools

These tools must be calibrated yearly.

- Fluke 87 DVM or equivalent
- Clamp on amp meter

4.3 Image Quality Calibration Tools

- GE Performance Phantom
- QA Phantom (2206352)
 - 20cc (2144721)
 - 40cc (2144721-2)
 - Phantom Holder

4.4 Detector Service Tools

These tools can be found in the install support kit.

- 2.5 mm hex bit
- Anti-Static Bags
- Electric Screw Driver
- Canned Air (Grounded) with metal nozzle (Miller Stephenson Aero Duster MS-222N or equivalent)
- Ground Wrist Strap
- Grounded Mat
- Dental Pick

4.5 Optional Tools

Needed for warranty recalibration:

- GE HV Bleeder
- Scope with 10X probes
- Beckman CT231 clamp-on Amp probe: 46-194427P228 (Fits up to #2 size wire)
- Beckman CT232 clamp-on Amp probe: 46-194427P270 (Fits #1 size wire and larger)
- Scope Probe-to-bleeder Cable 46-219921G1 (Belden #8422 cable, 30 ft)
- Polaroid type 52 film and developer
- Radial dial indicator (mm or inches) and mounting bracket
- Caliper Dial indicators (mm or inches)
- Ground Rod Tester Clamp on Amp Probe
- Torque Wrench kit

4.6 Safety Materials

- Lockout/Tagout kit, or equivalent
- Safety Glasses
- Nitrile Rubber Gloves

4.7 Cleanliness

Any dirt on the surface increases leakage current on the filter or converter cards and causes the DAS to fail the drift spec. Wear Nitrile gloves (part number 2207303-6 [large] and 2207303-7 [extra large]) when you handle the DAS, because fingerprints on the board can cause problems during humid conditions. Use only clean, new Nitrile gloves. Do not use hospital grade gloves.



NOTICE NEVER USE AN ERASER TO CLEAN ANY PART OF THE DAS.
DUST COVERS REQUIRED for installations on construction sites.

Section 5.0 Requirements/Assumptions

- All stations in a suite must have the same SUITE ID.
- You need the name of all hosts in the suite and their corresponding Internet/Ethernet numbers.
- You need the Internet (IP) addresses the first time you execute a reconfig on the system.
 - When you connect the system to a network, contact the system administrator to obtain the IP addresses for all the computers in the suite.
 - You also need an IP address for each gateway (second) ethernet board in any OC or IC.
 - You can use the default internet number on stand-alone systems (stand-alone = not connected to any network).
- For more detailed information and instructions regarding Network Integration, see [Chapter 12 on page 251](#).

Chapter 5

Electrical Power ON & Ground Checks



NOTICE
Potential for
Data Loss and/
or Equipment
Damage

To prevent potential data loss, please do the following:

- Record data collected from procedures in this chapter into Form F4879 when directed, located in [Chapter 13](#) of this book.
- Only use the Installation manual that arrives with your system for installation. Any other revisions of this manual may not exactly match your system.

Section 1.0

Compact PDU - Electrical Configuration



WARNING



THIS PROCEDURE MEASURES POTENTIALLY HAZARDOUS VOLTAGES. USE AND FOLLOW LOCKOUT/TAGOUT PROCEDURES.

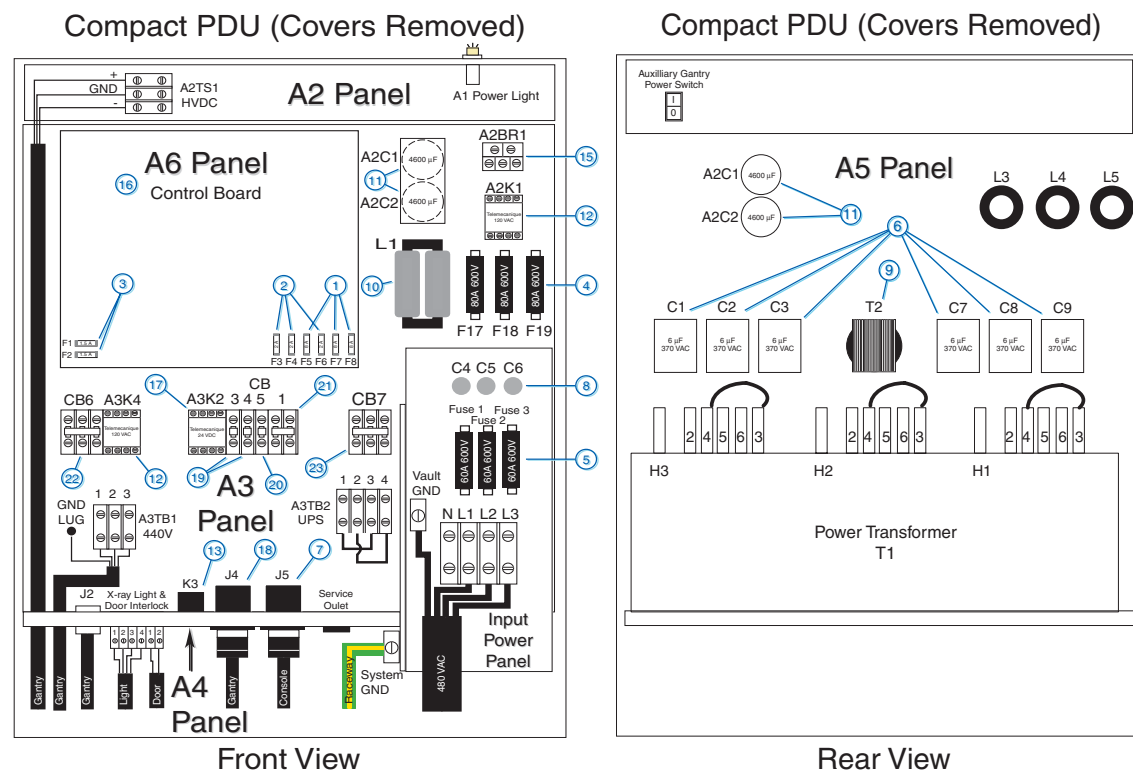
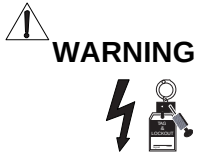


Figure 5-1 Compact PDU

1.1 Required Tools

- Multimeter with a rating of at least 1000 volts
- Multimeter leads with a rating of at least 1000 volts

1.2 Initial PDU Configuration



WARNING

THIS PROCEDURE MEASURES POTENTIALLY HAZARDOUS VOLTAGES. USE AND FOLLOW LOCKOUT/TAGOUT PROCEDURES.

1.2.1 Circuit Breakers

Set all circuit breakers to OFF

1.2.2 Relay Board

- 1.) Set SW1 and SW2 to the normal position.
- 2.) Set the Room Light jumper (JP1) to meet hospital requirements (refer to [Figure 5-2](#)).
 - Position A enables the room warning light during X-Ray only.
 - Position B enables the room warning light as soon as HVDC turns on.

Note:

Use position B unless the hospital requires otherwise.

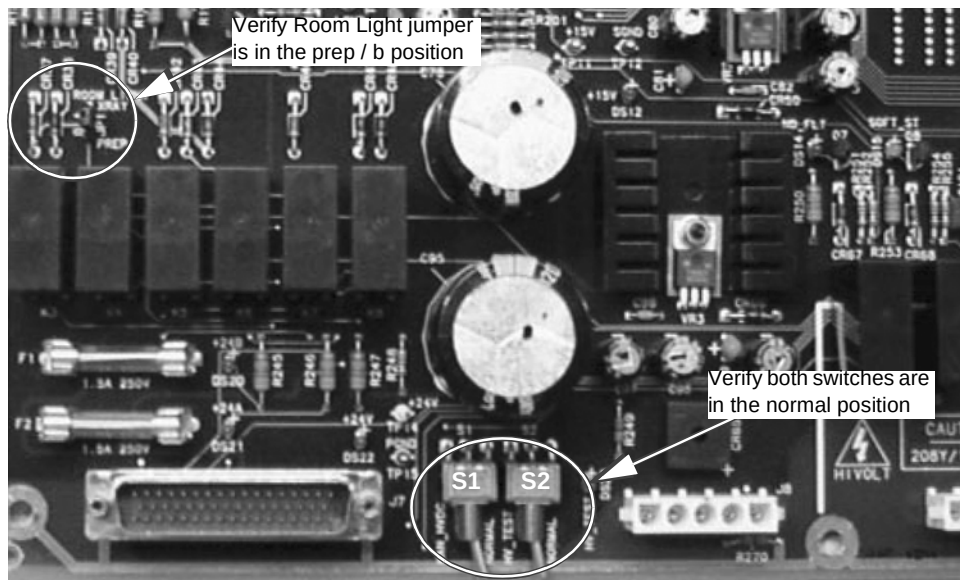


Figure 5-2 PDU Control Board (A6 Panel)

1.2.3 Power Switches

Turn OFF all power switches on all subsystems (gantry status display box, STC, DAS, table, console, computer, and SCU).

1.2.4 Hardware and Connection Check

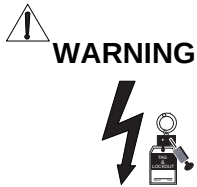
Use this step to check mechanical connections and tighten anything that may have shaken loose during shipment. Verify **all** hardware and connections in the PDU are securely fastened.

1.2.5 Covers

Install, or verify the presence of, all the lexan safety covers.

Section 2.0

NGPDU - Electrical Configuration



WARNING

THIS PROCEDURE MEASURES POTENTIALLY HAZARDOUS VOLTAGES. USE AND FOLLOW LOCKOUT/TAGOUT PROCEDURES.

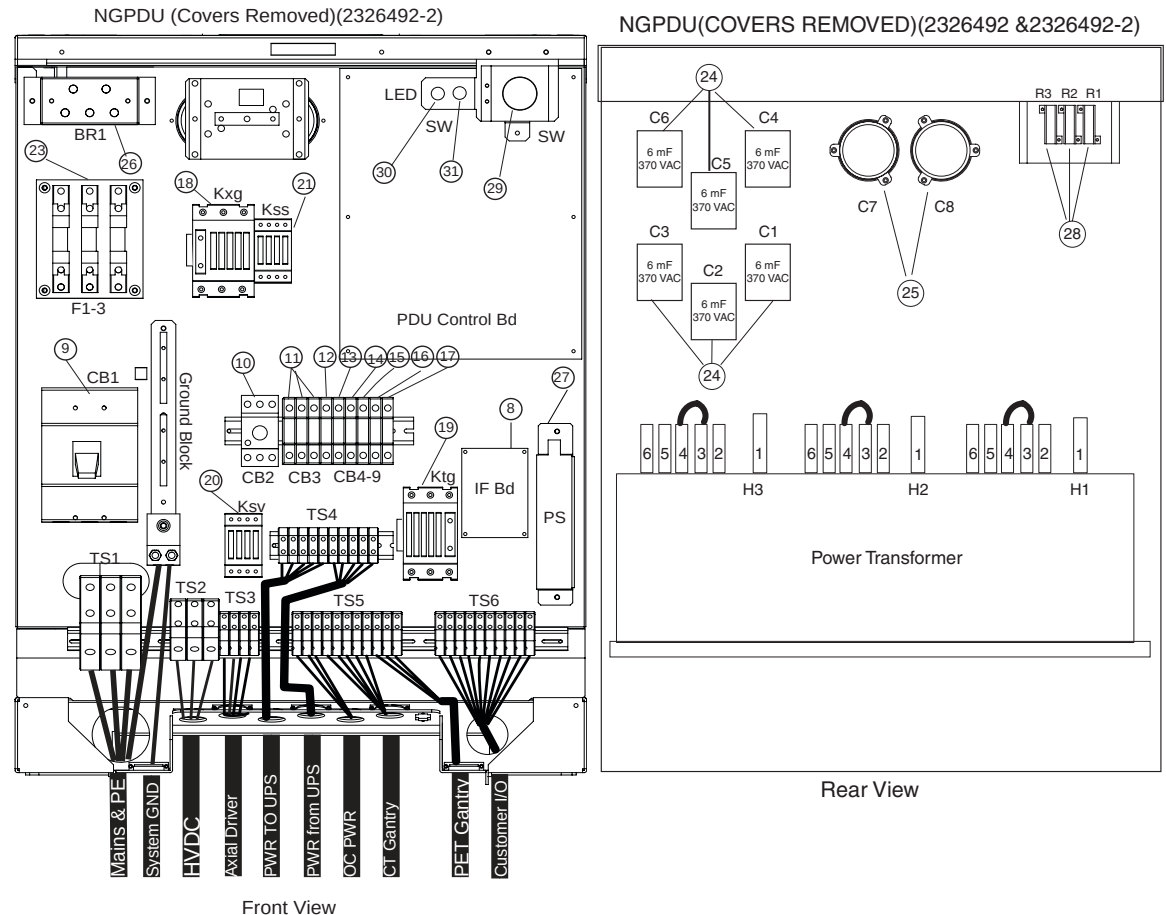


Figure 5-3 NGPDU

2.1 Required Tools

- Multimeter with a rating of at least 1000 volts
- Multimeter leads with a rating of at least 1000 volts

2.2 Initial PDU Configuration



WARNING

THIS PROCEDURE MEASURES POTENTIALLY HAZARDOUS VOLTAGES. USE AND FOLLOW LOCKOUT/TAGOUT PROCEDURES.

2.2.1 Circuit Breakers

Set all circuit breakers to OFF

2.2.2 Relay Board

- 1.) Set SW to the normal position.
- 2.) When system is already, three lamps are both lighting (refer to [Figure 5-4](#)).

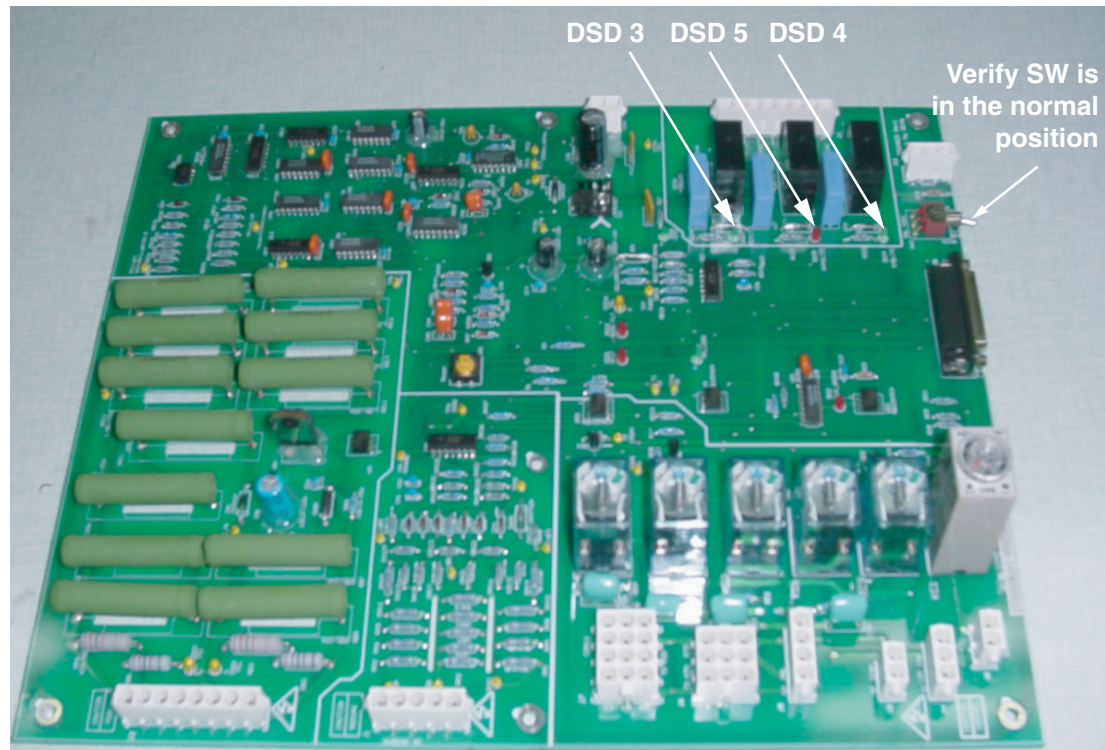


Figure 5-4 NGPDU Control Board

2.2.3 Power Switches

Turn OFF all power switches on all subsystems (gantry status display box, STC, DAS, table, console, computer, and SCU).

2.2.4 Hardware and Connection Check

Use this step to check mechanical connections and tighten anything that may have shaken loose during shipment. Verify **all** hardware and connections in the PDU are securely fastened.

2.2.5 Covers

Install, or verify the presence of, all the lexan safety covers.

Section 3.0

Suite Emergency Off Checks

This procedure is for systems with PDU 2133533 and 2326492-2 (NGPDU).



WARNING



VERIFY ALL PERSONNEL HAVE CLEARED THE SYSTEM BEFORE YOU TURN ON WALL POWER.

- 1.) Turn wall power ON to the PDU.
- 2.) **Press the suite emergency off button and verify it turns off wall power to the PDU.**
[Typically, this red palm button is located on the wall close to the console, within the scan suite.]
- 3.) Verify that all "Emergency Off" button are working properly.
- 4.) Leave power "OFF".

Section 4.0

Line Transformer Settings

4.1 Requirements

- 1.) The PDU is shipped configured for 480VAC.
- 2.) Complete only if your site uses a voltage other than 480VAC.
- 3.) If PDU is configured for 480VAC, go to Section 5.0. Otherwise, proceed to [Section 4.2](#).



WARNING

TURN OFF, TAG AND LOCK THE MAIN WALL POWER BEFORE CHANGING TAPS. FAILURE TO DISCONNECT POWER AT MAIN INPUT MAY RESULT IN ELECTROCUTION. TURN OFF WALL POWER TO CONNECT OR MOVE METER LEADS, OR TO REMOVE OR INSTALL COVERS.

4.2 Line Input Conditions

4.2.1 Compact PDU

- 1.) Remove metal cover over right A3 panel.
- 2.) Monitor the No Load Line to Line Voltage at A3 L1, L2, L3, during the workday. Do not record this data during “brown out” conditions.
- 3.) After you determine the nearest nominal line, verify the tap connections match (refer to [Table 5-1](#) and [Figure 5-5](#) for tap locations).

Transformer Taps and Jumpers

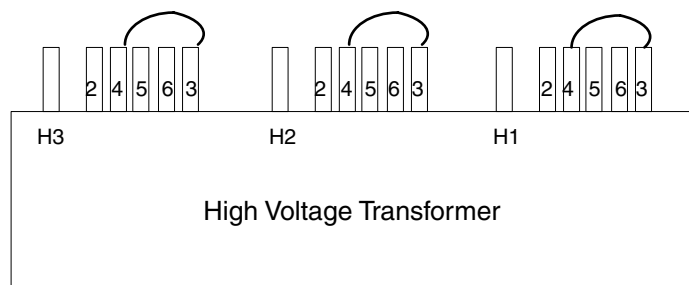
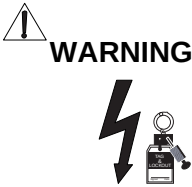


Figure 5-5 PDU Tap Positions (Rear)

Note: Taps should be shipped as shown for 480 VAC only. For all others, you must move the taps. The tap check should be completed by the mechanical installer.

- 4.) Verify that the No Load Line to Line Voltage never falls outside the corresponding minimum and maximum values listed in [Table 5-1](#).
- 5.) Use a 0-750 AC voltmeter of 3/4% accuracy to measure the line-to-line voltages at A3 L1, L2, and L3.
 - Verify the highest line-to-line voltage does not exceed 1.02 times the lowest voltage.
 - **Example:** If the lowest voltage equals 474, the highest voltage should not exceed $474 \times 1.02 = 483.5$ volts.



WARNING

THIS PROCEDURE MEASURES POTENTIALLY HAZARDOUS VOLTAGES. USE AND FOLLOW LOCKOUT/TAGOUT PROCEDURES.

6.) Replace the metal cover over the right A3 panel.

NO LOAD		TAP CONNECTIONS		
Line to Line Voltages		(All 3 phases must have same the configuration)		
Nominal	Maximum Range (8%)	Phase A Connection	Phase B Connection	Phase C Connection
480V*	442 to 518*	3-4*	3-4*	3-4*
460V	423 to 497	3-5	3-5	3-5
440V	405 to 475	3-6	3-6	3-6
420V	386 to 454	2-4	2-4	2-4
400V	368 to 432	2-5	2-5	2-5
380V	350 to 410	2-6	2-6	2-6
* Factory Default				

Table 5-1 PDU 2269902 Line Tap Connections

4.2.2 NGPDU

- 1.) Monitor the No Load Line to Line Voltage at L1, L2, L3, during the workday. Do not record this data during "brown out" conditions.
- 2.) After you determine the nearest nominal line, verify the tap connections match (refer to [Table 5-2](#) and [Figure 5-6](#) for tap locations).

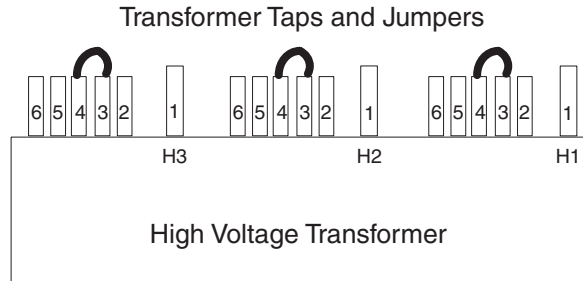


Figure 5-6 PDU Tap Positions (Rear)

Note: Taps should be shipped as shown for 480 VAC only. For all others, you must move the taps. The tap check should be completed by the mechanical installer.

- 3.) Verify that the No Load Line to Line Voltage never falls outside the corresponding minimum and maximum values listed in [Table 5-2](#).
- 4.) Use a 0-750 AC voltmeter of 3/4% accuracy to measure the line-to-line voltages at L1, L2, and L3.
 - Verify the highest line-to-line voltage does not exceed 1.02 times the lowest voltage.
 - **Example:** If the lowest voltage equals 474, the highest voltage should not exceed $474 \times 1.02 = 483.5$ volts.



WARNING

THIS PROCEDURE MEASURES POTENTIALLY HAZARDOUS VOLTAGES. USE AND FOLLOW LOCKOUT/TAGOUT PROCEDURES.

NO LOAD Line to Line Voltages		TAP CONNECTIONS (All 3 phases must have same the configuration)		
Nominal	Maximum Range (8%)	Phase A Connection	Phase B Connection	Phase C Connection
480V*	442 to 518*	3-4*	3-4*	3-4*
460V	423 to 497	3-5	3-5	3-5
440V	405 to 475	3-6	3-6	3-6
420V	386 to 454	2-4	2-4	2-4
400V	368 to 432	2-5	2-5	2-5
380V	350 to 410	2-6	2-6	2-6
240V	221 to 259	1-4	1-4	1-4
220V	202 to 238	1-5	1-5	1-5
200V	184 to 216	1-6	1-6	1-6
* Factory Default				

Table 5-2 PDU 2326492-2 Line Tap Connections

Section 5.0

System Power Up

- 1.) Turn OFF wall power.
- 2.) Turn ON all PDU A3 circuit breakers.



CAUTION Verify all personnel have cleared the system before you turn on wall power.

- 3.) Turn ON wall power.
- 4.) Check A1 Emergency Off push-button.

SUB-SYSTEM POWER-UP

- 1.) Turn ON switch S3 in the table (120vac 24hr power).
- 2.) Turn ON the gantry 120VAC enable switch on the STC. Tube fans turn on.
- 3.) Leave AXIAL DRIVE ENABLE OFF.
- 4.) Leave HV DC ENABLE OFF.
- 5.) Turn ON the DAS power switch located on the outside of the DAS power supply assembly.
- 6.) Turn ON Table switches S1 and S2.
- 7.) Turn ON console power.
- 8.) Make certain that the switch on the back of the PDU is ON.
- 9.) Turn ON HEMRC Power ON.
- 10.) Verify that the gantry 210VAC is ON at the Gantry Power Pan.

Section 6.0

Test HVDC Enable Switch



1.) If you are not on the Service Desktop, click on the SERVICE DESKTOP icon.



2.) Click on the UTILITIES icon.

3.) Select INSTALL.

4.) Select TABLE/GANTRY INTEGRATION.

5.) Select HV DC ENABLE SWITCH.

6.) Select START.

7.) Select the following parameters in the Rotor Functional window:

PARAMETER	MX200
Test Selection:	HEMRC
Rotor Speed:	LOW
Rotor Loop:	OPEN
Rail Diff Voltage:	Normal
Test Duration:	10 seconds
OBC Service Input:	Disabled

Table 5-3 Rotor Functional Window Parameters

8.) Verify all personnel have cleared the system, *before* you enable the HVDC.

9.) Verify the following:

- a.) HVDC enable switch on the STC is ON
- b.) "HVDC ENABLE ON" light on the STC is ON

10.) Select RUN. If the system asks for a diagnostic download at this time, answer YES, and wait 90 seconds.

11.) Verify the "HVDC BACKUP CONTACTOR CLOSED LIGHT" on the STC status display turns ON.

12.) When the test finishes, verify the system displays the following message along the bottom of the screen: Diagnostics have completed with zero errors reported.



Figure 5-7 STC Service Switches

VERIFY/PERFORM

- 1.) Turn OFF the HVDC enable switch on the gantry service switch box.
- 2.) Select RUN to run the test again.
- 3.) Verify the test aborts.
- 4.) Verify the message log contains the following error: *Gentry I/O backup contactor error*.
- 5.) Select FILE → EXIT.
- 6.) Select DISMISS.
- 7.) Select CLEANUP.
- 8.) Wait 90 seconds for the message "Scanning Hardware Reset Successful" to appear on the screen.

AXIAL ENABLE SWITCH TEST

- Note: For the initial condition, do not leave the tube at the 2:30 position.
- 1.) Turn OFF axial drive enable switch S2 on the STC backplane.
 - 2.) Clear the gantry area for rotation.
 - 3.) Press the alignment light push button.
 - 4.) Verify that the gantry did not rotate.
 - 5.) Turn ON axial drive enable switch S2 on the STC backplane.



WARNING

MAKE SURE THAT THERE IS NO OBSTRUCTIONS AROUND THE GANTRY. PRESSING THE ALIGNMENT LIGHT PUSHBUTTON WILL CAUSE THE GANTRY TO ROTATE.

- 6.) Press the alignment light push button.
- 7.) Verify that the gantry rotates.

ROTATION SAFETY CHECKLIST

- 1.) Turn OFF enable switch S2 on the gantry.
- 2.) Turn OFF HV enable switch S3.
- 3.) Press red E-STOP button.
- 4.) Manually rotate the gantry 360 degrees. (Keep one finger at the console E-STOP button.)
 - Listen for any interference between the rotating and stationary parts.
(Correct any interference problems.)
 - Listen for any loose parts.
(Tighten parts as needed.)
- 5.) Turn ON the enable switch S2 on the gantry.
- 6.) Turn ON the Drive power switch S1.
- 7.) Leave the large gantry bay door half open. (This will make it easier to hear for any loose or interfering parts.)
- 8.) Using DDC, prescribe a 4 second X-ray OFF scan. (Keep one finger at the console E-STOP button.)
 - Listen for any interference between the rotating and stationary parts.
(Correct any interference problems.)
 - Listen for any loose parts.
(Tighten parts as needed.)
- 9.) After completing the 4 second scan, repeat Step 8 with 2.0, 1.0, 0.7, and 0.5 second scans

Section 7.0

Emergency Stop Check

- 1.) Use the gantry push-buttons to advance the cradle about 0.5m (2ft) from the home position.
- 2.) Press one of the E-STOP buttons on the gantry.
- 3.) Make sure the TABLE POWER shuts off, and the green LED flashes.
- 4.) Depress one of the table elevation buttons, to verify the emergency stop disabled table elevation.
- 5.) Depress one of the cradle drive buttons, to verify the emergency stop disabled the cradle drive.
- 6.) Press one of the RESET buttons to turn on X-RAY DRIVES POWER.
- 7.) Press the other E-STOP button on the gantry.
 - a.) Make sure the Table Power shuts off.
 - b.) Manually move the cradle to the home position to make sure the cradle clutch released.
 - c.) Make sure the cradle latches securely in the home position.
- 8.) Press one of the RESET buttons to turn on X-RAY DRIVES POWER.
- 9.) Press one of the four table tape switches to make sure the X-RAY DRIVES POWER shuts off.
 - a.) Turn on TABLE POWER.
 - b.) Repeat with the three remaining table tape switches.
- 10.) Press the console emergency stop switch; make sure the Table Power shuts off.
- 11.) Turn on STC 120 Power.
- 12.) Exit.
- 13.) Return to Service Desktop home page.

Chapter 6

Computer Integration



NOTICE
Potential for
Data Loss and/
or Equipment
Damage

To prevent potential data loss and equipment damage, please do the following:

- Record data collected from procedures in this chapter into Form F4879 when directed, located in [Chapter 13](#) of this book.
- Only use the Installation manual that arrives with your system for installation. Any other revisions of this manual may not exactly match your system.

Section 1.0

Introduction

This chapter describes the reconfiguration, system state restore, options, and monitor adjustment procedures.

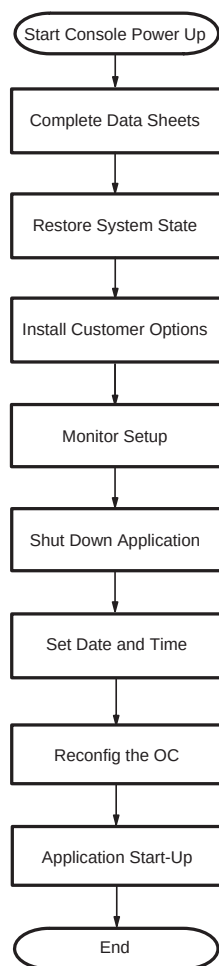


Figure 6-1 Computer Integration Process Overview

Section 2.0

System Configuration Data Sheets

2.1 Requirement

Record valuable system information in the data sheets that follow. Consult with your customer or network administrator to obtain the information. Understanding how the customer plans to use their HiSpeed QX/i scanner, and their network and filming expectation reduces the time required to reconfigure the system.

- [Table 6-1 Manual Film Composer Options, on page 171](#)
- [Table 6-2 System Network Configuration, on page 172](#)
- [Table 6-3 Networking Application \(Image transfer\) Configuration, on page 173](#)
- [Table 6-4 DASM Laser Camera Configuration, on page 173](#)
- [Table 6-5 DICOM Print Camera Configuration, on page 174](#)
- [Table 6-6 DICOM Print Camera Advanced Configuration, on page 174](#)

2.2 Manual Film Composer Options

MANUAL FILM COMPOSER OPTIONS	
Slide Format (if available):	
Greyscale:	
Auto Printing:	
Auto Clear Page:	
Icon Labels:	
Expose Order:	
No. of Copies:	

Table 6-1 Manual Film Composer Options

2.3 System Network Configuration

SYSTEM NETWORK CONFIGURATION			
	FIELD NAME:	SETENV NAME:	FIELD VALUE:
System Settings:	Service ID	SERVICE_ID	
	Hospital Name	HOSPITAL_NAME	
	Exam Number *	* ask customer or checklog	
	DAS Type	DASTYPE	
	PDU Type	PDUTYPE	
Network Settings:	Gateway Host Name	GATEWAY_HOSTNAME	
	Gateway IP	GATEWAY_IP	
	Gateway Net Mask	GATEWAY_NETMASK	
	Gateway Broadcast Mask	GATEWAY_BROADCAST	
	Suite Name	SUITEID	
Option	Network Printer IP Address		
Option	HIS Server IP Address		
Option	HIS Server AE Title		
Option	HIS server AE Port		
Option	CT Server AE Title		
Option	Connect Pro IP Address		

Table 6-2 System Network Configuration

2.4 Host Ethernet Address

_____ : _____ : _____ : _____ : _____

2.5 Network Application (Image Transfer) Configuration

Record the network application (image transfer) configuration.

NETWORKING APPLICATION (IMAGE TRANSFER) CONFIGURATION				
AE TITLE OR HOST NAME	NETWORK ADDRESS	NETWORK PROTOCOL	PORT NUMBER	COMMENTS

Table 6-3 Networking Application (Image transfer) Configuration

2.6 Camera Application Configuration

Record the camera application configuration for the DASM or DICOM print camera.

DASM LASER CAMERA CONFIGURATION	
Camera Type:	
DASM Type:	
Film Smooth/Sharp Setting:	
Options:	
Valid Film Formats:	
Default Film Formats:	

Table 6-4 DASM Laser Camera Configuration

DICOM PRINT CAMERA CONFIGURATION

Camera Type:	
Host Name:	
IP Address:	
AE Title:	
TCP/IP Listen Port:	
Comments (Optional):	
Valid Film Formats:	
Default Film Formats:	
Destination:	
Orientation:	
Medium Type:	
Magnification Type:	

Table 6-5 DICOM Print Camera Configuration

DICOM PRINT CAMERA ADVANCE CONFIGURATION

Smoothing Type:	
Configuration:	
Minimum Density:	
Maximum Density:	
Empty Density:	
Border Density:	
Association Timeout:	
Session Timeout:	
N-Set Timeout:	
N-Action Timeout:	
N-Create Timeout:	
N-Delete Timeout:	
N-Get Timeout:	

Table 6-6 DICOM Print Camera Advanced Configuration

Section 3.0 Configuration

Note: Type the text shown in **boldface**, and press the ENTER key on the keyboard.

3.1 Restore System State

Tools Required: None

Your system should have a system state MOD, located in the software collector box.
The system state MOD contains:

- Collimator Characterization
- Phantom Calibrations

The installation process uses all the system state files. At this time, use the system state MOD to restore the phantom calibrations and system characterization files.

If you cannot locate an existing system state MOD, manually characterize the system and perform phantom calibrations.

- 1.) If you are not on the Service Desktop, click on the SERVICE DESKTOP icon.



- 2.) Click on the PM icon.
- 3.) Select SYSTEM STATE.
- 4.) Insert the MOD in the MOD drive.
- 5.) Select CHARACT.
- 6.) Select CALS.
- 7.) Select RESTORE to restore the system characterization and phantom calibration files to the system.

Note: **Restore State can take as long as ten minutes, although typical times average about three minutes. When Restore State completes, dismiss the tool, and proceed to the next section.**

If any error should occur during the restore process, see the Software Load Procedure manual (2211220-100) for information regarding possible error messages and their recovery.

- 8.) Click NO for Reset Scan Hardware popup message.
- 9.) Select DISMISS.

3.2 Install Customer Options

Tools Required: None

3.2.1 Software

Note: Your system may have a MOD that contains customer purchased options. If your system has an options MOD, install it at this time.

Ensure that the options MOD is NOT write protected at this time. The initial install requires that the MOD be write enabled; subsequent installs can be done with the MOD write protected.

- 1.) If you are not on the Service Desktop, click on the SERVICE DESKTOP icon.
- 2.) Click on the UTILITIES icon.
- 3.) Select INSTALL.
- 4.) Select INSTALL OPTIONS.

An Options Window appears (Figure 6-2):

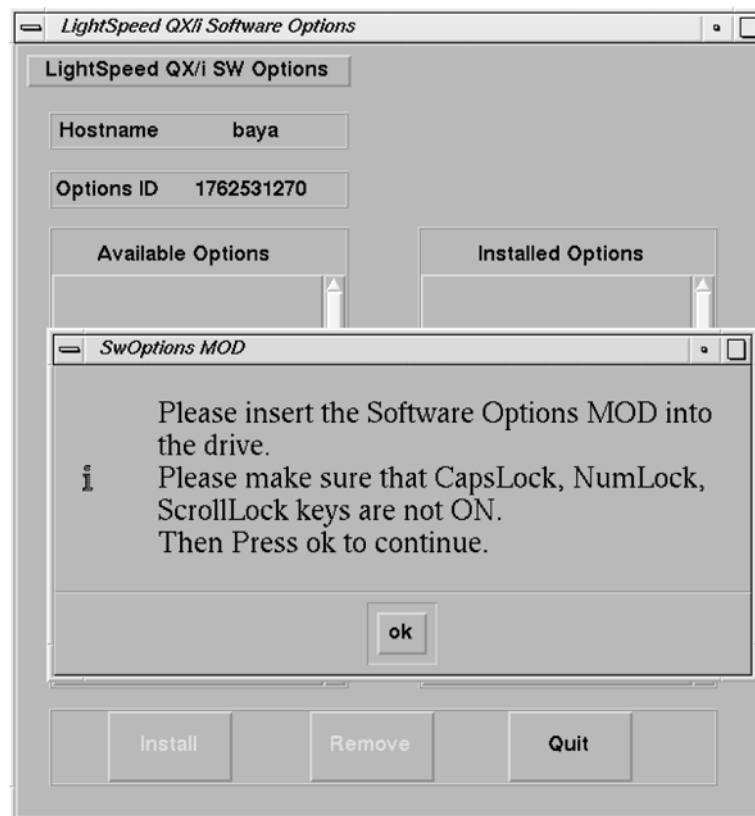


Figure 6-2 Options Window when First Selected

- 5.) Insert the options MOD into the MOD drive and click on OK. (If you do not have an options MOD, click on OK anyway, wait for the abort pop up, then abort the process.)

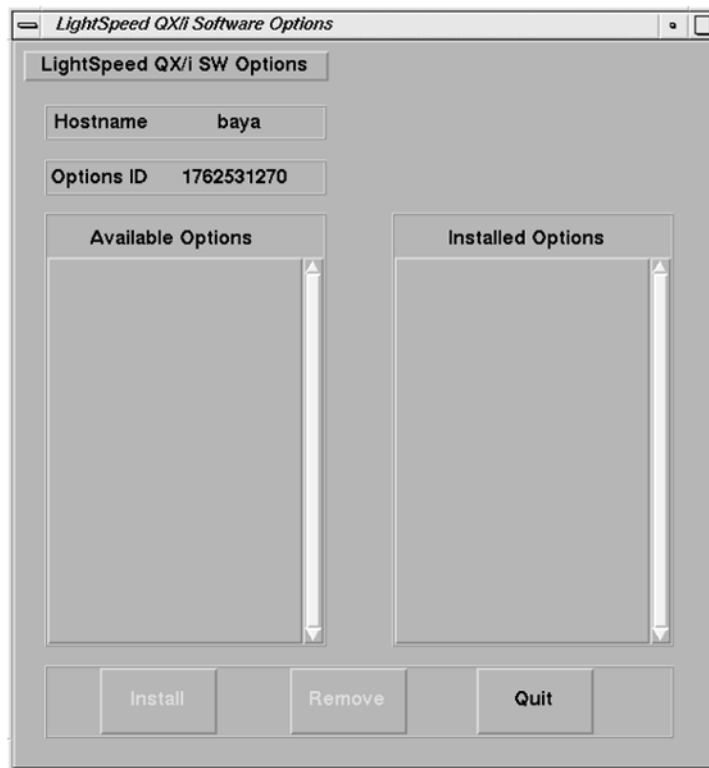


Figure 6-3 Example: Options Window

- 6.) Select all of the options in the left-hand column to install the corresponding software.
- 7.) Select INSTALL. A box may appear while the options are loading. When an option is displayed in the *Installed Options* list, then installation of that option is complete. Note that some options take a fraction of a second to install, while options like 3D may take a half minute (due to the fact that they are installing software).
- 8.) After the options are installed, select QUIT.
- 9.) Select OK.
- 10.) Remove the MOD and write protect the side with options.
- 11.) When the system prompts to Reboot, click YES, and reboot the system to complete the installation.

3.2.2 Camera

Tools Required: Small flat blade screw driver

Note: DASM hardware must be installed before proceeding.

For details on camera configuration, refer to the Software Load Procedures manual.

For details on troubleshooting the camera, refer to the System Service manual.

- 1.) Click on the SERVICE DESKTOP icon.
- 2.) Select UTILITIES.
- 3.) Select INSTALL CAMERA.
- 4.) Select ADD.
- 5.) Select DASM or DICOM.
- 6.) Follow procedures on the screen.
- 7.) Return to Home Page.

3.3 Monitor Setup

Tools Required: Sony Service Manual (located in monitor box)

3.3.1 Overview

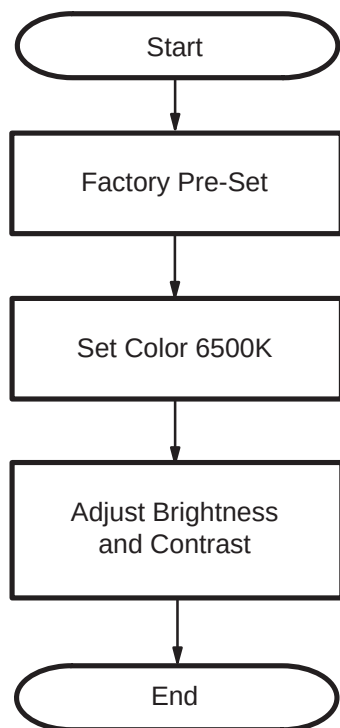


Figure 6-4 Monitor Setup Flowchart

3.3.2 Color Adjustments

- 1.) Adjust the ambient room light to the level used by the operator during image review.
- 2.) Select the COLOR menu to adjust the picture's color temperature. (You can use this to match the monitor's colors to a printed picture's colors.

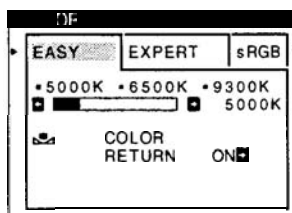


Figure 6-5 Color Menu

- 3.) Set monitor color temperature to 6500K. Refer to the system service manual for this procedure, if necessary.
- 4.) Use one of the following methods:



- a.) Click the IMAGEWORKS icon and proceed to step 3.

OR

- b.) If you are not on the Service Desktop, click on the SERVICE DESKTOP icon.



- c.) Click on the IMAGE QUALITY icon.
- d.) Select INSTALL SMPTE IMAGE and proceed to step 3. (Approximately 3-4 minutes.)
- 5.) Display the SMPTE pattern. Use the browser to select Exam 1000, which contains the SMPTE pattern, and enlarge the image to full screen display.
- 6.) Increase the Contrast to maximum.

Note: Adjust monitor contrast until the operator sees the anatomical structure (window width) at the expected window levels.

- 7.) Increase the Brightness to maximum.

Note: Adjust the monitor brightness until the operator sees the anatomical structure (window width) at the expected window levels.

- 8.) Decrease the Brightness, until the raster just fades into, and matches, the monitor screen background. At this point, the 5% and 95% patches should be just visible.

If additional tweaking is required to attempt to match the monitor image to the filmed image, use only the brightness control.

If the CRT image exhibits any tearing or smearing of the alphanumeric characters, then reduce the contrast setting slightly until the tearing/smearing is just eliminated. The optimum setting for contrast is the highest setting that does not cause tearing/smearing of the alphanumeric characters.

You should always finish up by displaying and filming images of anatomy (typical heads and bodies), and asking the technologist to compare the CRT image to the film image.

- 9.) To view SMPTE on text monitor, swap video cable.

3.3.3 Using Menu Buttons

- 1.) Press the Main Menu button to display the main menu on your screen.



Figure 6-6 Main Menu Button

- 2.) Press the / buttons to highlight the menu you want to adjust. Press the Main Menu button to select the menu item.

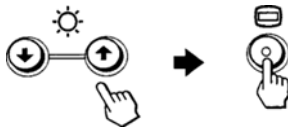


Figure 6-7 Select Menu Item

- 3.) Press the / buttons to select the desired adjustment item. Press the / buttons to make the adjustment.



Figure 6-8 Adjust the Menu

- 4.) Press the Main Menu button once to return to the main menu, and twice to return to normal viewing. If no buttons are pressed, the menu closes automatically after about 30 seconds.



Figure 6-9 Main Menu Button

- 5.) Press the reset button to reset the adjustments back to the factory settings.



Figure 6-10 Reset Button

3.3.4 Adjusting Brightness and Contrast

Brightness and contrast adjustments are made using a separate BRIGHTNESS/CONTRAST menu. These settings are stored in memory for the signals from the currently selected input connector.

- 1.) Press either one of the brightness or contrast buttons. The BRIGHTNESS/CONTRAST menu appears on the screen.

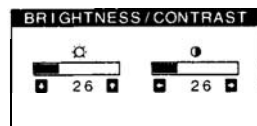


Figure 6-11 Brightness/Contrast Menu

- 2.) Press the / buttons to adjust the brightness and the / buttons to adjust the contrast.

3.4 Shut Down Application

- 1.) If Applications is currently running, you must shutdown system applications to the CT user desktop.
 - a.) Click on the SERVICE DESKTOP button.
 - b.) On the desktop toolbar select UTILITIES.
 - c.) Select APPLICATIONS SHUTDOWN (to bring down applications only).
- 2.) Open a UNIX Shell window.
 - a.) Type: `su` - ENTER at the prompt.
 - b.) Enter the root (super user) password: `#bigguy`

3.5 Check/Set Date and Time

Check the date and time.

- 1.) Type/enter `setdate` ENTER.
- 2.) Follow the instructions to set the correct time and date.

3.6 Reconfig the OC

Note: The document collector box that arrived with your system contains the GRE Service Information CD-ROM containing the GRE Software Installation Procedures manual, which documents the reconfiguration procedure in more detail.

3.6.1 Overview

On the following screens, you should make the changes necessary, pressing the corresponding button at the top of the screen to move from screen to screen. When you are done, you can either press the ACCEPT button to start the reconfiguration process, or press the QUIT button to exit without changing the system configuration.

While the reconfiguration is going on, messages are displayed in a shell window that closes when reconfiguration is complete. Should you later want to review the reconfiguration output, it is logged to the following file:

```
/var/adm/install.log.YYYYMMDDWWHHMMSS
```

Where

`YYYYMMDDWWHHMMSS` is the Date/Time that the reconfiguration was started.

To view the file, type: `more /var/adm/install.log.YYYYMMDDWWHHMMSS`

It is possible to abort the reconfiguration while entering information on the reconfiguration screens. Simply press the QUIT button at the top of the screen. There is NO safe way to abort the reconfiguration after pressing the ACCEPT button. If the entries made in the screens were incorrect, DO NOT try to stop the reconfiguration, instead wait for it to complete, and rerun reconfig, entering the correct parameters.

3.6.2 Procedure

- 1.) Change directory to scripts:
Type: `cd /usr/g/scripts` ENTER at the prompt.
- 2.) Launch the LightSpeed Install utility:
Type: `reconfig` ENTER at the prompt.
The OC displays the LightSpeed Install Utility Window as shown in [Figure 6-12](#).



Figure 6-12 Install Utility Window

- 3.) Using the mouse, click on the CONFIG button.
The OC displays the HiSpeed QX/i System Configuration - System Settings Screen as shown in [Figure 6-13](#).

Comment: The following pages show the screens that are used to change the configuration of the system. These screens are the same as those used for the Software Configuration during Load From Cold. The actual screens will vary depending on the current configuration of your system.

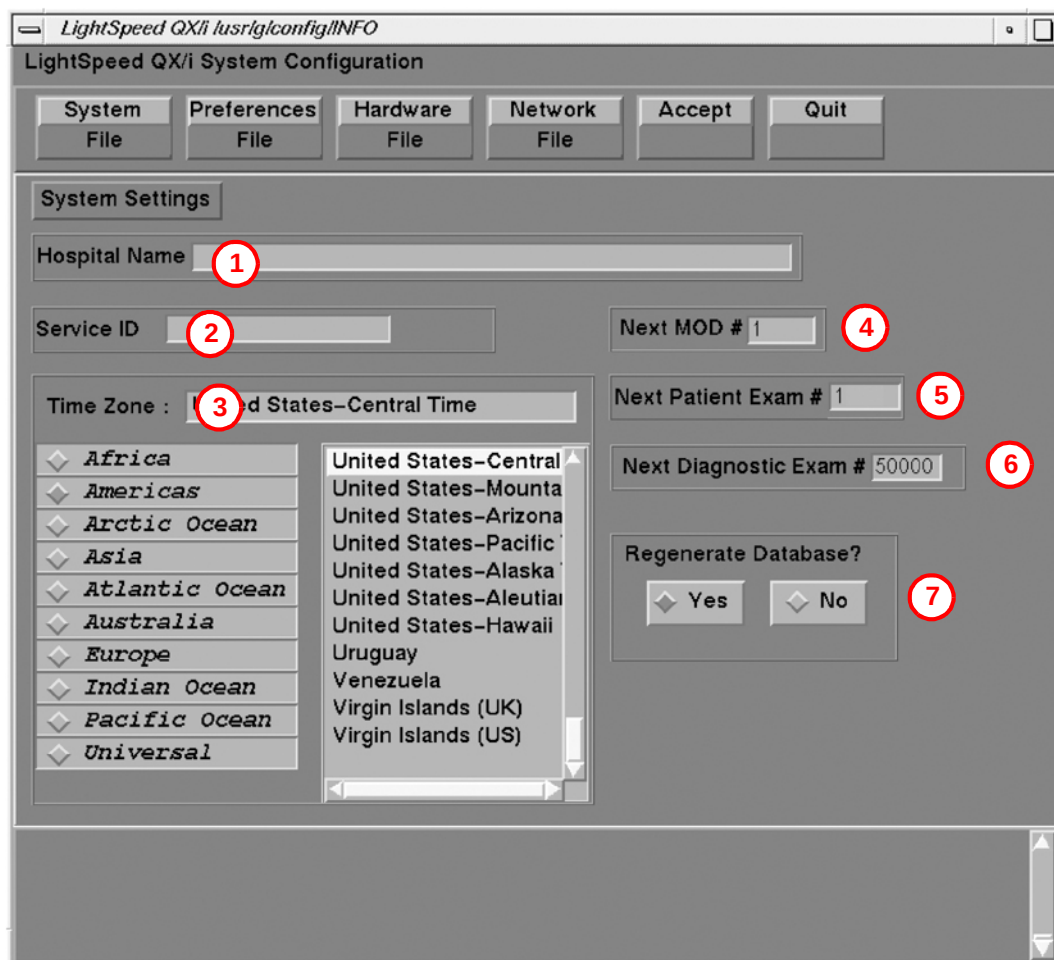


Figure 6-13 System Settings Screen

4.) Configure System Settings

- a.) Hospital Name (Figure 6-13, item 1) configures the name that will show up on images produced by this scanner. *Example:* ST MARYS HOSPITAL
- b.) Service ID (Figure 6-13, item 2) is issued by the Service organization.
Example: 262785CT2 (no spaces)
- c.) Select the Time Zone for the site.

Note: Use the scrollbar at the bottom of the time-zone selection list to view the entire description of the time-zone you are about to select, to ensure that you are selecting the correct time-zone for your location.

If the time-zone of your location is not in the list above, select one of the universal times in the selection menu. In this case, automatic changes for daylight savings time will not take effect. See Load from Cold manual, if you require more information regarding time-zone setting & selection.

- d.) Next MOD # *Customer selected.*
 - e.) Next Patient Exam # configures the next Exam number the scan user interface will use. At initial system installation, type: 1
 - f.) Next Diagnostic Exam # *Customer Selected.*
- 5.) Select the PREFERENCES button to display the Preference Settings Screen as shown in Figure 6-14.
- 6.) Regenerate Database (refer to Figure 6-13) determines whether the Scan Database will be regenerated during reconfiguration.

Note: If YES is selected, then all scan data will be deleted on the Scan Data Disk. To avoid accidental regen, NO is selected by default each time reconfig is run. Regenerating the Scan Database destroys all the scan and calibration files on the Scan Data Disk. Make sure that all images have been reconstructed before regenerating the Scan Database.

This procedure is normally only needed when a Scan Data Disk is replaced or reformatted. See Load from Cold manual, if you require more information regarding Scan Database Regeneration.

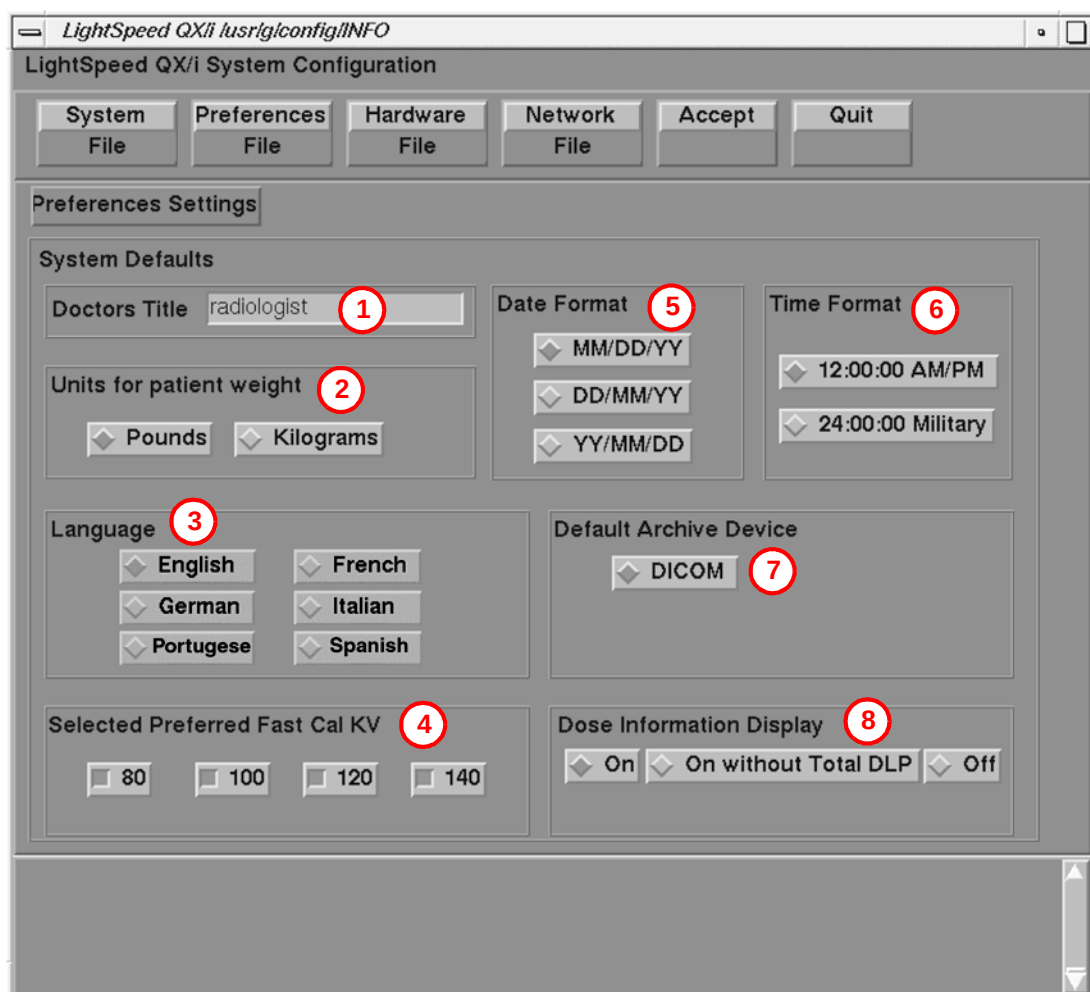


Figure 6-14 Preferences Setup Screen

7.) Configure Preferences Settings

a.) Language configures the language to be displayed on the Application screens (ENGLISH, FRENCH, GERMAN, ITALIAN, PORTUGESE, SPANISH).

b.) Preferred Fast Cal KV configures the preferred kV that the Fast Cal Routine will calibrate (80, 100, 120, 140 in the Selected Preferred Fast Cal KV field). The default selections are 120 and 140.

Comment:

These kVs should include all kVs that the site uses for patient scanning. Deselecting All Preferred FastCal KVs is the same as selecting ALL the Preferred FastCal KVs

c.) Date Format and Time Format configure the format in which the date and time will be displayed on the images.

d.) Select the site preferred Dose Information Display option for the site to use in monitoring calculated Patient Dose:

- * Select ON (full CTDiw Display)
- * Select ON WITHOUT TOTAL DLP (no Dose Length Product Display), or
- * Select OFF (no CTDiw Display)

- 8.) Select the HARDWARE button to display the Hardware Settings Screen (Figure 6-15).

Figure 6-15 Hardware Settings Screen

- 9.) Configure Hardware Settings
- DAS Type configures which DAS the software should expect.
 - Tube Type configures which X-Ray Tube the software should expect.
 - PDU Type configures which PDU the software should expect.
 - Collimator/Filter Type configures which Collimator/Filter the software should expect. Presently, HiSpeed QX/i systems only support HELIOS G1.
 - Gantry Type selects the type of Gantry connected to this console.
 - Select Network Image Printer only if the site plans to use a network connected postscript printer (such as Codonics printers) with this system. When selected:
 - * Enter the printer Hostname in the Name field.
 - * Enter the printer IP Address in the IP Address field.

Note: A "PRINT SERVER" IS NOT SUPPORTED (i.e., a computer with a printer attached); a Network Image Printer (which is supported) is a printer attached directly to the network.

- g.) Camera Setup:

Comment: Camera Setup is not performed using Reconfig, but rather is performed from the Service Desktop once Applications have been started. See "Declaring DICOM Filming Devices on the System" on page 264.

- 10.) Select the **NETWORK** button to display the Network Settings Screen as shown in Figure 6-16.

Figure 6-16 Network Settings Screen

- 11.) Configure Network Settings

Comment: This screen provides the ability to declare the HiSpeed QX/i system on a hospital network. Key information such as Host Name, IP Address, Net Mask (for HiSpeed QX/i systems on a subnet) must be obtained from the hospital network administrator.

See Chapter 12 for more information and complete details of setting the Hospital/System Network Configuration.

- a.) Enter the Suite Name.

The Suite Name must start with a letter, followed by three alphanumeric characters. Total must be four characters long. The name of the OC interface will be <Suite Name>, within the scanner's subnet.

Typically, you should use su01 or ct01 ("su" and "ct" must be lowercase), unless the customer prefers a different suite name.

- b.) Enter the hospital provided Host Name.

The Host Name identifies the network hostname and AE Title of the HiSpeed QX/i system to the hospital's network. The Host Name:

- * **MUST NOT** exceed 16 Characters.
- * **MUST** only contain the following characters: A through Z, a through z, 0 through 9, - and _.

Comment: The Host Name is typically stmary or ct01.

- c.) Enter the hospital provided IP Address for the system.
d.) Enter the hospital provided Net Mask (if the HiSpeed QX/i system is on a subnet).
e.) Enter the Broadcast Address.

- Comment:* *The Broadcast Address should be the same as the IP Address except for the bits of the host id portion (last digit group) set to 1's or 0's depending on the configuration of the network. The standard default is 1's, but older Sun OS machines used 0's.*
- For example:*
- If the IP Address is 192.100.9.17, the Broadcast Address should be 192.100.9.255 if the network is configured to use 1's to specify the Broadcast Address.*
- If the network contains genesis based scanners or other SunOS 3.5 or 4.1 computers, the Broadcast Address should be 192.100.9.0.*
- f.) Enter the hospital provided Default Gateway IP Address in the Default Gateway field (if applicable). If the site network does not use a default gateway, leave the field blank.
- g.) Select NIS (a.k.a. Yellow Pages database) Advanced Option—only if requested by the hospital network administrator—as follows:
- * Select ADVANCED OPTIONS button on the Network Settings Screen.
 - * Select USE NIS button.
 - * Enter the hospital provided Domain Name (from the hospital network administrator).
- 12.) Select ACCEPT on the HiSpeed QX/i System Configuration Screen.

- Comment:* *The system loads the CT Application Software, OS patches, kernel changes and configures the system on the OC.*
- The loading process takes approximately 15 minutes. While the load is going on, the results are displayed in a shell window that closes when the loading process is complete. All the window output is logged to a file named:*
- 13.) When the loading process and configuration changes are complete, the system displays a prompt to reboot. Click on YES. (See [Figure 6-17.](#))



Figure 6-17 Reboot Screen

- 14.) The system will automatically login as ctuser after the reboot. Select OK on the Autostart Disabled popup message.

3.7 Applications Start-Up

Open the Console shell window, and type: `st` ENTER. The applications desktop appears on the monitor.

Chapter 7

Table and Gantry Integration



NOTICE
Potential for
Data Loss and/
or Equipment
Damage

To prevent potential data loss and equipment damage, please do the following:

- Record data collected from procedures in this chapter into Form F4879 when directed, located in [Chapter 13](#) of this book.
- Only use the Installation manual that arrives with your system for installation. Any other revisions of this manual may not exactly match your system.

Section 1.0 Introduction

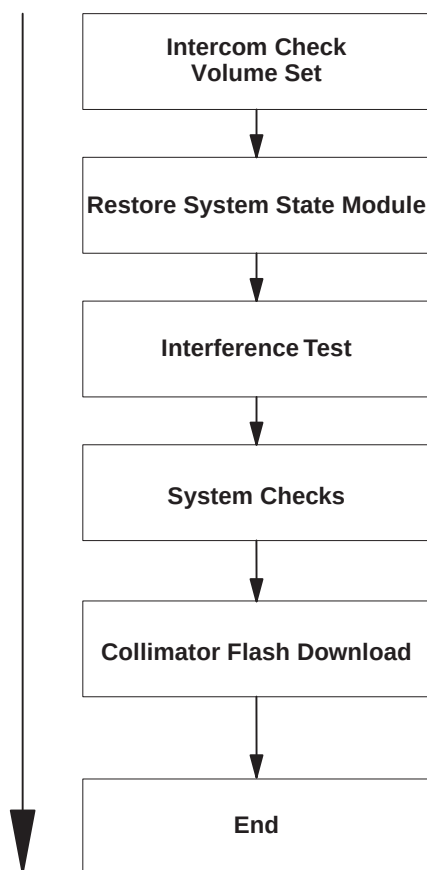


Figure 7-1 Table/Gantry Integration Flow Chart

REQUIRED TOOLS

- Multimeter
- Scope (or Multimeter with Frequency Measurement Capability, such as the FLUKE 87.)

Section 2.0

Autovoice/Intercom Check

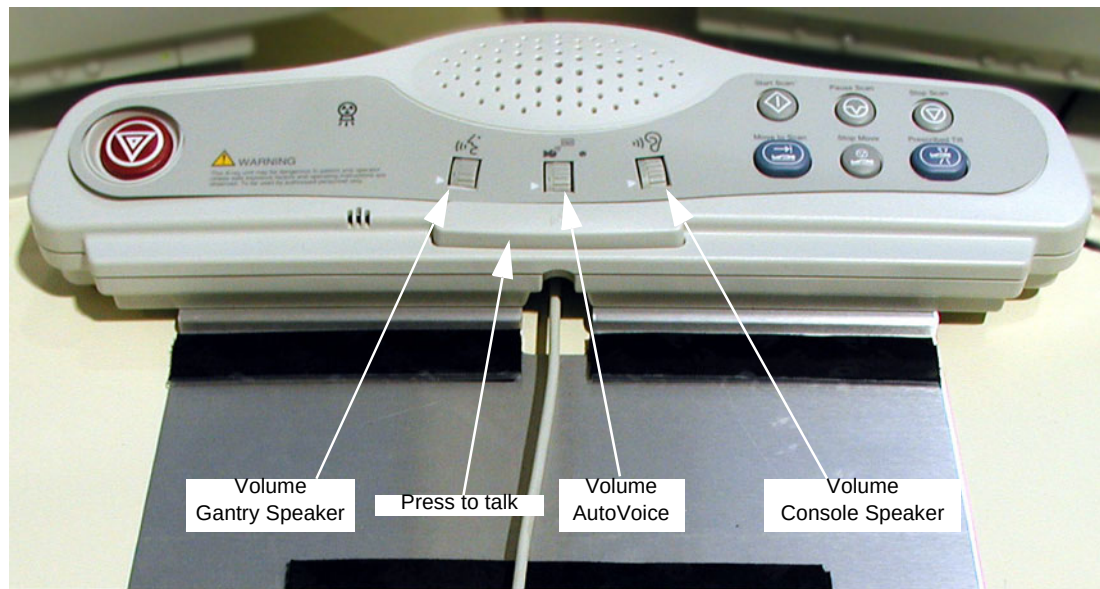


Figure 7-2 SCIM Volume Controls

2.1 Gantry Speaker

To adjust the gantry speaker volume, adjust the left-most volume thumb wheel while speaking into the console microphone (press the bar on the SCIM).

2.2 Console Speaker

To adjust the console speaker volume, adjust the right volume thumb wheel until a person speaking near the gantry can be clearly heard.

2.3 Autovoice Volume

- 1.) On the Scan Desktop, select PROTOCOL MANAGEMENT.
- 2.) Select AUTO VOICE RECORD.
- 3.) Click on the 3.4 button, to the right of "FF2. Inspiration".
- 4.) Click on the PLAY button, to play the Inspiration AutoVoice message.
- 5.) Adjust the center volume thumb wheel, while Autovoice is playing, to set the volume for the gantry speaker.
- 6.) Repeat steps 4 and 5, as necessary to achieve satisfactory volume.
- 7.) Select DONE, then Select QUIT.

Section 3.0

Table/Gantry Characterizations

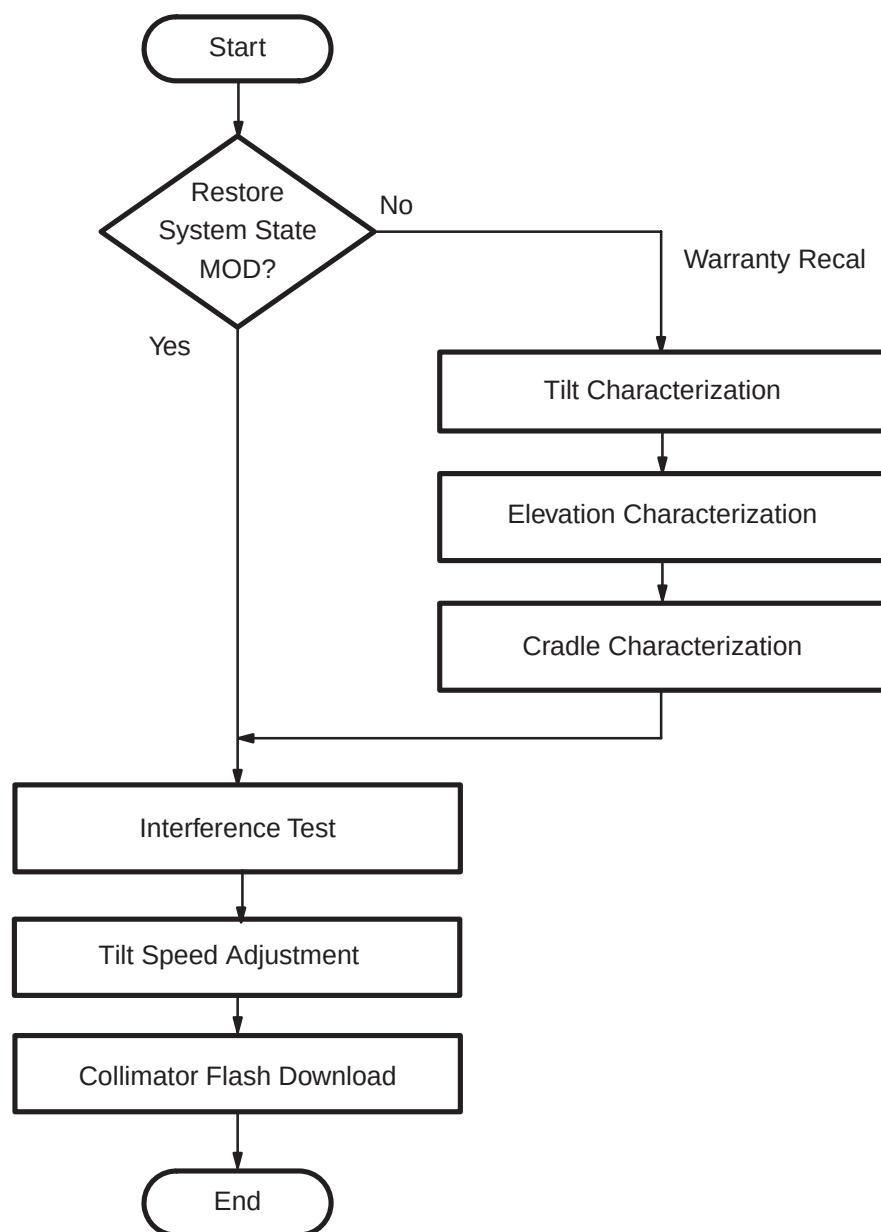


Figure 7-3 Characterization Flowchart

Section 4.0

Interference Test

Reset the hardware to download the new characterization values to the ETC, ICE, and STC before proceeding to the table/gantry interference test on this page.

If the following elevation and/or tilt display readings fail to meet specifications, DO NOT adjust the limit switches. Instead, re-characterize elevation and/or tilt.



NOTICE **Interference Test Requirement: No tilt motion shall cause the gantry tilting frame to touch the stationary base covers for any tilt angle.**

- 1.) Select HARDWARE RESET.

Press only the **fast** push-button to display the elevation limits for the present elevation/tilt combination.

- a.) Move the cradle to the home position.
- b.) Use the DOWN pushbutton to lower the table to the minimum height.
- c.) Make sure the elevation display reads 500.0 ± 3 mm.
- d.) Use the UP pushbutton to raise the table to the maximum height.
- e.) Make sure the elevation display reads 25.0 ± 3 mm.

Note: If this step fails, check table/gantry level, and run table elevation characterization.

“I” tilt means the top of the gantry tilts away from the table. “S” tilt means the top of the gantry tilts toward the table.

- 2.) Advance the cradle about 36 inches from the home position.
 - a.) Tilt the gantry to ± 23 degrees.
 - b.) Use the DOWN pushbutton to lower the table until it automatically stops.
 - c.) Make sure the elevation display reads 115.5 ± 3 mm.
 - d.) Make sure the table clears the gantry covers by at least $\frac{1}{2}$ inch.
- 3.) Tilt the gantry to ± 20 degrees.
 - a.) Use the DOWN pushbutton to lower the table until it automatically stops.
 - b.) Make sure the elevation display reads 210.0 ± 3 mm.
 - c.) Make sure the table clears the gantry covers by at least $\frac{1}{2}$ inch.
- 4.) Use the UP push-button to raise the table until the elevation display reads between 115.5mm and 210mm.
 - a.) Tilt the gantry to ± 10 degrees.
 - b.) Tilt the gantry away from the table; make sure it stops at $\pm 22.0 \pm 0.5$ degrees.
 - c.) Tilt the gantry toward the table; make sure it stops at $\pm 18.5 \pm 0.5$ degrees.
- 5.) Return the cradle to home position.
 - a.) Make sure you can tilt the gantry to 30.0 ± 0 degrees.
 - b.) Leave gantry tilted to ± 30 degrees for now.
- 6.) Raise the table until you can move the cradle.
 - a.) Advance the cradle about 12 inches from the home position.
 - b.) Use the HOME Button to return the cradle to the home position, tilt the gantry to 0 degrees, and lower the table to minimum height.

Make sure you cannot move the cradle at table heights below 210.0.
 - c.) Verify that the *cradle lock* indicator on the gantry display is lit.
- 7.) Select FILE $\rightarrow$ EXIT from the pull-down menu.
- 8.) Select CLEANUP.

Section 5.0

Tilt Speed Adjustment Procedure

- 1.) Remove rear gantry base covers.
- 2.) Using a stop watch or suitable time piece adjust for one (1) degree per second motion.



NOTICE Adjustments are very sensitive. Improper speed adjustments will result in overspeed error generation.

Note: Tilt speed will vary based upon hydraulic fluid temperature. Adjustments should be made at normal scan room temperature settings.

- 3.) From zero (0) degrees tilt, press and hold the tilt forward button,  until gantry display reads S25. Observe tilt speed using time piece.
- 4.) Press and hold backward tilt button,  until gantry display reads I25. Observe tilt speed using time piece.
- 5.) Adjust forward speed control valve for one (1) degree per second motion. See [Figure 7-4](#).
- 6.) Adjust backward speed control valve for one (1) degree per second motion. See [Figure 7-4](#).
- 7.) Repeat steps 3 through 6 until tilt speed for both directions is correct.
- 8.) Now tilt the gantry forward to S30. Observe the S25 to S30 speed. Do this several times. If a noticeable difference is observed, then check the hydraulic fluid levels.

Note: This is a self bleeding system. Trapped air can cause slowed or limited tilt range. Exercising the full range of motion several times should purge any trapped air from the hydraulic system.

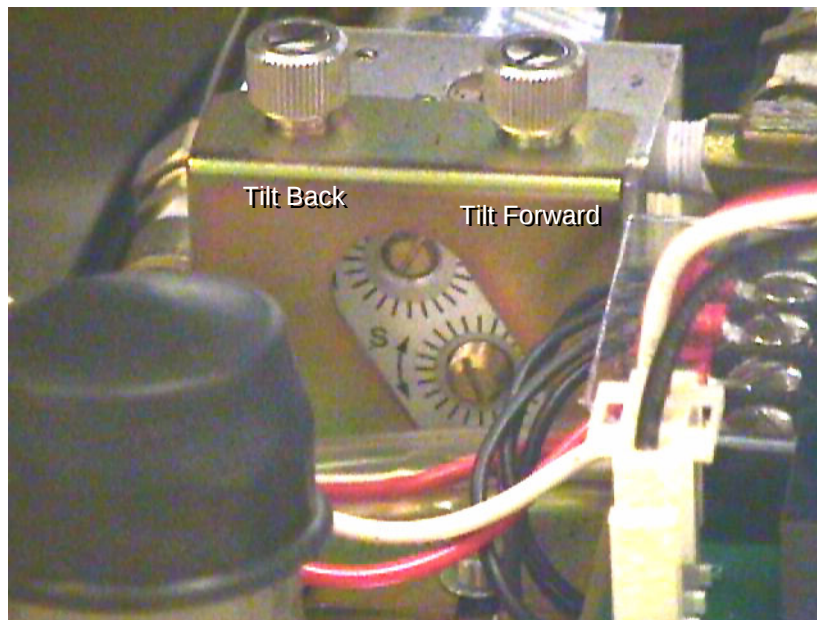


Figure 7-4 Tilt Speed Adjustment Screws

Section 6.0 System Checks

6.1 Alignment Lights

6.1.1 Turn Alignment Lights ON

-  **CAUTION** Verify all personnel have cleared the system. The Gantry rotates during this check.
- 
- 1.) Turn ON the Axial Drive Enable and HVDC Enable switches (located on the STC backplane – refer to [Figure 5-7, on page 166](#)).
 - 2.) Press the alignment light button on the gantry control panel.
The gantry will rotate and the lights will turn ON.

-  **WARNING** **NEVER STARE DIRECTLY INTO THE LASER BEAMS WHEN YOU OPERATE THE ALIGNMENT LIGHTS. STARING INTO THE BEAMS CAN CAUSE PERMANENT EYE INJURY DAMAGE.**

6.1.2 Internal Axial Lights

- 1.) Place a sheet of plain white paper over the output port of each light.
 - 2.) Verify that the two laser lines coincide and appear as a single line.
- Note: **GE designed the internal axial lasers on the current CT system to shine *down* on the collimator. Do NOT adjust the internal alignment lights at this time. The tomographic plane tests use the QA phantom to check the internal axial lasers alignment to the collimator.**

6.1.3 External Axial to Internal Axial Distance

- 1.) Raise the table to its highest elevation.
- 2.) Extend the cradle until you see both the internal and external laser lights shining on the cradle.
- 3.) Place a metric rule on the right edge of the cradle, and measure the distance from the internal axial laser line to the external axial line. Verify this distance equals 240.0 mm ± 1.0 mm.
- 4.) Place the rule on the left edge of the cradle, and measure again.
- 5.) Leave the cradle in its current position, and lower the table to the minimum elevation.
- 6.) Measure the distance between the internal and external lights on both edges of the cradle, as above. Verify the distance remains equal to 240.0 mm ± 1.0 mm.

6.1.4 Coronal Lights

- 1.) Place a sheet of plain white paper at the left side of the patient opening, in front of the coronal laser light. Verify the two coronal lines coincide.
- 2.) Move the paper to the right side of the patient opening. Verify the two coronal lines coincide.
- 3.) Place the paper in the center of the gantry opening. Use a level to verify that the coronal lines are horizontal.

6.1.5 Turn Lights OFF

- 1.) Press the alignment light button on the gantry control panel, again, to turn the lights off.
- 2.) Turn OFF the axial drive enable and HVDC enable switches.

6.2 Laser Warning Labels

- 1.) Make sure all laser warning labels appear in the correct location on the outside of the gantry.
- 2.) Obtain and install replacements for any missing labels.



Figure 7-5 Laser Warnings and Precautions

Section 7.0 Collimator Characterization & Flash Download

The Collimator Flash Download diagnostic compares the values in flash memory on the collimator control board with the values stored on the system, and updates if necessary. If the memory values don't match, the diagnostic prompts you to type/enter the collimator serial number to complete the update.

To prevent invalid test results, remove any obstructions from the gantry opening.

- 1.) If you are not on the Service Desktop, click on the SERVICE DESKTOP icon, .
- 2.) Click on the UTILITIES icon, .
- 3.) Select INSTALL.
- 4.) Select FLASH DOWNLOAD TOOL.
- 5.) Select UPDATE.
- 6.) Look for "Download Successful" message.
- 7.) Select DISMISS when finished.

Note: New installations upload values from FLASH.

Chapter 8

DAS Integration



NOTICE
Potential for
Data Loss and/
or Equipment
Damage

To prevent potential data loss and equipment damage, please do the following:

- Record data collected from procedures in this chapter into Form F4879 when directed, located in [Chapter 13](#) of this book.
- Only use the Installation manual that arrives with your system for installation. Any other revisions of this manual may not exactly match your system.
- Refer to DAS Cleaning in the GE Medical Systems *HiSpeed QX/i System Service Manual*

Section 1.0 Introduction

1.1 Failure Protocol

Follow standard troubleshooting protocol to verify all failures.

- If you are moving boards, make sure you follow special handling procedures (e.g., use ESD protection).
- Record all board moves on the DAS board tracking sheet.
- When you verify a failure, identify the defective part and remove it from the system.

Before running any verification tests, turn on the DAS and let it warm up. It takes two (2) minutes of warm-up time for every minute that the DAS was turned off, up to two (2) hours.

If you encounter failures in any section, restart the procedure at the DASTools.

1.2 Special Handling Precautions



NOTICE Potential for Static Electricity Damage



DAS circuit cards are more sensitive to both STATIC DAMAGE and DIRT than any other circuit cards in the HiSpeed QX/i system. Please read Sections [1.3](#) and [1.5](#), and observe all precautions.

1.3 Anti-static Handling

You **MUST** wear a grounding wrist strap anytime you remove the boards from the DAS or a conductive bag. **NEVER** place the DAS boards on non-conductive surfaces, such as white bench tops, plastic covered books, or non-conductive plastic bags.

When removing the covers from the DAS, put on a wrist strap, and push all the DAS boards firmly into their slots on the DAS backplane. Do **NOT** remove and re-seat any boards at this time.

1.4 Cleanliness

Any dirt on the surface increases leakage current on the converter cards and causes the DAS to fail the drift spec. Dirt also causes microphonic failures, which cause image artifacts. Wear anti-static

finger cots (part number 2221235 [small] and 2221235-2 [large]) when you handle the DAS, because fingerprints on the board can cause problems during humid conditions. Use only clean, new anti-static finger cots.

Follow cleaning and de-ionizing procedures in the System Manual.



NOTICE NEVER USE AN ERASER TO CLEAN ANY PART OF THE DAS.

1.5 Required Tools

TOOLS REQUIRED

- Ground Wrist Strap
- Aero Duster, Part #MS-222N (if cleaning is needed) with static-free metal nozzle inserted
- Lint-free towels, Part #46-307237P1
- Amax Contact and Circuit Cleaner, Part #2274104
- DAS/DET Interface Kit 2227089 (if cleaning is needed)

1.6 Process Overview

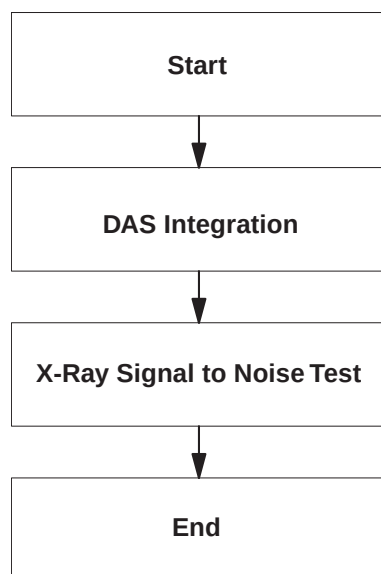


Figure 8-1 DAS Integration Flow Chart

Section 2.0

DAS Tools



1.) If you are not on the Service Desktop, click on the SERVICE DESKTOP icon.



- 2.) Click on the UTILITIES icon.
- 3.) Select INSTALL.
- 4.) Select DAS INTEGRATION.
- 5.) Select DAS TOOLS to open the corresponding menu window.
- 6.) Select ACCEPT.

DAS Tests

Auto Test

Manual Test

Interconnect Test

Pop Noise Microphonics Test

Auxillary Channel Test

X-ray Verification Test

DAS Viewers

DAS Architecture

View Log

View Specification

Dismiss

Test Name	Number of Iterations	Remaining Iterations	Total Successes	Total Failures
DC CAL Absolute	1	1	0	0
Offset Drift	1	1	0	0
Pop Noise & microphonics	10	10	0	0
DC Noise & DC Offsets	1	1	0	0

Options

Number of repetitions for Group

Delay between groups (msecs.)

Total Failure before stopping

Remaining

1

1

200

128

128

Accept

Stop Tests

Figure 8-2 DAS Tools Screen (Auto Test Selected)

- 7.) Select AUTO TEST to run all available DAS Tools.
- Use the default iteration values.
 - The auto test displays **comfort dots** to show activity, and displays iteration pass/fail information on the screen.

-or-

Manually select the following tests:

- DCCal: 1 iteration
- Offset Drift: 1 iteration
- Pop Noise: 10 iterations
- Microphonics: 10 iterations (See [Figure 8-3.](#))
- Offset Means: 1 iteration
- Offset Noise: 1 iteration

Auto Test takes approximately 35 minutes to complete 1 pass.

- 8.) After DAS Tools successfully completes one pass of the auto test, or the designated number of manual test iterations, select DISMISS to exit the test. If the Total Failures column lists any failures, troubleshoot before proceeding.

Test Name	Number of Iterations	Remaining Iterations	Total Successes	Total Failures
Pop Noise & microphonics	10	10	0	0

Options

Pop Noise & microphonics Options

- ◆ Rotor On
- ◆ Rotating
- ◆ Rotor Off
- ◆ Non-Rotating

DAS Tests

- Auto Test
- Manual Test
- Interconnect Test
- Pop Noise Microphonics Test
- Auxillary Channel Test
- X-ray Verification Test

DAS Viewers

- DAS Architecture
- View Log
- View Specification
- Dismiss

Number of repetitions for Group: 1 (Remaining: 1) [Accept]

Delay between groups (msecs.): 200

Total Failure before stopping: 128 (Remaining: 128) [Stop Tests]

Figure 8-3 DAS Microphonics Test screen

Section 3.0

Verify X-Ray/Signal to Noise Test

This section uses X-Ray to test the DAS for opens, lows, and shorts.

System Configuration:

- All covers ON
- Nothing in detector FOV

- 1.) If you are not on the Service Desktop, click on the SERVICE DESKTOP icon,



- 2.) Click on the UTILITIES icon, .
- 3.) Select INSTALL.
- 4.) Select DAS INTEGRATION.
- 5.) Select START.
- 6.) Select X-RAY VERIFICATION.
- 7.) Select X-RAY VERIFICATION (DDC).
- 8.) Follow on-screen instructions to acquire each set of X-ray verification scans, for a total of four sets.
- 9.) Select ACCEPT RX.
- 10.) Press **START SCAN** when it flashes.
- 11.) Select DISMISS to close the X-ray verification window.
- 12.) Select X-RAY VERIFICATION ANALYSIS.
- 13.) If X-ray verification scans do not pass, troubleshoot with DAS Tools and System Service Manual before continuing.
- 14.) Complete the F4879 Tube Signal/Output check.

Chapter 9

System Alignments



NOTICE Potential for Data Loss and/ or Equipment Damage

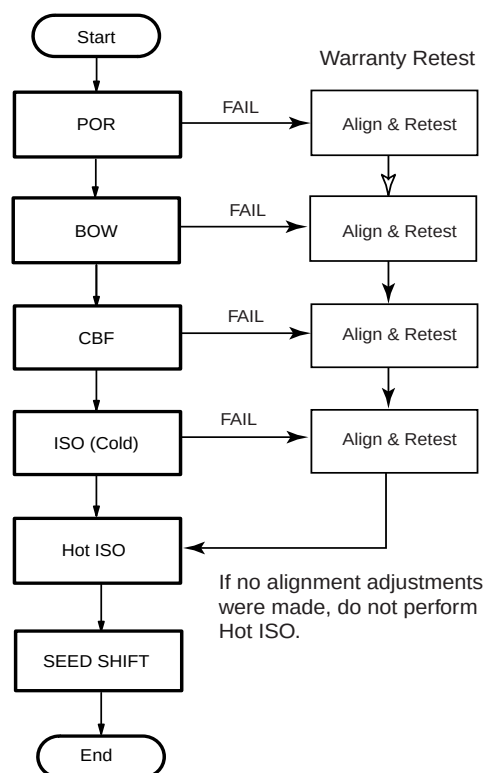
To prevent potential data loss or equipment damage, please do the following:

- Record data collected from procedures in this chapter into Form F4879 when directed, located in [Chapter 13](#) of this book.
- Locate the HHS data sent with your HiSpeed QX/i scanner. It can be found in the software collector carton. Retain this data with the HHS data collected at the site in a safe place.
- Only use the Installation manual that arrives with your system for installation. Any other revisions of this manual may not exactly match your system.

Note: For each adjustment procedure, do not loosen anything until after you have run your first scan and calculated your first adjustment. Before starting this chapter, the tube must have cooled at least two hours.

Section 1.0 Introduction

Use the procedures in this chapter to make sure the x-ray beam focuses directly upon the detector window. The flowchart in [Figure 9-1](#) shows an overview of this chapter.



IMPORTANT:

If alignment adjustments are made during this procedure, a FULL SET of ALL detail calibrations must be completed before proceeding with the system tests in Chapter 11.

The procedure for completing detail calibrations can be found in the *System Manual*, in the *Tube Change* section.

This should also be Code 10: Install Warranty.

Figure 9-1 System Alignment Overview

1.1 Required Tools

- Polaroid type 52 film
- Film developer
- Caliper Dial indicators
- Masking tape
- Socket set
- Torque wrench
- 48cm Phantom
- Dial indicator mount
- Radial dial indicator

1.2 Gantry Geometry Definitions

Theta: (Also known as ISO) Along the tangent to gantry rotation, on the rotating structure. Move the X-Ray tube to the 180° (six o'clock) position to equal 0° theta. You can adjust the tube unit and collimator in the theta direction.

Z-Axis: Parallel to the axis of gantry rotation, in the direction of cradle motion.

- You can adjust the tube unit and detector in the Z-Axis direction.
- Positive Z-Axis direction: Move the tube/detector toward the table.
- Negative Z-Axis direction: Move the tube/detector away from the table.

Radial: Along the radius, from the center of gantry rotation, in and out from gantry isocenter. You cannot adjust the HiSpeed QX/i detector in the radial direction.



Figure 9-2 Gantry Geometry

1.3 Tube Heat Definitions

Cold Tube:	No exposures for more than 6 hr. (overnight)
Cool Tube:	No exposures for more than 2 hr., but less than 6 hr.
Warm Tube:	No exposures for more than 1 hr., but less than 2 hr.
Hot Tube:	Exposures in less than 1 hr.

Section 2.0

Plane of Rotation (POR)

2.1 Overview

The purpose of Plane-of-Rotation measurement/alignment is to put the X-ray tube in the correct physical relationship to the detector and verify it. This is normally only necessary when X-ray tube has been replaced or moved.

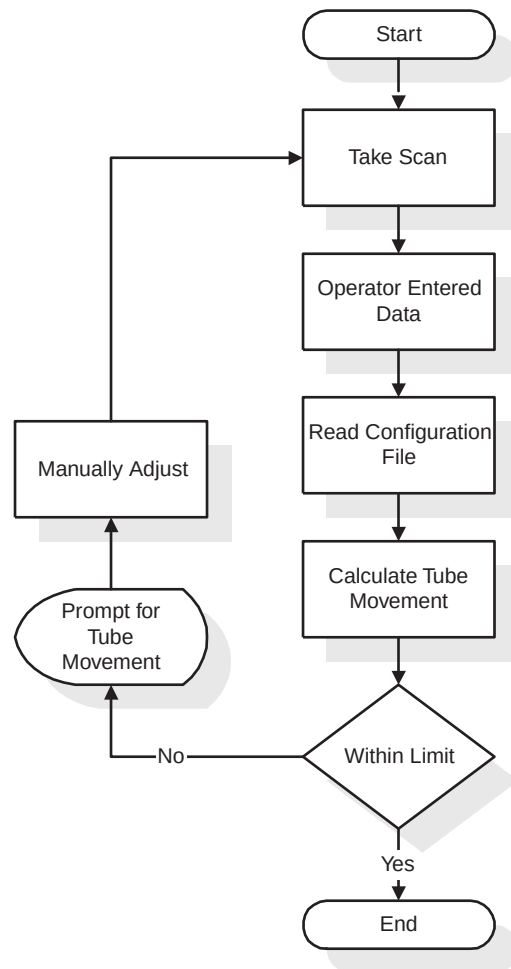


Figure 9-3 POR Procedural Flow

2.2 Tools Required



Figure 9-4 Vernier Caliper & Dial Gauge Indicator

- Calibrated Vernier Caliper (millimeters or inches).
- Dial Gauge Indicator (millimeters or inches)
- Type 52 Polaroid Film (see [Figure 9-7](#))
- Type 52 Polaroid Film Developer

2.3 Procedure

Remember to always perform POR measurements when the X-Ray tube unit is cool.

- If you have acquired any low technique scans prior to beginning the Plane-of-Rotation (POR) alignments, wait at least 5 minutes before starting the following procedure.
- If you recently acquired image scans, and want to re-check the Plane-of-Rotation, wait at least 90 minutes to allow the tube unit to cool. (*Wait 90 minutes, if the tube unit had more than 25 Kilo joules exposure, $[KV \times mA \times Sec \div 1000]$ within the last 30 minutes.*)

2.3.1 Start the POR Software

Note: Correct tube temperature is critical.

- 1.) If you are not on the Service Desktop, click on the SERVICE DESKTOP icon.



- 2.) Click on the UTILITIES icon.
- 3.) Click on INSTALL.
- 4.) Click on the SYSTEM ALIGNMENTS option.
- 5.) Click on the PLANE OF ROTATION option.



Plane of Rotation		System Messages Instructions Log
Scan Execute PLANE-OFROTATION scan	Mount the phantom holder and 48cm phantom on the cradle. Tape a type 52 film at 3:00 o'clock position on the outside edge of the 48 cm phantom, so the film projects into the Gantry. Advance the cradle and use the alignment lights to position the film for an exposure near the image center mark on the film pack.	
Calculate Calculate the amount of tube movement		
△ ▽ Prior Next Restart Dismiss		Save Log

Figure 9-5 Plane Of Rotation Alignment Screen

2.3.2 Measure Tube Alignment

- 1.) Get the system's phantom holder and attach the 48cm phantom to it. Refer to [Figure 9-6](#).

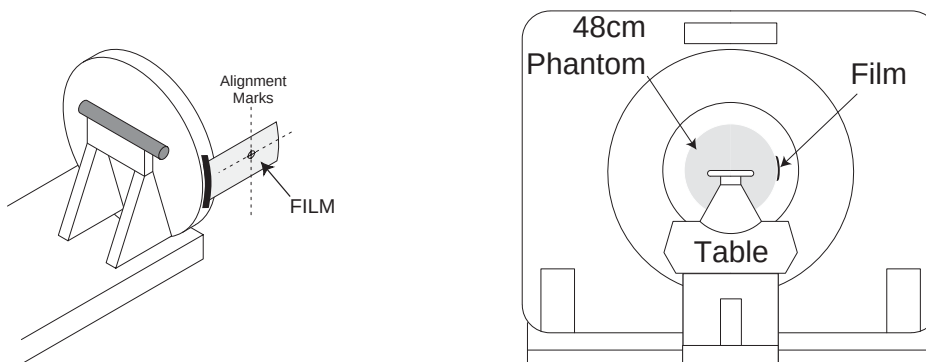


Figure 9-6 48CM Phantom and Film on Table

- 2.) Attach one *Polaroid type 52* film on the outside edge of the 48cm phantom, at the 3 o'clock position.

Note:

Only the film should be projecting into the Gantry bore when complete. The phantom is used only to position and hold the film in the gantry bore. Refer to [Figure 9-6](#).

Orient the side of the film side marked “**This side toward lens**” towards iso center (see [Figure 9-7](#)). When exposed and developed later, the film will show the alignment of the x-ray beam with respect to the table, as viewed from the X-Ray tube in the 3:00 o'clock position.

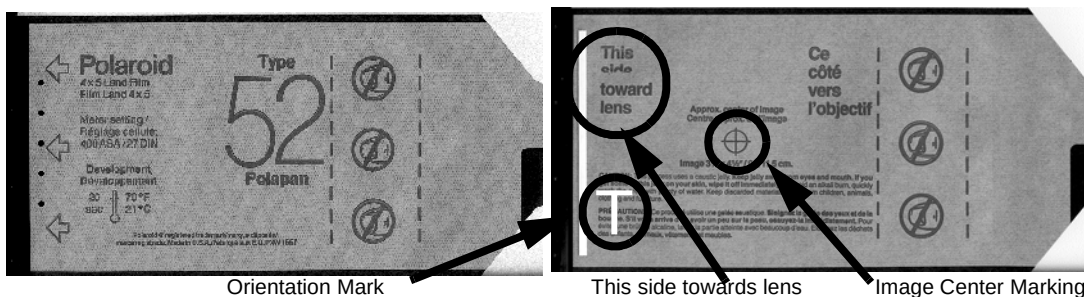


Figure 9-7 Polaroid Film, Type 52

- 3.) Advance the cradle, while using the alignment lights, to position the film's center marking on the alignment light marks.
- 4.) Take a scan.
- 5.) Remove the exposed film and immediately mark the outside of the film nearest the table. For example, use a pen and print the letter **T** on the film nearest the table ([Figure 9-8](#)). Go and develop the exposed film.



Figure 9-8 Placement of “T” on Film

- 6.) After the film is developed, transfer the table marking to the film itself while keeping the orientation correct (Figure 9-9).

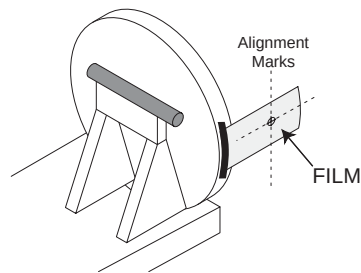


Figure 9-9 Alignment Marks on Film

- 7.) Verify the following (refer to Figure 9-10):
- Film's narrow (white beam) slit lies within (between) the wider (gray) X-Ray slit.
 - Film's edges in both Z direction are equally well defined by the exit slit of the collimator. The edges of the narrow beam should be much sharper than the wide beam.

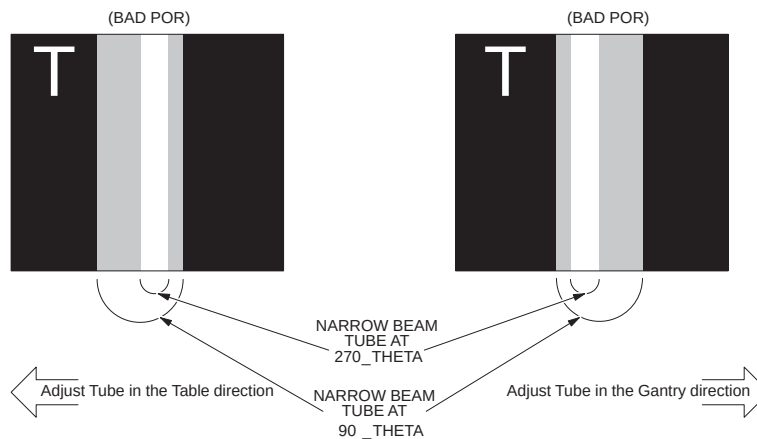


Figure 9-10 Plane of Rotation Film Interpretation

- 8.) Refer to Figure 9-10 and Figure 9-11 during the following instructions. Use a vernier calipers to measure the width of the two wider (dark gray) slits. They're the dark gray slits that extend past the edges of the narrower (off-white) slit and to the blackest part of the film.

- a.) Take three measurements to obtain an average value for X_F . X_f is the side of the film closest to the table. Using the same film measurement tool, take one X_f measurement at the top of the film, another near the middle and another near the bottom. Add these three X_f distances together and divide this sum by 3 or n . Where,

$$X_F = (X_{f1} + X_{f2} + X_{f3}) / (n)$$

Note:

It is important that you take multiple measurements, to reduce the likelihood of a measurement error and consequently increase the accuracy of the alignment.

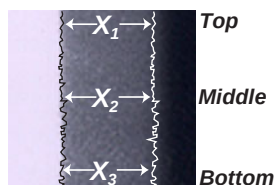


Figure 9-11 XF and XR measurement points

- b.) Now take three measurements to obtain an average value for X_R . Take one X_R measurement at the top of the film, another near the middle and another near the bottom of the film. Add these three values together and divide the sum by 3 or n to obtain an average for X_R .

$$\bar{X}_R = (X_{R1} + X_{R2} + X_{R3}) / (n)$$

- c.) Use the values obtained for X_F (front distance) and X_R (rear distance) in the calculation that follows.

- 9.) Click on the CALCULATE button (refer to [Figure 9-5](#)). Enter the values for X_F and X_R obtained in the steps above. The software will do the distance calculation.

Note: **The specification limits are 0.059 - 0.082 inches or 1.50 - 2.083 millimeters.**

- 10.) Check the results

- If the tube is out of specification, perform steps 1 and 2 ([page 207](#)), and move the tube. See [Figure 9-12](#) for more details.
- If adjustment is necessary, clockwise rotation (in) of adjustment bolt moves the tube towards the table side. Repeat Steps 1 through 9 again if you have moved the tube, to check accuracy of adjustment. See [Figure 9-10](#) for more details.

Note: Mount the dial indicator onto the collimator/tube assembly as shown [Figure 9-12](#). Make sure you zero the dial indicator when it's securely in place.

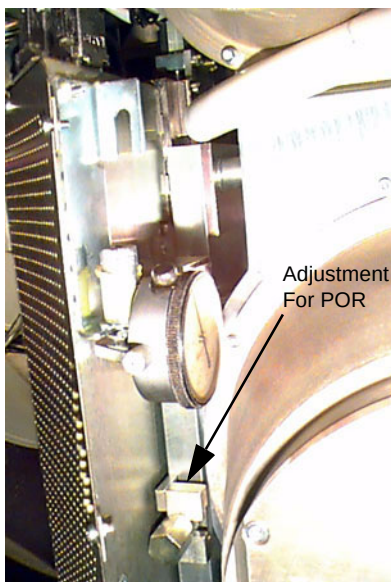


Figure 9-12 Dial Indicator Attached to Collimator/Tube Assembly

- When the tube is within specification, tighten the (4) M-12 Bolts on the tube and torque them to 50 ft-lbs (69 kg-cm) and you're done.
- 11.) Enter the results in the HHS table.

Section 3.0

Beam on Window Alignment (BOW)

3.1 Overview



NOTICE
Potential for
Image
Artifacts

Make sure the tube is cool before continuing. Wait 15-30 minutes after the completion of Plane-of-Rotation.

Purpose: Bring the detector into alignment with the x-ray beam of the now correctly positioned tube. There are two specifications for BOW Alignment. The software expects the user to evaluate the raw vectors to determine if it meets the requirements.

- The first specification is for the initial check of BOW. The limits are $-1.5 \pm 0.25\text{mm}$.
- The second specification is for verification scans when the system is warmer. The limit for verification scans is $-1.5 \pm 0.5\text{mm}$.

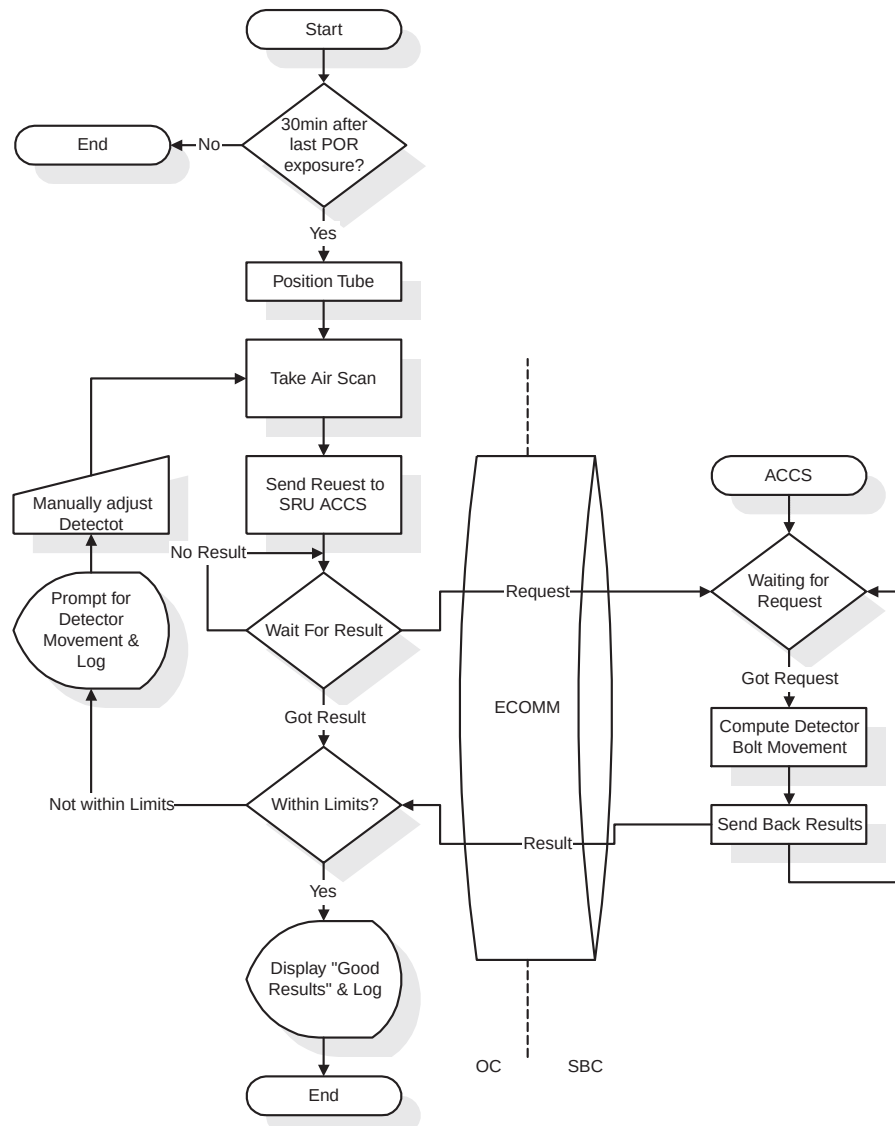


Figure 9-13 BOW Alignment

3.2 Cool Tube

Make sure the tube is cool before continuing. Wait 10 minutes per shot taken after the completion of Plane-of-Rotation.

3.3 Access the Software

Never position the 1/8 inch metal shaft directly on the ISO center - always offset it. The software calculates tube movement for IOS center and looks for a sinusoidal waveform generated by the off center of the metal shaft. A metal shaft, positioned at IOSO center, generates a waveform consisting of a straight line.

- 1.) If you are not on the Service Desktop, click on the SERVICE DESKTOP icon.



- 2.) Click on the UTILITIES icon.
- 3.) Select INSTALL.
- 4.) Select SYSTEM ALIGNMENTS.
- 5.) Select BEAM ON WINDOW.

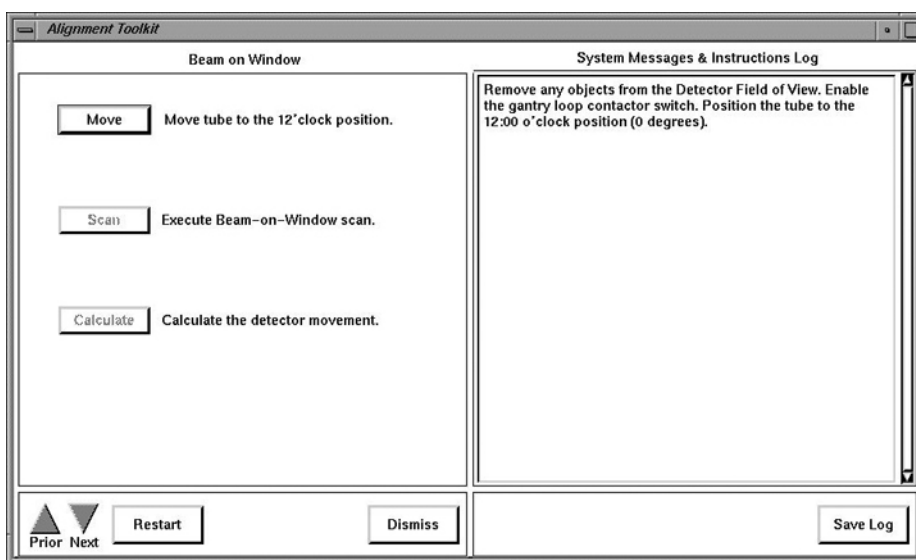


Figure 9-14 BOW screen

3.4 Adjustment Procedure

Tools Required: 21mm wrench
5mm 6"long hex bit
5mm thin wrench

Note: To make the adjustment, you must remove the front and rear covers (step 4).

Note: **Do not loosen anything until after you have run your first scan and calculated your first adjustment.**

- 1.) Select MOVE to send the tube to the twelve o'clock position.
- 2.) Acquire a Beam on Window Scan.

- 3.) Select CALCULATE and make adjustments as indicated by the program.
You'll get screen output similar to the following:
Upon initial inspection: spec = -1.5 +- 0.25mm
Upon verification scan: spec = -1.5 +- 0.50mm
Compare the raw alignment vectors to these specifications, make alignments suggested if specification is not met.
- 4.) Remove the front and rear covers.
- 5.) Refer to [Figure 9-15](#). Loosen the M-6 cap head screws, located at each end of the detector. (total of two cap head screws).
- 6.) Loosen the middle nut (jam nut) with a 21 mm thin profile wrench. Make adjustments, using a 21mm thin open end wrench, as requested by software. Tighten jam nut and cap screws after adjustment.
- 7.) Acquire a Beam on Window Scan.
- 8.) Select CALCULATE.
- 9.) If the adjustments pass the calculation, proceed to step 10, otherwise return to step 4.
- 10.) Tighten all three M-6 cap head screws, using a 5mm x 6" long hex bit, for 3/8" drive. (Torque to 10 ft-lb., or 14 N/m).

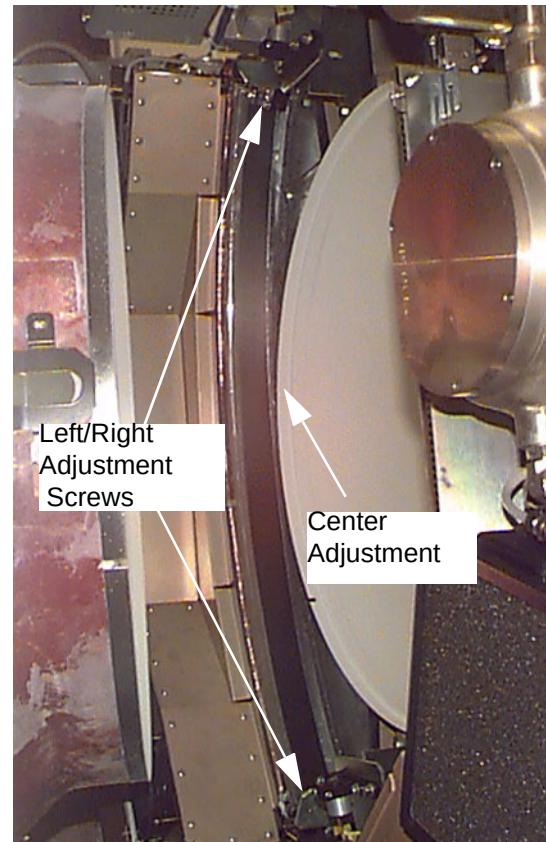
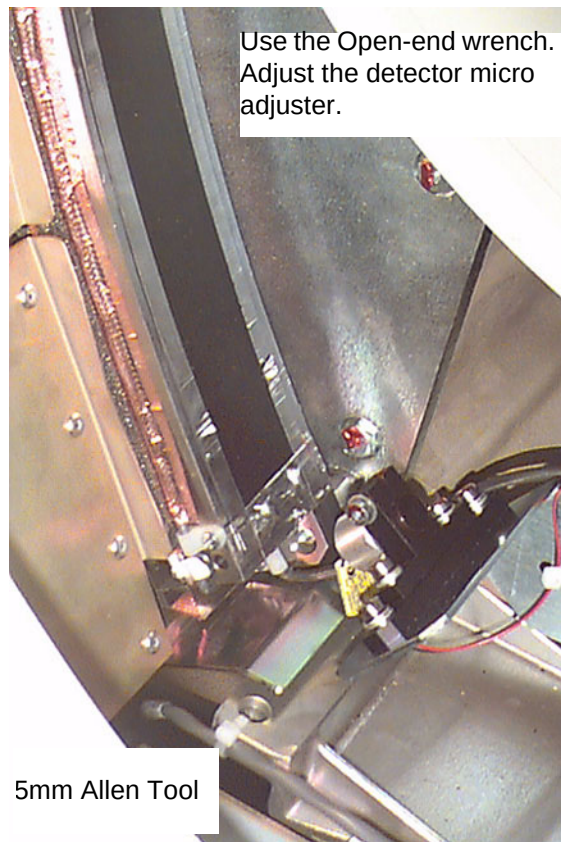


Figure 9-15 Detector Cap Head Screws

Note: **CW (Clockwise)** moves the detector towards the mounting plate. **CCW (Counter-clockwise)** moves the detector away from the mounting plate. **Right=Low Chnl, Center=Medium Chnl, Left=High Chnl**

- 11.) Enter results in the HHS table.

Section 4.0

CBF/SAG Alignment Process

Purpose: CBF/SAG Alignment ensures the focal spot is accurate, the bowtie filter is centered, and center of rotation is in a straight line.

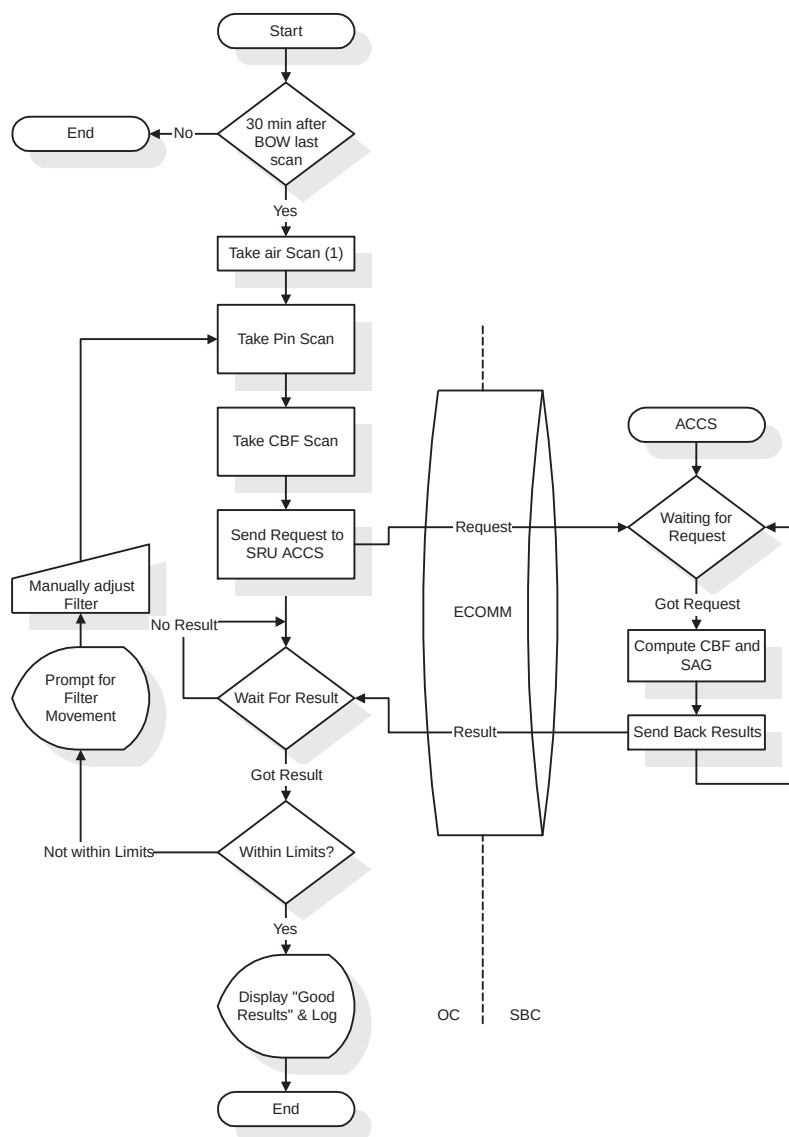


Figure 9-16 CBF Procedure

Never position the 1/8 inch metal shaft directly on the ISO center - always offset it. The software calculates tube movement for IOS center and looks for a sinusoidal waveform generated by the off center of the metal shaft. A metal shaft, positioned at IOSO center, generates a waveform consisting of a straight line.

4.1 Cool Tube

Make sure the tube is cool before continuing. Wait 10 minutes per shot taken after the completion of Beam On Window.

4.2 Access the Software

- 1.) If you are not on the Service Desktop, click on the SERVICE DESKTOP icon.



- 2.) Click on the UTILITIES icon.
- 3.) Select INSTALL.
- 4.) Select SYSTEM ALIGNMENT.
- 5.) Select CBF/SAG ALIGNMENT.

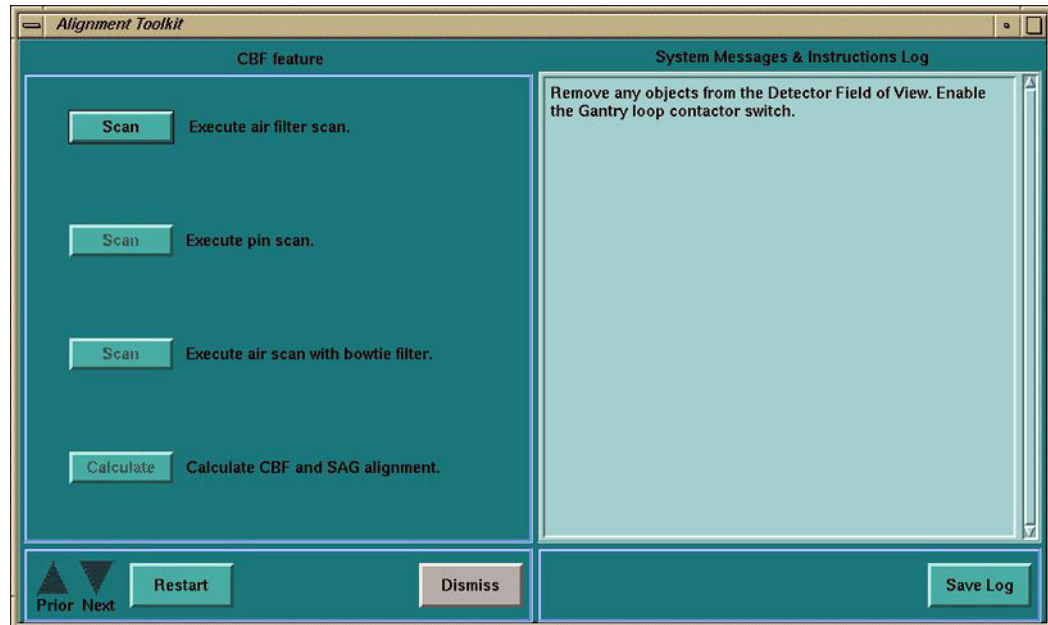


Figure 9-17 CBF/SAG Alignment Screen

4.3 Adjustment Procedure

Tools Required: 1/8 inch PIN (#0 Phillips) screw driver

Note: **Do not loosen anything until after you have run your first scan and calculated your first adjustment.**

- 1.) Press the flashing START SCAN button to acquire an air filter scan.
- 2.) Position the 1/8 inch Pin (#0 Phillips round) screw driver on the phantom holder, parallel to the Z-Axis to the end of the table nearest the gantry or on the phantom holder.
- 3.) Turn On the laser lights.
- 4.) Advance and adjust the table to position the metal shaft right of ISO center.
- 5.) Verify that the shaft remains level (perpendicular to the scan plane).
- 6.) Acquire a pin scan.
- 7.) Acquire an air scan with bowtie filter in place.
- 8.) Select CALCULATE to calculate the CBF and SAG alignment.
- 9.) Mount the dial indicator onto the collimator.
Remember to zero the dial indicator.
- 10.) Loosen the collimator bolts (4 in front, 2 in the rear). Use a swivel adapter on a ratchet wrench to loosen the bolt behind the cable connections.
- 11.) Adjust the collimator as indicated by the calculation results. Ignore the negative sign in front of the adjustments.
- 12.) Tighten the collimator bolts (4 in front, 2 in the rear). (Torque to 50 ft.-lb.)
- 13.) Wait 15 minutes for the tube to cool.
- 14.) Rescan and calculate.
- 15.) Repeat steps 10 – 11 until the adjustment falls within the limit, then proceed to the next section.
- 16.) Enter results in the HHS table.

Section 5.0 ISO Alignment

5.1 Overview

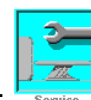
Never position the 1/8 inch metal shaft directly on the ISO center - always offset it. The software calculates tube movement for IOS center and looks for a sinusoidal waveform generated by the off center of the metal shaft. A metal shaft positioned at ISO center generates a straight line waveform.

5.2 Cool Tube

Make sure the tube is cool before continuing. Wait 10 minutes per shot taken after the completion of CBF/SAG Alignment.

5.3 Access the Software

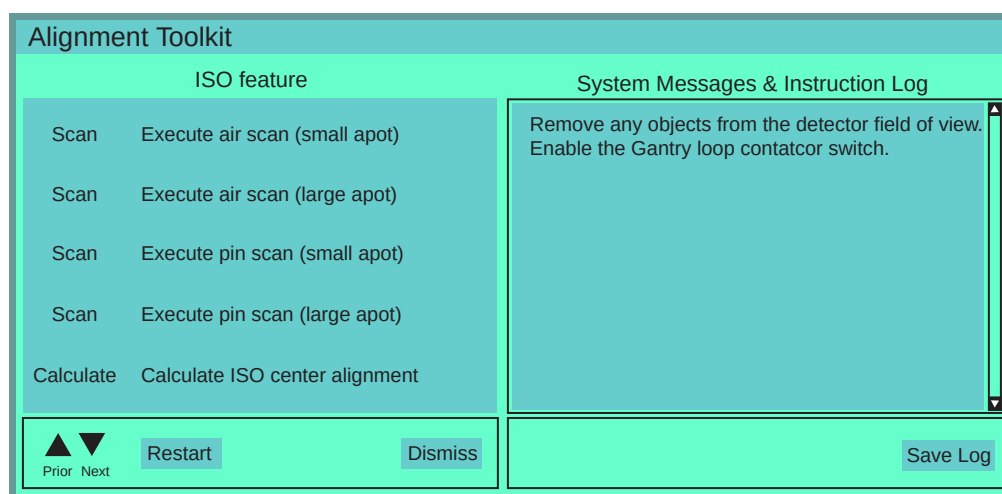
- 1.) If you are not on the Service Desktop, click on the SERVICE DESKTOP icon.



- 2.) Click on the UTILITIES icon.



- 3.) Select INSTALL.
- 4.) Select SYSTEM ALIGNMENTS.
- 5.) Select ISO.



↑ Use the Prior/Next Buttons to Access the Calculate Button

Figure 9-18 ISO Alignment Menu

5.4 ISO Adjustment Procedure

Tools Required: 1/8 inch PIN (#0 Phillips) screw driver

Note: Do not loosen anything until after you have run your first scan and calculated your first adjustment.

- 1.) Acquire Air scan (small spot). (Refer to [Figure 9-18.](#))
- 2.) Acquire Air scan (large spot). (Refer to [Figure 9-18.](#))
- 3.) Position the 1/8 inch Pin (#0 Phillips round) screw driver on the phantom holder in parallel with the Z-axis to the end of the table nearest the gantry or on the phantom holder.
- 4.) Turn On the laser lights.
- 5.) Advance and adjust the table to position the metal shaft right of ISO center.
- 6.) Verify that the shaft remains level (perpendicular to the scan plane).
- 7.) Acquire a small spot pin scan.
- 8.) Acquire a large spot pin scan.
- 9.) Select CALCULATE ISO. (If ISO falls outside 389.75 channels, no wait between tube adjustment and the test scan is required. If ISO falls between +/- 0.02 and +/- 0.1 channels, wait 15 minutes after you adjusted the tube before you scan.)
 - a.) If adjustments are required, proceed to step 10
 - b.) If adjustments are not required (pass), skip to step 15
- 10.) Zero the dial indicator, and mount it on the tube assembly.
- 11.) Using a 10 mm hex, loosen the four (4) M-12 bolts on the tube assembly.
- 12.) Refer to [Figure 9-19.](#)
 - a.) Locate the adjustment bolt on the top of the tube.
 - b.) Adjust the tube UP/DOWN as indicated by the calculation.
- 13.) Repeat steps 1 through 7.
 - If adjustments fall within the acceptable range, proceed to step 13.
 - If adjustments fall outside the limits, wait 15 minutes for the tube to cool and then return to step 9.
- 14.) Tighten the 4 M-12 bolts using a 10mm hex (Torque to 50 ft.-lb.).
- 15.) Enter results in the HHS table.

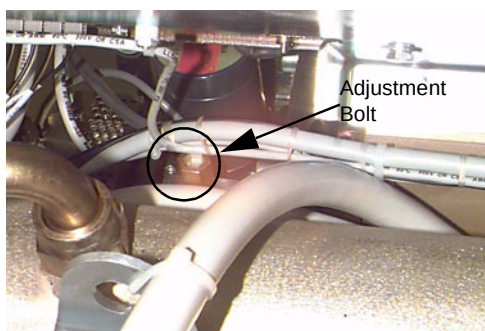


Figure 9-19 ISO Adjustment Bolt

- 16.) Wait 15 minutes, then recheck BOW to verify that the Beam-on-Window values are within the drift spec of -1.5mm +/- 0.5 mm.
 - If BOW is within spec, DONOT adjust anything
 - If BOW is out of spec, alignments must be done starting with POR and ending with BOW recheck.

Section 6.0

HOT ISO Alignment

THIS PROCEDURE SHOULD BE COMPLETED IF AND ONLY IF ANY ADJUSTMENTS WERE MADE.

6.1 Accessing the Software

- 1.) If you are not on the Service Desktop, click on the SERVICE DESKTOP icon.



- 2.) Click on the UTILITIES icon.
- 3.) Select INSTALL.
- 4.) Select SYSTEM ALIGNMENTS.
- 5.) Select HOT ISO ALIGNMENT.

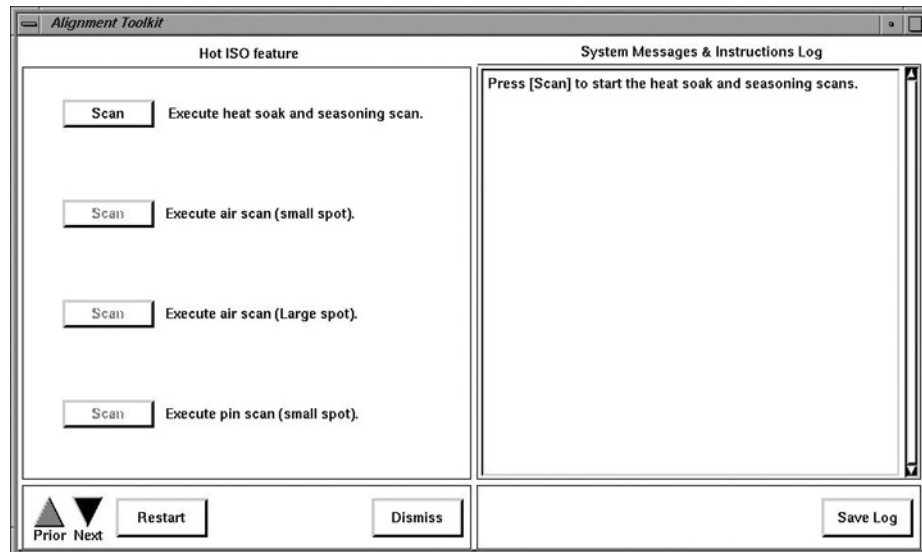


Figure 9-20 HOT ISO Screen

6.2 Adjustment Procedure

Note: No movement of the tube is required. This correction is a software correction done automatically.

- 1.) Execute the Heat Soak and Seasoning scan.
- 2.) Execute Air Scan (small spot).
- 3.) Execute Air Scan (large spot).
- 4.) Place the 1/8 inch screw driver on the phantom holder (should be pointing into the Z direction).
- 5.) Execute Pin Scan (small spot).
- 6.) Execute Pin Scan (Large spot).
- 7.) Calculate correction factor with Calculate button.

Chapter 10

Generator Calibration



NOTICE
Potential for
Data Loss and/
or Equipment
Damage

To prevent potential data loss and equipment damage, please do the following:

- Record data collected from procedures in this chapter into Form F4879 when directed, located in [Chapter 13](#) of this book.
- Only use the Installation manual that arrives with your system for installation. Any other revisions of this manual may not exactly match your system.

Section 1.0

Introduction

1.1 Process Overview

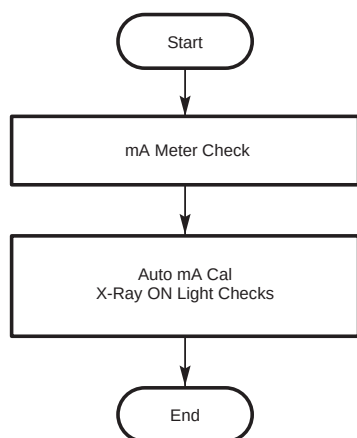


Figure 10-1 Chapter Flowchart

IMPORTANT:

- You must record the data collected in this chapter on **Form F4879**, located in Chapter 13 of this book.
- Locate the HHS data, sent from the factory, in the software collector box.
- Retain this data with the HHS data collected at the site.
- A complete generator RECAL is not required.

Note: If any one of the calibration procedures fail, data collected is different, or you replace any component in the High Voltage section, YOU MUST RECALIBRATE THE SYSTEM. To re-calibrate the system, refer to procedures in the *HiSpeed QX/i System Service Manual*.

1.2 Required Tool

DVM with meter leads

Section 2.0

mA Meter Check With Verification



1.) If you are not on the Service Desktop, click on the SERVICE DESKTOP icon,



2.) Click on the UTILITIES icon,

3.) Select INSTALL.

4.) Select GENERATOR CALIBRATION.

5.) Switch OFF the HVDC (550) ENABLE inside the gantry.

6.) Switch OFF the AXIAL DRIVE inside the gantry.

7.) Rotate the anode tank to the two o'clock position.

8.) Pin the gantry.

9.) Remove the OBC cover (if necessary).

10.) Use a DVM as a DC mA meter. Set the meter range to one decimal place.

11.) Connect the DVM black lead to TP8 (ACAL1) on the mA board.

12.) Connect the DVM red lead to TP11 (ACAL2) on the mA board.

MA Control
HEMRC Control
KV Control
Gentry I/O
LSCOM
CPU

Figure 10-2 OBC Board Slot Assignment

13.) Select MA METER CHECK.

14.) Select ACCEPT.

15.) Record the displayed and measured Anode mA values in the Circuit OFF and Circuit ON table in space provided below. Acceptable Circuit ON values: 140 to 200mA.

Note: **The cathode side reads approximately 19 mA during Circuit ON. The anode should read 140 to 200mA, as stated above.**

16.) Repeat the test for the cathode side:

a.) Use a DVM as a DC mA meter; set the meter range to one decimal place.

b.) Connect the DVM red lead to TP9 (CCAL1) on the mA board.

c.) Connect the DVM black lead to TP14 (CCAL2) on the mA board.

17.) Select ACCEPT.

- 18.) Record the displayed and measured cathode mA values in the Circuit OFF and Circuit ON table in space provided below. Acceptable Circuit ON values: 140 to 200mA.
- Note: **The anode side reads approximately 19 mA during Circuit ON. The cathode should read 140 to 200mA, as stated above.**
- 19.) Disconnect the test equipment.
- 20.) Switch ON the HVDC (550) enable.

2.1 mA Meter Verification - HHS Requirement

Circuit ON		
	A/D	Range
Anode mA		15 - 25
Cathode mA		15 - 25
DVM - A/D = Delta Compare Delta to Limits		

Table 10-1 mA Meter Verification (Table 1)

2.2 Auto mA Cal

Use this procedure to verify and complete Auto mA Cal on the system. If you have just run a Heat Soak and Seasoning, wait 30 minutes before you run the Auto mA Cal.

Note: **The system automatically warms up the tube.**

- 1.) Select AUTO MA CAL.
- 2.) Press START SCAN when it flashes. The program automatically runs Ductility and Auto mA Cal.
- 3.) During this test, verify the X-Ray on indicators, located on the gantry display and the rear gantry shroud, illuminate during X-Ray on.

If the system cannot complete the Auto mA Cal because HV aborts, run the "Tube Heat Soak and Seasoning" scan protocol (see the *HiSpeed QX/i System Service Manual* for details). Place a 48cm phantom in the X-Ray beam to protect the detector from radiation damage during this procedure. Do not worry about the image quality of the scans reconstructed during the Tube Heat Soak and Seasoning.

☐ Check box when complete Initial: _____

The MX200 tube uses a metal/ceramic frame, which means some electrons hit the target and actually bounce off, and are then captured by the frame. This causes a frame current, which results in an anode mA measurement of slightly less than the cathode mA, typically 5 to 10% less. Because the cathode mA is the true indicator of radiation output, the CT QX/i system controls cathode mA rather than anode mA.

2.3 X-ray Lights

- 1.) Perform several scans (see below).
- 2.) Verify that the X-ray ON lights are on during the scans. Check boxes in [Table10-2](#)
- 3.) Record data on **FORM 4879** (see [Chapter 13 on page 281](#)).

PERFORMING A SCAN



- 1.) If you are not on the Service Desktop, click on the SERVICE DESKTOP icon.
- 2.) Select DIAGNOSTICS.
- 3.) Select DIAGNOSTIC DATA COLLECTION
- 4.) Set the scan time to 2.00
- 5.) Set the Kv to 80
- 6.) Set the ma to 40.

Light On?	Light Locations
☐	SCIM
☐	Gantry Front
☐	Gantry Back (use a mirror to view)
☐	Room Light

Table 10-2 X-ray light chart

Chapter 11

System Tests

**NOTICE**
Potential for
Data Loss and/
or Equipment
Damage

To prevent potential data loss, please do the following:

- Record data collected from procedures in this chapter into Form F4879 when directed, located in [Chapter 13](#) of this book.
- Only use the Installation manual that arrives with your system for installation. Any other revisions of this manual may not exactly match your system.

Section 1.0

Introduction

This chapter contains descriptions of the procedures used to test system operation after installation, alignment and calibration, but prior to customer turnover and acceptance. The Image Series sections instruct you to scan system phantoms, record the means and standard deviations, and evaluate the image quality of the resulting images.

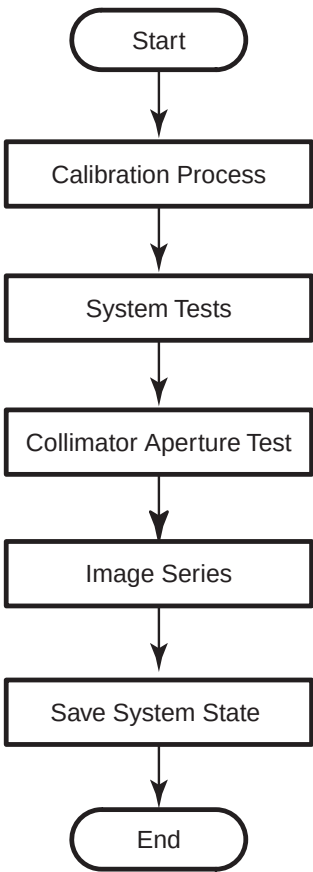


Figure 11-1 System Test Overview

Section 2.0 Calibration Process

2.1 Reference Procedures

Do NOT perform these procedures now. They are placed here as Reference for later procedures. You will be directed to follow these instructions at several points throughout this chapter.

2.1.1 Scanning with Service Protocols

Locate the Manufacturing and Installation protocols under Infant area 20.

Note: **Manufacturing and Service share this Protocol list. Different product option offerings also use this list. Carefully follow the scan section instructions, and verify you acquired the images with the correct technique before filling out the data sheet. Otherwise you may troubleshoot an image problem that only exists because you used the wrong technique.**

- 1.) Select the NEW PATIENT icon.
- 2.) Enter a Patient ID (e.g., getest).
- 3.) Click on the box labeled PEDIATRICS.
- 4.) Select a service protocol from the list, to display the corresponding view edit screen.

Optional Method: Enter the protocol number into the Protocol Number Field on the Exam Rx Screen.

2.1.2 Center Phantom

- 1.) Select SCANNER UTILITIES.
- 2.) Select CENTER PHANTOM.
- 3.) Follow the on-screen procedures.
- 4.) The phantom center spec is $\pm 1\text{mm}$.
- 5.) Select QUIT, when the phantom is within spec.

2.2 Prepare the QA Phantom

Note: **The QA phantom is shipped water filled.**

- 1.) Locate the multi-language sticker packet in the QA phantom shipping box.
- 2.) Attach the sticker with the customer's language to the face of the phantom hanger bracket.

2.3 Calibration Process Introduction

If your system has a factory supplied state MOD, you used it to load the system calibration files during the Restore System State ([3.1 on page 175](#) in Chapter 6 of this manual).

2.4 Fast Cal



- 1.) Select  to warm up the tube.
- 2.) Select FAST CALIBRATION from the Daily Prep menu.

Note: Use the default Fast Cal selections determined by the system configuration. (The system defaults to all four kV stations, but you can choose kV stations to calibrate during reconfig.)

- 3.) Run the selected air calibrations.
- 4.) When the calibration process completes, click on QUIT.

2.5 Table/Gantry Alignment Procedure

2.5.1 Conditions

This procedure may require the removal of the table bottom, side and rail covers.
This procedure requires that mechanical final alignment was completed and passed.

2.5.2 Procedure

- 1.) If this is a Non-Fixed Site, check the Van level. Correct, if necessary.
- 2.) Check Table level. Correct, if necessary.
- 3.) Drive the table to its highest elevation (phantom holder removed).
- 4.) Drive the table cradle in and out five times to seat the rollers.
- 5.) Turn on the alignment lights. Check the scan window for proper installation.
- 6.) Advance the end of the cradle (black dot on cradle) to the internal light.
- 7.) Complete [Section 2.1.1, "Scanning with Service Protocols"](#) now.
- 8.) Select the service protocol, PERPENDICULAR ALIGNMENT. Protocol should appear as shown in [Table 11-1](#). Add two scouts to this procedure. 120kv 40 ma tube at zero, 120kv 40 ma tube at ninety.

Scan Type	Direction	kV	mA	SFOV	Thickness	Scan Time	Start Loc.	End Loc.	Algorithm
Axial	Head First	120	40	Large	5mm/1i	2sec	S0	S0	Bone
Axial	Head First	120	40	Large	5mm/1i	2sec	I1000	I1000	Bone

Table 11-1 Perpendicular Alignment Scan Parameters

Set Internal Landmark. The red boxes will disappear from the screen.

- 9.) Click on CONFIRM, then press the START SCAN button when lighted, to perform the scan.
- 10.) Use over/under multi-image display to view both images at the same time:
 - a.) Click on IMAGE WORKS
 - b.) Highlight the scans just performed and click on VIEWER.
 - c.) Click on FORMAT, and select the "over/under" view (two boxes aligned vertically).
- 11.) Use the upper image (cradle front) to verify that table is centered:
 - a.) Click on the "grid" icon.
 - b.) Click on MEASURE, and select the straight (angled) line.
 - c.) Using the mouse, position one end of the line on the vertical axis.
 - d.) Position the other end at one outside edge of the cradle, forming a perfectly horizontal line (angle = 90). Read the distance displayed on screen.
 - e.) Repeat step d to measure the distance from the vertical axis to the other edge.
- 12.) Repeat step 11 for the lower image (cradle at 1000 mm).
- 13.) If the edges are off by greater than 1 mm (front to back), perform the following:

- a.) Loosen the table anchors at locations #5, #6 and #7.
 - b.) Move the table half the measured distance.
 - c.) Return to Step 8. Repeat the procedure until the table edges line up (± 1 mm). If needed a scout can be completed to check table horizontal presentation image.
Check the two scout scans
 - i.) Check the tube at zero scan for table gantry perpendicularity. Make table adjustments as needed
 - ii.) Check the tube at 90 scan for cradle level and visual irregularity with the white side molding. Molding strips should be parallel within ± 3 mm. Cradle should be level within ± 3 mm.
 - d.) Tighten the anchors and torque to 74.6 N-m (55 ft.-lb.) for fixed sites only. (75 ft.-lb.) for non-fixed systems.
 - e.) Reinstall all table parts removed to access the anchors, including covers, ground straps, elevation motor leads and the center support bar.
- 14.) Install all table and gantry covers.
- 15.) Use ERASE ALL to remove lines and grids from the images.
- 16.) Click on END EXAM to exit the program.

2.6 Auto CT N Number Adjustment

Note: **If it has been more than 30 minutes since calibration scans, rerun FASTCAL (approx 15 min). N Number Adjustment needs the tube to be warm, or CT numbers will change.**

Run the small and large phantom CT N number adjustment before you acquire the image series, or any time you doubt the N# values. The calibration procedure modifies the pcal (phantom cal) vector to assign water a CT number equal to zero. After a successful N number adjustment, any water phantom scanned against a small or large CAL has a CT number near zero.

- 1.) Place the QA Phantom in the holder.
- 2.) From the application desktop, select SCANNER UTILITIES.
- 3.) Select AUTO CT NUMBER ADJUSTMENT.
- 4.) Follow directions on the screen.
 - a.) When centering the phantom:
 - * Centering may be easier to perform from the gantry right side.
 - * Center on the water portion of the phantom (not over the internal block).
 - b.) The program will automatically make adjustments. If the program is unable to make adjustments, it will stop at the technique it is unable to adjust. Troubleshoot with manual CT # Adjust before continuing.

Section 3.0 System Tests

3.1 Introduction

Use the system tests in the following sections to exercise all aspects of the system, to assure system integrity, before turnover to the customer. Although the means, standard deviation, and resolution specifications do not apply during system functional tests, treat any artifact or image anomaly as a failure.

If you encounter a failure during the system tests:

- Record any evidence of artifacts, such as rings, streaks, shading, cupping, noise, or center artifacts.
- Correct artifacts, system test, or image series failures when they occur. Any delay in repairs could increase the number of retests.

3.2 Tomographic Plane Indication

- 1.) Place the QA phantom on the phantom holder.
- 2.) Turn ON the internal alignment lights, and drive the phantom into the gantry opening, until the line on the phantom lines up with the internal laser lights.
- 3.) Verify that BOTH internal axial lasers line up along the line on the QA phantom. If not, check table/gantry, cradle, and/or laser alignment.
- 4.) Center the phantom in the scan plane with the calibration program. (See [2.1.2 on page 224](#), for details on the phantom centering procedure.)
- 5.) Select the service protocol, TOMO PLANE INDICATION. (See [2.1.1 on page 224](#), for details on scanning with service protocols.)

or

Manually select the scan parameters in [Table 11-2](#).

Scan Type	kV	mA	SFOV	Thickness	Scan Time	Start Loc.	End Loc.	Algorithm	Interval
Helical	120	200	Small	1.25HQ	1.0sec	I3.0	S3.0	Bone detail	0.2

Table 11-2 Tomographic Plane Indication Scan Parameters

- 6.) Display the image series, and locate the scan plane indicator, the longest bar in the bar pattern on the right side of the phantom. The right side of the phantom corresponds to the side of the image labeled **L** on the display screen.
- 7.) On the HHS Data Sheet, record the scan location (shown on the image annotation) of the image with the darkest scan plane indicator (darkest long bar).
- 8.) If your system meets all the installation and alignment specifications, the image at scan location zero (S0.0) should contain the scan plane indicator. If scan location S1.0 or scan location I1.0 has the darkest bar, the system still meets the specification. The scan plane deviation should equal $S0.0 \pm 1.0\text{mm}$. If necessary, adjust the internal alignment light position to meet the $S0.0 \pm 1.0\text{mm}$ requirement.

- 9.) Repeat the Tomographic Plane Indication test with the external alignment lights.
 - a.) Use the external alignment light, and press the external landmark.
 - b.) Verify the external light lines up along the black line on BOTH the left and right sides of the QA phantom.
 - c.) The scan plane indication must fall within the $S0.0 \pm 1.0\text{mm}$ specification.
- 10.) Check the box on **Form 4879**.

3.3 System Scanning Test

Use the System Scanning Test to verify hardware functionality. Review images for visible artifacts, and review the message log for unacceptable errors.

- 1.) Place the QA phantom on the cradle.
 - Drive the table to an elevation of 100.
 - Align the line on the phantom with the internal laser lights.

Never scan above 50mA without first placing a phantom in the field of view. Levels in excess of 50mA can cause temporary radiation damage to the detector that lasts several hours. If you acquire image series cals with a radiation damaged detector, the cals may cause artifacts in subsequent image series scans.

Note: 2.) Select the service protocol, SYSTEM SCAN/CUST QUAL RELI.
Stop the service protocol after the second helical series. It is not necessary to proceed past that point in the protocol.

-or-

Manually select the scan parameters specified in [Table 11-3](#).

Scan Type	kV	mA	SFOV	Thickness	Rotation Time	Start Loc.	End Loc.	Tilt/Table Speed
Scout	120	40	-	-	-	S200	I800	0°
Scout	120	40	-	-	-	S200	I800	90°
Axial (2 scans)	120	50	Large	10mm/1i	1.0sec	S0	S0	I30
Axial (2scans)	120	50	Large	10mm/1i	1.0sec	S0	S0	S30
Helical HQ	120	50	Large	5mm	1.0sec	S70	I75	15mm/rot
Helical HS	120	50	Large	5mm	0.8sec	S70	I75	22.5mm/rot

Table 11-3 System Scanning Test Scan Parameters

- 3.) Complete the scans.

Section 4.0

Collimator Aperture Test

4.1 Small Aperture Test

- 1.) Clear all obstructions out of the gantry SFOV.
- 2.) If you are not on the Service Desktop, click on the SERVICE DESKTOP icon, .
- 3.) Click on the UTILITIES icon. .
- 4.) Select INSTALL.
- 5.) Select COLLIMATOR APERTURE TEST.
- 6.) Select SMALL APERTURE SCAN to display a scan protocol window, and start gantry activity (refer to [Figure 11-2](#)).
- 7.) Press the START SCAN key when it flashes, to initiate the first scan.
- 8.) When the test completes, the screen updates to display the Small Aperture values.

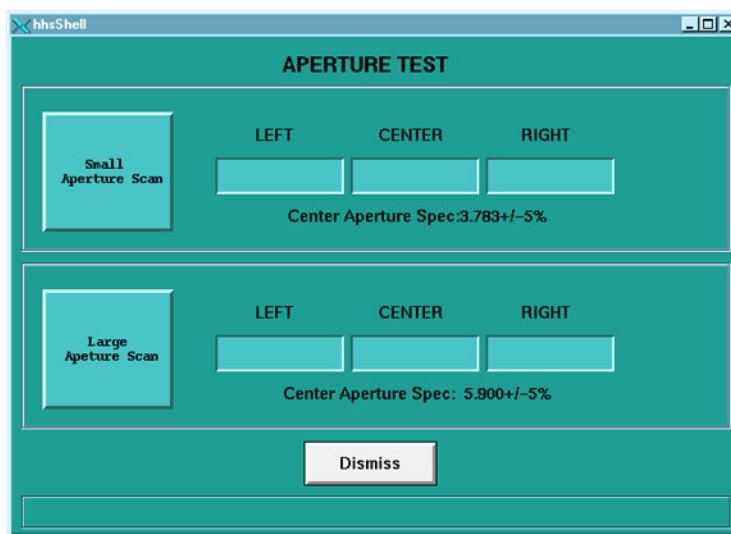


Figure 11-2 Collimator Aperture Test Screen

4.2 Large Aperture Test

Note: The center channel is the only channel with a spec. The left and right channels may or may not pass. Small aperture center spec is $3.783 \pm 5\%$. Large aperture center spec is $5.900 \pm 5\%$.

- 1.) Select LARGE APERTURE to display a scan protocol window, and start gantry activity (Refer to [Figure 11-2](#)).
- 2.) Press the START SCAN key when it flashes, to initiate the first scan.
- 3.) When the test completes, the screen updates to display the large aperture values.
- 4.) Record the large and small aperture values on Form 4879, located in [Chapter 13](#).

Section 5.0 Image Series

5.1 Scan Protocol

The person who acquires the image series has the responsibility to review the images, and verify they meet the specifications listed on data sheets. Responsibilities also include means and standard deviation measurements, and keeping a record of failures that occur during the image series.

Unless otherwise stated, use the following scan parameters during the image series acquisition:

- Scan FOV equal to display FOV (Field of View)
- 512x512 matrix size

Note: **Consider any image series scan that does not meet specifications as failing.**

For means and standard deviations, 90% of the slices must pass. Any failure on a particular technique requires at least ten additional slices to evaluate effectively.

Systems with metal-free cradles have a phantom holder with a perpendicular adjustment (Z-axis) knob on it. **Each time you change phantoms**, make sure you use a bubble level, and the Z-axis knob on the phantom holder, to level the phantom.

Note: For systems without the 440 mA option, scan protocols that use more than 350 mA techniques will not be executed as they are. Instead, 350 mA technique is used for those mA stations. However, the test results will still be valid.

5.2 Data Recording: Means and Standard Deviation

Any failure on a particular technique requires at least a ten additional slices to evaluate effectively. For means and standard deviations, 90% of the slices must pass.

- Record means to two decimal places, and round to the nearest one-tenth, (one decimal place) when you compare the resulting values to the spec.
- Record standard deviations to two decimal places, then round off to one decimal place, to compare it to the spec.
- Average standard deviations: Use two decimal places to average the values, then round off to one place.

Before you record the means and standard deviations, check the image data sheets to determine whether to average the means and standard deviations, or record them slice by slice. Make sure you record all the required image data on the HHS data sheets.

5.3 Term Definitions & Screens

Xc - Mean CT number for the specified center coordinates of the phantom image.

AvXc - Average Mean CT number for the center of the phantom image: Average the mean CT value for all specified center coordinates of all slices in an exam.

Xo - Mean CT number for the outside of the phantom image: Average the mean CT value for all specified outside coordinates of one slice.

AvXo - Average outside mean CT number for the number of slices in an exam.

AvSDc - Average image noise about the center image coordinate (measured as the standard deviation) of all slices in an exam.

AvSDo - Average image noise (standard deviation) for the outside of a phantom: Average of all outside coordinates of all the slices in an exam.

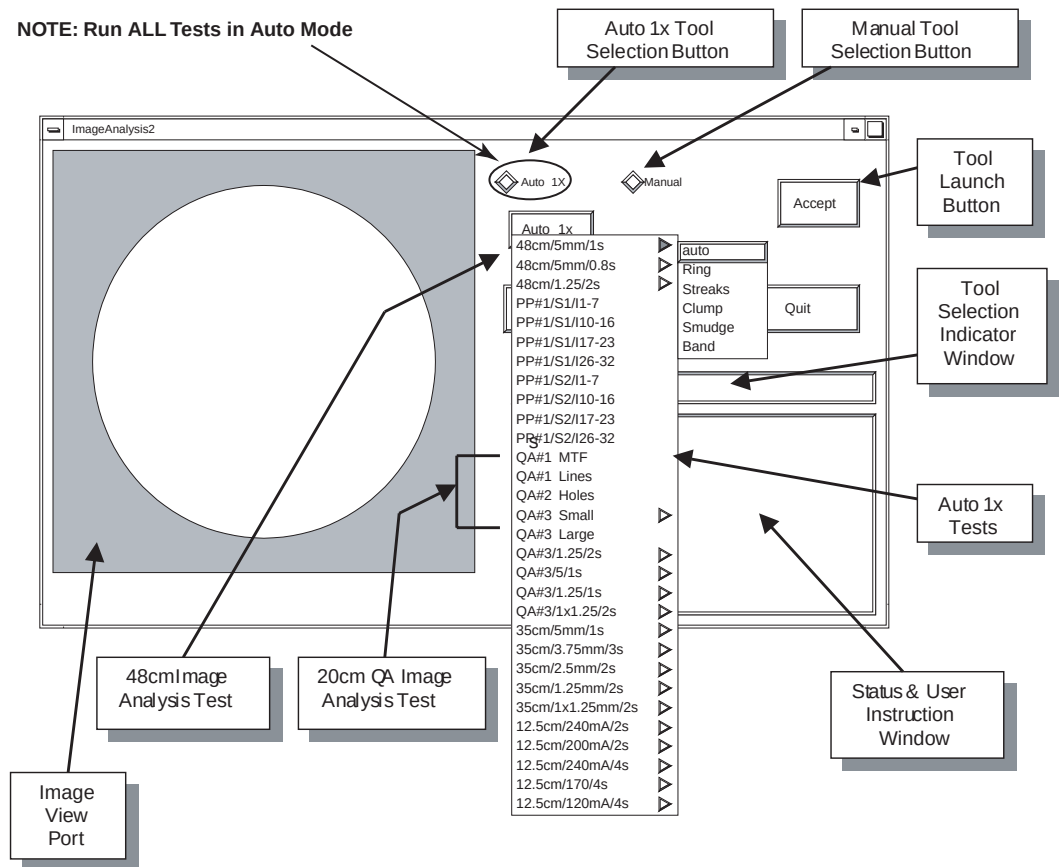


Figure 11-3 Image Analysis Tool User Interface – Auto 1x Test Pull Down Menu

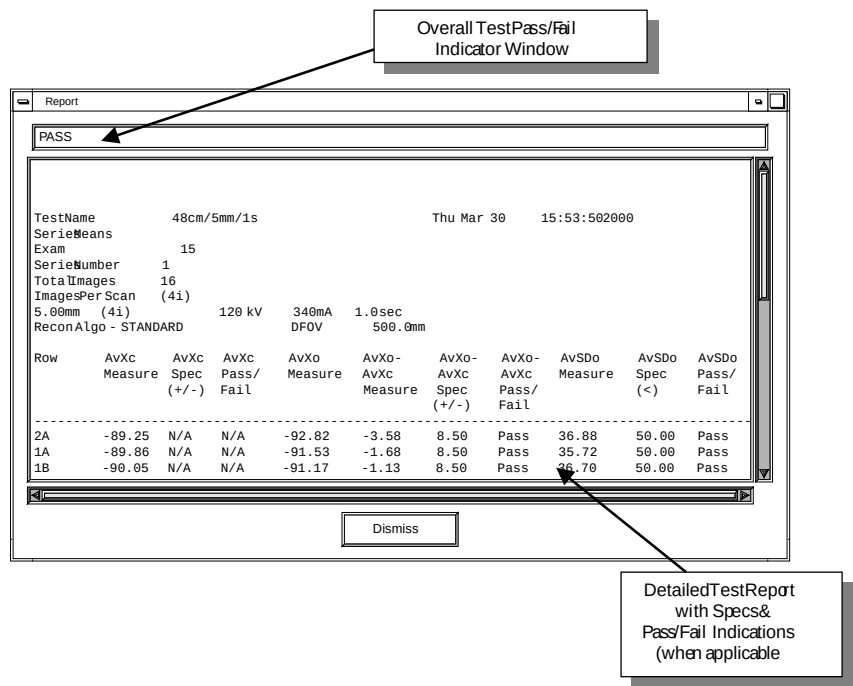


Figure 11-4 Analysis Tool User Interface – Test Results Report Window

5.4 48cm Phantom Image Series Image Performance Verification

5.4.1 Acquiring the 48cm Phantom Image Series

- 1.) Mount the Phantom Holder on the head-end of the table.
- 2.) Mount the 48cm Phantom on the Phantom Holder.
- 3.) Align, level, & center the 48cm Phantom.
 - Align phantom using the internal laser lights.
 - Level phantom using bubble level and the Z Axis knob on the Phantom Holder.
 - Center phantom using the CENTER PHANTOM utility in the left head SCANNER UTILITIES selection and the X and Y Axis knobs on the Phantom Holder.
- 4.) Set up the system to scan a four scan, 16 image, 48cm Phantom image series.

MANUAL SCAN PROTOCOL SETUP

Refer to [Table 11-4](#) and set-up an Axial scan with the parameters as shown.

Series Description	Scan Type	Start Loc.	End Loc.	Total # of Images	Thick Speed	Interval (mm)	Gantry Tilt	SFOV	kV	mA	Total Exposure Time	DFOV (cm)	Recon Type
48 Lg Series 1	Axial Full 1.0 sec.	I7.50	S7.50	16	5.0 4i	0.00	S0.0	Large	120	340	4.0 sec.	50.0	Std

Table 11-4 48cm Phantom Image Series Scan Parameters

AUTO SCAN PROTOCOL SETUP

- a.) On the Exam Rx desktop, select NEW PATIENT.
- b.) Type the following entries in the two listed Patient Information fields:

Patient ID: Service

Name: 48cm Phantom Image Series
- c.) From the Protocol display, click on the infant box.
- d.) On the infant display window, click on the area below the infant's right foot to display the Miscellaneous menu.
- e.) Click on the 20:10 IMAGE SERIES 48 CM button.
- f.) Select the first series (Series Description filed should display 48cm Lg Series 1)
- 5.) Set internal Landmark.
- 6.) Acquire a four-scan, 16 image, 48 cm Phantom image series.

5.4.2 Brightness Uniformity and Noise

5.4.2.1 Image Performance Verification

- 1.) Select the 48cm Phantom image series exam acquired in the previous section.
 - a.) From the Global Control Palette, click on the IMAGE WORKS Desktop.
 - b.) From the Image Works Desktop, select the IMAGE WORKS BROWSER window.
 - c.) Select the exam and series acquired in the previous section.
 - d.) Select the VIEWER button on the Image Works Browser window. Set up the viewer window for four-image viewing.
- 2.) Build a 45 x 45 pixel reference ROI Box using the Image Works Viewer tools.
 - a.) Click on the grid button to place a grid on the first image.
 - b.) Click on the MEASURE button and select the box ROI icon.
 - c.) Adjust the size of the ROI box to be a square 45 mm x 45 mm (2025 mm²) box. Tolerance: 45 mm +/- 5 mm (1600 mm² to 2500 mm²).

If required, magnify the image to adjust to proper dimensions. See [Figure 11-5](#) for additional ROI size and placement information.
- 3.) Collect Mean and Standard deviation values for five reference ROI Box positions on the sixteen 48cm Phantom images.

If required, magnify the image to adjust to proper dimensions. See [Figure 11-5](#) for additional ROI placement information.

 - a.) Position the reference ROI Box built in step 2 directly over the center of the image using the grid cross-hairs as a guide.
 - b.) Type **prop a** in the Image Works Accelerator Line followed by <Return> to propagate Box # 1 ROI (size and position) on the remaining images in the series.
 - c.) Click on the MEASURE button and select the box ROI icon to display Box # 2. The system places an ROI box labeled # 2 at the center of the image with the exact same dimensions as Box # 1.
 - d.) Reposition Box # 2 to the left center portion on the first image (Box # 2 position in [Figure 11-5](#)).
 - e.) Type **prop a** in the Image Works Accelerator Line followed by <Return> to propagate Box # 2 ROI (size and position) on the remaining images.
 - f.) Repeat steps 3.c. through 3.e. for each of the remaining ROI box positions shown in [Figure 11-5](#) (Box # 3 through 5).
 - g.) Record the Mean and Standard Deviation values of the 16 images in the series for each of the five box positions in [Table 11-5](#).

Each image can only display text for the mean, standard deviation, and box area for three images at a time. To view the data for a particular box, select the box on the image and the system displays the data for the box number selected.

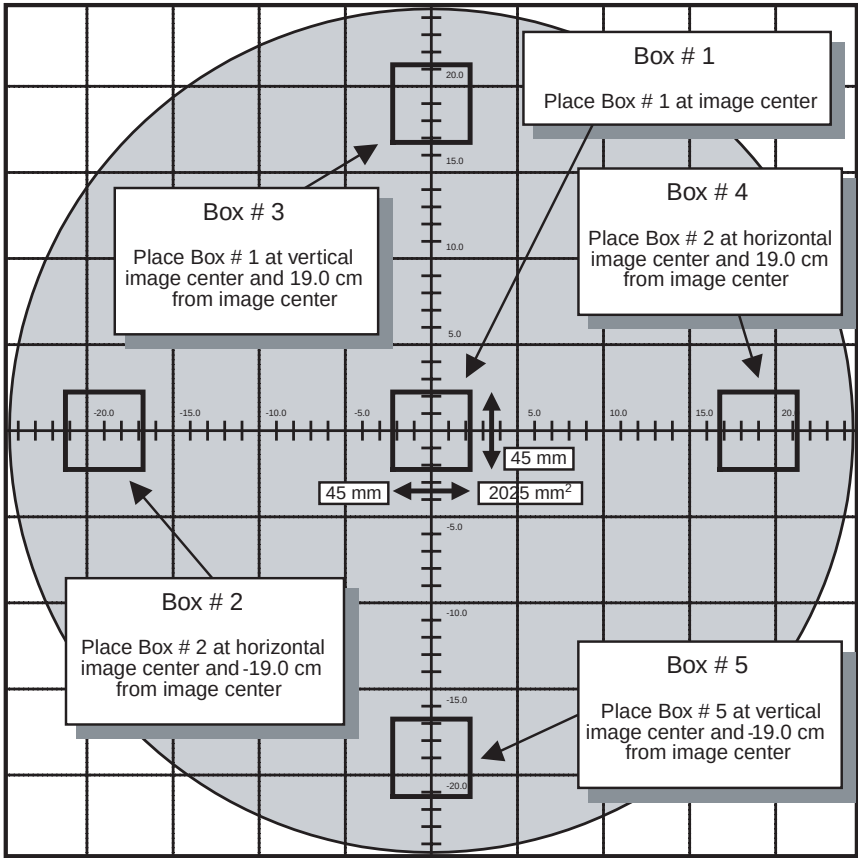


Figure 11-5 48cm Phantom Brightness Uniformity & Noise Measurement – Building Placing Reference ROI Boxes

Image	Box 1 (Center)	Box 2 (Left Center)		Box 3 (Top Center)		Box 4 (Right Center)		Box 5 (Bottom Center)		Outside Boxes Avg	
	Means (Xc)	Means	Std dev	Means	Std dev	Means	Std dev	Means	Std dev	Means (AvXo)	Std dev (AvSDo)
1											
2											
3											
4											
5											
6											
7											
8											
9											
10											
11											
12											
13											
14											
15											
16											

Table 11-5 48cm Phantom CT# Brightness Uniformity & Noise Image Performance Worksheet

Box Size = 1600 mm² to 2500 mm²
45 mm (+/- 5 mm) x 45 mm (+/- 5 mm)
45 (+/- 4 pixels) x 45 (+/- 4 pixels)

Box Positions: Box 1 = 0 mm x 0 mm
Box 2 = 0 mm x -190 mm
Box 3 = 190 mm x 0 mm
Box 4 = 0 mm x 190 mm
Box 5 = -190 mm x 0 mm

Row	Images	Center Box Means (AvXc)	Outer Boxes Means Averages (AvXo)	Outer Boxes Means Averages minus Center Box Means (AvXo - AvXc)	Outer Boxes Standard Deviation Averages (AvSDo)	Comments
2A	1, 5, 9, 13					
1A	2, 6, 10, 14					
1B	3, 7, 11, 15					
2B	4, 8, 12, 16					
Specifications		n/a	n/a	± 8.5	< 70	

Table 11-6 48cm Phantom CT# Brightness Uniformity & Noise Row Performance Worksheet

- 4.) Calculate the Brightness Uniformity and Noise values for each image in the series, record values in [Table 11-6](#), and compare to specifications.
 - a.) Calculate and record the average means and average standard deviation values for the four outside Boxes (Boxes 2 through 5) for each of the images and record in [Table 11-5](#).
 - b.) Calculate and record the center box (Box 1) average means values (AvXc) for each row in [Table 11-6](#).
 - c.) Record the outside boxes (Box 2 through 5) average means values (AvXo) and average standard deviation values (AvSDo) for each row in [Table 11-6](#).
 - d.) Calculate the Brightness Uniformity (AvXo - AvXc) value for each row and record in [Table 11-6](#).
 - e.) Verify Brightness Uniformity (AvXo - AvXc) value and Noise value (AvSDo) for each row meets specifications listed in [Table 11-6](#).
 - f.) Record the 48cm Phantom Brightness Uniformity (AvXo - AvXc) value and Noise value (AvSDo) for each row in the HHS Record Tables.

5.4.2.2 Failure Recovery

Specifications

Each Row (2A, 1A, 1B, and 2B) of the series *must* pass 48cm Brightness Uniformity and Noise (for the first series scan parameters) specifications:

- AvXo - AvXc: < +/- 8.5
- AvSDo: < 70.0

Recommended Recovery

- A.) Perform DETAILED CAL.
- B.) Perform AUTO CT# ADJUST.
- C.) Repeat this procedure to verify 48cm Phantom Image Performance.

5.5 20cm QA Phantom Image Series Image Performance Verification

5.5.1 Acquiring the 20cm QA Phantom Image Series

- 1.) Mount the Phantom Holder on the head-end of the table.
- 2.) Mount the 20cm QA Phantom on the Phantom Holder.
- 3.) Align, level, & center the 20cm QA Phantom.
 - Align the etched line (QA#1 position) on phantom using the internal laser lights.
 - Level phantom using bubble level and the Z Axis knob on the Phantom Holder.
 - Center phantom using the CENTER PHANTOM procedure in the left head SCANNER UTILITIES selection and the X and Y Axis knobs on the Phantom Holder.
- 4.) Set up the system to scan the QA#1, QA#2, and QA#3 positions on the 20cm QA Phantom.

MANUAL SCAN PROTOCOL SETUP

Refer to [Table 11-7](#) and set-up an Axial scan with the parameters shown

Series Description	Scan Type	Start Loc.	End Loc.	Total # of Images	Thick Speed	Interval (mm)	Tilt	SFOV	kV	mA	Total Exposure Time	DFOV (cm)	Recon Type
QA#1	Axial Full 1.0 sec.	I5.00	S5.00	4	10.0 2i	0.00	S0.0	Small	120	260	2.0 sec.	25.0	Std
Recon 2: Q	Recon #2 of Series #1–4 images											15.0	Bone
QA#2	Axial Full 1.0 sec.	S40.00	S50.00	8	10.0 2i	0.00	S0.0	Small	120	260	4.0 sec.	25.0	Std
QA#3	Axial Full 1.0 sec.	S55.00	S65.00	8	10.0 2i	0.00	S0.0	Small	120	260	4.0 sec.	25.0	Std

Table 11-7 20cm QA Phantom Image Series Scan Parameters

AUTO SCAN PROTOCOL SETUP

- a.) On the Exam Rx desktop, select NEW PATIENT.
- b.) Type the following entries in the listed Patient Information following fields:
 - * Patient ID: **Service**
 - * Name: **20cm QA#3 Phantom Image Series**
- c.) From the Protocol display, click on the infant box.
- d.) On the Infant display window, click on the area below the infant's right foot to display the Miscellaneous menu.
- e.) Click on the 20:12 IMAGE SERIES QA button.
- f.) When the scan prescription appears on the left monitor, select the 1st series.
- 5.) Set internal Landmark.
- 6.) Acquire the QA#1, QA#2 and QA#3 positions of the 20cm QA Phantom.

5.5.2 High Contrast Spatial Resolution

5.5.2.1 Image Performance Verification

- 1.) Select the first 20cm QA Phantom image series (QA#1) from the exam acquired in the previous section.
 - a.) From the Global Control Palette, click on the IMAGE WORKS Desktop.
 - b.) From the Image Works Desktop, select the IMAGE WORKS BROWSER window.
 - c.) Select the exam and the first series acquired in the previous section.
 - d.) Select the VIEWER button on the Image Works Browser window. Set up the viewer window for four-image viewing.
- 2.) Build a 17 x 17 pixel reference ROI Box using the Image Works Viewer tools.
 - a.) Click on the grid button to place a grid on the first image.
 - b.) Click on the MEASURE button and select the box ROI icon.
 - c.) Adjust the size of the ROI box to be a square of approximately 8 mm x 8 mm (64 mm²) box. If required, magnify the image to adjust to proper dimensions. See [Figure 11-6](#) for additional ROI size and placement information.
- 3.) Collect Mean and Standard deviation values for three reference ROI Box positions on the four 20cm QA#1 Phantom images.

If required, magnify the image to adjust to proper dimensions. See [Figure 11-6](#) for additional ROI placement information.

 - a.) Position the reference ROI Box built in step 2 directly over the 1.6 mm Line Pair Pattern on the QA#1 phantom image. Fine adjust size of Box #1 ROI to ensure that the box does not exceed the line pair boundaries. See [Figure 11-6](#).
 - b.) Type **prop a** in the Image Works Accelerator Line followed by <Return> to propagate Box # 1 ROI (size and position) on the remaining images in the series.
 - c.) Click on the MEASURE button and select the box ROI icon to display Box # 2. The system places an ROI box labeled # 2 at the center of the image with the exact same dimensions as Box # 1.
 - d.) Reposition Box # 2 over the Plexiglas portion of the QA#1 phantom image (Box # 2 position in [Figure 11-6](#)).
 - e.) Type **prop a** in the Image Works Accelerator Line followed by <Return> to propagate Box # 2 ROI (size and position) on the remaining images.
 - f.) Click on the MEASURE button and select the box ROI icon to display Box # 3. The system places an ROI box labeled # 3 at the center of the image with the exact same dimensions as Boxes # 1 and 2.
 - g.) Reposition Box # 3 over the water portion of the QA#1 phantom image (Box # 3 position in [Figure 11-6](#)).
 - h.) Record the Mean and Standard Deviation values of the four images in the series for each of the three box positions in [Table 11-8](#).

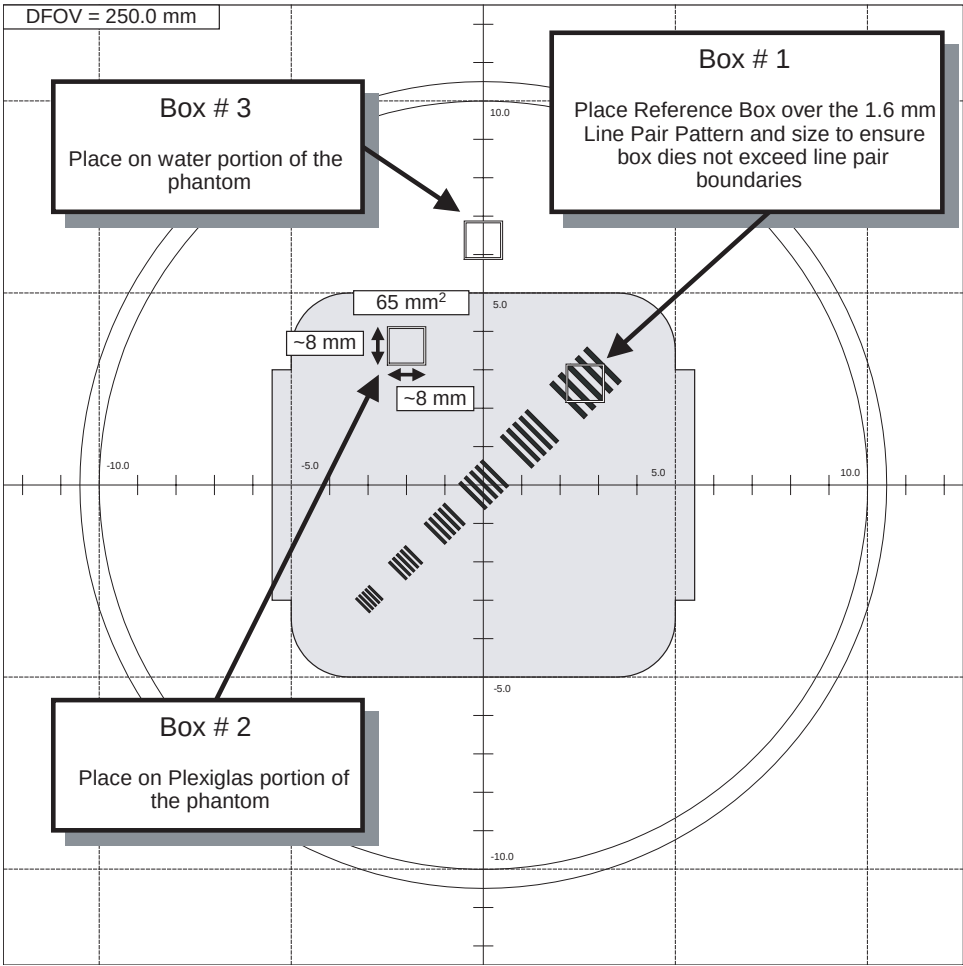


Figure 11-6 20cm QA#1 Phantom High Contrast Spatial Resolution Testing - Building And Placing Reference ROI Boxes

Image	Box 1		Box 2		Box 3		Contrast Scale (Box 2 Means – Box 3 Means)	Std Dev Average (Box 2 Std Dev – Box 3 Std Dev / 2)
	Means	Std Dev	Means	Std Dev	Means	Std Dev		
1								
2								
3								
4								

Table 11-8 20cm QA#1 Phantom High Contrast Spatial Resolution Image Performance Worksheet #1

Image	MTF	MTF 4-slice average	Contrast Scale	Comments
1		n/a		
2		n/a		
3		n/a		
4		n/a		
Specifications	n/a	0.65 to 1.0	110.0 to 130.0	n/a

Table 11-9 20cm QA#1 Phantom High Contrast Spatial Resolution Image Performance Worksheet #2

- 4.) Calculate the Contrast Scale value and four-image average MTF values for each image in the series, record values in [Table 11-9](#), and compare to specifications.
 - a.) Calculate and record the Contrast Scale value (Means value of Box 2 - Means value of Box 3) for each image and record in [Table 11-8](#) and [Table 11-9](#).
 - b.) Calculate and record the MTF value for each image in [Table 11-9](#) using the following formula. Record Standard Deviation Average (Std Dev AVE) values in both [Table 11-8](#) and [Table 11-9](#).

$$\text{Std Dev AVE} = (\text{Std Dev}_{\text{Box 2}} - \text{Std Dev}_{\text{Box 3}}) + 2$$

$$\text{Modulation} = \sqrt{\text{StdDev}_{\text{Box 1}}^2 - \text{StdDev}_{\text{AVE}}^2}$$

$$\text{MTF} = (2.2 \times \text{Modulation}) + \text{Contrast Scale}$$
 - c.) Calculate the average MTF value for each image and record in [Table 11-9](#).
 - d.) Verify Contrast value for each image and the four-image MTF value meets specifications listed in [Table 11-9](#).
 - e.) Record the 20cm QA Phantom High Contrast Spatial Resolution values (Contrast Scale for each image and four-image MTF value) in the HHS Record Tables.
- 5.) View each image of the Recon 2 Series, record all the visible Line Pair Patterns in [Table 11-10](#), and verify each image meets specifications.
 - a.) From the Image Works Desktop, select the IMAGE WORKS BROWSER window.
 - b.) Select the exam and the second series (Recon 2: Q) acquired in the previous section.
 - c.) Select the VIEWER button on the Image Works Browser window. Set up the viewer window for four-image viewing.
 - d.) While viewing each image, indicate in [Table 11-10](#) all the Line Pair Pattern groups that can be visually distinguished. See [Figure 11-7](#).
 - e.) Verify Line Pair Pattern visual check meets specifications and record in the HHS Record Tables.

Image	Line Patterns Visible	Comments
1		
2		
3		
4		
Specifications	B, C, D, E, F	n/a

Table 11-10 20cm QA#1 Phantom High Contrast Spatial Resolution Image Performance Worksheet #3

5.5.2.2 Failure Recovery

Specifications

Each image of the series must pass 20cm QA#1 High Contrast Spatial Resolution parameter (Contrast Scale, Four-Image MTF Average and Visible Line Pair (for the first and second series scan parameters) specifications:

- Contrast Scale: 110.0 to 130.0
- MTF Average: 0.65 to 1.0
- Visible Lines: B, C, D, E, F

Recommended Recovery

- A.) Check Phantom Alignment (leveling is critical) and repeat this scanning and High Contrast Spatial Resolution Performance Test.
- B.) Perform Alignment Procedures (POR Alignment, BOW Alignment, CBF/SAG Alignment, ISO Alignment, and Hot ISO Alignment), perform Full Calibration (Detailed Calibration and Auto CT# Adjust, and repeat this scanning and High Contrast Spatial Resolution Performance Test.

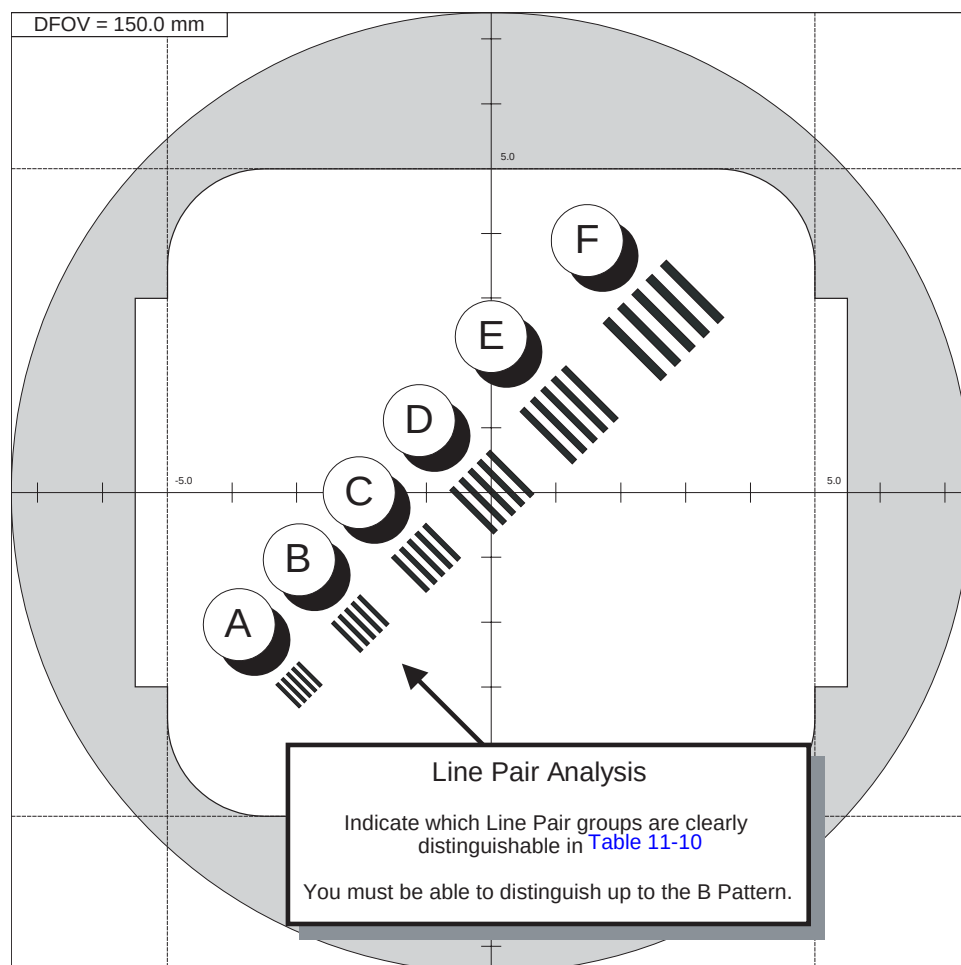


Figure 11-7 20cm QA#1 Phantom High Contrast Spatial Resolution (Visible Line Verification)

5.5.3 Low Contrast Detectability

5.5.3.1 Image Performance Verification

- 1.) Select the third 20cm QA Phantom image series (QA#2 Holes) from the exam acquired in the previous section.
 - a.) From the Global Control Palette, click on the IMAGE WORKS Desktop.
 - b.) From the Image Works Desktop, select the IMAGE WORKS BROWSER window.
 - c.) Select the exam and the third series acquired in the previous section.
 - d.) Select the VIEWER button on the Image Works Browser window. Set up the viewer window for four-image viewing.
- 2.) Build a 10 mm x 100 mm reference ROI Box using the Image Works Viewer tools.
 - a.) Click on the grid button to place a grid on the first image.
 - b.) Click on the MEASURE button and select the box ROI icon.
 - c.) Adjust the size of the ROI box to be a rectangle of approximately 10 mm x 50 mm (500 mm²) box.

If required, magnify the image to adjust to proper dimensions. See [Figure 11-8](#) for additional ROI size and placement information.
- 3.) Collect Mean values for two reference ROI Box positions on four (1st, 3rd, 5th, and 7th) of the eight 20cm QA#2 Phantom images.

If required, magnify the image to adjust to proper dimensions. See [Figure 11-8](#) for additional ROI placement information.

 - a.) Adjust the Window setting of the image to 20 by simultaneously holding the center mouse button down while dragging the cursor to the left. Adjust level for a viewable image as shown in [Figure 11-8](#).
 - b.) Position the reference ROI Box built in step 2 on the Plexiglas portion of the QA#2 phantom image without touching the water portion or the hole pattern. See [Figure 11-8](#).
 - c.) Type **prop a** in the Image Works Accelerator Line followed by <Return> to propagate Box # 1 ROI (size and position) on the remaining images in the series.
 - d.) Click on the MEASURE button and select the box ROI icon to display Box # 2. The system places an ROI box labeled # 2 at the center of the image with the exact same dimensions as Box # 1.
 - e.) Reposition Box # 2 over the water portion of the QA#1 phantom image and directly above Box # 1 (Box # 2 position in [Figure 11-8](#)).
 - f.) Type **prop a** in the Image Works Accelerator Line followed by <Return> to propagate Box # 2 ROI (size and position) on the remaining images.
 - g.) Record the Mean values of the 1st, 3rd, 5th, and 7th images in the series for each of the two box positions in Table 8.

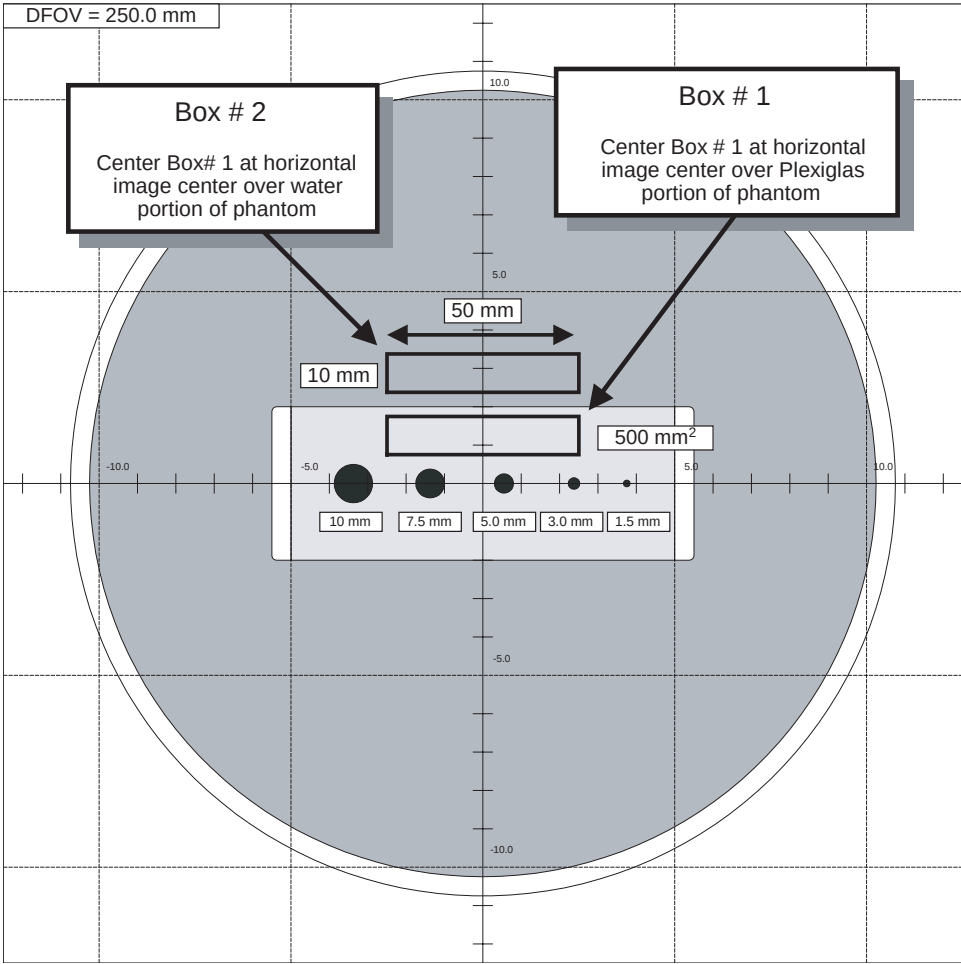


Figure 11-8 20cm QA#2 Phantom Low Contrast Detectability - Building and Placing Reference ROI Boxes

Image	Visible Holes Viewable at Window 20	Box 1 Means (Plexiglas)	Box 2 Means (Water)	Contrast Factor (Box 1 Means – Box 2 Means)	Comments
1					
3					
5					
7					
Specifications	See Table 11-12	n/a	n/a	2.0 to 12.0	n/a

Table 11-11 20cm QA#2 Phantom Low Contrast Detectability Image Performance Worksheet #1

- 4.) View the 1st, 3rd, 5th, and 7th images of the QA#2 Holes Series, record the number of visible holes in Table 11-11, and verify each image meets specifications.
- a.) While viewing the 1st, 3rd, 5th, and 7th images, indicate in Table 11-11 the number of holes that can be visually distinguished. See Figure 11-8.
 - b.) Verify Visible Hole visual check meets specifications listed in Table 11-12 for the calculated Contrast Factor and record in the HHS Record Tables.

Contrast Factor Range (Box 1 Means – Box 2 Means)	Visible Number of Holes		Smallest Visible Hole Size
	Lower Limit*	Upper Limit*	
2.00 to 3.99	2	5	7.5mm
4.00 to 7.99	3	5	5.0mm
8.00 to 12.00	4	5	3.0mm

*Required number of visible holes depends on the contrast factor.

Table 11-12 20cm QA#2 Phantom Visible Hole Specifications

5.5.3.2 Failure Recovery

Specifications

At least two out of the four images of the series must pass the 20cm QA#2 Low Contrast Detectability parameter (Visible Hole Size for a calculated Contrast Factor) for the third series scan parameters specifications.

Recommended Recovery

- A.) Check Phantom Alignment (leveling is critical) and repeat this scanning and High Contrast Spatial Resolution Performance Test.
- B.) Perform Alignment Procedures (POR Alignment, BOW Alignment, CBF/SAG Alignment, ISO Alignment, and Hot ISO Alignment), perform Full Calibration (Detailed Calibration and Auto CT# Adjust, and repeat this scanning and Low Contrast Detectability Performance Test.

5.5.4 QA#3 Phantom Brightness Uniformity and CT#

5.5.4.1 Performance Verification

- 1.) Select the fourth 20cm QA#3 Phantom exam acquired in the previous section.
 - a.) From the Global Control Palette, click on the IMAGE WORKS Desktop.
 - b.) From the Image Works Desktop, select the IMAGE WORKS BROWSER window.
 - c.) Select the exam and fourth (QA#3) series acquired in the previous section.
 - d.) Select the VIEWER button on the Image Works Browser window. Set up the viewer window for four-image viewing.
- 2.) Build a 31 x 31 pixel reference ROI Box using the Image Works Viewer tools.
 - a.) Click on the grid button to place a grid on the first image.
 - b.) Click on the MEASURE button and select the box ROI icon.
 - c.) Adjust the size of the ROI box to be a square 15 mm x 15 mm (225 mm²) box. Tolerance: 15 mm +/- 1 mm (196 mm² to 256 mm²).
If required, magnify the image to adjust to proper dimensions. See [Figure 11-9](#) for additional ROI size and placement information.
- 3.) Collect Mean values for five reference ROI Box positions on the eight 20cm QA#3 Phantom images.
If required, magnify the image to adjust to proper dimensions. See [Figure 11-9](#) for additional ROI placement information.
 - a.) Position the reference ROI Box built in step 2 directly over the center of the image using

the grid cross-hairs as a guide.

- b.) Type **prop a** in the Image Works Accelerator Line followed by <Return> to propagate Box # 1 ROI (size and position) on the remaining images in the series.
- c.) Click on the **MEASURE** button and select the box ROI icon to display Box # 2. The system places an ROI box labeled # 2 at the center of the image with the exact same dimensions as Box # 1.
- d.) Reposition Box # 2 to the left center portion on the first image (Box # 2 position in [Figure 11-5](#)).
- e.) Type **prop a** in the Image Works Accelerator Line followed by <Return> to propagate Box # 2 ROI (size and position) on the remaining images.
- f.) Repeat steps 3.c. through 3.e. for each of the remaining ROI box positions shown in [Figure 11-6](#) (Box # 3 through 5).
- g.) Record the Mean values of the eight images in the series for each of the five box positions in [Table 11-13](#).

Each image can only display text for the mean, standard deviation, and box area for three images at a time. To view the data for a particular box, select the box on the image and the system displays the data for the box number selected.

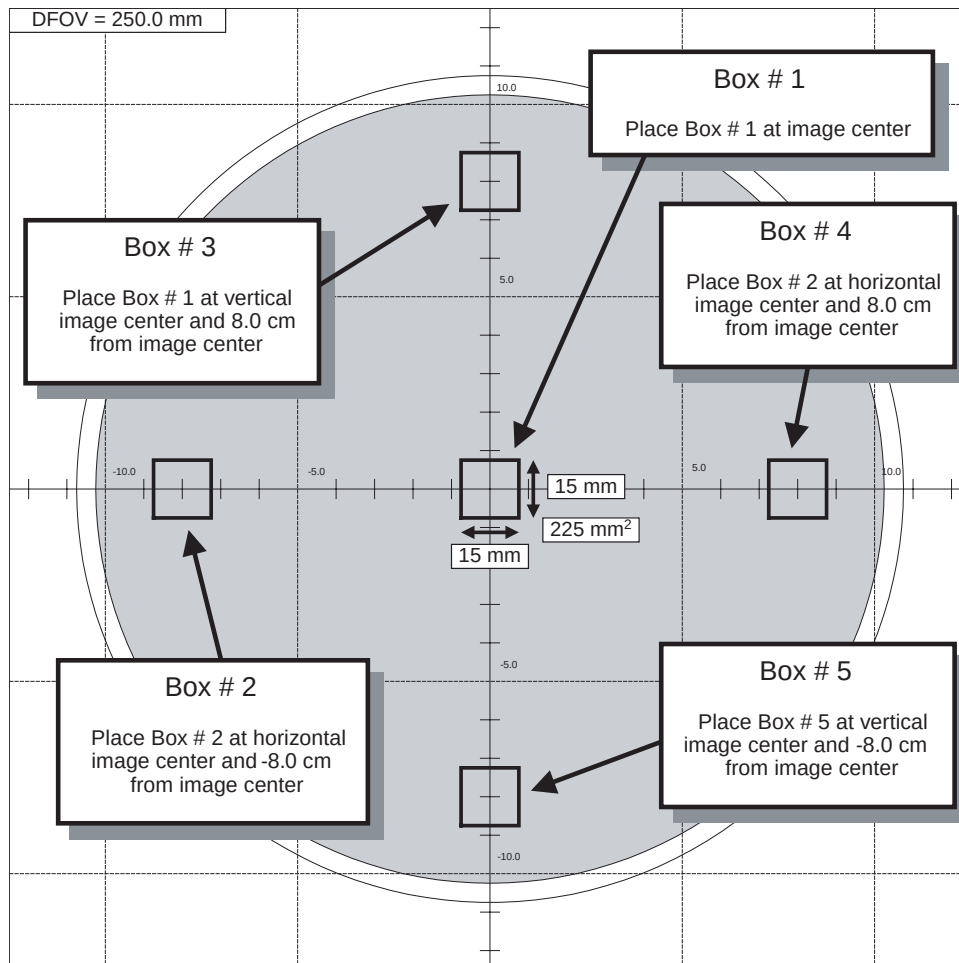


Figure 11-9 20cm QA#3 Phantom Brightness Uniformity & CT# Measurement - Building And Placing Reference ROI Boxes

Image	Box 1 (Center)	Box 2 (Left Center)		Box 3 (Top Center)		Box 4 (Right Center)		Box 5 (Bottom Center)		Outside Boxes Avg	
	Means (Xc)	Means	Std dev	Means	Std dev	Means	Std dev	Means	Std dev	Means (AvXo)	Std dev (AvSDo)
1			n/a		n/a		n/a		n/a		n/a
2			n/a		n/a		n/a		n/a		n/a
3			n/a		n/a		n/a		n/a		n/a
4			n/a		n/a		n/a		n/a		n/a
5			n/a		n/a		n/a		n/a		n/a
6			n/a		n/a		n/a		n/a		n/a
7			n/a		n/a		n/a		n/a		n/a
8			n/a		n/a		n/a		n/a		n/a

Table 11-13 20cm QA#3 Phantom CT# Brightness Uniformity & CT# Image Performance Worksheet

Box Size = 196 mm² to 256 mm²
 15 mm (+/- 1 mm) x 15 mm (+/- 1 mm)
 31 (+/- 2 pixels) x 31 (+/- 2 pixels)

Box Positions: Box 1 = 0 mm x 0 mm
 Box 2 = 0 mm x -80 mm
 Box 3 = 80 mm x 0 mm
 Box 4 = 0 mm x 80 mm
 Box 5 = -80 mm x 0 mm

Row	Images	Center Box Means (AvXc)	Outer Boxes Means Averages (AvXo)	Outer Boxes Means Averages minus Center Box Means (AvXo - AvXc)	Outer Boxes Standard Deviation Averages (AvSDo)	Comments
2A1A	1, 3, 5, 7				n/a	
1B2B	2, 4, 6, 8				n/a	
Specifications		0.0 ± 3.0	n/a	<± 3.0	n/a	

Table 11-14 48cm Phantom CT# Brightness Uniformity & Noise Row Performance Worksheet

- 4.) Calculate the Brightness Uniformity and CT# values for each image in the series, record values in [Table 11-14](#), and compare to specifications.
 - a.) Calculate and record the average means values (AvXo) for the four outside Boxes (Boxes 2 through 5) for each of the images and record in [Table 11-13](#).
 - b.) Calculate and record the average center box (Box 1) means values (AvXc) for each row (2A1A: Images 1, 3, 5, and 7; 1B2B: Images 2, 4, 6, 8) in [Table 11-14](#).
 - c.) Calculate and record the average outside boxes (Box 2 through 5) means values (AvXo) for each row (2A1A: Images 1, 3, 5, and 7; 1B2B: Images 2, 4, 6, 8) in [Table 11-14](#).
 - d.) Calculate the Brightness Uniformity (AvXo - AvXc) value for each row and record in [Table 11-14](#).
 - e.) Verify Brightness Uniformity (AvXo - AvXc) value and the average CT# value (AvXc) for each row meets specifications listed in [Table 11-14](#).
 - f.) Record the 20cm QA#3 Phantom Brightness Uniformity (AvXo - AvXc) value and average CT# value (AvXc) for each row in the HHS Record Tables.

5.5.4.2 Failure Recovery

Specifications

Each Row (2A1A, 1B2B) of the series must pass 20cm QA#3 Phantom Brightness Uniformity and average CT# specifications:

- AvXo - AvXc: < +/- 3.0
- AvXc: < +/- 3.0

Recommended Recovery

- A.) Perform DETAILED CAL.
- B.) Perform AUTO CT# ADJUST.
- C.) Repeat this procedure to verify 20cm QA#3 Phantom Image Performance.

5.5.5 QA#3 Phantom Noise

5.5.5.1 Performance Verification

- 1.) Select the fourth 20cm QA#3 Phantom exam acquired in the previous section.
 - a.) From the Global Control Palette, click on the IMAGE WORKS Desktop.
 - b.) From the Image Works Desktop, select the IMAGE WORKS BROWSER window.
 - c.) Select the exam and fourth (QA#3) series acquired in the previous section.
 - d.) Select the VIEWER button on the Image Works Browser window. Set up the viewer window for four-image viewing.
- 2.) Build a 51 x 51 pixel reference ROI Box using the Image Works Viewer tools.
 - a.) Click on the grid button to place a grid on the first image.
 - b.) Click on the MEASURE button and select the box ROI icon.
 - c.) Adjust the size of the ROI box to be a square 25 mm x 25 mm (625 mm²) box. Tolerance: 25 mm +/- 1 mm (576 mm² to 676 mm²).
If required, magnify the image to adjust to proper dimensions. See [Figure 11-10](#) for additional ROI size and placement information.
- 3.) Collect Standard Deviation values for the single reference ROI Box position on the eight 20cm QA#3 Phantom images.
If required, magnify the image to adjust to proper dimensions. See [Figure 11-10](#) for additional ROI placement information.
 - a.) Position the reference ROI Box built in step 2 directly over the center of the image using the grid cross-hairs as a guide.
 - b.) Type **prop a** in the Image Works Accelerator Line followed by <Return> to propagate Box # 1 ROI (size and position) on the remaining images in the series.
 - c.) Record the Standard Deviation values of the eight images in the series for centered box position in [Table 11-15](#).
- 4.) Calculate the average Noise values for each image in the series, record values in [Table 11-15](#), and compare to specifications.
 - a.) Calculate and record the average Noise values (AvSDc) for the inside Boxes for each of the two rows (2A1A and 1B2B) and record in [Table 11-15](#).
 - b.) Verify Noise (AvSDc) values for each row meets specifications listed in [Table 11-15](#).
 - c.) Record the 20cm QA#3 Phantom Noise (AvSDc) for each row in the HHS Record Tables.

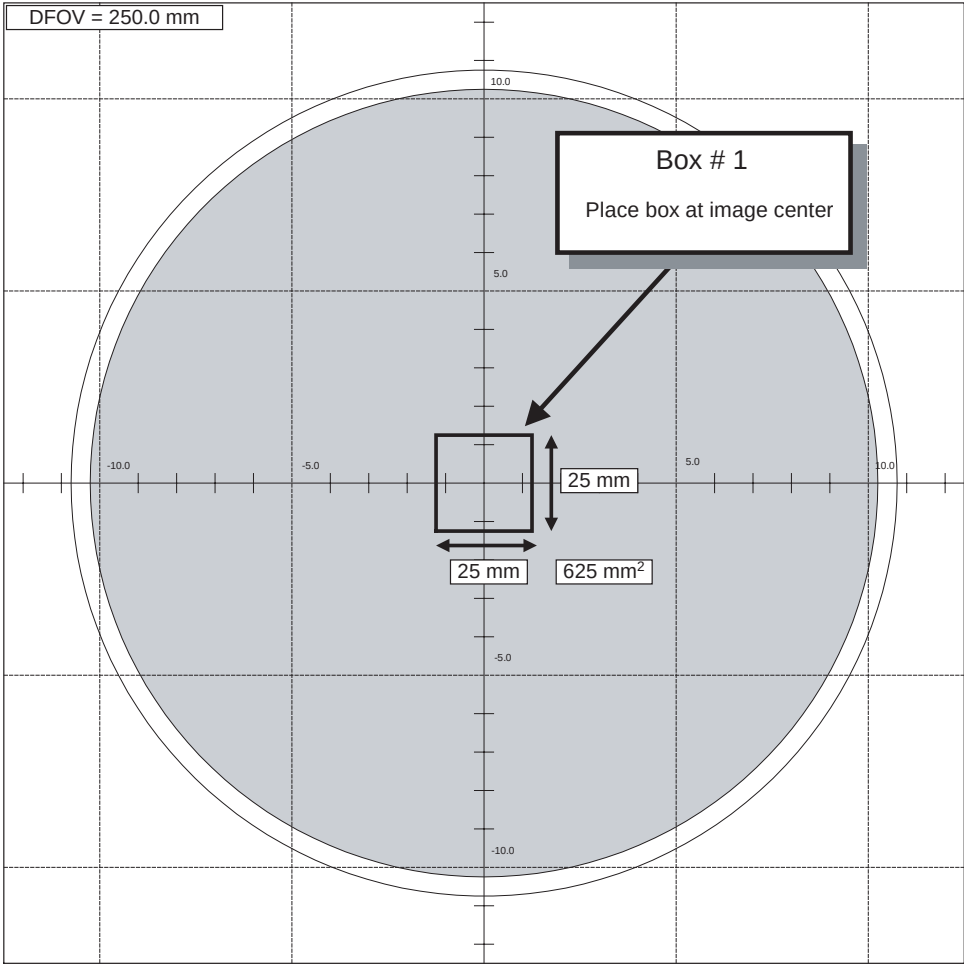


Figure 11-10 20cm QA#3 Noise Measurement - Building And Placing Reference ROI Box

Image	Row	Box 1 Std Dev (SDc)	Average Std Dev (AvSDc)	AvSDc Specifications		Comments
				Systems with less than 5000 scans	Systems with more than 5000 scans	
1	2A1A			3.2 ± 0.3	3.2 ± 0.4	
3						
5						
7						
2	1B2B					
4						
6						
8						

Table 11-15 20cm QA#3 Phantom Noise Performance Worksheet

Box Size = 576 mm2 to 676 mm2
25 mm (+/- 1 mm) x 25 mm (+/- 1 mm)
51 (+/- 2 pixels) x 51 (+/- 2 pixels)

Box Position: Box 1 = 0 mm x 0 mm

5.5.5.2 Failure Recovery

Specifications

Each Row (2A1A, 1B2B) of the series must pass 20cm QA#3 Phantom Noise specifications shown in [Table 11-15](#).

Recommended Recovery

- A.) Perform DETAILED CAL.
- B.) Perform AUTO CT# ADJUST.
- C.) Repeat this procedure to verify 20cm QA#3 Phantom Image Performance.

Section 6.0

Save System State

Use the following commands to create the System State MOD.

- 1.) Load a MOD into the mod drive on the front of the console.

- 2.) If you are not on the Service Desktop, click on the SERVICE DESKTOP icon, .

- 3.) Click on the PM icon. .

- 4.) Select SYSTEM STATE to open the System State Save/Restore menu.

- 5.) Select ALL

- 6.) Select SAVE

- 7.) When the save operation completes, select FILE and QUIT from the pull down menu.

- 8.) Remove the MOD from the drive

Chapter 12

Hospital/System Network Integration



NOTICE
Potential for
Data Loss and/
or Equipment
Damage

To prevent potential data loss and equipment damage, please do the following:

- Record data collected from procedures in this chapter into Form F4879 when directed, located in [Chapter 13](#) of this book.
- Only use the Installation manual that arrives with your system for installation. Any other revisions of this manual may not exactly match your system.

Section 1.0 Overview

HiSpeed QX/i systems support two basic Networking Protocols:

- Advantage NET (IC, Signa 4.X, CT-HLA, CT/I ...)
- DICOM (CT/I, CT Synergy, Advantage Workstations, ...)

DICOM NETWORK INTRODUCTION

DICOM networks basically operate on the **tasks** or services that various devices on the network use or provide. These services are labeled as Application Entity Titles (AE Titles). The HiSpeed QX/i system is a **user** of six DICOM Network Services and is a **provider** of two DICOM Services:

AS A DICOM SERVICE USER:

- Send or **Push** images to another network device.
- Send or **Push** images to a DICOM Printer.
- Review image database on another device and retrieve or **Pull** selected images from that device (Query/Retrieve User).
- Send or **Push** images to a an image storage device and obtain confirmation that the images have been archived (Storage Commitment).
- Obtain Patient Worklist Information from the Hospital HIS/RIS System.
- Store images on MOD media.

AS A DICOM SERVICE PROVIDER:

- Receive **Pushed** images from another network device
- Allow another network device to review the HiSpeed QX/i image database and to retrieve or **Pull** selected images (Query/Retrieve Provider)

For each DICOM Service that the HiSpeed QX/i system will be a **User** (except for storing images on MOD media), you must **declare** this device on the HiSpeed QX/i system using three menu selections. For some devices, you must declare not only the device, but each service (AE Title) that the device provides. For example, you may be required to declare a PACS System twice on the HiSpeed QX/i system: once as a destination to **push** images and, second, as destination that provides storage commitment capability after images have been **pushed**.

For each DICOM Service that the HiSpeed QX/i system will be a **Provider**, you must **declare** the HiSpeed QX/i system on the network device that will be using these services.

Information required to complete configuring a hospital DICOM network is provided by the hospital network administrator (hostnames, IP Addresses) and the DICOM Conformance Statement document (AE Titles, Port Numbers) provided with each DICOM compatible network device on the network.

Section 2.0 Before You Start

2.1 Network Physical Requirements

Before setting up the HiSpeed QX/i system on the hospital network, verify the following physical items are complete:

- HiSpeed QX/i console, monitor, keyboard, and mouse are installed and connected.
- HiSpeed QX/i system power is on
- Hospital Ethernet network RJ45 Class IV twisted pair cable is connected to the HiSpeed QX/i console network receptacle.
- Hospital network connection is operational and is running 10baseT or 100baseT.

2.2 HiSpeed QX/i Network Identity Information

To declare the HiSpeed QX/i system on the network, ensure the following network identity information is available from the Hospital Network Administrator:

- Hostname (No more than 16 Characters)
- Internet Protocol (IP) Address
- Subnet Net Mask IP Address (if applicable)
- Broadcast Address (if applicable)

2.3 HiSpeed QX/i to DICOM Remote Hosts Network Information

To declare DICOM remote hosts (PACS systems, archival devices, review workstations) on the HiSpeed QX/i system, ensure the following information is available for each remote host:

From the Hospital Network Administrator:

- Hostname
- Internet Protocol (IP) Address
- Network Protocol (DICOM for HiSpeed QX/i Systems)

From the Remote Host Device DICOM Conformance Statement Document:

- DICOM Application Entity Title or AE Title (DICOM service that remote host provides or uses)
- DICOM Listening Port Number

2.4 HiSpeed QX/i to DICOM HIS/RIS Interface Network Information

To declare DICOM Hospital HIS/RIS Interface devices (Mitra and others) on the HiSpeed QX/i System, ensure the following information is available:

From the Hospital Network Administrator: *Internet Protocol (IP) Address*

From the HIS/RIS Interface Device DICOM Conformance Statement Document:

- DICOM Application Entity Title or AE Title (DICOM Service that the HIS/RIS interface provides)
- DICOM Listening Port Number

2.5 HiSpeed QX/i to DICOM Printers Network Information

To declare DICOM on the HiSpeed QX/i System, ensure the following information is available for each printer:

From the Hospital Network Administrator:

- Hostname
- Internet Protocol (IP) Address

From the Printer DICOM Conformance Statement Document:

- DICOM Application Entity Title or AE Title (DICOM service that remote host provides or uses)
- DICOM Listening Port Number

Section 3.0

Declaring the System on the Hospital Network

3.1 Enter Configuration Routine

- 1.) On the OC, open a Unix shell window.
 - 2.) Enter root as a superuser:
Type: `su -` **ENTER** at the prompt.
Type: `#bigguy` **ENTER** at the password prompt.
 - 3.) Change directory to scripts:
Type: `cd /user/g/scripts` **ENTER** at the root prompt.
 - 4.) Launch the Install Utility:
Type: `reconfig` **ENTER** at the prompt.
- The OC displays the Install Utility Window as shown in [Figure 12-1](#).



Figure 12-1 Install Utility Window

- 5.) Enter the Configuration Routine:
Using the mouse, click on the CONFIG button.
The OC displays the System Configuration - System Settings screen, as shown in [Figure 12-2](#).

Figure 12-2 System Settings Screen

3.2 Configure Network Settings

This screen provides the ability to declare the system on a hospital network. Key information such as Host Name, IP Address, Net Mask (for systems on a subnet) must be obtained from the hospital network administrator.

- 1.) Select the NETWORK button to display the Network Settings screen as shown in [Figure 12-3](#).
- 2.) Enter the Suite Name.

The Suite Name is a means of identifying this particular HiSpeed QX/i system as a part of a group of CT Systems in a suite configuration. This Suite Name will appear on all image headers.

The Suite Name must start with a letter, followed by three alphanumeric characters (total MUST be four characters long). The name of the OC interface will be <Suite Name>_oc and the SBC interface will be <Suite Name>_sbc.

Figure 12-3 Networks Settings Screen

- 3.) Enter the hospital provided Host Name.
The Host Name identifies the network hostname and AE Title of the HiSpeed QX/i system.
The Host Name:
 - **MUST NOT** be <Suite Name>_oc or <SUITE NAME>_OC.
 - **MUST NOT** exceed 16 Characters.
 - **MUST** only contain the following characters: A through Z, a through z, 0 through 9, - and _
- 4.) Enter the hospital provided IP Address.
- 5.) Enter the hospital provided Net Mask (if the system is on a subnet).
- 6.) Enter the Broadcast Address:
The Broadcast Address should be the same as the IP Address except for the bits of the host id portion (last digit group) set to 1s or 0s depending on the configuration of the network. The standard default is 1s but older SunOS machines used 0s.
For example:
If the IP Address is 192.100.9.17, the Broadcast Address should be 192.100.9.255 if the network is configured to use 1's to specify the broadcast address.
If the network contains genesis based scanners or other SunOS 3.5 or 4.1 computers, the Broadcast Address should be 192.100.9.0.
- 7.) Enter the hospital provided Default Gateway IP Address in the Default Gateway field (if applicable). If the site network does not use a default gateway, leave the field blank.
- 8.) Select NIS (Yellow Pages database) Advanced Option only if requested by the hospital network administrator as follows:
 - a.) Select ADVANCED OPTIONS button on the Network Settings screen.
 - b.) Select Use NIS? button.

- c.) Enter the hospital provided Domain Name.
- 9.) Record all the Network parameters in the *HiSpeed QX/i Software Installation Procedures* Document, or on the worksheet in Chapter 6.

3.3 Initiate System Reconfiguration

- 1.) Select ACCEPT on the System Configuration Screen.
The system loads the application software, OS patches, and kernal changes, and configures the system on both the OC and the SBC.
This loading process takes approximately 15 minutes. While the load is going on, the results are displayed in a shell window, which closes when the loading process is complete. All the window output is logged to a file named:
`/var/adm/install.log.YYYYMMDDWWHHMMSS.`
(Where `YYYMMDDWWHHMMSS` is the Date/Time that the loading process was started.)
- 2.) When the loading process and configuration changes are complete, the system displays a prompt to reboot. Click on YES.
- 3.) The system will automatically login as ctuser after the reboot. Select OK on the Autostart Disabled popup message.
- 4.) To startup Applications, in the console shell window, type **startup** ENTER.

Section 4.0

Declaring Remote Hosts on the System

4.1 Enter Remote Host Configuration Screen

- 1.) On the OC, select the IMAGE WORKS icon.
- 2.) Select NETWORK.



4.2 Declaring Advantage NET Remote Hosts on the HiSpeed QX/i

Use Advantage NET Protocol networks to communicate with older CT or MR Systems (MR Signa 4.x, CT-HLA, CT/I Systems, and Workstations that support the Advantage NET protocol). Advantage NET Protocol does not offer full compatibility with HiSpeed QX/i DICOM formats.

Repeat the following procedure for each Advantage NET Remote Host device that the customer expects to have communicating with this system.

- 1.) Select REMOTE HOSTS from the pull down menu. The system displays the Remote Host Parameter Screen as shown in [Figure 12-4](#).

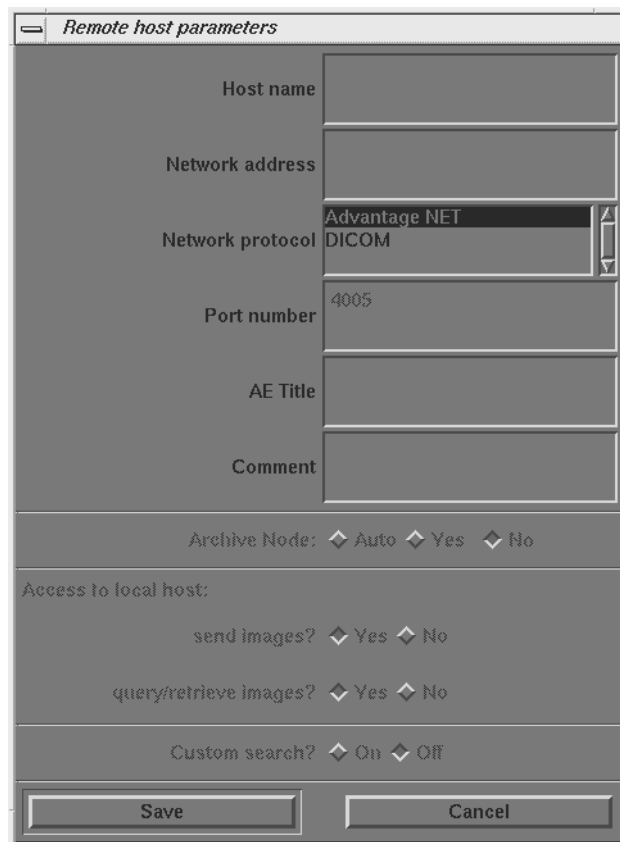


Figure 12-4 Advantage Net Network Protocol Parameter Settings

- 2.) Enter the hospital provided Host name.
- 3.) Enter the hospital provided Network Address (IP Address).

- 4.) Select ADVANTAGE NET as the Network Protocol.
The systems automatically de-highlights the remaining parameter fields on the Remote Host parameter selection screen. These are dedicated DICOM protocol parameters and do not apply to Advantage NET type devices.
- 5.) Select SAVE to store the parameter settings of the remote host.

4.3 Declaring DICOM Remote Hosts on the HiSpeed QX/i

Use DICOM protocol networks to communicate to DICOM devices such as CT/i, CT Synergy, DLX, MR Lx, and third party hosts.

Repeat the following procedure for each DICOM remote host device that the customer expects to have communicating with this system.

- 1.) Select REMOTE HOSTS from the pull down menu. The system displays the Remote Host Parameter screen as shown in [Figure 12-5](#).

Figure 12-5 DICOM Network Setting Protocol Parameter Settings

- 2.) Enter the hospital provided Host name.
- 3.) Enter the hospital provided Network Address (IP Address).
- 4.) Select DICOM as the Network Protocol.
The system automatically highlights the remaining parameter fields on the Remote Host parameter selection screen. These are dedicated DICOM Protocol parameters.
- 5.) Enter the TCP/IP Listening Port Number from the DICOM Conformance Statement provided with the device.

- 6.) Enter the AE Title from the DICOM Conformance Statement provided with the device.
Application Entity Titles (also known as ACR-Nema or Dicom Name) refer to the DICOM Network Services that a device provides to the HiSpeed QX/i System. For most devices, the AE Title is the same as the hostname (HiSpeed QX/i systems are equipped with this feature). However, some devices such as PACS systems may have separate AE Titles and port numbers for each of the services that the PACS system provides. In these cases, you must enter a separate remote host (same hostname and IP Address) for each of the independent AE Title Services that the host provides (one host as an image **push-to** destination, another host as a **query/retrieve** provider, and another host as a **storage/commitment** provider).
Be sure to review the DICOM Conformance Statement for each device that will provide a remote host network service for the HiSpeed QX/i system (image **push-to** or store destination, Query/Retrieve, and Storage Commitment) to ensure that each service is correctly configured.
- 7.) Select the correct Archive Node choice for the device. The Archive Node selection field defines the ability of the remote host to act as a DICOM Storage/Commitment provider and indicate to the operator that a study/series/image was archived. Select:
 - AUTO to have the HiSpeed QX/i system automatically check to see if the designated remote host is a DICOM Storage/Commitment Provider.
 - YES if the device is the hospital designated DICOM Storage/Commitment Provider. During an Application Study Archive process, the local browser screen will indicate Archive Status = Y to the operator.
 - NO if the device is not a DICOM Storage/Commitment Provider.
- 8.) Select the correct Access to local host: settings. These two selections allow you to selectively block the remote host from using the HiSpeed QX/i DICOM services as a provider (image **push-to** destination, and a **Query/Retrieve** provider).
 - Send Images? Set to YES if the customer wants the HiSpeed QX/i system to be able to have images **pushed** to the system from the applicable remote host. Set to NO if the customer wants to block an image **push** from the applicable remote host.
 - Query/retrieve images? Set to YES if the customer wants the remote host to be able to review the HiSpeed QX/i image database (query) and **pull** selected images from the database. Set to NO if the customer does not want the remote host to have this ability.
- 9.) Select the correct Custom search? setting. This selection allows the HiSpeed QX/i to selectively search through the remote host's image database when the operator is using remote browser screen to **query** the remote host. The search parameters that the HiSpeed QX/i system allows the customer to use are: last name contains, patient ID, exam number, accession number, and exam date.
 - Select ON if the device supports custom searches as part of the devices **Query/Retrieve** DICOM Provider service.
 - Select OFF if the device does not support custom searches.
- 10.) Record all the remote host network parameters for each remote host in the *HiSpeed QX/i Software Installation Procedures* Document.
- 11.) Select SAVE to store the parameter settings of the remote host.

Section 5.0

Declaring the System on Remote Hosts

5.1 Declaring HiSpeed QX/i on Advantage NET Protocol Devices/Systems

Refer to the appropriate service manual provided with the Advantage NET Protocol device or system to find instructions how to declare the System as an Advantage NET remote host.

5.2 Declaring HiSpeed QX/i on DICOM Protocol Devices/Systems

Refer to the appropriate Service Manual provided with the DICOM protocol device or system to find instructions how to declare the System as a DICOM remote host.

The HiSpeed QX/i System provides two DICOM Services as a provider to remote hosts:

- A remote host can **push** images to the HiSpeed QX/i image database.
- A remote host can review the HiSpeed QX/i image database (query) and **pull** selected images (retrieve).

Use the following HiSpeed QX/i parameter information to configure the DICOM device/system to either **push** images to the HiSpeed QX/i and/or perform a **Query/Retrieve** operation:

- Hostname: Provided by the Hospital Network Administrator. Exactly the same HiSpeed QX/i assigned hostname entered in the Network Configuration Screen.
- Application Entity Title: Exactly the same entry as the Hostname.
- Network Address: Provided by the Hospital Network Administrator. Exactly the same HiSpeed QX/i assigned IP Address entered in the Network Configuration Screen.
- Network Protocol: DICOM 3.0.
- Port Number: For all DICOM service that the HiSpeed QX/i System provides, use 4006.
- Provider Type: This field concerns the HiSpeed QX/i DICOM **Query/Retrieve** provider capability. All HiSpeed QX/i systems are study root systems, which allow queries at the exam, series, and image level.
- Support Worklist: This field concerns whether a DICOM **Query/Retrieve** provider capable device or system supports a filter search of the image database. All HiSpeed QX/i systems support a filtered search of the image database as part of the HiSpeed QX/i DICOM **Query/Retrieve** provider capability.

Section 6.0

Declaring DICOM HIS/RIS Interfaces on the System

6.1 Prerequisites

Most hospital HIS/RIS systems are not DICOM compatible and require a DICOM HIS/RIS Worklist Interface to provide patient scheduling information to the HiSpeed QX/i system. Contact your local HNS support engineer to determine exactly what DICOM HIS/RIS Interface is appropriate for the customer.

In addition, the HiSpeed QX/i system must have the ConnectPRO software option installed to utilize the DICOM Protocol Worklist capability.

6.2 Loading ConnectPRO Software Option on the System

- 1.) If you are not on the Service Desktop, click on the SERVICE DESKTOP icon.



- 2.) Click on the UTILITIES icon.



- 3.) Click on INSTALL.

- 4.) Select INSTALL OPTIONS and click START. The console displays the Software Options window as shown in Figure 12-6.

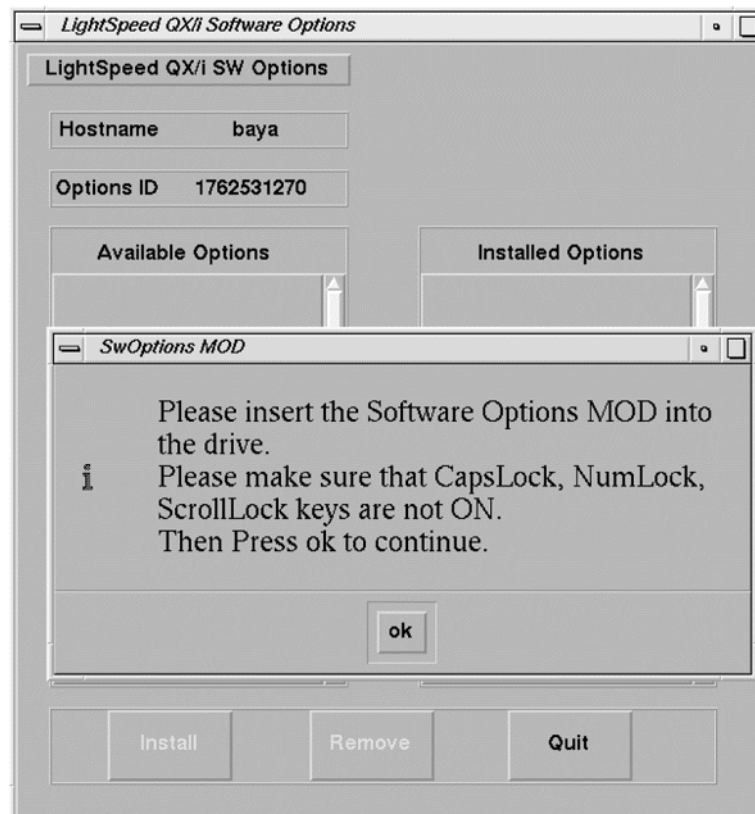


Figure 12-6 Options Window when First Selected

- 5.) Insert the ConnectPRO Options MOD into the MOD drive.
- 6.) Click on OK to continue.
- 7.) On the Software Option window, select the CONNECTPRO option in the Available Options window.
- 8.) Click on the INSTALL button.

The console may display a message box during the software loading operation. When the system has completed loading the software:

- the Installed Options window displays the ConnectPRO option, and
- the console displays the ConnectPro Setup window as shown in [Figure 12-7](#).

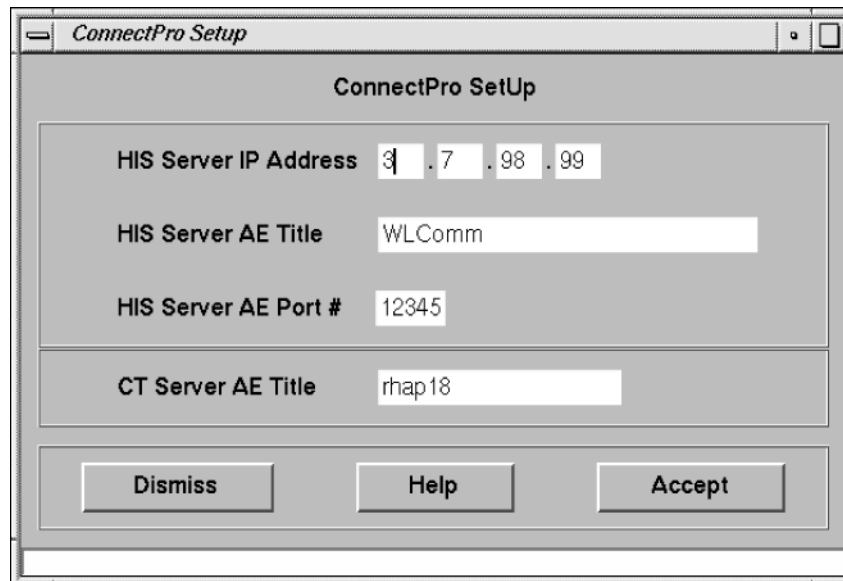


Figure 12-7 ConnectPro Setup Screen

- 9.) Enter the hospital provided HIS Server IP Address.
- 10.) Enter the HIS Server AE Title from the DICOM HIS/RIS Interface device DICOM Conformance Statement document.
- 11.) Enter the HIS Server AE Port # from the DICOM HIS/RIS Interface device DICOM Conformance Statement document.
- 12.) Record all the ConnectPRO HIS/RIS Network parameters in the *HiSpeed QX/i Software Installation Procedures* Document.
The system will automatically load the HiSpeed QX/i CT Server AE Title (Hostname of the system).
- 13.) Click on the ACCEPT button. The system will check valid setup entries, execute a perl script that makes changes to the configuration files for HIS/RIS, and return to the Software Option window shown in [Figure 12-6](#).
- 14.) Click on the QUIT button and the subsequent OK button in the message window.
- 15.) Remove the MOD from the MOD drive, and write protect the side containing the ConnectPRO option.

Section 7.0

Declaring DICOM Filming Devices on the System

7.1 Prerequisites

Before configuring DICOM filming devices (cameras, printers) on the HiSpeed QX/i System, ensure the following are complete:

- Filming Device Service Representative to assist in camera/printer setup for best image quality film presentation.
- Hospital DICOM network is operational.
- Filming device is connected to the DICOM network with the correct filmer DICOM interface.
- Filming device is DICOM protocol compatible.
- Filming device DICOM Conformance Statement document is available.

Note: **Filmer DICOM Application Entity Titles may be site specific. Make sure that you check with the Filming Device Service Representative and the hospital network administrator to ensure you are using the correct AE Title for the destination filming device.**

7.2 Declaring DICOM Filming Devices on the System



NOTICE Potential For Data Loss

Empty all filming queues before modifying camera parameters.

This section contains procedures for recording important Camera setup information. Use the table(s) at the end of this section to record information from the setup screens.

- 1.) Click on the SERVICE DESKTOP button.
- 2.) On the Desktop Toolbar select UTILITIES -> INSTALL -> INSTALL CAMERA. The Install Camera window appears, along with a warning message pop-up box, to remind you that all filming queues must be empty before you begin to update or delete a camera.
- 3.) The Graphical User Interface displayed shows a list of cameras installed (See Figure 12-9). First, you must click OK in the warning message box. See Figure 12-8.

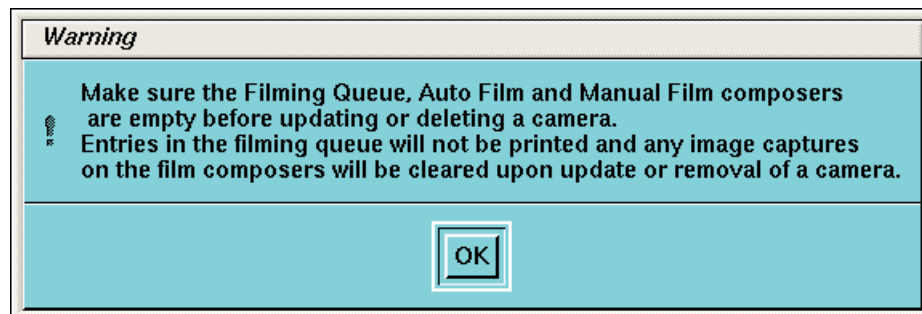


Figure 12-8 Warning Screen

- 4.) Now you are asked a series of questions.
 - a.) To add a new camera, click the ADD button (See Figure 12-9).
 - b.) A dialog window for the camera type (DASM/DICOM) appears. If no DASM is detected during the OC boot, the DASM button will be disabled (Figure 12-10). If a DASM is present

and has not been detected, reboot the OC and run the camera configuration tool again.

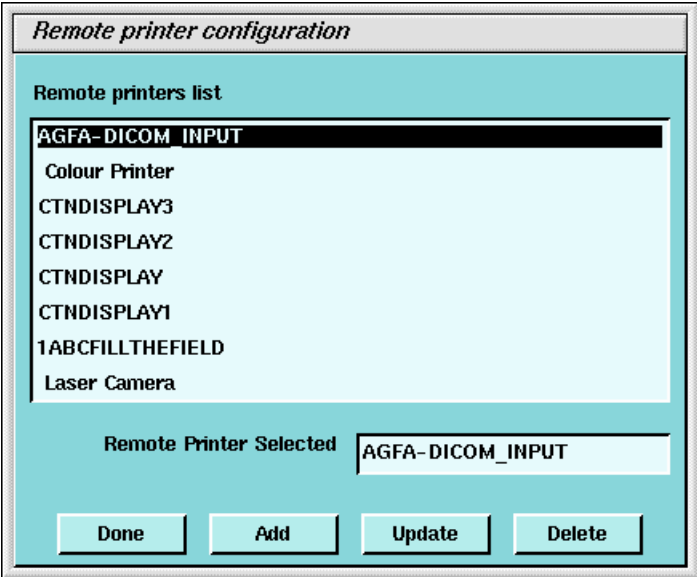


Figure 12-9 Printer Configuration GUI

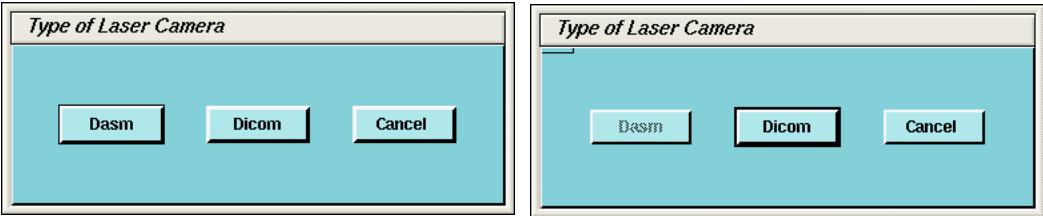


Figure 12-10 Dialog Box for camera Type (with and without DASM)

- 5.) To add a new laser camera, click DASM in the camera type dialog box. This brings up a list of available camera models. Select the appropriate model form the list and click SELECT (See [Figure 12-11](#)). Now you must configure it.

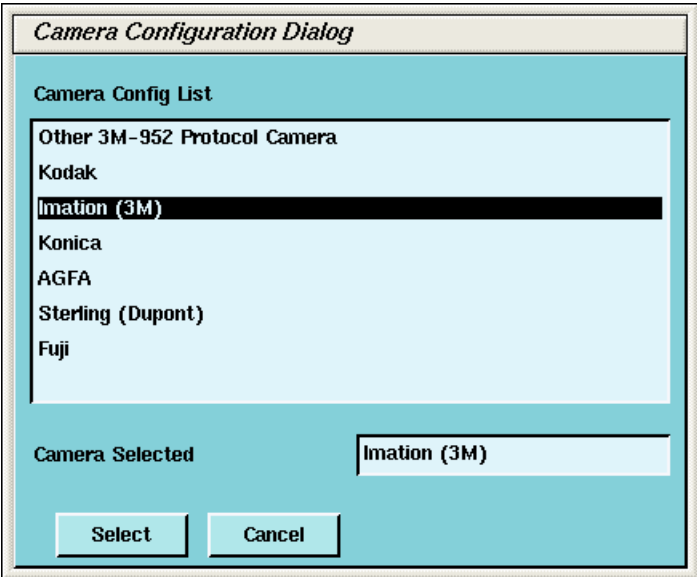


Figure 12-11 Camera Model Dialog (with DASM)

- a.) DASM Interface is automatically detected as either Analogue or Digital
- b.) Two Laser Options are available for laser cameras: SLIDES and ZOOM. Set this option only if the camera being installed supports slides and zoom. Setting the option allows it to be enabled or disabled at the application level.
- c.) Camera manufactures provide two (2) Magnification Type options for cameras. The SMOOTH resolution blurs the image, while the SHARP resolution makes the image pixels more pronounced. The default is smooth.

Comment:

To film good images, and have them look like images filmed by other GE CT products, the following camera settings are suggested:

Kodak: SMOOTH
 Dupont/Sterling: SMOOTH
 3M/Imation (Laser Camera): SHARP
 3M/Imation (Dry View): SMOOTH
 Agfa: SMOOTH

- d.) Select the appropriate File Format. Select ON from the drop down list boxes on the menu. Valid film formats are determined by the camera manufacture. IMATION for example, doesn't support 4x4, 2x4 or 1x2 and AGFA does not support 2x4) The DICOM print convention designates film formats by column and row (e.g. 12 on 1 film is 3x4). When finished setting parameters, click on DONE and proceed to step 8.

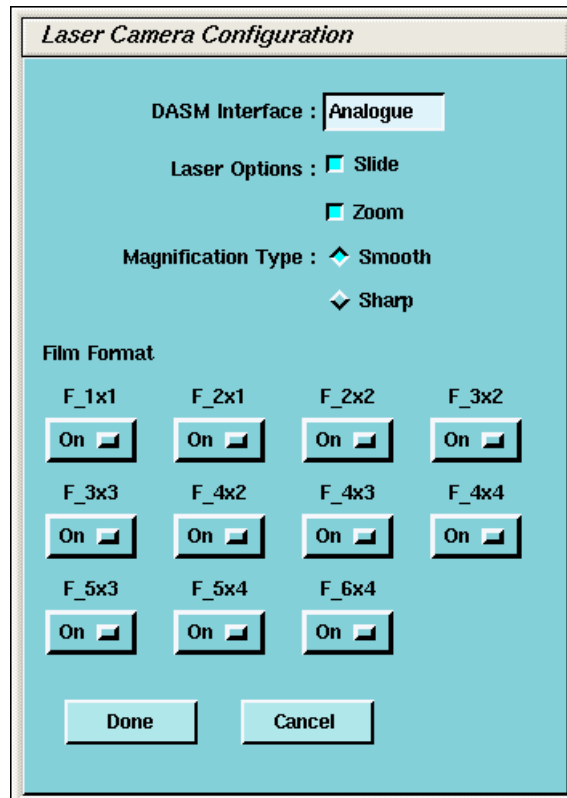


Figure 12-12 Laser Camera Configuration

- 6.) To add a new DICOM camera, click on ADD and then DICOM in the dialog box that appears.
 - a.) A list of camera models appears (See Figure 12-13). Select the appropriate model from the list and click SELECT. Clicking SELECT presets all the parameters to that models

except the Network parameters.

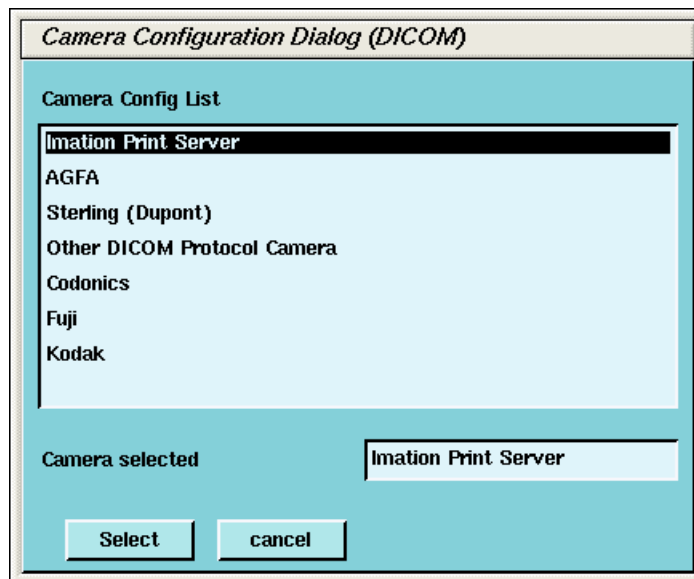


Figure 12-13 Camera Model Dialog (DICOM)

*Comment:
It's advised to
recheck the
preset
information with
the camera
manufacturer's
representative.*

Selection of a different camera model clears the Image Quality parameters, because these are camera manufacture dependent.

b.) Enter the Network Parameters (See Figure 12-14)

- > Device Name A unique name used to identify the camera.
- > Host Name DICOM print server host name, as defined by the hospital.
- > IP Address DICOM print server IP address, as defined by the hospital.
- > AE Title DICOM print server application entity title, as defined by the print server. *You should consult the manufacturers DICOM Conformance Statement.*

Note: The Application Entity Title for the Camera may be site specific. Make sure that you check with the Camera Manufacturer's Representative and the hospital network administrator to ensure you are using the correct AE Title for the destination DICOM Print Camera.

- > TCP/IP Listen Port DICOM print server TCP/IP listen port, as defined by the server. *You should consult the manufacturers DICOM Conformance Statement.*
- > Comments Optional comments used by the DICOM print server.

Figure 12-14 DICOM Camera Configuration

- c.) Medium Type specifies the type of film being used. Currently, only BLUE FILM and CLEAR FILM are supported.
- d.) Set Destination to the final location for film output: either MAGAZINE or PROCESSOR.
- e.) Orientation selects film orientation. Only PORTRAIT is currently supported.
- f.) Set the Magnification Type. This parameter selects the algorithm used to interpolate pixels for proper film resolution. Set this parameter after consulting the camera manufacture to ensure optimal image quality. Choices are describe below:
 - > None No interpolation. This option is not supported by all cameras.
 - > Replicate Adjacent pixels are interpolated. This can result in "pixelized" images. *This algorithm is not normal preferred.*
 - > Bilinear A 1st order interpolation of pixels. Results in images usually described as blurred. *This algorithm is not normal preferred.*
 - > Cubic A 3rd order interpolation. Used with a large number of possible formulations. Camera manufactures define parameters called "smoothing type" to set coefficients used in this algorithm. The implementation of these "smoothing soefficients" is camera dependent.
- g.) Select the Standard Film Formats. Select the film format by choosing ON in the pull-down menu box located below each selection. See Figure 12-14. Valid film formats are set by the camera manufacture. IMATION for example, doesn't support 4x4, 2x4 or 1x2 and

Comment: For most Camera Manufacturers, the preferred selection is CUBIC.

AGFA does not support 2x4) The DICOM print convention designates film formats by column and row (e.g. 12 on 1 film is 3x4).

- h.) After the camera has been configured, click the ADVANCED button. This creates the camera device file for you automatically and pops up the Advanced Parameters screen. See Figure 12-15.

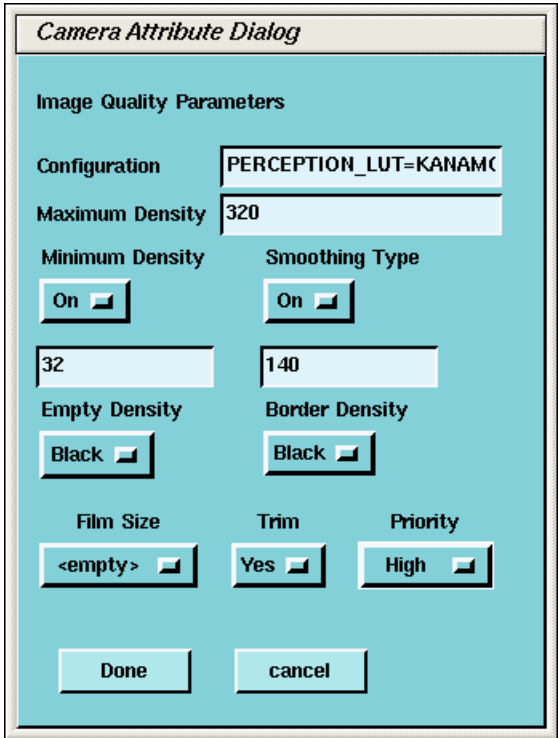


Figure 12-15 Advanced Parameters (Camera Attribute Dialog)

- 7.) Advanced camera parameters control the image quality of films.

Note: For more information on the proper settings for these parameters, please see the Camera’s DICOM Print Device Conformance Statement or the Camera Manufacturer Representative. You may need to refer to a copy of the Conformance Statement as you are working with the Camera Manufacturer’s Representative, to correctly set up the DICOM Print Camera I/Q and Time-out Settings.

- a.) Configuration - This parameter is camera manufacturer dependent as is typically used to specify the image contrast. The Configuration field may be up to 1024 characters long. The field will scroll automatically as text is entered. To review your entry, simply click and hold the middle mouse button, while the cursor is in the field, and drag the mouse towards the right (or left) as needed.

Note: **Recommended Configuration Setting Values:**

Agfa Drystar (MG3000)	PERCEPTION_LUT=KANAMORI (100)
Imation Dryview (8700)	LUT=0,7
Kodak Laser Printer 190	CS434\CN0\PD1.20

- b.) Smoothing Type - Set Smoothing Type to ON, and input the selected value. This parameter is used when the magnification type is CUBIC. It represents the coefficient for the image resolution algorithm. This parameter is camera manufacturer dependent, and should be re-verified with your radiology department.

Note: **Recommended Smoothing Type Starting Values and Ranges:**

Agfa DryStar (MG3000)	Start Value: 140	Range: 137-150
Imation Dryview (8700)	Start Value: 3	Range: 3-13
Kodak Laser Printer 190	Start Value: Enhanced	Range: Normal

- c.) Minimum and Maximum Density - Used to set brightness of the images on film. The range of values is 0-4095, although the valid range for a specific camera is manufacture dependent. For Maximum Density, input the correct value into the text box. For Minimum Density, set it to ON and input the correct value in the text box.

Note:

Recommended Minimum and Maximum Density Starting Values:

Agfa Drystar (MG3000)	Min.: 20 or 23	Max: 300
Imation Dryview (8700)	Min.: (Blank)	Max: 300
Kodak Laser Printer 190	Min.: 20	Max: 300

- d.) Empty Image Density - This parameter sets the density for empty film viewports. Typically, BLACK is used but WHITE is an available option. The minimum and maximum density values are used as the representation.
- e.) Border Density - This sets the density for the border used around the film viewports. Typically, BLACK is used but WHITE is an available option. The minimum and maximum density values are used as the representation.
- f.) Film Size - Allows the system to specify a particular film size, if selected.
- g.) Trim - YES produces a white (clear) box surrounding each image.
- h.) Priority - This sets the print priority.
- i.) If you have completed entry of advanced parameters, click DONE.
- 8.) **Camera Data Tables:** To locate Install Camera Information: Click on the SERVICE DESKTOP button. On the Desktop Toolbar select UTILITIES -> INSTALL -> INSTALL CAMERA. The Install Camera window appears. Select each of the cameras that are installed from the list of installed cameras, and click on UPDATE to view the camera's settings. Record the values used to set up each camera in the tables that follow. Extra tables are provided for multiple cameras

Note: You can determine this information by looking at the contents of the following files:

- For a DASM Camera: `/usr/g/ctuser/app-defaults/devices/camera.dev`
- For a DICOM Print Camera: `/usr/g/ctuser/app-defaults/devices/name.cfg`
where, *name.cfg* is the camera device name from the printer configuration GUI.

Example: `more <filename from above> ENTER`

DASM CAMERA #1

GUI SETTING	SELECTIONS	VALUE
Camera Type	Model Type of Camera	
DASM Type	Digital or Analog	☐ Digital ☐ Analog
Options	Slides or Zoom	☐ Slides ☐ Analog
Film	Smooth or Sharp	☐ Smooth ☐ Sharp
Film Format Available	1x1, 2x1, 2x2, 3x2, etc.	
Film Format Default	1x1, 2x1, 2x2, 3x2, etc.	

Table 12-1 DASM Camera #1

DICOM CAMERA #1

GUI SETTING	SELECTIONS	VALUES
DICOM Camera Type	Model Type of Camera	
Film Format Available	1x1, 2x1, 2x2, 3x2, etc.	
Network Parameters	Host Name	
	IP Address	
	AE Title	
	TCP Listen Port	
	Comments	
	Destination	☐ Magazine ☐ Processor
Special Set Up	Orientation	☐ Portrait ☐ Landscape
	Medium Type	☐ Blue ☐ Clear
	Magnification Type	☐ None ☐ Replicate ☐ Bilinear ☐ Cubic
*Advanced Parameters - IQ	Smoothing Type	☐ ON ☐ OFF Value:
	Configuration	
	Minimum Density	☐ ON ☐ OFF Value:
	Maximum Density	
	Empty Density (Black/White)	☐ Black ☐ White
	Border Density (Black/White)	☐ Black ☐ White
	TRIM	☐ YES ☐ NO
	Priority	☐ HI ☐ MED ☐ LOW
	Film Size	

NOTES

*To view Advanced DICOM Camera Settings, you must click on ADVANCED.

Table 12-2 DICOM Camera #1

DICOM CAMERA #2

GUI SETTING	SELECTIONS	VALUES
DICOM Camera Type	Model Type of Camera	
Film Format Available	1x1, 2x1, 2x2, 3x2, etc.	
Network Parameters	Host Name	
	IP Address	
	AE Title	
	TCP Listen Port	
	Comments	
	Destination	☐ Magazine ☐ Processor
Special Set Up	Orientation	☐ Portrait ☐ Landscape
	Medium Type	☐ Blue ☐ Clear
	Magnification Type	☐ None ☐ Replicate ☐ Bilinear ☐ Cubic
*Advanced Parameters - IQ	Smoothing Type	☐ ON ☐ OFF Value:
	Configuration	
	Minimum Density	☐ ON ☐ OFF Value:
	Maximum Density	
	Empty Density (Black/White)	☐ Black ☐ White
	Border Density (Black/White)	☐ Black ☐ White
	TRIM	☐ YES ☐ NO
	Priority	☐ HI ☐ MED ☐ LOW
	Film Size	

NOTES

*To view Advanced DICOM Camera Settings, you must click on ADVANCED.

Table 12-3 DICOM Camera #2

DICOM CAMERA #3

GUI SETTING	SELECTIONS	VALUES
DICOM Camera Type	Model Type of Camera	
Film Format Available	1x1, 2x1, 2x2, 3x2, etc.	
Network Parameters	Host Name	
	IP Address	
	AE Title	
	TCP Listen Port	
	Comments	
	Destination	☐ Magazine ☐ Processor
Special Set Up	Orientation	☐ Portrait ☐ Landscape
	Medium Type	☐ Blue ☐ Clear
	Magnification Type	☐ None ☐ Replicate ☐ Bilinear ☐ Cubic
*Advanced Parameters - IQ	Smoothing Type	☐ ON ☐ OFF Value:
	Configuration	
	Minimum Density	☐ ON ☐ OFF Value:
	Maximum Density	
	Empty Density (Black/White)	☐ Black ☐ White
	Border Density (Black/White)	☐ Black ☐ White
	TRIM	☐ YES ☐ NO
	Priority	☐ HI ☐ MED ☐ LOW
	Film Size	

NOTES

*To view Advanced DICOM Camera Settings, you must click on ADVANCED.

Table 12-4 DICOM Camera #3

7.3 Troubleshooting Tips

The following is a summary of troubleshooting information for DICOM print that was gathered during software testing and validation of the DICOM print feature.

There is also a significant amount of additional troubleshooting procedures, and the theory of the DICOM print feature in the System Service Manual. Should you have problems installing a DICOM print camera, first read the information in System Service Manual.

ERROR BRINGING UP THE CAMERA INSTALLATION/FILMING APPLICATION

- **Symptom:** After creating/modifying the DICOM print camera the install camera interface does not come up and the filming application indicates it cannot bring up the film composer.
Cause: The configuration field contains invalid information such as a backslash (\) as the final character in the field or brackets ({}).
- **Solution:** The camera.dev file in ~ctuser/app-defaults/devices must be manually edited to remove the offending characters in the set configuration line. Invalid characters include \{ }

CONFIGURATION INFORMATION FIELD

- **Symptom:** Cannot view the entire configuration field (> 25 characters)
- **Solution:** Hold down the middle mouse button and move the field contents

NEED TO SET DICOM PRINT ATTRIBUTES NOT SUPPORTED BY SOFTWARE

- **Symptom:** User wants the white border around each image box ON/OFF permanently for this system and it cannot be set as the default for the camera.
- **Solution:** Using your favorite editor, add the following line to the camera.dev file located in ~ctuser/app-defaults/devices after the DICOM print device has been otherwise configured.
For Trim Off - set TRIM NO
For Trim On - set TRIM YES
- **Symptom:** DICOM print camera supports multiple film sizes and the user only wants to print if the film size is correct for HiSpeed QX/i (14x17). [Otherwise the camera will queue the films or return an error causing the queue to pause (based upon the DICOM print camera specifications).]
- **Solution:** Using your favorite editor, add the following line to the camera.dev file located in ~ctuser/app-defaults/devices after the DICOM print device has been otherwise configured.
To force a 14x17 film size - set filmSize 14INX17IN

NEED TO PREVENT DICOM PRINT ATTRIBUTES FROM BEING SENT TO DICOM PRINT CAMERA

- **Symptom:** Some DICOM print attributes are optional, and may result in fatal errors. For example, the Fuji camera does not support the Empty Image Density parameter for the film box.
- **Solution:** Using your favorite editor, add the following line(s) to the camera.dev file located in ~ctuser/app-defaults/devices after the DICOM print device has been otherwise configured.
 - To prevent sending the Smoothing Parameter set FB_Smooth FALSE
 - To prevent sending the Border Density set FB_Border FALSE
 - To prevent sending the Empty Image Density set FB_EID FALSE
 - To prevent sending the Minimum Density set FB_MinD FALSE

- To prevent sending the Trim Parameter set FB_Trim FALSE

ERROR TRYING TO CONNECT TO THE DICOM PRINT CAMERA

- **Symptom:** DICOM print server can be reached (ping), but Application error indicates "Unable to start filming interface" and the help message talks about running the install.dasm (Association Error)
- **Solution:** The system is unable to complete the association. Check the AE Title and the Port number of the DICOM print server and correct them through the Install Camera procedure.

FILM COMPOSER ERROR NOT USABLE

- **Symptom:** Film composer error says "unrecognized status - code 0"
- **Solution:** Review the log file, the attention and status windows. These areas have the correct filming status (for example, **film jam** and **supply empty**).

DEBUGGING CONNECTION ISSUES DIFFICULT

- **Symptom:** The timeouts for the DICOM print are very long, which means one needs to wait a long time before you know the application is not working.
- **Solution:** The timeouts for the DICOM print were setup to ensure that the system would work regardless of whether the DICOM print camera was on a LAN or a WAN halfway around the world. The DICOM print timeouts for the association and DIMSE classes (for example, **N-GET**, **N-DELETE**) can be modified within the DICOM print camera installation. They can be reduced down to 90 seconds.

DICOM PRINT ERROR ON N-GET TIMEOUT CONFUSING

- **Symptom:** When the N-GET timeout goes off, the error message in the prslog file will be "Could not get printer status, invalid command sequence for N-GET".
- **Solution:** When the user sees the above error they may want to consider that the issue may be an inactivity timer on the N-GET DIMSE service.

DICOM PRINT CAMERA SLIDE SUPPORT

- **Symptom:** Current implementation of DICOM print does not allow selection of slide format.
- **Solution:** Feature not currently supported. Possibly in future releases.

CONFUSION ON FILM FORMAT NOTATION

- **Symptom:** GEMS Laser Camera and DICOM Print film format notations are opposite.
- **Solution:**
 - GEMS Laser Camera film format notation has always been row x col (for example, 12 on 1 = 4x3)
 - DICOM Print Standard film format notation is col x row (for example, 12 on 1 = 3x4)

Section 8.0

Teleradiology (Framegrabber Type) Systems & Aux. Monitors

8.1 Introduction

Some Genesis based systems have teleradiology (TR) systems that framegrab the Genesis GFB video (512 x 512 50/60Hz). HiSpeed QX/i DOES NOT directly support this type of TR. The HiSpeed QX/i RGB color display video is a much larger format at a much higher pixel frequency. GE does not promise any direct compatibility with framegrabbing TR systems (DICOM 3.0 TR systems may work depending on the DICOM implementation but GEMS does not and cannot validate all the various TRs.)

In the framegrabber case, a high quality (300Mhz bandwidth) video splitter/amplifier (as listed above) is needed to intercept and re-drive the display CRT RGB video. Composite greyscale would then be available on Green #2 (1280x1024 pixels at 72Hz). Any framegrabber hardware attempting to capture this signal must be capable of a 140Mhz pixel rate. This also involves TR system configuration parameters. The TR capture software may also need upgrading to deal with 1280x1024 and/or **crop** the signal. The TR remote display software may need upgrading to view the larger format. The image transmission times to the remote TR may be up to 4 times as long. GEMS will supply all technical information necessary to assist TR suppliers in making their systems work with HiSpeed QX/i, but GEMS cannot be responsible for this third party TR equipment, software, or compatibility with HiSpeed QX/i.

The following common parts are available from GE Medical Systems:

PART NUMBER	DESCRIPTION
2237018-2	Four-way Video Splitter (BNC Converter)
2154425	BNC to BNC 7' Splitter Cable
2142221	BNC to Octane 7' video Cable
2256482	DB15 to Octane 7' Video Cable
2256485	DB15 to BNC 7' Adapter Cable

Table 12-5 Common Parts Available From GEMS

8.2 Auxiliary Monitors

Figure 12-16 illustrates the **only approved GE CT configuration** for remote monitors, including the part numbers for the only approved components. The use of other configurations and/or components should be strictly avoided.

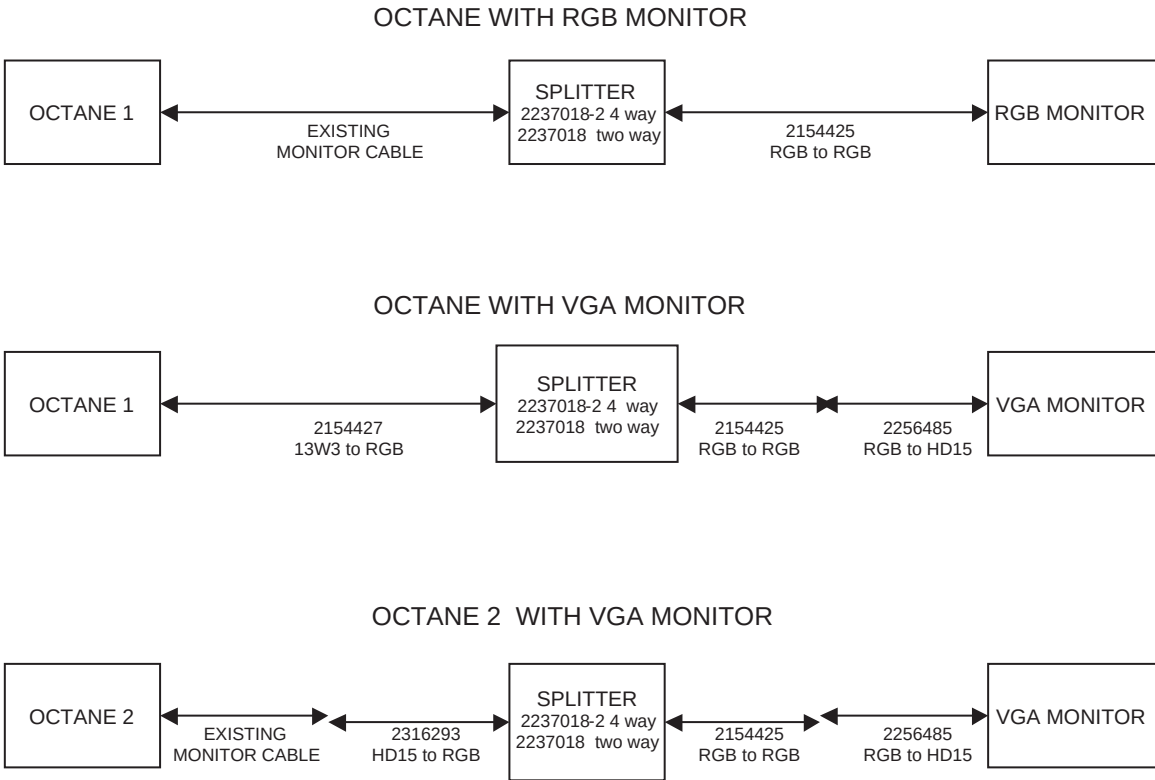


Figure 12-16 Video Splitter / Monitor Cabling

8.3 GEMS Analog Filming Interface Specs (Video & Serial)

Note: The analog (VDB) video output specs and serial interface were specified from, and are the same as, the Genesis interface output specs. This does NOT apply to the digital (LCAM) filming interface.

OUTPUT	VALUE
amplitude	1 volt peak-to-peak
video	0.643V $\pm$ 10%
setup	0.071V $\pm$ 10%
sync	0.286V $\pm$ 10%
DAC resolution	8 bits
diff. linearity	$\pm$ 1 LSB max
glitch area	80 picovolt seconds max, for any step size
rise/fall times	> 10 nsec, $\pm$ 10-90%
FS settling time	7.5 nsec typical to 1 LSB
transfer function	guaranteed monotonic
noise level	> 5.0 millivolt peak-t~peak, combined sync/async noise
DC offset	$\pm$ 1 VDC referenced to ground

Table 12-6 Video Output (Measured Into 75Q at BNC Output)

OUTPUT	VALUE
logic family	F series TTL
output low level	0.8VDC max
output high level	2.0VDC mm
output period	41.336 nsecs $\pm$ 10%
transition times	10 nsec max, $\pm$ 10-90%

Table 12-7 Pixel Clock Output

VIDEO TIMING CHARACTERISTIC	60HZ	50 HZ
pixel frequency	24.192 Mhz	24.192 Mhz
pixel period	41.336 nsec	41.336 nsec
horizontal line frequency	33.6 Khz	33.6 khz
horizontal line width	720 pixels	720 pixels
horizontal active	544 pixels	544 pixels
horizontal blanking	176 pixels	176 pixels
horizontal front porch	26 pixels	26 pixels
horizontal sync	76 pixels	76 pixels
horizontal back porch	74 pixels	74 pixels

Table 12-8 Video Timing Characteristics

VIDEO TIMING CHARACTERISTIC	60HZ	50 HZ
vertical frame frequency	60Hz	50Hz
vertical frame time	560 lines	672 lines
vertical active	524 lines	524 lines
vertical blanking	36 lines	148 lines
vertical sync	3 lines	3 lines
vertical back porch	30 lines	86 lines
vertical front porch	3 lines	59 lines
scanning format	non-interlaced	non-interlaced

Table 12-8 Video Timing Characteristics (Continued)

FORMAT	VALUE
visible field	544 pixels by 524 lines
image field	512 pixels by 512 lines
greyscale field	32 pixels by 16 level grey bar on left side of image
greyscale	software selectable on/off
greyscale off value	0 (black)
initial greyscale	255 (white) at upper left corner
border field	12 lines at bottom of visible field
border field value	any 8bit value, software programmable

Table 12-9 Video Display Format

ATTRIBUTE	VALUE
interface	R5422
25D conn pinout	pin 8 (RX+), pin 21 (RX-), pin 9 (TX+), pin 22 (TX-), pin 7 (GND)
baud rate	1200 baud
word length	8 bit, 1 start bit, 1 stop bit
parity	even
type	asynchronous

Table 12-10 Host Communications/Control Serial Port (analog interface only)

Note: The digital DASM/LCAM serial control has standard R5232 on pins 2, 3, & 7.

Chapter 13

FORM 4879 (Data Record for HiSpeed QX/i)



NOTICE
Potential for
Data Loss and/
or Equipment
Damage

To prevent potential data loss, please do the following:

- Record data collected from procedures into Form F4879 if directed, located in this chapter.
- Only use the Installation manual that arrives with your system for installation. Any other revisions of this manual may not exactly match your system.

Section 1.0

Site History

SYSTEM HISTORY

Where Installed _____

City _____ State _____

Customer's Telephone Number _____ Room No. _____

System I.D. No. _____

Operator's Console (Master Control)

Model Number _____ Serial Number _____

System Gantry

Model Number _____ Serial Number _____

Dates of Tests _____ Performed by _____

Table 13-1 Data Record

TEST EQUIPMENT HISTORY

Test Equipment Used (If Applicable)				
Name	Manufacturer	Model Number	Serial Number	Calib. Due
HV Bleeder				
Oscilloscope				
DMM				

Table 13-2 Test Equipment Used

COMPONENT TRACEABILITY

The Component Locator Installation Report has been completed and sent to the Product Locator File.

Certified Component Changes				
Equipment	Model Number	Serial Number	Install Date	Exp. Ctr. Reading
Old Tube Casing				
Old Tube Insert				
New Tube Casing				
New Tube Insert				
Old Detector				
New Detector				
Old Collimator				
New Collimator				
Old Anode Tank				
New Anode Tank				
Old Cathode Tank				
New Cathode Tank				

Table 13-3 Certified Component Changes

Section 2.0 System Tests

MA METER VERIFICATION - HHS REQUIREMENT

Initials: _____

Circuit ON		
	A/D	Range
Anode mA		15 - 25
Cathode mA		15 - 25
DVM - A/D = Delta Compare Delta to Limits		

Table 13-4 mA Meter Verification (Table 1)

AUTO MA CAL

☐ Check box when complete

The MX200 tube uses a metal/ceramic frame, which means some electrons hit the target and actually bounce off, and are then captured by the frame. This causes a frame current, which results in an anode mA measurement of slightly less than the cathode mA, typically 5 to 10% less. Because the cathode mA is the true indicator of radiation output, the CT QX/i system controls cathode mA rather than anode mA.

OPERATOR INDICATORS - HHS REQUIREMENT

Initials: _____

X-Ray Indicators on Front and Back of Gantry Illuminate During X-Ray

SIGNAL TO NOISE TEST (X-RAY VERIFICATION)

Initials: _____ ☐ Check box when complete

CT NUMBER CHECK

All CT N numbers for all calibrated techniques fall within 0 $\pm$ 1.5 HU. All Values 0 $\pm$ 1.5 HU? Yes: ☐

TUBE Z-AXIS ALIGNMENT (FORMERLY TUBE PLANE OF ROTATION (POR))

☐ Check box when complete

Z-Align Used: Adjusted In Specification. (Go to Detector Z-Axis, step A),

-OR-

Tube Plane of Rotation (POR) using Film. XF - XR = 1.8mm $\pm$ 0.3mm

BEAM ON WINDOW

☐ Check box when complete

- 1.) If you used Z-Align (**Future**) for the Z-Axis Alignment, no additional adjustment is necessary
-OR-

Run BOW on a cold tube. If you have to check BOW after completing the system alignments, let at least three hours elapse from the last scan, to let the tube completely cool.

- The channel should equal $-1.5\text{mm} \pm 0.25\text{mm}$ on initial checks.
- The channel should equal $-1.5\text{mm} \pm 0.50\text{mm}$ on verification checks.
- This program computes the position of the detector in three places: Right, Center and Left

- 2.) Record the displayed position values in the following table:.

Right/Low	Center/Medium	Left/High

Table 13-5: Beam On Window

ISOCENTER

☐ Check box when complete

Average of Large and Small Focal Spots. Spec: 389.75 ± 0.02 channels

CENTER BODY FILTER (CBF)

☐ Check box when complete

Adjust CBF Value. Spec: Target Value ± 0.2 channels

COLLIMATOR APERTURE CHECK - HHS REQUIREMENT

Initials: _____ ☐ Check box when complete

Aperture	Center	Spec - Center only
Small		$3.783 \pm 5\%$
Large		$5.9 \pm 5\%$

Table 13-6 Collimator Aperture Check

X-RAY BEAM LOCATION - HHS REQUIREMENT

Initials: _____

TOMO PLANE INDICATION ACCURACY

Refer to the system service manual, and verify that the relationship of the alignment lights to the x-ray beam meets compliance.

- Internal Lights (1mm)
- External Lights (1mm)
- All alignment lights remain visible with room lights at maximum settings.

IMAGE CALIBRATION

☐ Check box when complete

Detailed Calibrations completed and CT N Number adjustment completed. The Number of Phantom Cals vary with site-specific scanning protocols.

IMAGE QUALITY CHECK - HHS REQUIREMENT

Initials: _____

48CM POLY PHANTOM

48cm/L Scan parameters: 5mm (4i)/ 340mA /1 sec/16 Images

Exam/Series/ Image	AvXc	AvXo	AvXo - AvXc	AvSDo	Comments	Scan Tech Signature/ Date
___/___/ 1, 5, 9, 13						
2, 6, 10, 14						
3, 7, 11, 15						
4, 8, 12, 16						
SPECS	---	---	±8.5	<40.0	---	---

Table 13-7 48cm Poly Phantom

*If your AvSDo fails, check to make sure you did not scan this technique at 1mm.

Box Size: 45 x 45 pixels

Center Coordinates: 256, 256

Outside Coordinates: 256, 60 452, 256 256, 452 60, 256

Image Acceptance/Date: _____

Certified Image Reviewer: _____

ARTIFACT LIMITS:

Rings:	Smudge:
48/L; 30 to 36 counts	48/L; 6.8 counts
1.25mm	1.25mm; 12.0 counts
Band:	Streaks:
8.0 counts	4.0 counts
Band Radius:	Center Artifact:
0 to 23.5cm	N/A
Clump:	Center Artifact:
48/L; 3.0 SIGMA	1.25mm; 14.0 counts
1.25mm; N/A	
Center Spot: N/A	

Table 13-8 Artifact Limits

IMAGE QUALITY CHECK - HHS REQUIREMENT

Initials: _____

20CM QA#1 PHANTOM

QA#1 scan parameters: S/10mm (2i)/120kV/260mA/1 sec/4 images. Use this scan data for Bone Retro. Record data on **Form 4879**.

Exam/Series/ Image	MTF	MTF 2 slice avg.	Contrast Scale	Comments /Artifacts	Scan Tech Signature/ Date
____/____/1		MTF1 + MTF3 ÷ 2 =			
3					
2		MTF2 + MTF4 ÷ 2 =			
4					
SPECS	---	0.65 to 1.0	110.0 to 130.0	---	---

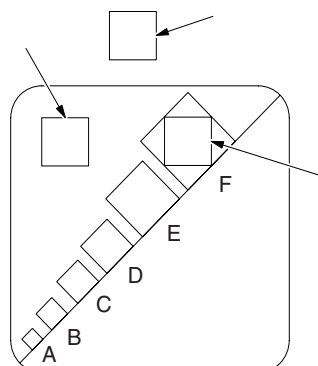
Table 13-9 QA#1 Data (Prospective Images)

QA#1 scan parameters: S/10mm (2i)/120KV/260mA/1 sec/4 images Bone Retro/Display FOV = 15cm Record data on **Form 4879**

Exam/Series/Image	Line Patterns Visible	Comments/Artifacts	Scan Tech Signature/ Date
____/____/1			
2			
3			
4			
SPECS	B, C, D, E, F	---	---

Table 13-10 QA#1 Data (Retrospective Images)

Box Size: 17 x 17 pixels



Contrast Scale = mean of B—mean of C.

SD = SD of ROI Box over F slits.

SD = (SD + SC) / 2

Modulation = _____

MTF = 2.2 (Modulation/Contrast Scale)

IMAGE QUALITY CHECK - HHS REQUIREMENT

Initials: _____

20CM QA#2 PHANTOM

QA#2 scan parameters: S/10mm (2i)/120KV/260mA/1 sec/Peristaltic OFF

Scan from S40.0 to S50.0 - Record results on **Form 4879**

Exam/Series/ Image at location S40.0	Visible Hole View w/ Window 20	Contrast Factor	Comments/ Artifacts	Scan Tech Signature/ Date
____/____/1				
3				
5				
7				
SPECS	See Table, below	2.0 to 12.0	- - -	- - -

Table 13-11 QA#2 Data (Part 1)

Box size: 49 x 9 pixels

Contrast Factor	Visible Number of Holes: Lower Limit*	Visible Number of Holes: Upper Limit*	Smallest Visible Hole: Size
2.00 to 3.99	2	5	7.5mm
4.00 to 7.99	3	5	5.0mm
8.00 to 12.00	4	5	3.0mm

Table 13-12 QA#2 Data (Part 2)

*Required number of visible holes depends on the contrast factor.

2 out of the 4 scans taken must meet this specification.

Image Acceptance/Date: _____

Certified Image Reviewer: _____

FOR REFERENCE ONLY:

Use the manual method for calculating Contrast Factor.

ROI D

ROI E				
10mm	7.5mm	5.0mm	3.0mm	1.5mm

ROI D 49 X 9 pixels x=256, y=200
ROI E 49 X 9 pixels x= 256, y=235
Contrast Factor = Mean E – Mean D

IMAGE QUALITY CHECK - HHS REQUIREMENT

Initials: _____

20CM QA#3 PHANTOM

QA#3 scan parameters: S/10mm 2i/120KV/260mA/1 sec/8 Images

Scan from S55.0 to S65.0

Record data on **Form 4879**.

Exam/Series/ Image	AVXc 4 scan avg.	Std Dev 4 scan avg.	AvXo - AvXc	Comments/ Artifacts	Scan Tech Signature/ Date
____/____/1,3,5,7					
2,4,6,8					
SPECS	0 ±3.0	2.6 to 3.4	±3.0	---	---

Table 13-13 QA#3 Data

Note: The average value of each row in [Table13-13](#) must meet the specifications technique listed at the bottom of the corresponding column.

Box Size: 31 x 31 pixels at Coordinate (256, 256)

Center Coordinates: (256, 256)

Outside Coordinates: (256, 95), (417, 257), (256, 417), (95, 256)

Visually Inspect images for rings and bands

Visual results acceptable?. Yes: ☐

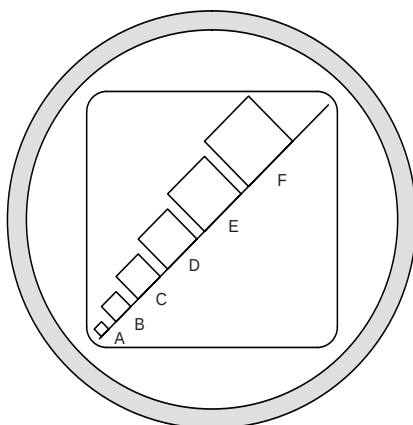
Artifact Limits:

- **Rings:** 4.0/4.8 cts
- **Band:** 2.8 cts
- **Band Radius:** 0-8.5 cm
- **Clump:** N/A
- **Center Spot:** N/A
- **Center Artifact:** 3.5 Std. Dev.
- **Smudge:** 2.2 counts 1mm; 4.0 counts
- **Streaks:** 4.0 cts

Image Acceptance/Date: _____

Certified Image Reviewer: _____

UNUSUAL ALIASING, OTHER ARTIFACTS



Visual inspection for unusual aliasing, and other artifacts. Visual results acceptable? Yes: ☐

Chapter 14

Install Completion Checklist

Complete the following checklist:

Section 1.0 CT System

- ☐ Functional tests completed.
- ☐ Power and ground audit completed.
- ☐ Seismic kit installed (if needed in your area).

Section 2.0 CT Options

- ☐ Injector functional tests completed.
- ☐ AWW functional tests completed.
- ☐ Filming/Camera/DASM functional tests completed.
- ☐ UPS functional tests completed.
- ☐ Network items installed and functional tests completed.
- ☐ Verify that all customer software options are installed and functional.
- ☐ Teleradiology Connections (Refer to [Section 8.0 on page 276](#))

Section 3.0 System Cleanup

- ☐ Clean system with supplies shipped with your system.
- ☐ Touch-up nicks and scratches (as needed).
- ☐ Install pads and stickers (as needed).
- ☐ Store customer materials appropriately.
- ☐ Store service documents, tools, and materials appropriately.

Section 4.0 Dollies Return

- ☐ Confirm that dollies have been picked up. If not, complete the necessary documentation.

Section 5.0

Install Completion Material Checks

- ☐ Complete all HHS documentation. Complete Install CQAs as needed.
- ☐ Complete all dispatching activities (03-04-10 codes).
- ☐ Complete customer acceptance checks/test.
- ☐ Complete system transfer and notify appropriate GE personnel.
- ☐ Communicate with the customer and address installation issues.

Appendix C

Additional Characterization Procedures

Section 1.0 Tilt Characterization

Tools Required: Hex (metric) set
Flat blade screw driver



1.) If you are not on the Service Desktop, click on the SERVICE DESKTOP icon.



2.) Click on the UTILITIES icon.

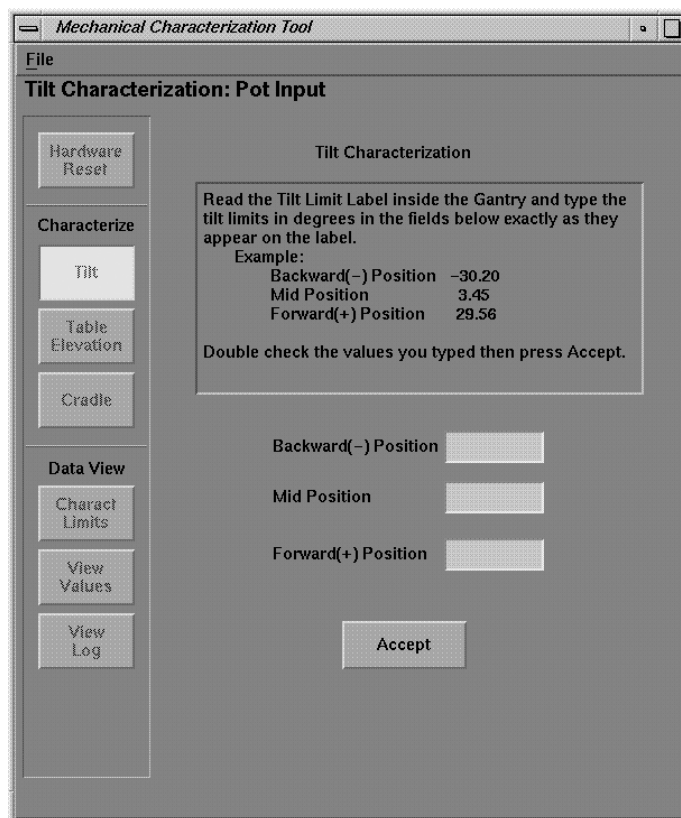
3.) Select INSTALL.

4.) Select TABLE GANTRY INTEGRATION.

5.) Select CHARACTERIZATION.

6.) Select TILT.

7.) Follow the Tilt Characterization instructions on the screen.



The screenshot shows the 'Mechanical Characterization Tool' window. The title bar reads 'Mechanical Characterization Tool'. The menu bar has 'File'. The main window is titled 'Tilt Characterization: Pot Input'. On the left, there is a sidebar with buttons: 'Hardware Reset', 'Characterize' (with a sub-menu containing 'Tilt', 'Table Elevation', and 'Cradle'), 'Data View' (with a sub-menu containing 'Charact Limits', 'View Values', and 'View Log'). The main area is titled 'Tilt Characterization' and contains the following text: 'Read the Tilt Limit Label inside the Gantry and type the tilt limits in degrees in the fields below exactly as they appear on the label. Example: Backward(-) Position -30.20, Mid Position 3.45, Forward(+) Position 29.56. Double check the values you typed then press Accept.' Below this text are three input fields: 'Backward(-) Position', 'Mid Position', and 'Forward(+) Position'. At the bottom center is an 'Accept' button.

Figure C-1 Gantry Tilt Characterization Screen

Section 2.0

Elevation Characterization

Tools Required: None

2.1 Prerequisite

The table to gantry MUST be level before performing this procedure. The following tools are required:

- Metric tape measure
- Flat blade screwdriver
- Metric hex key set
- 9" level

2.2 Procedure

- 1.) Remove upper right and lower right table covers.
- 2.) Locate the small L-shaped measurement block, stored on top of the bigger measurement block.
 - a.) Remove the small L-shaped measurement block from its storage position.
 - b.) Install it in its characterization position, at the right rear of the table.
 - c.) Move the measurement plate, on the right side of table, to the **out** position.
 - d.) Tighten the mounting screws for both plates.

- 1.) If you are not on the Service Desktop, click on the SERVICE DESKTOP icon.



- 2.) Click on the UTILITIES icon.



- 3.) Select INSTALL.
- 4.) Select TABLE GANTRY INTEGRATION.
- 5.) Select CHARACTERIZATION.
- 6.) Select TABLE ELEVATION.

- 7.) Follow the Elevation Characterization instructions displayed on the screen. *Important - Take care to measure the exact elevation distances. Incorrect measurement results in Elevation Characterization failure.* Table Low = 374.1 mm. Table High = 849.1 mm.

Refer to [Figure C-2](#). Measure the elevation distance as vertically as possible. Measure between the bottom surface of the each measurement block and the center of the rear leg upper pivot pin.

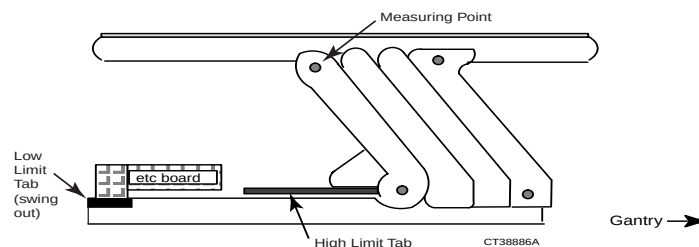


Figure C-2 Measure Elevation Distance

- 8.) Return the measurement plates to the storage position.

Section 3.0

Cradle Characterization

Tools Required: Hex (metric) set
Metric tape measure



1.) If you are not on the Service Desktop, click on the SERVICE DESKTOP icon.



2.) Click on the UTILITIES icon.

3.) Select INSTALL.

4.) Select TABLE GANTRY INTEGRATION.

5.) Select CHARACTERIZATION.

6.) Select CRADLE.

7.) Follow the cradle characterization instructions displayed on the screen.

Note:
A table/gantry
reset may be
required.

If the display flashes "888", then you must reset the table before proceeding. To reset the table, lower it to its lowest position and then raise it to its maximum position.

a.) In the alignment light procedure (3rd and 4th screens), use the BLACK DOT on the cradle right-side bumper.

Elevate the table to at least 40mm from isocenter, so the black dot reaches the external scan plane alignment light *before* the cradle reaches its home position.

b.) In the end-of-cradle position procedure (5th screen), use the same BLACK DOT on the cradle right-side bumper.

Note: Do not use buttons to drive table cradle.

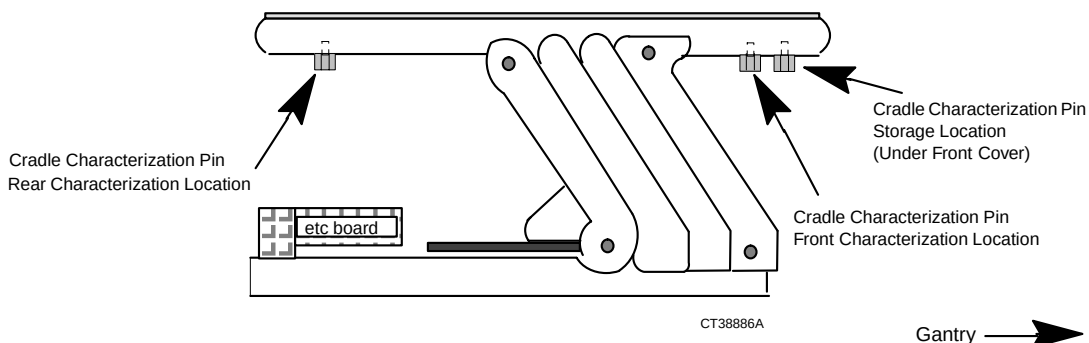


Figure C-3 Cradle Characterization Pin Locations



GE MEDICAL SYSTEMS

**GE MEDICAL SYSTEMS-AMERICAS: FAX 414.544.3384
P.O. BOX 414; MILWAUKEE, WISCONSIN 53201-0414, U.S.A.**

**GE MEDICAL SYSTEMS-EUROPE: FAX 33.1.40.93.33.33
PARIS, FRANCE**

GE MEDICAL SYSTEMS-ASIA: FAX 65.291.7006